Physics

Q. 1 The dimensional formula of $\frac{1}{2} \in_0 \mathrm{E}^2$ (ϵ_0 = permittivity of vacuum and E = electric field) is $\mathrm{M}^{\mathrm{a}}\mathrm{L}^{\mathrm{b}}\mathrm{T}^{\mathrm{c}}$. The value of $2a - b + c =$ _____

Option 1:
0

Option 2:
1

Option 3:
-1

Option 4:
2

Correct Answer:
1

Solution:

$$\frac{1}{2} \in_0 \mathrm{E}^2 = \text{ Energy density}$$

$$\therefore \left[\frac{1}{2} \in_0 \mathrm{E}^2\right] = \frac{\mathrm{ML}^2\,\mathrm{T}^{-2}}{\mathrm{L}^3} = \mathrm{ML}^{-1}\,\mathrm{T}^{-2}$$

$$\therefore \mathrm{a} = 1, \quad \mathrm{b} = -1, \quad \mathrm{c} = -2$$

$$\therefore 2\mathrm{a} - \mathrm{b} + \mathrm{c} = 2 + 1 - 2 = 1$$

Q. 2 The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm . The length measured by a scale of least count 0.1 cm is 150 cm . When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm , measured by a micrometer of least count 0.001 cm . The error in the measured Young's modulus is $\alpha \times 10^9\ \mathrm{N/m^2}$. The value of α is _____ .
(Ignore the contribution of the load to Young's modulus error calculation)

Option 1:
1.3

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Option 2:
1.65

Option 3:
0.13

Option 4:
0.25

Correct Answer:
1.65

Solution:

$$\mathrm{Y}=\frac{\mathrm{F}\ell}{\mathrm{A}\Delta\ell}$$

$$\therefore \frac{\Delta \mathrm{Y}}{\mathrm{Y}}=\frac{\Delta \mathrm{F}}{\mathrm{F}}+\frac{\Delta \ell}{\ell}+2\frac{\Delta\, \mathrm{d}}{\mathrm{d}}+\frac{\Delta(\Delta \ell)}{\Delta \ell}$$

$$\left[\mathrm{A}=\pi\frac{\mathrm{d}^2}{4} \Rightarrow \frac{\Delta\, \mathrm{A}}{\mathrm{A}}=\frac{2\Delta\, \mathrm{d}}{\mathrm{d}}\right]$$

Since F is known exactly $\Rightarrow \Delta F=0$

$$\frac{\Delta Y}{Y}=\frac{0.1}{150}+\frac{2\times 0.001}{0.08}+\frac{0.001}{0.5}=0.0277$$

$$\therefore \Delta Y=0.0277\mathrm{Y}$$

$$Y=\frac{100\times 1.5}{\frac{\pi}{4}\left(\frac{0.08}{100}\right)^2\times 0.5\times 10^{-2}}=5.97\times 10^{10}\ \mathrm{N/m^2}$$

$$\therefore \Delta Y=0.0277\times 5.97\times 10^{10}\ \mathrm{N/m^2}=1.65\times 10^{9}\ \mathrm{N/m^2}$$

Q. 3 The velocity of a particle is given as $\vec{v}=-x\hat{i}+2y\hat{j}-z\hat{k}\ \mathrm{m/s}$. The magnitude of acceleration at point ($1, 2, 4$) is _____ $\mathrm{m/s^2}$.

Option 1:
$\sqrt{6}$

Option 2:
9

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Option 3:
$\sqrt{33}$

Option 4:
0

Correct Answer:
9

Solution:
$\vec{v} = -x\hat{i} + 2y\hat{j} - z\hat{k}$
$\vec{a} = \frac{d\vec{v}}{dt} = -\frac{dx}{dt}\hat{i} + 2\frac{dy}{dt}\hat{j} - \frac{dz}{dt}\hat{k}$
$= -(-x)\hat{i} + 2(2y)\hat{j} - (-z)\hat{k} = x\hat{i} + 4y\hat{j} + z\hat{k}$
$\therefore \vec{a} \text{ at } (1, 2, 4) = \hat{i} + 8\hat{j} + 4\hat{k}$
$\therefore |\vec{a}| = 9 \text{ m/s}^2$

Q. 4 The position of an object having mass 0.1 kg as a function of time t is given as $\vec{r} = \left(10t^2\hat{i} + 5t^3\hat{j}\right)\text{m}$. At $\text{t} = 1$ s, which of the following statements are correct?

A. The linear momentum $\overrightarrow{\text{p}} = (2\hat{\text{i}} + 1.5\hat{\text{j}})\text{kg} \cdot \text{m/s}$.
B. The force acting on the object $\vec{F} = (2\hat{i} + 3\hat{j})N$.
C. The angular momentum of the object about its origin $\overrightarrow{\text{L}} = 15\hat{\text{k}}\text{Js}$.
D. The torque acting on the object about its origin $\vec{\tau} = 20\hat{\text{k}}\text{Nm}$.

Choose the correct answer from the option given below.

Option 1:
A, B and C only

Option 2:
B, C and D only

Option 3:
A, C and D only

Option 4:
A, B and D only

Correct Answer:
A, B and D only

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Solution:

Given $\vec{r} = 10t^2\hat{i} + 5t^3\hat{j} \Rightarrow \vec{r}_{t=1} = 10\hat{i} + 5\hat{j}$

$\vec{v} = \frac{d\vec{r}}{dt} = 20t\hat{i} + 15t^2\hat{j} \Rightarrow \vec{v}_{t=1sec} = 20\hat{i} + 15\hat{j}$

$\vec{a} = \frac{d^2\vec{r}}{dt^2} = 20\hat{i} + 30t\hat{j} \Rightarrow \vec{a}_{t=1sec} = 20\hat{i} + 30\hat{j}$

$\therefore \vec{P} = m\vec{v} = 2\hat{i} + 1.5\hat{j}$

$\vec{F} = m\vec{a} = 2\hat{i} + 3\hat{j}$

$\vec{\tau} = \vec{r} \times \vec{F} = 20\hat{k}$

$\vec{L} = \vec{r} \times \vec{P} = 5\hat{k}$

Q. 5 A planet (P_1) is moving around the star of mass 2 M in the orbit of radius R . Another planet (P_2) is moving around another star of mass 4 M in a orbit of radius 2R. Ratio of time periods of revolution of P_2 and P_1 is ____ .

Option 1:

$\frac{1}{2}$

Option 2:

2

Option 3:

4

Option 4:

$\frac{1}{4}$

Correct Answer:

2

Solution:

From Kepler's law,

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$$T^2 = \frac{4\pi^2}{GM_S} r^3$$

$$\therefore\ T_{P_1} = \frac{2\pi}{\sqrt{G(2M)}} R^{3/2}$$

$$T_{P_2} = \frac{2\pi}{\sqrt{G(4M)}} (2R)^{3/2}$$

$$\therefore\ \frac{T_{P_2}}{T_{P_1}} = \frac{(2)^{3/2}}{\sqrt{4}} \times \frac{\sqrt{2}}{1} = \frac{2\sqrt{2} \times \sqrt{2}}{2} = 2$$

Q. 6 A particle is rotating in a circular path and at any instant its motion can be described as $\theta = \frac{5t^4}{40} - \frac{t^3}{3}$. The angular acceleration of the particle after 10 seconds is _____ $\mathrm{rad/s^2}$.

Option 1:
150

Option 2:
120

Option 3:
130

Option 4:
170

Correct Answer:
130

Solution:

$$\theta = \frac{5t^4}{40} - \frac{t^3}{3}$$

$$\frac{d\theta}{dt} = \frac{5}{40}(4t^3) - \frac{3t^2}{3} = \frac{t^3}{2} - t^2$$

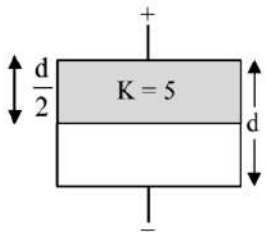
$$\alpha = \frac{d^2\theta}{dt^2} = \frac{3t^2}{2} - 2t$$

at $t = 10$

$$\alpha = \tfrac{3}{2}(10)^2 - 2 \times 10 = 130 \mathrm{rad/sec^2}$$

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Q. 7 A parallel plate air capacitor has a capacitance C . When it is half filled as show in figure with a dielectric constant $\mathrm{K}=5$, the percentage increase in the capacitance is _____ .



Option 1:
33.34

Option 2:
66.67

Option 3:
200

Option 4:
400

Correct Answer:
66.67

Solution:
Initial capacitance

$\mathrm{C_i}=\frac{\mathrm{A}\in_0}{\mathrm{d}}=\mathrm{C}$

Final capacitance

$C_1=\frac{A\in_0\times 5}{\frac{d}{2}}=10C$

$C_2=\frac{A\in_0}{\frac{d}{2}}=2C$

$C_f=\frac{C_1C_2}{C_1+C_2}=\frac{10\times 2}{10+2}C=\frac{5}{3}C$

$\%$ increase $=\frac{\frac{5}{3}C-C}{C}\times 100=66.67$

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Q. 8 Heat is supplied to a diatomic gas at constant pressure. Then the ratio of $\Delta Q : \Delta U : \Delta W$ is _____ .

Option 1:
$2 : 3 : 5$

Option 2:
$5 : 3 : 2$

Option 3:
$2 : 5 : 7$

Option 4:
$7 : 5 : 2$

Correct Answer:
$7 : 5 : 2$

Solution:

$$\Delta \mathrm{Q_P} = \mathrm{nC_P}\Delta \mathrm{T} = \frac{7}{2}\mathrm{nR}\Delta \mathrm{T}$$

$$\Delta \mathrm{U} = \mathrm{nC_v}\Delta \mathrm{T} = \frac{5}{2}\mathrm{nR}\Delta \mathrm{T}$$

$$\mathrm{W} = \mathrm{nR}\Delta \mathrm{T}$$

$$\Delta \mathrm{Q} : \Delta \mathrm{U} : \mathrm{W} :: 7 : 5 : 2$$

Q. 9 Two charged conducting spheres S_1 and S_2 of radii 8 cm and 18 cm are connected to each other by a wire. After equilibrium is established, the ratio of electric fields on $\mathrm{S_1}$ and $\mathrm{S_2}$ spheres are $\mathrm{E_{S1}}$ and $\mathrm{E_{S2}}$ respectively. The value of $\frac{\mathrm{E_{S1}}}{\mathrm{E_{S2}}}$ is _____ .

Option 1:
$\frac{3}{2}$

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Option 2:
$\frac{2}{3}$

Option 3:
$\frac{4}{9}$

Option 4:
$\frac{9}{4}$

Correct Answer:
$\frac{9}{4}$

Solution:



$$V_1 = V_2$$

$$\frac{kQ_1}{8} = \frac{kQ_2}{18} \left\{ v_{surface} = \frac{kQ}{R} \right\}$$

$$\frac{Q_1}{Q_2} = \frac{4}{9}$$

$$\frac{E_{S_1}}{E_{S_2}} = \frac{kQ_1}{r_1^2} \frac{r_2^2}{kQ_2} \left\{ E_{surface} = \frac{kQ}{R^2} \right\}$$

$$= \frac{Q_1}{Q_2} \left(\frac{r_2}{r_1} \right)^2 = \frac{4}{9} \times \left(\frac{9}{4} \right)^2 = \frac{9}{4}$$

Q. 10 The equation of a plane progressive wave is given by $y = 5 \cos \pi \left(200t - \frac{x}{150} \right)$ where x and y are in cm and t is in second. The velocity of the wave is _____ m/s

Option 1:
120

Option 2:
150

Option 3:
200

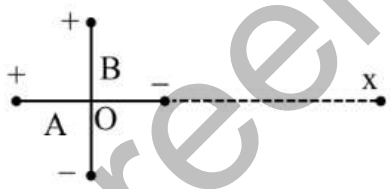
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Option 4:
300

Correct Answer:
300

Solution:
$v = \frac{\text{coeff of } t}{\text{coeff of } x} = \frac{200}{\frac{1}{150}\times 100} = 300 \text{ m/s}$

Q. 11 Two short electric dipoles A and B having dipole moment p_1 and p_2 respectively are placed with their axis mutually perpendicular as shown in the figure. The resultant electric field at a point x is making an angle of 60° with the line joining points O and x . The ratio of the dipole moments p_2/p_1 is ____ .



Option 1:
$\frac{\sqrt{3}}{2}$

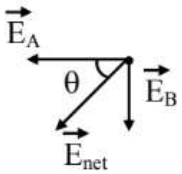
Option 2:
$2\sqrt{3}$

Option 3:
$\frac{1}{\sqrt{3}}$

Option 4:
$\sqrt{3}$

Correct Answer:
$2\sqrt{3}$

Solution:



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$$\overrightarrow{E}_A = \frac{2kP_1}{x^3}(-\hat{i})\{\text{ Axial point }\}$$

$$\overrightarrow{E}_B = \frac{kP_2}{x^3}(-\hat{j})\{\text{ Equatorial point }\}$$

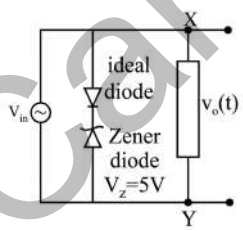
$$\text{Now : } \tan\theta = \frac{\left|\overrightarrow{E}_B\right|}{\left|\overrightarrow{E}_A\right|}$$

$$\Rightarrow \tan 60^\circ = \frac{P_2}{2P_1}$$

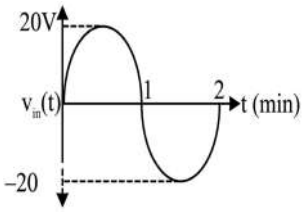
$$\Rightarrow \frac{P_2}{P_1} = 2\sqrt{3}$$

Q. 12 For the given circuit (shown in part (A)) the time dependent input voltage $v_{in}(t)$ and corresponding output $v_o(t)$ are shown in part (B) and part (C), respectively. Identify the components that are used in the circuit between points X and Y .

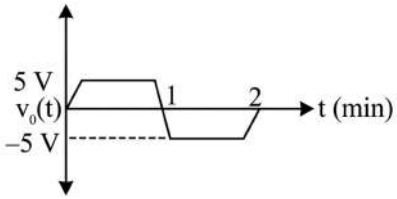
(A) Circuit used



Part(B) : Input voltage



Part(C) : Output voltage



Option 1:



Option 2:



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Option 3:



Option 4:



Correct Answer:



Solution:

For the +ve half cycle of input

Middle branch has ideal diode in forward bias and Zener diode in reverse bias Thus, we get constant 5V output due to Zener breakdown. For this situation component between $x\&y$ should act as open circuit.

For -ve half cycle of input middle branch has ideal diode in reverse bias, which makes it open branch without any current. for this situation component between $\mathrm{x\&y}$ should give a constant -5 V output due to Zener breakdown.

(Note : The input side must contain some resistance else V_{in} is equal to V_{out} for every cases)

Q. 13 When a coil is placed in a time dependent magnetic field the power dissipated in it is P . The number of turns, area of the coil and radius of the coil wire are N , A and r respectively. For a second coils number of turns, area of the coil and radius of the coil wire are $2\ \mathrm{N}, 2\ \mathrm{A}$ and 3 r respectively. When the first coil is replaced with second coil the power dissipated in it is $\sqrt{2}\alpha \mathrm{P}$. The value of α is _____ .

Option 1:

36

Option 2:

$128\sqrt{2}$

(

Option 3:

16

Option 4:

64

Correct Answer:

36

Solution:

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Power dissipated : $\mathrm{P}=\frac{\varepsilon^2}{\mathrm{R}}$
Induced emf $\varepsilon=-\mathrm{NA}\frac{\mathrm{dB}}{\mathrm{dt}}$
Resistance of coil : $\mathrm{R}=\rho\frac{\ell}{\pi \mathrm{r}^2}$
Now : $\ell=\mathrm{N}\left(2\pi \mathrm{R}_{\text{coil}}\right)$
Also : $A=\pi R_{\text{coil}}^2 \Rightarrow R_{\text{coil}}=\sqrt{\frac{A}{\pi}}$

Thus ; $\mathrm{R}=\frac{\rho \mathrm{N}(2\pi)}{\pi \mathrm{r}^2}\sqrt{\frac{\mathrm{A}}{\pi}}$

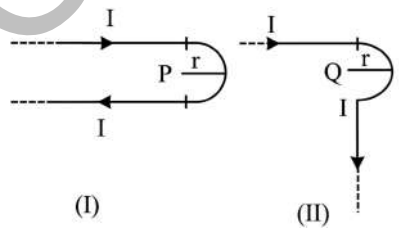
$$\mathrm{P}=\frac{\mathrm{N}^2\,\mathrm{A}^2\left(\frac{\mathrm{dB}}{\mathrm{dt}}\right)^2}{\frac{2\rho\,\mathrm{N}}{\mathrm{r}^2}\sqrt{\frac{\mathrm{A}}{\pi}}} \Rightarrow \mathrm{P}\propto \mathrm{NA}^{3/2}\mathrm{r}^2 \ldots\ldots\ldots(1)$$

$$\mathrm{P}'\propto(2\,\mathrm{N})(2\,\mathrm{A})^{3/2}(3\mathrm{r})^2 \ldots\ldots(2)$$

$$\frac{(2)}{(1)}:\frac{\mathrm{P}'}{\mathrm{P}}=36\sqrt{2}\Rightarrow \mathrm{P}'=(\sqrt{2})(36)\mathrm{P}$$

Thus : $\alpha=36$

Q. 14 Two identical long current carrying wires are bent into the shapes shown in the following figures. If the magnitude of magnetic fields at the centres P and Q of a semicircular arc are B_1 and B_2 respectively, then the ratio $\frac{B_1}{B_2}$ is _____ .



Option 1:
$\frac{2+\pi}{1+\pi}$

Option 2:
$\frac{1+\pi}{1-\pi}$

Option 3:
$\frac{2+\pi}{1-\pi}$

Option 4:
$\frac{1+\pi}{2-\pi}$

Correct Answer:
$\frac{2+\pi}{1+\pi}$

Solution:

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$$B_1 = \frac{\mu_0 I}{4\pi r} \times 2 + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4r}\left(\frac{2}{\pi} + 1\right)$$

$$B_2 = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4r}\left(\frac{1}{\pi} + 1\right)$$

$$\frac{B_1}{B_2} = \frac{2+\pi}{1+\pi}$$

Q. 15 For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _____

Option 1:
$3 : 4$

Option 2:
$3 : 2$

Option 3:
$2 : 1$

Option 4:
$1 : 2$.

Correct Answer:
$3 : 2$

Solution:
We have for thin prism minimum deviation

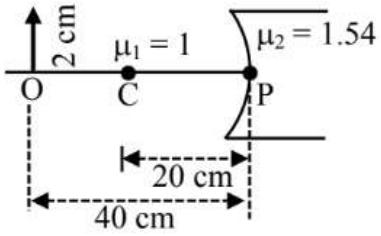
$$\delta_{\mathrm{m}} = (\mu - 1)\mathrm{A}$$

$$\Rightarrow \delta_{\mathrm{m}} = (1.5 - 1)\mathrm{A} \quad \Rightarrow \delta_{\mathrm{m}} = \frac{\mathrm{A}}{2}$$

Also, $i = \frac{A+\delta_m}{2} \Rightarrow i = \frac{A+\frac{A}{2}}{2} \Rightarrow i = \frac{3A}{4}$
Thus $\frac{\mathrm{i}}{\delta_{\mathrm{m}}} = \frac{3}{2}$

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Q. 16 Refer the figure given below. μ_1 and μ_2 are refractive indices of air and lens material. The height of image will be _____ cm .



Option 1:
1

Option 2:
0.5

Option 3:
1.2

Option 4:
0.25

Correct Answer:
1

Solution:

$$\frac{\mu_2}{\mathrm{v}} - \frac{\mu_1}{\mu} = \frac{\mu_2 - \mu_1}{\mathrm{R}}$$

$$\frac{1.54}{\mathrm{v}} - \frac{1}{(-40)} = \frac{1.54 - 1}{-20}$$

$$\frac{1.54}{\mathrm{v}} = -\frac{1}{40} - \frac{0.54}{20}$$

$$\frac{1.54}{\mathrm{v}} = \frac{-2.08}{40}$$

$$\mathrm{v} = -29.61 \text{ cm}$$

$$\frac{\mathrm{h_i}}{\mathrm{h_0}} = \frac{\mu_1 \mathrm{v}}{\mu_2 \mathrm{u}} = \frac{1(-29.61)}{1.54(-40)} = 0.48$$

$$\mathrm{h_i} = 0.48 \times 2 = 0.96 \approx 1 \text{ cm}$$

Q. 17 For a certain metal, when monochromatic light of wavelength λ is incident, the stopping potential for photoelectrons is $3\ \mathrm{V_0}$. When the same metal is illuminated by light of wavelength 2λ, then the stopping potential becomes $\mathrm{V_0}$. The threshold wavelength for photoelectric emission for the given metal is $\alpha\lambda$.The value of α is

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Option 1:
1

Option 2:
4

Option 3:
2

Option 4:
3

Correct Answer:
4

Solution:

$\frac{hc}{\lambda} - \phi = 3eV_0 \ldots\ldots\ldots$ (i)

$\frac{hc}{2\lambda} - \phi = eV_0 \ldots\ldots$.(ii)

$$\frac{hc}{\lambda} - \phi = 3\left[\frac{hc}{2\lambda} - \phi\right]$$

$$\phi = \frac{hc}{4\lambda}$$

$$\frac{hc}{\lambda_0} = \frac{hc}{4\lambda}$$

$$\lambda_0 = 4\lambda$$

$$\alpha = 4$$

Q. 18 An electromagnetic wave travelling in x -direction is described by field equation $E_y = 300 \sin\omega\left(t - \frac{x}{c}\right)$. If the electron is restricted to move in y -direction only with speed of $1.5 \times 10^6\ m/s$ then ratio of maximum electric and magnetic forces acting on the electron is _____ .

Option 1:
200

Option 2:
150

Option 3:
400

Option 4:
300

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Correct Answer:
200

Solution:

$E_y = 300 \sin \omega \left(t - \frac{x}{v}\right)$

$F_M = qvB \ldots\ldots \text{(i)}$

$F_e = qE \ldots\ldots\ldots \text{(ii)}$

$$\frac{F_e}{F_m} = \frac{qE}{qvB} = \frac{E}{vB} = \frac{c}{v} = \frac{3 \times 10^8}{1.5 \times 10^6} = 200$$

Q. 19 Angular momentum of an electron in a hydrogen atom is $\frac{3\,\mathrm{h}}{\pi}$, then the energy of the electron is _____ eV .

Option 1:
-1.51

Option 2:
-0.85

Option 3:
-0.38

Option 4:
-0.28

Correct Answer:
-0.38

Solution:

$$\mathrm{L} = \frac{\mathrm{nh}}{2\pi}$$

$$\frac{3\,\mathrm{h}}{\pi} = \frac{\mathrm{nh}}{2\pi}$$

$$\mathrm{n} = 6$$

$$\mathrm{E} = \frac{-13.6}{\mathrm{n}^2}\mathrm{eV}$$

$$= \frac{-13.6}{6^2} \approx -0.38\mathrm{eV}$$

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Q. 20 A liquid drop of diameter 2 mm breaks into 512 droplets. The change in surface energy is $\alpha \times 10^{-6}$ J. The value of α is _____ . (Take surface tension of liquid $= 0.08\ \mathrm{N/m}$)

Option 1:
10

Option 2:
7

Option 3:
8

Option 4:
11

Correct Answer:
7

Solution:

$\mathrm{V_1 = V_2}$

$\frac{4}{3}\pi \mathrm{R^3} = 512\left(\frac{4}{3}\pi \mathrm{r^3}\right)$

$\mathrm{R} = (512)^{1/3}\mathrm{r}$

$\mathrm{l} = 8\mathrm{r}$

$\mathrm{r} = \frac{1}{8}\ \mathrm{mm}$

$\Delta \mathrm{SE} = \mathrm{SE_2} - \mathrm{SE_1}$

$= 512\left[\ \mathrm{T} \times 4\pi \mathrm{r^2}\right] - \left[\mathrm{T} \times 4\pi \mathrm{R^2}\right]$

$= 4\pi\ \mathrm{T}\left[512\mathrm{r^2} - \mathrm{R^2}\right]$

$= 4\pi(0.08)\left[512\left(\frac{1}{8}\right)^2 - 1^2\right] \times 10^{-6} \approx 7 \times 10^{-6}$

$\alpha = 7$

Q. 21 In single slit diffraction pattern, the wavelength of light used is 628 nm and slit width is 0.2 mm , the angular width of central maximum is $\alpha \times 10^{-2}$ degrees. The value of α is _____

Correct Answer:
36

Solution:

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Angular width of central maxima $\theta_{\mathrm{cm}} = \frac{2\lambda}{\mathrm{d}}$
$= \frac{2\times628\times10^{-9}}{0.2\times10^{-3}}\mathrm{rad.}$
$= \left(628\times10^{-5}\right)\times\frac{180}{\pi}$ degrees
$\approx 36\times10^{-2}$ degrees
$\therefore$ Answer : $\alpha = 36$

Q. 22 A vessel contains $0.15\ \mathrm{m}^3$ of a gas at pressure 8 bar and temperature $140^{\circ}\mathrm{C}$ with $\mathrm{c_p} = 3\mathrm{R}$ and $\mathrm{c_v} = 2\mathrm{R}$. It is expanded adiabatically till pressure falls to 1 bar. The work done during this process is _____ k J . (R is gas constant)

Correct Answer:
-

Solution:
$\mathrm{r} = \dfrac{\mathrm{C_p}}{\mathrm{C_v}} = \dfrac{3}{2}$
$\because \mathrm{P_1\ V_1^{\gamma} = P_2\ V_2^{\gamma}}$
$\Rightarrow \mathrm{V_2} = \left(\dfrac{\mathrm{P_1}}{\mathrm{P_2}}\right)^{1/\gamma}\mathrm{V_1}$
$\Rightarrow\ \mathrm{V_2} = \left(\dfrac{8}{1}\right)^{2/3}\times 0.15$
$\Rightarrow\ \mathrm{V_2} = 0.6\ \mathrm{m}^3$
$\because\ \mathrm{W} = \dfrac{\mathrm{P_1\ V_1 - P_2\ V_2}}{\gamma - 1}$
$= \left(\dfrac{8\times0.15 - 1\times0.6}{1.5-1}\right)\times10^5 = 1.2\times10^5$
$\Rightarrow\ \mathrm{W} = 120\ \mathrm{kJ}$

Q. 23 $1\mu\mathrm{C}$ charge moving with velocity $\overrightarrow{\mathrm{v}} = (\hat{\mathrm{i}} - 2\hat{\mathrm{j}} + 3\hat{\mathrm{k}})\mathrm{m/s}$ in the region of magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 5\hat{k})T$. The magnitude of force acting on it is $\sqrt{\alpha}\times10^{-6}\ \mathrm{N}$. The value of α is _____ .

Correct Answer:
171

Solution:

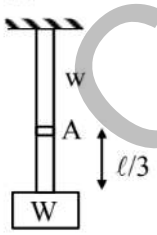
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$$\vec{F} = q(\vec{v} \times \vec{B})$$
$$= 1 \times 10^{-6} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & 3 & -5 \end{vmatrix}$$
$$= 10^{-6}[\hat{i} + 11\hat{j} + 7\hat{k}]$$
$$\Rightarrow \vec{F} = \sqrt{171} \times 10^{-6}\ \text{N}$$
$$\therefore \alpha = 171$$

Q. 24 A uniform wire of length l of weight w is suspended from the roof with a weight of W at the other end. The stress in the wire at $\frac{l}{3}$ distance from the top is $\left(\frac{\text{W}}{\text{A}} + \frac{2}{\gamma}\frac{\text{w}}{\text{A}}\right)$, where, A is the cross sectional area of the wire. The value of γ is _____ .

Correct Answer:
6

Solution:



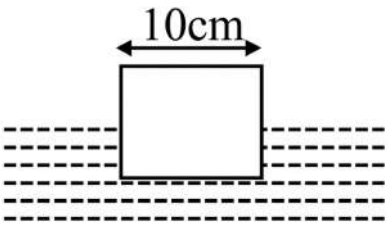
$$\sigma = \frac{T}{A} = \frac{W + \frac{W}{3}}{A}$$
$$\Rightarrow \sigma = \frac{W}{A} + \frac{2}{6}\frac{W}{A}$$
$$\therefore \gamma = 6$$

Q. 25 A tube is filled with water and a wooden cube $10\ \text{cm} \times 10\ \text{cm} \times 10\ \text{cm}$ is placed in the water. The wooden cube is found to float on the water with a part of it submerged in water. When a metal coin is placed on the wooden cube, the submerged part is increased by 3.87 cm . The mass of the metal coin is _____ gram.
(Take water density as $1\ \text{g/cm}^3$ and density of wood as $0.4\ \text{g/cm}^3$)

Correct Answer:
387

Solution:

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$\text{Weight of coin } = \text{ Additional upthrust}$
$\Rightarrow \mathrm{m_c\ g} = \rho_\ell \mathrm{V_{add}\ g}$
$\Rightarrow\ \mathrm{m_c} = 1 \times 3.87 \times (10)^2$
$\Rightarrow\ \mathrm{m_c} = 387\ \text{gram}$

Chemistry

Q. 1 The mass of iron converted into $\mathrm{Fe_3O_4}$ by the action of 18 g of steam is :
(Given : Molar mass of $\mathrm{H, O}$ and Fe are 1,16 and $56\ \mathrm{g\ mol^{-1}}$ respectively)
Assume iron is present in excess :

Option 1:
2.1 g

Option 2:
4.2 g

Option 3:
21 g

Option 4:
42 g

Correct Answer:
42 g

Solution:
$3\mathrm{Fe(\ s)} + 4\mathrm{H_2O(\ g)} \rightarrow \mathrm{Fe_3O_4(\ s)} + 4\mathrm{H_2(\ g)}$

$\mathrm{n_{steam}} = \dfrac{18}{18} = 1$

$\mathrm{n_{Fe}\ required} = \dfrac{3}{4}$

$\text{mass of Fe required} = \dfrac{3}{4} \times 56 = 42gm$

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Q. 2 What is the energy (in J atom $^{-1}$) required for the following process?

$\mathrm{Li}^{2+}(\mathrm{g}) \rightarrow \mathrm{Li}^{3+}(\mathrm{g}) + \mathrm{e}^{-}$

(Take the ionization energy for the H atom in the ground state as $2.18 \times 10^{-18}\ \mathrm{J}^{-1}$ atom $^{-1}$)

Option 1:
8.72×10^{-18}

Option 2:
1.962×10^{-18}

Option 3:
1.962×10^{-17}

Option 4:
6.54×10^{-17}

Correct Answer:
1.962×10^{-17}

Solution:
$\mathrm{E}_{\mathrm{Li}^{2+}} = \mathrm{E}_{\mathrm{H}} \times \mathrm{z}^2$

$= 2.18 \times 10^{-18} \times (3)^2$
$= 1.962 \times 10^{-17}\ \mathrm{Jatom}^{-1}$

Q. 3 Given below are two statements :
Statement (I) : The correct sequence of bond lengths in the following species is :

$\mathrm{O}_2^{+} < \mathrm{O}_2 < \mathrm{O}_2^{-} < \mathrm{O}_2^{2-}$

Statement (II) : The correct sequence of number of unpaired electrons in the following species is :

$\mathrm{O}_2 > \mathrm{O}_2^{+} > \mathrm{O}_2^{-} > \mathrm{O}_2^{2-}$

In the light of the above statements, choose the correct answer from the options given below :

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Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is true but Statement II is false

Solution:

Species	Bond order	unpaired electrons
$O_2^{\oplus}$	2.5	1
O_2	2	2
O_2^{-}	1.5	1
O_2^{2-}	1	0

Q. 4 Consider the following data

(i) $2Al(s) + 6HCl(aq) \rightarrow Al_2Cl_6(aq) + 3H_2(\ g) + 1200\ kJ/mol.$

(ii) $H_2(\ g) + Cl_2(\ g) \rightarrow 2HCl(g) + 164\ kJ/mol.$

(iii) $HCl(g) + aq \rightarrow HCl(aq) + 83\ kJ/mol.$

(iv) $Al_2Cl_6(\ s) + aq \rightarrow Al_2Cl_6(aq) + 663\ kJ/mol.$

The enthalpy of formation of anhydrous solid Al_2Cl_6 is :

Option 1:
$-648\ kJ\ mol^{-1}$

Option 2:
$-1350\ kJ\ mol^{-1}$

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Option 3:
$-2002 \text{ kJ mol}^{-1}$

Option 4:
$-1527 \text{ kJ mol}^{-1}$

Correct Answer:
$-1527 \text{ kJ mol}^{-1}$

Solution:
$2Al(s) + 6HCl(aq) \rightarrow Al_2Cl_6(aq) + 3H_2(g)$

$\Delta_r H_1 = -1200 \text{ kJ/mol}$
$[H_2(g) + Cl_2(g) \rightarrow 2HCl(g)] \times 3$
$\Delta_r H_2 = -163 \times 3 \text{ kJ/mol}$
$[HCl(g) + aq \rightarrow HCl(aq)] \times 6$
$\Delta_r H_3 = -83 \times 6 \text{ kJ/mol}$
$Al_2Cl_6(aq) \rightarrow Al_2Cl_6(s) + aq. \ \Delta_r H_4 = +663 \text{ kJ/mole}$
$2Al(s) + 3Cl_2 \rightarrow Al_2Cl_6(s) \quad \Delta_r H$
$\Delta_f H = \Delta_r H_1 + \Delta_r H_2 + \Delta_r H_3 + \Delta_r H_4$
$= -1200 - 163 \times 3 - 83 \times 6 + 663$
$= -1527 \text{ kJ/mole}$

Q. 5 19.5 g of fluoro acetic acid (molar mass $= 78 \text{ g mol}^{-1}$) is dissolved in 500 g of water at 298 K . The depression in the freezing point of water was 1°C. What is K_a of fluoro acetic acid ? (For water, $K_f = 1.86 \text{ K kg mol}^{-1}$). Assume molarity and molality to have same values.

Option 1:
10^{-6}

Option 2:
4×10^{-4}

Option 3:
3×10^{-5}

Option 4:
3×10^{-3}

Correct Answer:
3×10^{-3}

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Solution:

$\Delta T_f = i\, K_f\, m$

$\text{no. of moles } = \frac{19.5}{78} = \frac{1}{4} \text{ mole}$

$1 = i \times 1.86 \times \frac{1/4}{1/2}$

$i = \frac{2}{1.86}$

$i = 1 + (n-1)\alpha$

$i = 1 + \alpha$

$\alpha = \frac{2}{1.86} - 1 = 0.075$

$K_a = \frac{Ca^2}{1-\alpha}$

$= \frac{1}{2} \times \frac{(0.075)^2}{(1-0.075)}$

$= 3 \times 10^{-3}$

Q. 6 The solubility product constants of Ag_2CrO_4 and AgBr are 32 x and 4 y respectively at 298 K .

The value of $\left(\frac{\text{molarity of } Ag_2CrO_4}{\text{molarity of AgBr}}\right)$ can be expressed as:

Option 1:

$\frac{2\sqrt[3]{x}}{y}$

Option 2:

$2\sqrt{\frac{x}{y}}$

Option 3:

$\sqrt{\frac{x}{y}}$

Option 4:

$\frac{\sqrt[3]{x}}{\sqrt{y}}$

Correct Answer:

$\frac{\sqrt[3]{x}}{\sqrt{y}}$

Solution:

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For $Ag_2C_2O_4$

$4S_1^3 = 32x$
$\Rightarrow S_1 = 2x^{1/3}$

For AgBr

$S_2^2 = 4y$
$\Rightarrow S_2 = 2y^{1/2}$

Ratio $\left(\frac{S_1}{S_2}\right) = \frac{\sqrt[3]{x}}{\sqrt{y}}$

Q. 7 An electrochemical cell is constructed using half cells in the direction of spontaneous change

$Fe(OH)_2(s) + 2e^- \rightarrow Fe(s) + 2OH(aq) E^0 = -0.88\ V$
and $AgBr(s) + e^- \rightarrow Ag(s) + Br^-(aq) \quad E^0 = +0.07\ V$
Which of the following option is correct ?

Option 1:
Overall reaction $Fe(s) + 2OH^{-1}(aq) + 2AgBr(s)$
$\rightleftharpoons Fe(OH)_2(s) + 2Ag(s) + 2Br^-(aq)$

Option 2:
$E^0_{cell} = -0.95\ V$

Option 3:
Fe is reduced in the electrochemical cell

Option 4:
E^0_{cell} is an extensive property

Correct Answer:
Overall reaction $Fe(s) + 2OH^{-1}(aq) + 2AgBr(s)$
$\rightleftharpoons Fe(OH)_2(s) + 2Ag(s) + 2Br^-(aq)$

Solution:
Overall reaction

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$$\mathrm{Fe(s) + 2OH^-(aq) + 2AgBr(s) \rightleftharpoons Fe(OH)_2(s) + 2Ag(s) + 2Br^-(aq)}$$

$$\mathrm{E^\circ_{cell} = E^\circ_{R(cathode)} - E^\circ_{R(anode)}}$$

$$= 0.07 - (-0.88) = 0.95\ \mathrm{V}$$

Q. 8 $t_{100\%}$ is the time required for the 100% completion of the reaction while $t_{1/2}$ is the time required for 50% of the reaction to be completed. Which of the following option correctly represents the relation between $t_{100\%}$ and $t_{1/2}$ for zero and first order reactions respectively?

Option 1:
$t_{100\%} = \left(t_{1/2}\right)^2$ and $t_{100\%} = \left(t_{1/2}\right)^{-\infty}$

Option 2:
$t_{100\%} = 2t_{1/2}$ and $t_{100\%} = \left(t_{1/2}\right)^{\infty}$

Option 3:
$t_{100\%} = 2t_{1/2}$ and $t_{100\%} = \left(2t_{1/2}\right)^2$

Option 4:
$t_{100\%} = \left(t_{1/2}\right)^{\infty}$ and $t_{100\%} = 2t_{1/2}$

Correct Answer:
$t_{100\%} = 2t_{1/2}$ and $t_{100\%} = \left(t_{1/2}\right)^{\infty}$

Solution:
Zero -order

$$\mathrm{t_{100\%} = \frac{a}{k}; t_{1/2} = \frac{a}{2k}}$$

$$\mathrm{t_{100\%} = 2t_{1/2}}$$

Ist order

$$\mathrm{t_{100\%} = \infty}$$

$$\mathrm{t_{1/2} = \frac{\ell n 2}{k}}$$

$$\mathrm{t_{100\%} = \left(t_{1/2}\right)^{\infty}}$$

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Q. 9 Given below are two statements :

Statement (I) : The first ionisation enthalpy of the elements $\mathrm{Na}, \mathrm{Mg}, \mathrm{Cl}$ and Ar follows the order

$$\mathrm{Na} > \mathrm{Mg} > \mathrm{Cl} > \mathrm{Ar}$$

Statement (II) : Among $\mathrm{Ca}, \mathrm{Al}, \mathrm{Fe}$ and B , the third ionisation enthalpy is very high for Ca .

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

Solution:
(i) $\mathrm{Na} < \mathrm{Mg} < \mathrm{Cl} < \mathrm{Ar} \rightarrow \mathrm{I}^{\mathrm{st}}$ ionization energy
(ii) Among B, Ca, Al, Fe

Ca has maximum 3rd ionization energy among given due to inert gas configuration.

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Q. 10 Given below are two statements:

Statement (I) : Oxidising power of halogens decreases in the order $F_2 > Cl_2 > Br_2 > I_2$, which is the basis of "Layer test".

Statement (II) : "Layer test" to identify Br_2 and I_2 in aqueous solution involves the oxidation of bromide or iodide into Br_2 or I_2 respectively with Cl_2, which is a type of displacement redox reaction.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:
(I) $F_2 > Cl_2 > Br_2 > I_2$ (Oxidizing power order)
(II) $NaBr + Cl_2\text{-water} \rightarrow Br_2 + NaCl$
$NaI + Cl_2\text{-water} \rightarrow I_2 + NaCl$

Q. 11 Which of the following sets includes all the species that will change the orange colour of $K_2Cr_2O_7$ in acidic medium?

Option 1:
$Fe^{2+}, Sn^{2+}, I^-, S^{2-}$

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Option 2:
$S^{2-}, Fe^{3+}, I, C_2O_4^{2-}$

Option 3:
$Fe^{2+}, NO_2^-, SO_2, Sn^{4+}$

Option 4:
$Fe^{3+}, SO_4^{2-}, S^{2-}, Sn^{4+}$

Correct Answer:
$Fe^{2+}, Sn^{2+}, I^-, S^{2-}$

Solution:

$$\text{Reducing agent} + KmnO_4 \xrightarrow[\text{medium}]{\text{acidic}} Mn^{+2} + \text{Product}$$

Given Reducing Agent

$Fe^{+2} \rightarrow Fe^{+3}$
$Sn^{+2} \rightarrow Sn^{+4}$
$I^- \rightarrow I_2$
$S^{-2} \rightarrow S$

Q. 12 **Match List - I with List - II.**

List - I	List - II
Chromium (III) Complexes (en = ethylene diamine)	$\Delta_0\ (cm^{-1})$
A. $[Cr(CN)_6]^{3-}$	I. 15,060
B. $[CrF_6]^{3-}$	II. 17,400
C. $[Cr(H_2O)_6)]^{3+}$	III. 22,300
D. $[Cr(en)_3]^{3+}$	IV. 26,600

Choose the correct answer from the options given below :

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Option 1:
A-I, B-II, C-III, D-IV

Option 2:
A-II, B-III, C-IV, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-IV, B-I, C-II, D-III

Correct Answer:
A-IV, B-I, C-II, D-III

Solution:
CFSE is proportional to strength of ligand followed by chelation.
CN^- is stronger ligand than en

Q. 13 Given below are two statements :
Statement (I) : 1,2,3-Trihydroxypropane can be separated from water by simple distillation.

Statement (II) : An azotropic mixture cannot be separated by fractional distillation.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

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Correct Answer:
Statement I is false but Statement II is true

Solution:
Glycerol is separated by distillation under reduced pressure.

Q. 14 Given below are two statements :
Statement (I): Benzyl chloride reacts faster in S_N1 mechanism than ethyl chloride.
Statement (II) : Ethyl carbocation intermediate is less stabilized by hyperconjugation than benzyl carbocation by resonance.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:
Rate of $SN^1 \propto$ Stability of carbocation
Order of stability of carbocation :

$$Ph-\overset{\oplus}{C}H_2 > CH_3-\overset{\oplus}{C}H_2$$

Hyperconjugation in $CH_3-\overset{\oplus}{C}H_2$
Resonance in $Ph-\overset{\oplus}{C}H_2$

Q. 15 In IUPAC nomenclature, the correct order of decreasing priority of functional group is :

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Option 1:
$-\mathrm{CONH_2}, >_{\mathrm{C=O}, -\mathrm{CHO}, -\mathrm{NH_2}, -\mathrm{C\equiv C-}}$

Option 2:
$-\mathrm{CONH_2}, -\mathrm{COOCH_3}, -\mathrm{CHO}, -\mathrm{NH_2}, -\mathrm{OH}$

Option 3:
$-\mathrm{CONH_2}, -\mathrm{CHO}, > \mathrm{C=O}, -\mathrm{NH_2}, -\mathrm{C\equiv C-}$

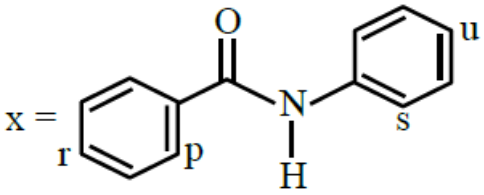
Option 4:
$-\mathrm{CONH_2}, -\mathrm{CHO}, -\mathrm{CN}, -\mathrm{NH_2}, -\mathrm{C\equiv C-}$

Correct Answer:
$-\mathrm{CONH_2}, -\mathrm{CHO}, > \mathrm{C=O}, -\mathrm{NH_2}, -\mathrm{C\equiv C-}$

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Solution:
Theory based

Q. 16 For the given molecule, " x ", the preferred site for the attack of the electrophile is :



Option 1:
Predominantly at " r "

Option 2:
" r " and " u "

Option 3:
"p" and "s"

Option 4:
Predominantly at "u"

Correct Answer:
Predominantly at "u"

Solution:
U is predominant site towards attack of electrophile because of +M nature of nitrogen.

Q. 17 Match List-I with List-II

List-I	**List-II**
Mixture of Compounds	**Reagent used to distinguish**
A.Diethyl amine + Ethyl amine	I.Bromine water
B.Acetaldehyde + Acetone	II. $\mathrm{CHCl_3 + KOH, \Delta}$
C. Ethanol + Phenol	III. Neutral $\mathrm{FeCl_3}$
D. Benzoic acid +Cinnamic acid	IV. Ammonical silver Cinnamic acid nitrate

Choose the correct answer from the options given below :

Option 1:
A-IV, B-II, C-I, D-III

Option 2:
A-IV, B-II, C-III, D-I

Option 3:
A-II, B-IV, C-I, D-III

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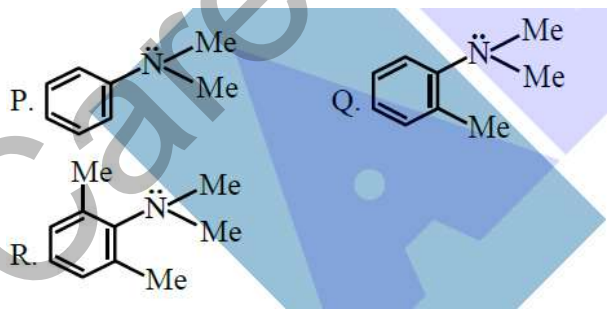
Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-II, B-IV, C-III, D-I

Solution:
- Phenol gives violet colour with neutral $\mathrm{FeCl_3}$ not ethanol.
- 1° amine gives positive test with $\mathrm{CHCl_3/KOH}$ not 2° amine.
- Aldehyde gives positive test with ammonical silver nitrate not ketone.
- Cinnamic acid gives bromine water test.

Q. 18 Consider the three aromatic molecules $(\mathrm{P}, \mathrm{Q}$ and R$)$ whose structures have been given below :

The correct order regarding the reactivity of these compounds with $\mathrm{Ph-N \equiv NCl^{(-)}}$ under optimum but (+) slightly acidic medium is :



Option 1:
P $>$ Q $>$ R

Option 2:
$\mathrm{R > P > Q}$

Option 3:
$R > Q > P$

Option 4:
$P > R > Q$

Correct Answer:
P $>$ Q $>$ R

Solution:
Rate of ESR $\propto$ nucleophilicity of benzene ring.

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Q. 19 Match List-I with List-II.

List-I	List-II
Vitamin	**Name**
A. Vitamin B B_1	I. Pyridoxine
B. Vitamin B_2	II. Ascorbic acid
C. Vitamin B_6	III. Thiamine
D. Vitamin C	IV. Riboflavin

Choose the correct answer from the options given below :

Option 1:
A-II, B-I, C-III, D-IV

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-I, B-III, C-II, D-IV

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
Theory based

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Q. 20 A salt with few drops of conc. HCl gives apple green colour in flame test. The group precipitate of the salt is dissolved in acetic acid and treated with K_2CrO_4 to give yellow precipitate. When the sodium carbonate extract of the salt solution is heated with conc. HNO_3 and ammonium molybdate, it resulted a canary yellow precipitate. The cation and anion present in the salt are respectively.

Option 1:
Ca^{2+} and SO_4^{2-}

Option 2:
Ba^{2+} and PO_4^{3-}

Option 3:
Mn^{2+} and PO_4^{3-}

Option 4:
Ba^{2+} and SO_4^{2-}

Correct Answer:
Ba^{2+} and PO_4^{3-}

Solution:
(i) Ba^{+2} gives apple green colour while Ca^{+2} and Sr^{+2} gives brick red and crimson red in flame test.
(ii) $Ba^{+2} + CrO_4^{2-} \rightarrow BaCrO_4$

Yellow ppt
(iii) PO_4^{3-} + Conc $HNO_3 + (NH_4)_2MoO_4 \longrightarrow \downarrow$

$(NH_4)_3\,[P(Mo_3O_{10})_4]$
Canery yellow ppt

Q. 21 5.33 g of $CrCl_3 \cdot 6H_2O$, which is a $1:3$ electrolyte, is dissolved in water and is passed through a cation exchanger. The chloride ions in the eluted solution, on treatment with $AgNO_3$ results in 8.61 g of AgCl . The ratio of moles of complex reacted and moles of AgCl formed is _____ $\times 10^{-2}$. (Nearest integer) [Molar mass in $gmol^{-1} Cr: 52, Ag: 108, Cl: 35.5, H: 1, O: 16$]

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Correct Answer:
33

Solution:

$$[Cr(H_2O)_6]Cl_3 + \underset{\text{excess}}{AgNO_3} \rightarrow \underset{\text{ppt}}{3AgCl(s)}$$

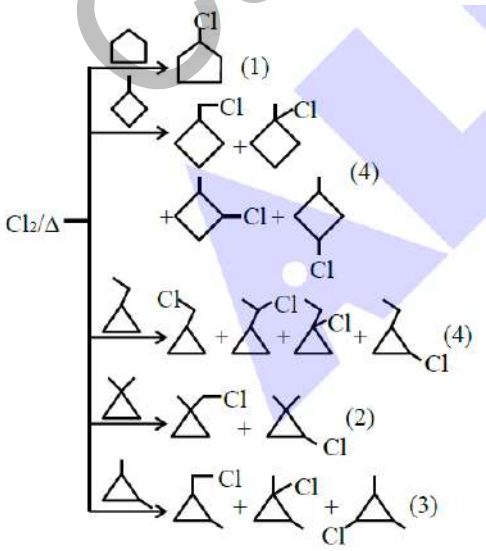
$$\frac{\text{no. of moles of complex reacted}}{\text{no. of moles of AgCl precipitated}} \times 100$$

$$= \frac{1}{3} \times 100 = 33.3$$

Q. 22 Consider the isomers of hydrocarbon with molecular formula C_5H_{10}. These isomers do not decolourise $KMnO_4$ solution. These isomers are subjected to chlorination with chlorine in presence of light to give monochloro compounds. The total number of monochloro compounds (structural isomers only) formed is _____ .

Correct Answer:
14

Solution:



1+4+4+2+3=14

Q. 23 One mole of an alkane (x) requires 8 mole oxygen for complete combustion. Sum of number of carbon and hydrogen atoms in the alkane (x) is _____ .

Correct Answer:
17

Solution:

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$$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \rightarrow nCO_2 + (n+1)H_2O$$

$$1 \quad \left(\frac{3n+1}{2}\right)$$

$$\frac{3n+1}{2} = 8 \Rightarrow 3n+1 = 16$$

$$\Rightarrow 3n = 15$$

$$n = 5$$

So alkane is $C_5H_{12} \Rightarrow 17$

Q. 24 For reaction $A \rightarrow P$, rate constant $k = 1.5 \times 10^3\ s^{-1}$ at $27^\circ C$. If activation energy for the above reaction is $60\ kJ\ mol^{-1}$, then the temperature (in $^\circ C$) at which rate constant, $k = 4.5 \times 10^3\ s^{-1}$ is _____.
(Nearest integer)
Given : $\log 2 = 0.30, \log 3 = 0.48, R = 8.3\ J\ K^{-1}\ mol^{-1}, \ln 10 = 2.3$

Correct Answer:
14

Solution:

$$\ell n\left(\frac{K_2}{K_1}\right) = \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_1\ T_2}\right]$$

$$\ell n\left(\frac{4.5 \times 10^3}{1.5 \times 10^3}\right) = \frac{60 \times 10^3}{8.3}\left[\frac{T_2 - 300}{300.\ T_2}\right]$$

$$\ell n3 = \frac{60 \times 10^3}{8.3}\left[\frac{T_2 - 300}{300.\ T_2}\right]$$

$$\Rightarrow T_2 = 314.4\ K$$

$$\Rightarrow t = 41.4^\circ C$$

Q. 25 At the transition temperature T, $A \rightleftharpoons B$ and $\Delta G^0 = 105 - 35 \log T$ where A and B are two states of substance X . The transition temperature in $^\circ C$ when pressure is 1 atm is _____ . (Nearest integer)

Correct Answer:
727

Solution:

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$\Delta G^{\circ}=0(\text{eqm. at standard pressure})$
$105-35 \log T=0$
$\log T=3$
$T=1000 K$
$T=727^{\circ} C$

Maths

Q. 1 Let $\alpha, \alpha+2, \alpha \in \mathbf{Z}$, be the roots of the quadratic equation
$x(x+2)+(x+1)(x+3)+(x+2)(x+4)+\ldots\ldots+(x+n-1)(x+n+1)=4n$
for some $n \in N$.

Then $(n+\alpha)$ is equal to :

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
3

Correct Answer:
2

Solution:
$nx^2+x(2+4+6+\ldots\ldots+2n)+(1.3+\ldots\ldots+(n-1)$
$(n+1))=4n$

$nx^2+n(n+1)x+\frac{n(n-1)(2n+5)}{6}=4n$

$x^2+(n+1)x+\frac{(n-1)(2n+5)}{6}=4$

D must be a perfect square

$D=\frac{122-2n^2}{6}=20-\left(\frac{n^2-1}{3}\right)$

If make perfect square $\Rightarrow \frac{n^2-1}{3}=16 \Rightarrow n=7$

So Equation is

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$\Rightarrow x^2+8x+\frac{8\times 15}{6}-5=0$

$x^2+8x+15=0$

$x=-3,-5$

$\alpha=-5, \alpha+2=-3$

$\alpha+n=7-5=2$

Q. 2 Let x and y be real number such that $50\left(\frac{2x}{1+3i}-\frac{y}{1-2i}\right)=31+17i, i=\sqrt{-1}$. Then the value of $10(x-3y)$ is :

Option 1:
20

Option 2:
31

Option 3:
35

Option 4:
75

Correct Answer:
75

Solution:

$50\left(\frac{2x(1-3i)}{10}-\frac{y(1+2i)}{5}\right)=31+17i$

$10x-(30x)i-10y-20yi=31+17i$

$\left.\begin{matrix}10(x-y)=31\\ -30x-20y=17\end{matrix}\right\}$

$30x-30y=93$

$-10(3x+2y)=17$

$x=0.9, y=-2.2$

$10(x-3y)=75$

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Q. 3 Let $\alpha, \beta \in \mathbf{R}$ be such that the system of linear equations

$$x + 2y + z = 5$$
$$2x + y + \alpha z = 5$$
$$8x + 4y + \beta z = 18$$

has no solution. Then $\frac{\beta}{\alpha}$ is equal to :

Option 1:
-4

Option 2:
4

Option 3:
8

Option 4:
-8

Correct Answer:
4

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Solution:
$x + 2y + z = 5$
$2x + y + \alpha z = 5$
$8x + 4y + \beta z = 18$

for no solution

$$\Delta = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & \alpha \\ 8 & 4 & \beta \end{vmatrix} = 0$$
$$\Delta = 1(\beta - 4\alpha) - 2(2\beta - 8\alpha) + 1(0) = 0$$
$$= \beta - 4\alpha - 4\beta + 16\alpha = 0$$
$$\Rightarrow 4\alpha = \beta$$

Q. 4

Let $A=\begin{bmatrix}1 & 2\\ 1 & \alpha\end{bmatrix}$ and $B=\begin{bmatrix}3 & 3\\ \beta & 2\end{bmatrix}$.

If $A^2-4A+I=O$ and $B^2-5B-6I=O$, then among the two statements :

(S1) : $[(\mathrm{B}-\mathrm{A})(\mathrm{B}+\mathrm{A})]^{\mathrm{T}}=\begin{bmatrix}13 & 15\\ 7 & 10\end{bmatrix}$

and

(S2) : $\det(\mathrm{adj}(\mathrm{A}+\mathrm{B}))=-5$,

Option 1:
only (S1) is correct

Option 2:
only (S2) is correct

Option 3:
both (S1) and (S2) are correct

Option 4:
both (S1) and (S2) are wrong

Correct Answer:
only (S2) is correct

Solution:

$\mathrm{A}=\begin{bmatrix}1 & 2\\ 1 & \alpha\end{bmatrix}$

$\mathrm{A}^2-4\,\mathrm{A}+\mathrm{I}=0$

trace of matrix

$\alpha+1=4$

$\alpha=3$

$\mathrm{B}=\begin{bmatrix}3 & 3\\ \beta & 2\end{bmatrix}$

$\mathrm{B}^2-5\,\mathrm{B}-6\mathrm{I}=0$

$\det(B)=6-3\beta=-6$

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$3\beta = 12$
$\beta = 4$
$\mathrm{B} - \mathrm{A} = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}; \mathrm{B} + \mathrm{A} = \begin{bmatrix} 4 & 5 \\ 5 & 5 \end{bmatrix}$
$(\mathrm{B} - \mathrm{A})(\mathrm{B} + \mathrm{A}) = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$
$\mathrm{adj}(\mathrm{A}) = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}, \mathrm{adjB} = \begin{bmatrix} 2 & -3 \\ -4 & 3 \end{bmatrix}$
$\mathrm{adj}(\mathrm{A} + \mathrm{B}) = \begin{bmatrix} 5 & -5 \\ -5 & 4 \end{bmatrix}$

Q. 5 Let A be the set of first 101 terms of an A.P., whose first term is 1 and the common difference is 5 and let B be the set of first 71 terms of an A.P., whose first term is 9 and the common difference is 7. Then the number of elements is $\mathrm{A} \cap \mathrm{B}$, which are divisible by 3 , is :

Option 1:
4

Option 2:
5

Option 3:
6

Option 4:
7

Correct Answer:
5

Solution:
AP_1 Set $\mathrm{A} = \{1, 6, 11, 16 \ldots\ldots\ldots\ldots 101 \text{ terms }\}$
AP_2 Set $\mathrm{B} = \{9, 16 \ldots\ldots\ldots\ldots 71 \text{ terms }\}$
$\mathrm{D} = \mathrm{L.C.M}\{\mathrm{d}_1,\ \mathrm{d}_2\} = 35$
1st Common term is 16

$16 + (\mathrm{n}-$
$\mathrm{n} \leq 14.8$
$\Rightarrow \mathrm{n} = 14$

$A \cap B = \{16, 51, 86, 121, 156, 191, 226, 261, 296,$ 331, 366,401,436,471\}
terms divisible by ' 3 '

$= \{51, 156, 261, 366, 471\}$

Ans. 5 terms

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Q. 6 The number of seven-digit numbers, that can be formed by using the digits $1, 2, 3, 5$ and 7 such that each digit is used at least once, is ;

Option 1:
15400

Option 2:
17800

Option 3:
16800

Option 4:
29400

Correct Answer:
16800

Solution:
Exactly one digit repeated $\rightarrow {}^{5}\mathrm{C}_{1} \times \frac{7!}{3!} = 4200$

Exactly two digits repeated $\rightarrow {}^{5}\mathrm{C}_{2} \times \frac{7!}{2!2!} = 12600$

Total numbers $= 16800$

Q. 7 The number of elements in the set $S = \left\{(r, k) : k \in \mathbf{Z} \text{ and } {}^{36}C_{r+1} = \frac{6\left({}^{35}C_{r}\right)}{(k^2-3)}\right\}$, is

Option 1:
2

Option 2:
4

Option 3:
8

Option 4:
16

Correct Answer:
4

Solution:

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$$\frac{36}{r+1} \cdot {}^{35}C_r = 6 \cdot \frac{{}^{35}C_r}{k^2-3}$$

$$k^2 - 3 = \frac{r+1}{6}$$

$$k^2 = \frac{r+19}{6}$$

$$0 \le r \le 35$$

$$19 \le r + 19 \le 54$$

$$3.1 \le \frac{r+19}{6} \le 9$$

$$\therefore k^2 = 4, 9$$

$$k = \pm 2, \pm 3$$

$$\therefore (r, k) \Rightarrow (5, \pm 2) \Rightarrow (35, \pm 3)$$

$$\therefore 4$$

Q. 8 If the mean of the data

Class	5-10	10-15	15-20	20-25	25-30	30-35
Frequency	2	k	28	54	k+1	5

is 21, then k is one of the roots of the equation :

Option 1:
$2x^2 - 23x - 10 = 0$

Option 2:
$4x^2 - 35x + 24 = 0$

Option 3:
$2x^2 - 19x - 10 = 0$

Option 4:
$2x^2 - 35x + 98 = 0$

Correct Answer:
$2x^2 - 19x - 10 = 0$

Solution:

Class	5-10	10-15	15-20	20-25	25-30	30-35
Freq.	2	k	28	54	k+1	5

mean

$$21 = \frac{15 + 12.5k + 490 + 1215 + 27.5 + 27.5k + 162.5}{90 + 2k}$$

$$1890 + 42k = 1910 + 40k$$

$$k = 10$$

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Q. 9 Let the mid points of the sides of a triangle ABC be $\left(\frac{5}{2}, 7\right), \left(\frac{5}{2}, 3\right)$ and $(4, 5)$. If its incentre is (h, k), then $3\,\mathrm{h} + \mathrm{k}$ is equal to :

Option 1:
11

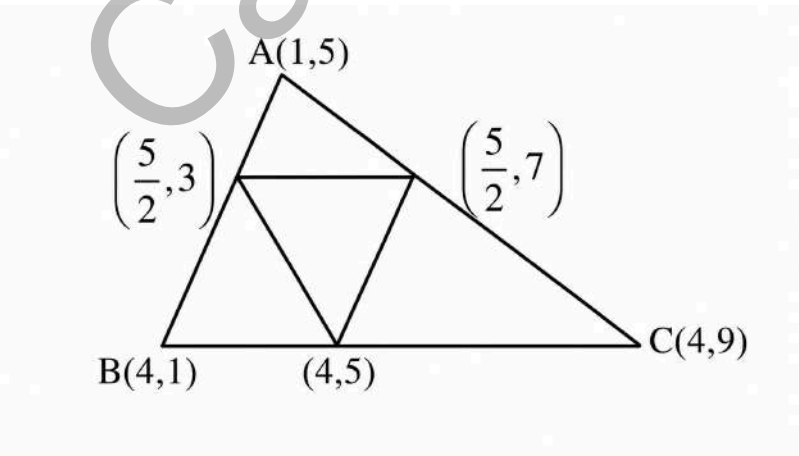
Option 2:
12

Option 3:
13

Option 4:
14

Correct Answer:
13

Solution:



$\mathrm{I} = \left(\frac{24}{9}, 5\right)$

$\therefore 3\,\mathrm{h} + \mathrm{k} = 8 + 5 = 13$

Q. 10 Let an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a < b$, pass through the point $(4, 3)$ and have eccentricity $\frac{\sqrt{5}}{3}$. Then the length of its latus rectum is :

Option 1:
$\frac{4\sqrt{5}}{3}$

Option 2:
$2\sqrt{5}$

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Option 3:
$\frac{7\sqrt{5}}{3}$

Option 4:
$\frac{8\sqrt{5}}{3}$

Correct Answer:
$\frac{8\sqrt{5}}{3}$

Solution:

$$e^2 = 1 - \frac{a^2}{b^2} \Rightarrow \frac{a^2}{b^2} = 1 - e^2 = 1 - \frac{5}{9} = \frac{4}{9}$$

$$\Rightarrow \frac{a^2}{b^2} = \frac{4}{9}$$

Passes through $(4, 3) \Rightarrow \frac{16}{a^2} + \frac{9}{b^2} = 1$

from (1) and (2)

$$a^2 = 20 \text{ and } b^2 = 45$$

$$LR = \frac{2a^2}{b} = \frac{2(20)}{3\sqrt{5}} = \frac{8\sqrt{5}}{3}$$

Q. 11 If $\sin\left(\frac{\pi}{18}\right)\sin\left(\frac{5\pi}{18}\right)\sin\left(\frac{7\pi}{18}\right) = K$, then the value of $\sin\left(\frac{10\,K\pi}{3}\right)$ is :

Option 1:
$\frac{\sqrt{3}+1}{2\sqrt{2}}$

Option 2:
$\frac{\sqrt{3}-1}{\sqrt{2}}$

Option 3:
$\frac{\sqrt{3}}{2}$

Option 4:
$\frac{1}{2}$

Correct Answer:
$\frac{\sqrt{3}+1}{2\sqrt{2}}$

Solution:

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$\mathrm{K} = \sin 10^\circ \sin 50^\circ \sin 70^\circ$

$= \frac{1}{4}\sin 30^\circ = \frac{1}{8}$

$\sin 10\ \mathrm{K}\frac{\pi}{3} = \sin\left(10 \times \frac{1}{8} \cdot \frac{\pi}{3}\right) = \sin\frac{5\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}}$

Q. 12 Let $S = \{x \in [-\pi, \pi] : \sin x(\sin x + \cos x) = a, \mathrm{a} \in \mathbf{Z}\}$. Then $\mathrm{n(S)}$ is equal to :

Option 1:
3

Option 2:
6

Option 3:
7

Option 4:
9

Correct Answer:
9

Solution:

$\sin x(\sin x + \cos x) \in \left[\frac{1-\sqrt{2}}{2}, \frac{1+\sqrt{2}}{2}\right]$

2 integer will be there $\Rightarrow \mathrm{a} = 0, 1$

If $\mathrm{a} = 0 \quad \sin \mathrm{x}(\sin \mathrm{x} + \cos \mathrm{x}) = 0$

$\Rightarrow \sin x = 0 \quad$ or $\quad \sin x + \cos x = 0$

$\mathrm{x} = -\pi, 0, \pi$

$= \tan x = -1$

3 solutions $x = -\frac{\pi}{4}, \frac{3\pi}{4}$ 2 solutions

If $a = 1 \quad \sin x(\sin x + \cos x) = 1$

$\sin^2 x + \sin x \cos x = 1$

$2\sin^2 x + 2\sin x \cos x = 2$

$1 - \cos 2x + \sin 2x = 2$

$\sin 2x - \cos 2x = 1 \rightarrow$ square

$\sin 4x = 0$

$4x = -4\pi, -3\pi, \ldots\ldots\ldots\ldots, 3\pi, 4\pi$

$x = -\pi, -\frac{3\pi}{4}, -\frac{\pi}{2}, -\frac{\pi}{4}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$

$\Rightarrow x = -\frac{3\pi}{4}, -\frac{\pi}{2}, \frac{\pi}{4}, \frac{\pi}{2}$ 4 solutions

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Total 9 solution

Q. 13 If the point of intersection of the lines $\frac{x+1}{3}=\frac{y+a}{5}=\frac{z+b+1}{7}$ and $\frac{x-2}{1}=\frac{y-b}{4}=\frac{z-2a}{7}$ lies on xy-plane, then the value of $a+b$ is :

Option 1:
2

Option 2:
5

Option 3:
7

Option 4:
9

Correct Answer:
7

Solution:
Line (L_1)

$\frac{x+1}{3}=\frac{y+a}{5}=\frac{z+b+1}{7}=r_1$

⇒ General point P on L_1 is

$(3r_1-1, 5r_1-a, 7r_1-b-1)$

Line (L_2)

$\frac{x-2}{1}=\frac{y-b}{4}=\frac{z-2a}{7}=r_2$

⇒ General point Q on L_2 is

$(\mathrm{r}_2+2, 4\mathrm{r}_2+\mathrm{b}, 7\mathrm{r}_2+2\mathrm{a})$

For point of intersection

$3r_1-1=r_2+2 \Rightarrow r_2=3r_1-3$
$5r_1-a=4r_2+b$
$7r_1-b-1=7r_2+2a$

Since point lies on XY plane

$\Rightarrow \mathrm{z}-\ \mathrm{coordinate}\ =0$

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From $L_1 : 7r_1 - b - 1 = 0 \Rightarrow 7r_1 = b + 1$
From $L_2 : 7r_2 + 2a = 0 \Rightarrow 2a = -7r_2$
Substitute $r_2 = 3r_1 - 3$

$$2a = -7(3r_1 - 3)$$
$$\Rightarrow a = \frac{-21r_1 + 21}{2}$$

Put in equation (2)

$$5r_1 - a = 4r_2 + b$$
$$5r_1 - \left(\frac{-21r_1 + 21}{2}\right) = 4(3r_1 - 3) + (7r_1 - 1)$$
$$\Rightarrow r_1 = \frac{5}{7}$$
$$\Rightarrow b = 7r_1 - 1 = 4$$
$$\Rightarrow a = 3 \therefore a + b = 7$$

Q. 14 If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 2$ and $|\vec{b}| = 3$, then the maximum value of $3|(3\vec{a} + 2\vec{b})| + 4|(3\vec{a} - 2\vec{b})|$ is :

Option 1:
30

Option 2:
36

Option 3:
60

Option 4:
72

Correct Answer:
60

Solution:

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$$
\begin{aligned}
\mathrm{E} &= 3\sqrt{9\mathrm{a}^2+4\,\mathrm{b}^2+12\mathrm{a}\vec{\mathrm{b}}}+4\sqrt{9\mathrm{a}^2+4\,\mathrm{b}^2-12\vec{\mathrm{a}}\cdot\vec{\mathrm{b}}} \\
&= 3\sqrt{36+36+12\times 6\cos\theta}+4\sqrt{36+36-72\cos\theta} \\
&= 3\sqrt{72+72\cos\theta}+4\sqrt{72-72\cos\theta} \\
&= 18\sqrt{2}\sqrt{1+\cos\theta}+24\sqrt{2}\sqrt{1-\cos\theta} \\
&= 18\sqrt{2}\sqrt{2}\cos\frac{\theta}{2}+24\sqrt{2}\left(\sqrt{2}\sin\frac{\theta}{2}\right) \\
&= 36\left(\cos\frac{\theta}{2}\right)+48\sin\left(\frac{\theta}{2}\right) \\
\mathrm{E}_{\max} &= \sqrt{(36)^2+(48)^2}=\sqrt{6^4+6^2\times 8^2}=60
\end{aligned}
$$

Q. 15 Let a line L passing through the point $(1,1,1)$ be perpendicular to both the vector $2\hat{i}+2\hat{j}+\hat{k}$ and $\hat{\mathrm{i}}+2\hat{\mathrm{j}}+\hat{\mathrm{k}}$. If $\mathrm{P}(\mathrm{a},\mathrm{b},\mathrm{c})$ is the foot of perpendicular from the origin on the line L , then the value of $34(\mathrm{a}+\mathrm{b}+\mathrm{c})$ is :

Option 1:
50

Option 2:
80

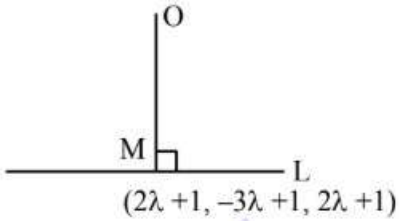
Option 3:
100

Option 4:
120

Correct Answer:
100

Solution:
Direction vector of line is

$$
=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & 2\\ 2 & 2 & 1\end{vmatrix}=-2\hat{i}+3\hat{j}-2\hat{k}
$$



$$\frac{x-1}{2}=\frac{y-1}{-3}=\frac{z-1}{2}$$

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So equation of line is:

$\frac{x-1}{2}=\frac{y-1}{-3}=\frac{z-1}{2}=\lambda$ let

let $(2\lambda+1,-3\lambda+1,2\lambda+1)$ be the foot of perpendicular from $(0,0,0)$ upon given line

$\Rightarrow \overrightarrow{\mathrm{OM}}\cdot(2\hat{\mathrm{i}}-3\mathrm{j}+2\hat{\mathrm{k}})=0$

$\Rightarrow 2(2\lambda+1)-3(-3\lambda+1)+2(2\lambda+1)=0$

$\Rightarrow \lambda=-\frac{1}{17}$

So $(\mathrm{a},\mathrm{b},\mathrm{c})\equiv\left(\frac{-2}{17}+1,\frac{3}{17}+1,\frac{-2}{17}+1\right)$

$\Rightarrow 34(a+b+c)=100$

Q. 16 If $\lim_{x\to 2}\frac{\sin\left(x^3-5x^2+ax+b\right)}{(\sqrt{x-1}-1)\log_e(x-1)}=m$, then $a+b+m$ is equal to:

Option 1:
5

Option 2:
6

Option 3:
8

Option 4:
10

Correct Answer:
6

Solution:

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$\because$ Denominator $= 0$ at $x = 2$

$\Rightarrow$ Numerator $= 0$ at $x = 2$

$\Rightarrow 2^3 - 5(2)^2 + a(2) + b = 0$

$\Rightarrow 2a + b = 12 \ldots (1)$

$\therefore m = \text{Lim}_{x \to 2} \dfrac{\sin\left(x^3 - 5x^2 + ax + b\right)}{(x^3 - 5x^2 + ax + b)} \cdot \left(x^3 - 5x^2 + ax + b\right)$

$\Rightarrow m = \text{Lim}_{x \to 2}\, 2\left(\dfrac{x^3 - 1 - 1}{\sqrt{x-1}+1}\right) \cdot \dfrac{\log_e(1 + (x-2))}{(x-2)} \cdot (x-2)$

$\Rightarrow m = \text{Lim}_{x \to 2}\, 2\left(\dfrac{3x^2 - 10x + a}{2(x-2)}\right)$

$\because$ Denominator $= 0$ at $x = 2$

$\Rightarrow$ Numerator $= 0$ at $x = 2$

$\Rightarrow 3(2)^2 - 10(2) + a = 0 \Rightarrow a = 8$

put $a = 8$ in (1), we get $b = -4$

$\therefore\ m = \text{Lim}_{x \to 2}\, 2\left(\dfrac{6x - 10}{2}\right) \Rightarrow m = 2$

$\therefore a + b + m = 6$

Q. 17 If the curve $y = f(x)$ passes through the point $(1, e)$ and satisfies the differential equation $dy = y\,(2 + \log_e x)dx,\ x > 0$, then $f(e)$ is equal to :

Option 1:
e^e

Option 2:
e^{e^2}

Option 3:
e^{2e}

Option 4:
e^{2^e}

Correct Answer:
e^{2e}

Solution:

$\dfrac{dy}{dx} = y(2 + \ell nx)$

$\displaystyle\int \frac{dy}{y} = \int (2 + \ell nx)dx$

$\ell ny = 2x + x\ell nx - x + C$

$\ell ny = x + x\ell nx + C$

Since it passes through $(1, e)$

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$\mathrm{l} = \mathrm{l} + 0 + \mathrm{C} \Rightarrow \mathrm{C} = 0$

$\ell \mathrm{ny} = \mathrm{x} + \mathrm{x}\ell \mathrm{nx}$

$\mathrm{f(x)} = \mathrm{y} = \mathrm{e}^{\mathrm{x}+\mathrm{x}\ln \mathrm{x}}$

$\Rightarrow \mathrm{f(e)} = \mathrm{e}^{\mathrm{cec}} = \mathrm{e}^{2\mathrm{c}}$

Q. 18

The number of critical points of the function $f(x) = \begin{cases} \left|\frac{\sin x}{x}\right|, & x \neq 0 \\ 1, & x = 0 \end{cases}$ in the interval $(-2\pi, 2\pi)$ is equal to:

Option 1:
1

Option 2:
3

Option 3:
5

Option 4:
7

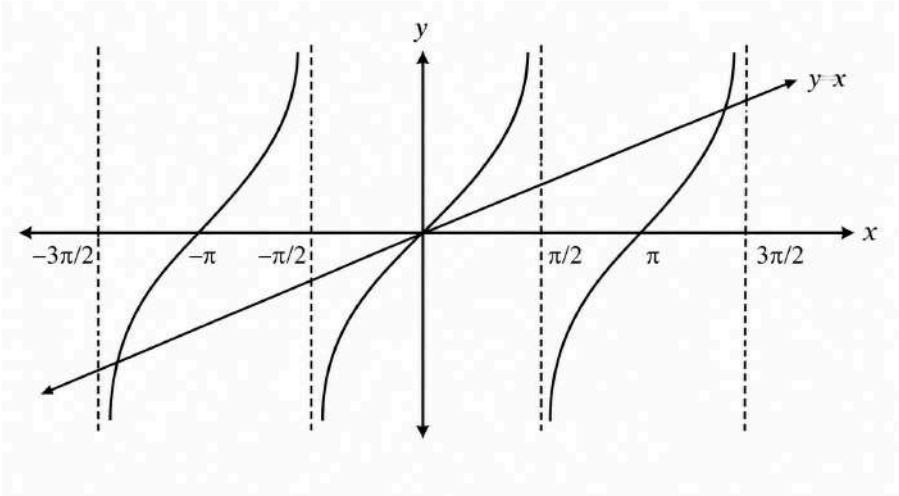
Correct Answer:
5

Solution:

$\lim_{x\to 0}\left|\frac{\sin x}{x}\right| = 1 = f(0) \to f(x)$ is continuous

Now $\frac{d}{dx}\left(\frac{\sin x}{x}\right) = \frac{x\cos x - \sin x}{x^2}$

$\Rightarrow \mathrm{f'(x)} = 0 \Rightarrow \tan \mathrm{x} = \mathrm{x}$



3 solution in $(-2\pi, 2\pi)$

also $\mathrm{f'(0)}$ does not exist at $\mathrm{x} = -\pi, \pi$

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total 5 points

Q. 19 Let [•] denote the greatest integer function. Then the value of $\int_0^3 \left(\frac{e^x+e^{-x}}{[x]!}\right)dx$ is:

Option 1:
$e^2+e^3-\frac{1}{e^2}-\frac{1}{e^3}$

Option 2:
$\frac{1}{2}\left(e^2+e^3-\frac{1}{e^2}-\frac{1}{e^3}\right)$

Option 3:
$e^2+e^3-\frac{1}{2e^2}-\frac{1}{2e^3}$

Option 4:
$\frac{1}{2}\left(e^2+e^3\right)-\frac{1}{e^2}-\frac{1}{e^3}$

Correct Answer:
$\frac{1}{2}\left(e^2+e^3-\frac{1}{e^2}-\frac{1}{e^3}\right)$

Solution:

$$\int_0^1 \frac{e^x+e^{-x}}{1}dx+\int_1^2 \frac{e^x+e^{-x}}{1}dx+\int_2^3 \frac{e^x+e^{-x}}{2}dx$$

$$\int_0^2 e^x dx+\int_0^2 e^{-x}dx+\frac{1}{2}\int_2^3 \left(e^x+e^{-x}\right)dx$$

$$\left(e^2-1\right)+\frac{\left(e^{-2}-1\right)}{-1}+\frac{1}{2}\left(\left(e^3-e^2\right)-\left(e^{-3}-e^{-2}\right)\right)$$

$$\frac{e^3+e^2}{2}+1-1-\frac{e^{-2}}{2}-\frac{e^{-3}}{2}$$

$$\frac{e^3+e^2-e^{-2}-e^{-3}}{2}$$

Q. 20 Let $y=y(x)$ be the solution curve of the differential equation $(1+\sin x)\frac{dy}{dx}+(y+1)\cos x=0, y(0)=0$. If the curve $y=y(x)$ passes through the point $\left(\alpha,\frac{-1}{2}\right)$, then a value of α is :

Option 1:
$\frac{\pi}{6}$

Option 2:
$\frac{\pi}{4}$

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Option 3:
$\frac{\pi}{3}$

Option 4:
$\frac{\pi}{2}$

Correct Answer:
$\frac{\pi}{2}$

Solution:

$$\int \frac{dy}{y+1} = -\int \frac{\cos x}{1+\sin x}dx$$

$$\Rightarrow \ln(y+1) = -\ln(1+\sin x) + C$$

$$\because y(0) = 0 \Rightarrow C = 0$$

$$\Rightarrow y+1 = \frac{1}{1+\sin x}$$

passes through $\left(\alpha, -\frac{1}{2}\right)$

$$\Rightarrow \sin x = 1$$

$$\Rightarrow x = \frac{\pi}{2} \text{ (as per options)}$$

Q. 21 If the domain of the function $f(x) = \sqrt{\log_{(0.6)}\left(\left|\frac{2x-5}{x^2-4}\right|\right)}$ is $(-\infty, a] \cup \{b\} \cup [c, d) \cup (e, \infty)$, then the value of $a + b + c + d + e$ is ____ .

Correct Answer:
4

Solution:

For domain $\log_{0.6}\left|\frac{2x-5}{x^2-4}\right| \geq 0$

$$\left|\frac{2x-5}{x^2-4}\right| \leq 1 \& x \neq \frac{5}{2} \ldots\ldots$$

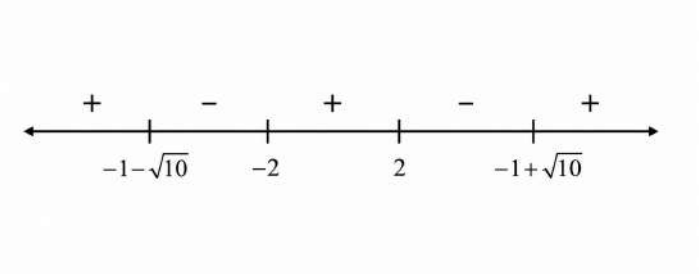
$$-1 \leq \frac{2x-5}{x^2-4} \leq 1$$

$$\frac{2x-5}{x^2-4} + 1 \geq 0$$

$$\frac{x^2+2x-9}{x^2-4} \geq 0$$

$$\frac{(x+1)^2-10}{(x-2)(x+2)} \geq 0$$

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$x \in (-\infty, -1-\sqrt{10}] \cup (-2, 2) \cup [-1+\sqrt{10}, \infty) \ldots\ldots (2)$

$$\frac{2x-5}{x^2-4} - 1 \leq 0$$

$$\frac{2x-5-x^2+4}{x^2-4} \leq 0$$

$$\frac{x^2-2x+1}{x^2-4} \geq 0$$

$$\frac{(x-1)^2}{(x-2)(x+2)} \geq 0$$

$x \in (-\infty, -2) \cup (2, \infty) \cup \{1\} \quad \ldots . (3)$

(1) $\cap (2) \cap (3)$

$$x \in (-\infty, -1-\sqrt{10}] \cup \{1\} \cup \left[-1+\sqrt{10}, \frac{5}{2}\right) \cup \left(\frac{5}{2}, \infty\right)$$

$a + b + c + d + e = -2 + 1 + 5 = 4$

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Q. 22 If $\sum_{k=1}^{n} a_k = 6n^3$, then $\sum_{k=1}^{6} \left(\frac{a_{k+1}-a_k}{36}\right)^2$ is equal to_____.

Correct Answer:
91

Solution:

$a_1 + a_2 + \ldots\ldots + a_n = 6n^3.$

$a_1 + a_2 + \ldots\ldots + a_n + a_{n+1} = 6(n+1)^3$

$a_{n+1} = 6(n+1)^3 - 6n^3$

$= 6((n+1) - n)\left((n+1)^2 + n^2 + n(n+1)\right)$

$a_{n+1} = 6(1)\left(3n^2 + 3n + 1\right)$

$a_n = 6\left(3(n-1)^2 + 3(n-1) + 1\right)$

$= 6\left(3\left(n^2 - 2n + 1 + n - 1\right) + 1\right)$

$= 6\left(3\left(n^2 - n\right) + 1\right)$

$a_n = 6\left(3n^2 - 3n + 1\right)$

$$\sum_{k=1}^{6}\left(\frac{6\left(3k^2+3k+1\right)-6\left(3k^2-3k+1\right)}{36}\right)^2$$

$$\sum_{k=1}^{6}\left(\frac{36k}{36}\right)^2 = \sum_{k=1}^{6} k^2 = \frac{6\times 7\times 13}{6} = 91$$

Q. 23 Let $a, b, c \in \{1, 2, 3, 4\}$. If the probability, that $ax^2 + 2\sqrt{2}bx + c > 0$ for all $x \in \mathbf{R}$, is $\frac{m}{n}$, $\gcd(\mathrm{m}, \mathrm{n}) = 1$, then $\mathrm{m} + \mathrm{n}$ is equal to _____ .

Correct Answer:
81

Solution:
$8b^2 - 4ac < 0 \Rightarrow 2b^2 < ac$

b	a	c		
1	1	3,4	$\rightarrow$	2
1	2	2,3,4	$\rightarrow$	3
1	3	1,2,3,4	$\rightarrow$	4
1	4	1,2,3,4	$\rightarrow$	4
2	3	3,4	$\rightarrow$	2
2	4	3,4	$\rightarrow$	2

$$\text{Required probability } = \frac{17}{64} \equiv \frac{m}{n}$$

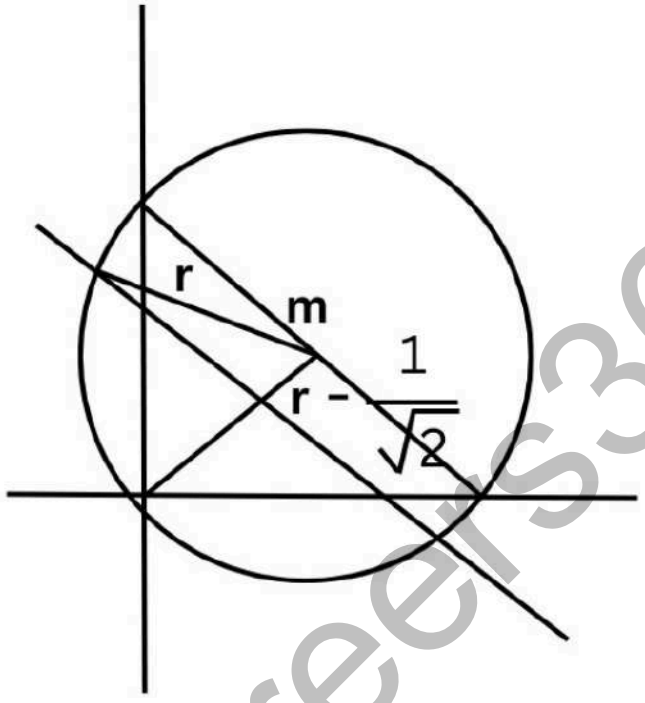
$\therefore m + n = 17 + 64 = 81$

Q. 24 Let a circle C have its centre in the first quadrant, intersect the coordinate axes at exactly three points and cut off equal intercepts from the coordinate axes. If the length of the chord of C on the line $x + y = 1$ is $\sqrt{14}$, then the square of the radius of C is _____ .

Correct Answer:
8

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Solution:



$$r^2 = \left(\frac{\sqrt{14}}{2}\right)^2 + \left(r - \frac{1}{\sqrt{2}}\right)^2$$
$$r^2 = \frac{7}{2} + r^2 - \sqrt{2}r + \frac{1}{2}$$
$$\sqrt{2}r = 4$$
$$r = 2\sqrt{2}$$
$$r^2 = 8$$

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Q. 25 If $\alpha = \int_0^{2\sqrt{3}} \log_2\left(x^2+4\right)dx + \int_2^4 \sqrt{2^x - 4}dx$, then α^2 is equal to _____ .

Correct Answer:
192

Solution:
As we know that

$$\int_a^b f(x)dx + \int_{f(a)}^{f(b)} f^{-1}(x)dx = bf(b) - af(a)$$
$$\therefore \alpha = (2\sqrt{3})(4) - (0)(2)$$
$$= 8\sqrt{3}$$
$$\Rightarrow \alpha^2 = 192$$

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JEE MAINS

2026

2nd April

2nd Shift

Question Paper

Physics

Q. 1 Dimensions of universal gravitational constant (G) in terms of Planck's constant (h), distance (L), mass (M) and time (T) are _____ .

Option 1:
$\left[\mathrm{hTLM}^{-2}\right]$

Option 2:
$\left[\mathrm{hT}^{-1}\mathrm{LM}^{-2}\right]$

Option 3:
$\left[\mathrm{hTL}^{2}\mathrm{M}^{-2}\right]$

Option 4:
$\left[\mathrm{h}^{-1}\ \mathrm{T}^{-1}\mathrm{LM}^{-2}\right]$

Correct Answer:
$\left[\mathrm{hT}^{-1}\mathrm{LM}^{-2}\right]$

Solution:

$$\text{Energy } = \frac{\mathrm{hc}}{\lambda} = \frac{\mathrm{GM_1M_2}}{\mathrm{r}}$$

$$\left[\mathrm{ML}^2\ \mathrm{T}^{-2}\right] = [\mathrm{h}]\left[\mathrm{T}^{-1}\right] = [\mathrm{G}]\left[\mathrm{M}^2\ \mathrm{L}^{-1}\right]$$

$$[\mathrm{G}] = \left[\mathrm{M}^{-2}\ \mathrm{L}^{+1}\ \mathrm{T}^{-1}\ \mathrm{h}^{+1}\right]$$

Q. 2 A 0.5 kg mass is in contact against the inner wall of a cylindrical drum of radius 4 m rotating about its vertical axis. The minimum rotational speed of the drum to enable the mass to remain stuck to the wall (without falling) is $5\mathrm{rad/s}$. The coefficient of friction between the drum's inner wall surface and mass is _____ (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Option 1:
0.1

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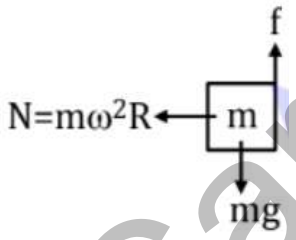
Option 2:
0.5

Option 3:
0.7

Option 4:
0.3

Correct Answer:
0.1

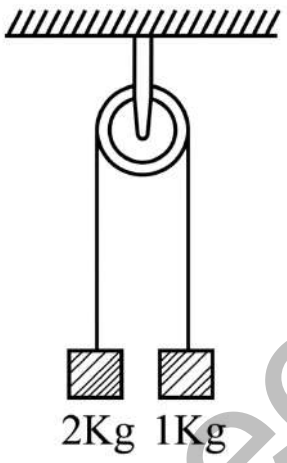
Solution:



$$N = m\omega^2 R$$
$$f_s = mg$$
$$f_s \leq f_\ell$$
$$mg \leq \mu N$$
$$mg \leq \mu m\omega^2 R$$
$$\mu \geq \frac{g}{\omega^2 R}$$
$$\mu_{min} = \frac{g}{\omega^2 R} = \frac{10}{10^2 \times 1} = 0.1$$

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Q. 3 Two blocks of masses 2 kg and 1 kg respectively, are tied to the ends of a string which passes over a light frictionless pulley as shown in the figure below. The masses are held at rest at the same horizontal level and then released. The distance traversed by the centre of mass in 2 s is _____ m.
(Take $g = 10\ m/s^2$)



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Option 1:

3.33

Option 2:
3.12

Option 3:
2.22

Option 4:
1.42

Correct Answer:
2.22

Solution:

$$a=\frac{(m_2-m_1)}{m_1+m_2}g=\frac{(2-1)}{3}\times 10=\frac{10}{3}\ m/s^2$$

$$a_{cm}=\frac{(1)(10/3)-(2)(10/3)}{3}=\frac{-10}{9}\ m/s^2(\downarrow)$$

$$S_{cm}=u_{cm}t+\frac{1}{2}a_{cm}t^2$$

$$=0+\frac{1}{2}\left(\frac{10}{9}\right)(4)=\frac{20}{9}\ m(\text{ downward })$$

$$S_{COM}=\frac{20}{9}(\text{ Downward })=2.22$$

Q. 4 A particle having charge $10^{-9}C$ moving in $x-y$ plane in fields of $0.4\hat{j}N/C$ and $4\times 10^{-3}kT$ experiences a force of $(4\hat{i}+2\hat{j})\times 10^{-10}$ N. The velocity of the particle at that instant is _____ m/s.

Option 1:
$50\hat{i}+100\hat{j}$

Option 2:
$100\hat{i}+50\hat{j}$

Option 3:
$-50\hat{i}+100\hat{j}$

Option 4:
$50\hat{i}-100\hat{j}$

Correct Answer:
$50\hat{i}+100\hat{j}$

Solution:

$$\vec{F}=q\vec{E}+q(\vec{V}\times\vec{B})$$

$$(4\hat{i}+2\hat{j})\times 10^{-10}$$

$$=10^{-9}\left[0.4\hat{j}+\left(v_x\hat{i}+v_y\hat{j}\right)\times\left(4\times 10^{-3}\right)\hat{k}\right]$$

$$(4\hat{i}+2\hat{j})\times 10^{-10}$$

$$=10^{-9}\left(0.4\hat{j}-4\times 10^{-3}v_x\hat{j}+4\times 10^{-3}v_y\hat{i}\right)$$

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On comparing

$v_y = 100, v_x = 50$

$\vec{V} = 50\hat{i} + 100\hat{j}$

Q. 5 The dimensional formula of $\frac{1}{2} \in_0 \mathrm{E}^2$ (ϵ_0 = permittivity of vacuum and E = electric field) is $\mathrm{M^aL^bT^c}$. The value of $2a - b + c =$ _____

Option 1:
0

Option 2:
1

Option 3:
-1

Option 4:
2

Correct Answer:
1

Solution:

$\frac{1}{2} \in_0 \mathrm{E}^2 = \text{ Energy density}$

$\therefore \left[\frac{1}{2} \in_0 \mathrm{E}^2\right] = \frac{\mathrm{ML^2\ T^{-2}}}{\mathrm{L^3}} = \mathrm{ML^{-1}\ T^{-2}}$

$\therefore \mathrm{a} = 1, \quad \mathrm{b} = -1, \quad \mathrm{c} = -2$

$\therefore 2\mathrm{a} - \mathrm{b} + \mathrm{c} = 2 + 1 - 2 = 1$

Q. 6 The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm . The length measured by a scale of least count 0.1 cm is 150 cm . When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm , measured by a micrometer of least count 0.001 cm . The error in the measured Young's modulus is $\alpha \times 10^9\ \mathrm{N/m^2}$. The value of α is _____ .
(Ignore the contribution of the load to Young's modulus error calculation)

Option 1:
1.3

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Option 2:
1.65

Option 3:
0.13

Option 4:
0.25

Correct Answer:
1.65

Solution:

$$\mathrm{Y}=\frac{\mathrm{F}\ell}{\mathrm{A}\Delta\ell}$$

$$\therefore \frac{\Delta \mathrm{Y}}{\mathrm{Y}}=\frac{\Delta \mathrm{F}}{\mathrm{F}}+\frac{\Delta \ell}{\ell}+2\frac{\Delta\, \mathrm{d}}{\mathrm{d}}+\frac{\Delta(\Delta \ell)}{\Delta \ell}$$

$$\left[\mathrm{A}=\pi\frac{\mathrm{d}^2}{4} \Rightarrow \frac{\Delta\, \mathrm{A}}{\mathrm{A}}=\frac{2\Delta\, \mathrm{d}}{\mathrm{d}}\right]$$

Since F is known exactly $\Rightarrow \Delta F=0$

$$\frac{\Delta Y}{Y}=\frac{0.1}{150}+\frac{2\times 0.001}{0.08}+\frac{0.001}{0.5}=0.0277$$

$$\therefore \Delta Y=0.0277\mathrm{Y}$$

$$Y=\frac{100\times 1.5}{\frac{\pi}{4}\left(\frac{0.08}{100}\right)^2\times 0.5\times 10^{-2}}=5.97\times 10^{10}\ \mathrm{N/m^2}$$

$$\therefore \Delta Y=0.0277\times 5.97\times 10^{10}\ \mathrm{N/m^2}=1.65\times 10^{9}\ \mathrm{N/m^2}$$

Q. 7 The velocity of a particle is given as $\vec{v}=-x\hat{i}+2y\hat{j}-z\hat{k}\ \mathrm{m/s}$. The magnitude of acceleration at point ($1, 2, 4$) is _____ $\mathrm{m/s^2}$.

Option 1:
$\sqrt{6}$

Option 2:
9

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Option 3:
$\sqrt{33}$

Option 4:
0

Correct Answer:
9

Solution:
$\vec{v} = -x\hat{i} + 2y\hat{j} - z\hat{k}$
$\vec{a} = \frac{d\vec{v}}{dt} = -\frac{dx}{dt}\hat{i} + 2\frac{dy}{dt}\hat{j} - \frac{dz}{dt}\hat{k}$
$= -(-x)\hat{i} + 2(2y)\hat{j} - (-z)\hat{k} = x\hat{i} + 4y\hat{j} + z\hat{k}$
$\therefore \vec{a} \text{ at } (1, 2, 4) = \hat{i} + 8\hat{j} + 4\hat{k}$
$\therefore |\vec{a}| = 9 \text{ m/s}^2$

Q. 8 The position of an object having mass 0.1 kg as a function of time t is given as $\vec{r} = \left(10t^2\hat{i} + 5t^3\hat{j}\right)\text{m}$. At $\text{t} = 1$ s, which of the following statements are correct?

A. The linear momentum $\overrightarrow{\text{p}} = (2\hat{\text{i}} + 1.5\hat{\text{j}})\text{kg} \cdot \text{m/s}$.
B. The force acting on the object $\vec{F} = (2\hat{i} + 3\hat{j})N$.
C. The angular momentum of the object about its origin $\overrightarrow{\text{L}} = 15\hat{\text{k}}\text{Js}$.
D. The torque acting on the object about its origin $\vec{\tau} = 20\hat{\text{k}}\text{Nm}$.

Choose the correct answer from the option given below.

Option 1:
A, B and C only

Option 2:
B, C and D only

Option 3:
A, C and D only

Option 4:
A, B and D only

Correct Answer:
A, B and D only

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Solution:

Given $\vec{r} = 10t^2\hat{i} + 5t^3\hat{j} \Rightarrow \vec{r}_{t=1} = 10\hat{i} + 5\hat{j}$

$\vec{v} = \frac{d\vec{r}}{dt} = 20t\hat{i} + 15t^2\hat{j} \Rightarrow \vec{v}_{t=1sec} = 20\hat{i} + 15\hat{j}$

$\vec{a} = \frac{d^2\vec{r}}{dt^2} = 20\hat{i} + 30t\hat{j} \Rightarrow \vec{a}_{t=1sec} = 20\hat{i} + 30\hat{j}$

$\therefore \vec{P} = m\vec{v} = 2\hat{i} + 1.5\hat{j}$

$\vec{F} = m\vec{a} = 2\hat{i} + 3\hat{j}$

$\vec{\tau} = \vec{r} \times \vec{F} = 20\hat{k}$

$\vec{L} = \vec{r} \times \vec{P} = 5\hat{k}$

Q. 9 A planet (P_1) is moving around the star of mass 2 M in the orbit of radius R . Another planet (P_2) is moving around another star of mass 4 M in a orbit of radius 2R. Ratio of time periods of revolution of P_2 and P_1 is ____ .

Option 1:

$\frac{1}{2}$

Option 2:

2

Option 3:

4

Option 4:

$\frac{1}{4}$

Correct Answer:

2

Solution:

From Kepler's law,

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$$T^2 = \frac{4\pi^2}{GM_S} r^3$$

$$\therefore\ T_{P_1} = \frac{2\pi}{\sqrt{G(2M)}} R^{3/2}$$

$$T_{P_2} = \frac{2\pi}{\sqrt{G(4M)}} (2R)^{3/2}$$

$$\therefore\ \frac{T_{P_2}}{T_{P_1}} = \frac{(2)^{3/2}}{\sqrt{4}} \times \frac{\sqrt{2}}{1} = \frac{2\sqrt{2} \times \sqrt{2}}{2} = 2$$

Q. 10 A particle is rotating in a circular path and at any instant its motion can be described as $\theta = \frac{5t^4}{40} - \frac{t^3}{3}$. The angular acceleration of the particle after 10 seconds is _____ $\mathrm{rad/s^2}$.

Option 1:
150

Option 2:
120

Option 3:
130

Option 4:
170

Correct Answer:
130

Solution:

$$\theta = \frac{5t^4}{40} - \frac{t^3}{3}$$

$$\frac{d\theta}{dt} = \frac{5}{40}(4t^3) - \frac{3t^2}{3} = \frac{t^3}{2} - t^2$$

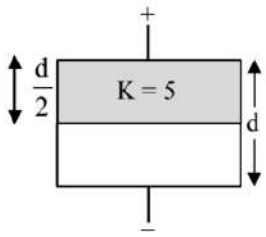
$$\alpha = \frac{d^2\theta}{dt^2} = \frac{3t^2}{2} - 2t$$

at $t = 10$

$$\alpha = \tfrac{3}{2}(10)^2 - 2 \times 10 = 130 \mathrm{rad/sec^2}$$

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Q. 11 A parallel plate air capacitor has a capacitance C . When it is half filled as show in figure with a dielectric constant $\mathrm{K}=5$, the percentage increase in the capacitance is _____ .



Option 1:
33.34

Option 2:
66.67

Option 3:
200

Option 4:
400

Correct Answer:
66.67

Solution:
Initial capacitance

$\mathrm{C_i}=\frac{\mathrm{A}\in_0}{\mathrm{d}}=\mathrm{C}$

Final capacitance

$C_1=\frac{A\in_0\times 5}{\frac{d}{2}}=10C$

$C_2=\frac{A\in_0}{\frac{d}{2}}=2C$

$C_f=\frac{C_1C_2}{C_1+C_2}=\frac{10\times 2}{10+2}C=\frac{5}{3}C$

$\%$ increase $=\frac{\frac{5}{3}C-C}{C}\times 100=66.67$

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Q. 12 Heat is supplied to a diatomic gas at constant pressure. Then the ratio of $\Delta Q : \Delta U : \Delta W$ is _____ .

Option 1:
$2 : 3 : 5$

Option 2:
$5 : 3 : 2$

Option 3:
$2 : 5 : 7$

Option 4:
$7 : 5 : 2$

Correct Answer:
$7 : 5 : 2$

Solution:

$$\Delta \mathrm{Q_P} = \mathrm{nC_P}\Delta \mathrm{T} = \frac{7}{2}\mathrm{nR}\Delta \mathrm{T}$$

$$\Delta \mathrm{U} = \mathrm{nC_v}\Delta \mathrm{T} = \frac{5}{2}\mathrm{nR}\Delta \mathrm{T}$$

$$\mathrm{W} = \mathrm{nR}\Delta \mathrm{T}$$

$$\Delta \mathrm{Q} : \Delta \mathrm{U} : \mathrm{W} :: 7 : 5 : 2$$

Q. 13 Two charged conducting spheres S_1 and S_2 of radii 8 cm and 18 cm are connected to each other by a wire. After equilibrium is established, the ratio of electric fields on $\mathrm{S_1}$ and $\mathrm{S_2}$ spheres are $\mathrm{E_{S1}}$ and $\mathrm{E_{S2}}$ respectively. The value of $\frac{\mathrm{E_{S1}}}{\mathrm{E_{S2}}}$ is _____ .

Option 1:
$\frac{3}{2}$

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Option 2:
$\frac{2}{3}$

Option 3:
$\frac{4}{9}$

Option 4:
$\frac{9}{4}$

Correct Answer:
$\frac{9}{4}$

Solution:



$$V_1 = V_2$$

$$\frac{kQ_1}{8} = \frac{kQ_2}{18} \left\{ v_{surface} = \frac{kQ}{R} \right\}$$

$$\frac{Q_1}{Q_2} = \frac{4}{9}$$

$$\frac{E_{S_1}}{E_{S_2}} = \frac{kQ_1}{r_1^2} \frac{r_2^2}{kQ_2} \left\{ E_{surface} = \frac{kQ}{R^2} \right\}$$

$$= \frac{Q_1}{Q_2} \left(\frac{r_2}{r_1} \right)^2 = \frac{4}{9} \times \left(\frac{9}{4} \right)^2 = \frac{9}{4}$$

Q. 14 The equation of a plane progressive wave is given by $y = 5 \cos \pi \left(200t - \frac{x}{150} \right)$ where x and y are in cm and t is in second. The velocity of the wave is _____ m/s

Option 1:
120

Option 2:
150

Option 3:
200

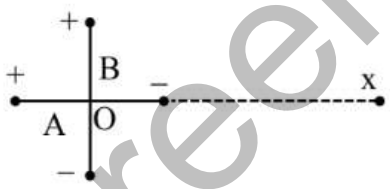
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Option 4:
300

Correct Answer:
300

Solution:
$v = \frac{\text{coeff of } t}{\text{coeff of } x} = \frac{200}{\frac{1}{150}\times 100} = 300 \text{ m/s}$

Q. 15 Two short electric dipoles A and B having dipole moment p_1 and p_2 respectively are placed with their axis mutually perpendicular as shown in the figure. The resultant electric field at a point x is making an angle of 60° with the line joining points O and x . The ratio of the dipole moments p_2/p_1 is ____ .



Option 1:
$\frac{\sqrt{3}}{2}$

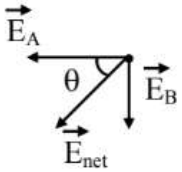
Option 2:
$2\sqrt{3}$

Option 3:
$\frac{1}{\sqrt{3}}$

Option 4:
$\sqrt{3}$

Correct Answer:
$2\sqrt{3}$

Solution:



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$$\overrightarrow{E}_A = \frac{2kP_1}{x^3}(-\hat{i})\{\text{ Axial point }\}$$

$$\overrightarrow{E}_B = \frac{kP_2}{x^3}(-\hat{j})\{\text{ Equatorial point }\}$$

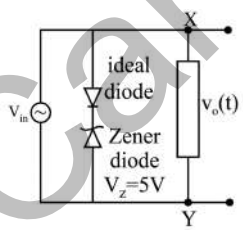
$$\text{Now : } \tan\theta = \frac{\left|\overrightarrow{E}_B\right|}{\left|\overrightarrow{E}_A\right|}$$

$$\Rightarrow \tan 60^\circ = \frac{P_2}{2P_1}$$

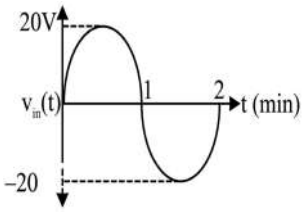
$$\Rightarrow \frac{P_2}{P_1} = 2\sqrt{3}$$

Q. 16 For the given circuit (shown in part (A)) the time dependent input voltage $v_{in}(t)$ and corresponding output $v_o(t)$ are shown in part (B) and part (C), respectively. Identify the components that are used in the circuit between points X and Y .

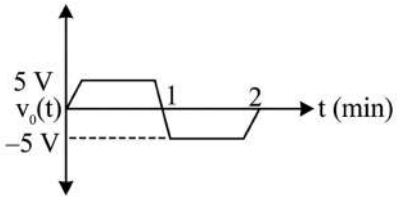
(A) Circuit used



Part(B) : Input voltage



Part(C) : Output voltage



Option 1:



Option 2:



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Option 3:



Option 4:



Correct Answer:



Solution:

For the +ve half cycle of input

Middle branch has ideal diode in forward bias and Zener diode in reverse bias Thus, we get constant 5V output due to Zener breakdown. For this situation component between $x \& y$ should act as open circuit.

For -ve half cycle of input middle branch has ideal diode in reverse bias, which makes it open branch without any current. for this situation component between $\mathrm{x \& y}$ should give a constant -5 V output due to Zener breakdown.

(Note : The input side must contain some resistance else V_{in} is equal to V_{out} for every cases)

Q. 17 When a coil is placed in a time dependent magnetic field the power dissipated in it is P . The number of turns, area of the coil and radius of the coil wire are N , A and r respectively. For a second coils number of turns, area of the coil and radius of the coil wire are $2\ \mathrm{N}, 2\ \mathrm{A}$ and 3 r respectively. When the first coil is replaced with second coil the power dissipated in it is $\sqrt{2}\alpha \mathrm{P}$. The value of α is _____ .

Option 1:

36

Option 2:

$128\sqrt{2}$

(

Option 3:

16

Option 4:

64

Correct Answer:

36

Solution:

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Power dissipated : $\mathrm{P}=\frac{\varepsilon^2}{\mathrm{R}}$
Induced emf $\varepsilon=-\mathrm{NA}\frac{\mathrm{dB}}{\mathrm{dt}}$
Resistance of coil : $\mathrm{R}=\rho\frac{\ell}{\pi \mathrm{r}^2}$
Now : $\ell=\mathrm{N}\left(2\pi \mathrm{R}_{\text{coil}}\right)$
Also : $A=\pi R_{\text{coil}}^2 \Rightarrow R_{\text{coil}}=\sqrt{\frac{A}{\pi}}$

Thus ; $\mathrm{R}=\frac{\rho \mathrm{N}(2\pi)}{\pi \mathrm{r}^2}\sqrt{\frac{\mathrm{A}}{\pi}}$

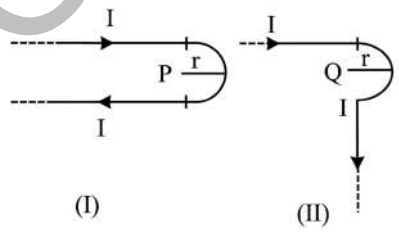
$$\mathrm{P}=\frac{\mathrm{N}^2\,\mathrm{A}^2\left(\frac{\mathrm{dB}}{\mathrm{dt}}\right)^2}{\frac{2\rho\,\mathrm{N}}{\mathrm{r}^2}\sqrt{\frac{\mathrm{A}}{\pi}}} \Rightarrow \mathrm{P}\propto \mathrm{NA}^{3/2}\mathrm{r}^2 \ldots\ldots\ldots(1)$$

$$\mathrm{P}'\propto(2\,\mathrm{N})(2\,\mathrm{A})^{3/2}(3\mathrm{r})^2 \ldots\ldots(2)$$

$$\frac{(2)}{(1)}:\frac{\mathrm{P}'}{\mathrm{P}}=36\sqrt{2}\Rightarrow \mathrm{P}'=(\sqrt{2})(36)\mathrm{P}$$

Thus : $\alpha=36$

Q. 18 Two identical long current carrying wires are bent into the shapes shown in the following figures. If the magnitude of magnetic fields at the centres P and Q of a semicircular arc are B_1 and B_2 respectively, then the ratio $\frac{B_1}{B_2}$ is _____ .



Option 1:
$\frac{2+\pi}{1+\pi}$

Option 2:
$\frac{1+\pi}{1-\pi}$

Option 3:
$\frac{2+\pi}{1-\pi}$

Option 4:
$\frac{1+\pi}{2-\pi}$

Correct Answer:
$\frac{2+\pi}{1+\pi}$

Solution:

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$$B_1 = \frac{\mu_0 I}{4\pi r} \times 2 + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4r}\left(\frac{2}{\pi} + 1\right)$$

$$B_2 = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{4r}\left(\frac{1}{\pi} + 1\right)$$

$$\frac{B_1}{B_2} = \frac{2+\pi}{1+\pi}$$

Q. 19 For a thin symmetric prism made of glass (refractive index 1.5), the ratio of incident angle and minimum deviation will be _____

Option 1:
$3:4$

Option 2:
$3:2$

Option 3:
$2:1$

Option 4:
$1:2$.

Correct Answer:
$3:2$

Solution:
We have for thin prism minimum deviation

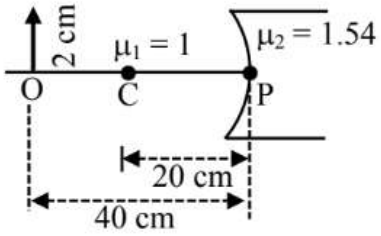
$$\delta_{\mathrm{m}} = (\mu - 1)\mathrm{A}$$

$$\Rightarrow \delta_{\mathrm{m}} = (1.5 - 1)\mathrm{A} \quad \Rightarrow \delta_{\mathrm{m}} = \frac{\mathrm{A}}{2}$$

Also, $i = \frac{A+\delta_m}{2} \Rightarrow i = \frac{A+\frac{A}{2}}{2} \Rightarrow i = \frac{3A}{4}$
Thus $\frac{\mathrm{i}}{\delta_{\mathrm{m}}} = \frac{3}{2}$

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Q. 20 Refer the figure given below. μ_1 and μ_2 are refractive indices of air and lens material. The height of image will be _____ cm .



Option 1:
1

Option 2:
0.5

Option 3:
1.2

Option 4:
0.25

Correct Answer:
1

Solution:

$$\frac{\mu_2}{\mathrm{v}} - \frac{\mu_1}{\mu} = \frac{\mu_2 - \mu_1}{\mathrm{R}}$$

$$\frac{1.54}{\mathrm{v}} - \frac{1}{(-40)} = \frac{1.54 - 1}{-20}$$

$$\frac{1.54}{\mathrm{v}} = -\frac{1}{40} - \frac{0.54}{20}$$

$$\frac{1.54}{\mathrm{v}} = \frac{-2.08}{40}$$

$$\mathrm{v} = -29.61 \text{ cm}$$

$$\frac{\mathrm{h_i}}{\mathrm{h_0}} = \frac{\mu_1 \mathrm{v}}{\mu_2 \mathrm{u}} = \frac{1(-29.61)}{1.54(-40)} = 0.48$$

$$\mathrm{h_i} = 0.48 \times 2 = 0.96 \approx 1 \text{ cm}$$

Q. 21 In a Young's double slit experiment, the intensity at some point on the screen is found to be $\frac{3}{4}$ times of the maximum of the interference pattern. The path difference between the interfering waves at this point is $\frac{\lambda}{x}$ where λ is wavelength of the incident light. The value of x is _____ .

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Correct Answer:
6

Solution:

$I_r = I_{\max} \cos^2 \frac{\phi}{2}$

$\frac{3I_{\max}}{4} = I_{\max} \cos^2 \frac{\phi}{2}$

$\cos \frac{\phi}{2} = \frac{\sqrt{3}}{2}$

$\frac{\phi}{2} = 30^\circ = \frac{\pi}{6}$

$\left(\phi = \frac{\pi}{3}\right)$

Path difference $\Delta x = \frac{\phi}{2\pi} \times \lambda$

$\Rightarrow \frac{\pi}{3 \times 2\pi} \times \lambda$

$\Delta x = \frac{\lambda}{6}$

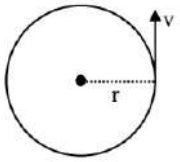
$x = 6$

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Q. 22 Using Bohr's model, calculate the ratio of the magnetic fields generated due to the motion of the electrons in the 2nd and 4th orbits of hydrogen atom _____ .

Correct Answer:
64

Solution:



$$B = \frac{\mu_0 I}{2\pi r}$$
$$I = \frac{ev}{2\pi r}$$
$$B \propto \frac{V}{r^2}$$
$$V \propto \frac{Z}{n}$$
$$r \propto \frac{n^n}{Z}$$
$$B \propto \frac{\left(\frac{Z}{n}\right)}{\left(\frac{n^2}{Z}\right)^2}$$
$$B \propto \frac{Z^3}{n^5}$$
$$\frac{B_1}{B_2} = \frac{n_2^5}{n_1^5} = \frac{4^5}{2^5} = 64$$

Q. 23 5 moles of unknown gas is heated at constant volume from $10^\circ C$ to $20^\circ C$. The molar specific heat of this gas at constant pressure $c_p = 8 cal/mol.\ ^\circ C$ and $R = 8.36\ J/mol.\ ^\circ C$. The change in the internal energy of the gas is _____ calorie.

Correct Answer:
300

Solution:
$C_p - C_v = R \Rightarrow C_v = (C_p - R)$
$C_v = 6 cal/mol^\circ C$
$\Delta U = nC_v \Delta T$
$= 5 \times 6 \times 10 = 300 cal$

Q. 24 If sunlight is focused on a paper using convex lens, it starts burning the paper in shortest time when the lens is kept at 30 cm above the paper. If the radius of curvature of the lens is 60 cm then the refractive index of the lens material is $\frac{\alpha}{10}$. The value of α is _____

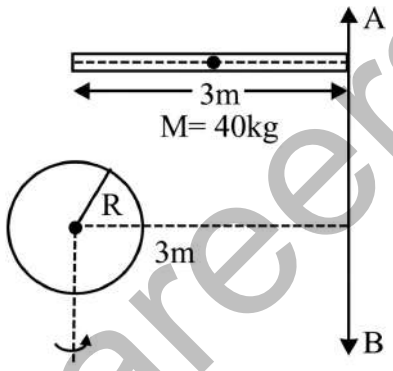
Correct Answer:
20

Solution:
To burn in minimum time distance of paper must be at focus.
So $f = 30\ cm$

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$$\frac{1}{30} = (\mu - 1)\left(\frac{1}{60} - \frac{1}{-60}\right)$$
$$\frac{1}{30} = \frac{\mu - 1}{30}$$
$$\mu = 2$$
$$\alpha = 20$$

Q. 25 Moment of inertia about an axis AB for a rod of mass 40 kg and length 3 m is same as that of a solid sphere of mass of 10 kg and radius R about an axis parallel to AB axis with separation of 3 m as shown in figure below. The value of R is given as $\sqrt{\frac{\alpha}{2}}$. The value of α is _____ .



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Correct Answer:
15

Solution:
$$\mathrm{I_{Rod}} = \mathrm{I_{ss}}$$
$$\frac{\mathrm{M_1\,L^2}}{3} = \left(\frac{2}{5}\,\mathrm{m_2 R^2} + \mathrm{m_2\,L^2}\right)$$
$$\frac{40 \times (3)^2}{3} = \frac{2}{5} \times 10 \times \mathrm{R^2} + 10 \times (3)^2$$
$$120 = 4\mathrm{R^2} + 90$$
$$4\mathrm{R^2} = 30$$
$$\mathrm{R} = \sqrt{\frac{15}{2}} = \sqrt{\frac{\alpha}{2}}$$
$$\alpha = 15$$

Chemistry

Q. 1 The ratio of mass percentage ($\mathrm{w/w}$) of $\mathrm{C:H}$ in a hydrocarbon is $12:1$. It has two carbon atoms. The weight (in g) of $\mathrm{CO_2(\,g)}$ formed when 3.38 g of this hydrocarbon is completely burnt in oxygen is :
(Given : Molar mass in $\mathrm{gmol^{-1}}$ $\mathrm{C:12, H:1, O:16}$)

Option 1:
5.68

Option 2:
11.44

Option 3:
22.74

Option 4:
17.05

Correct Answer:
11.44

Solution:
Molecular formula of hydrocarbon is $\mathrm{C_2H_2}$
Moles of $\mathrm{C_2H_2} = \frac{3.38}{26}$

$\mathrm{C_2H_2(\ g) + \frac{5}{2}O_2(\ g) \rightarrow 2CO_2(\ g) + H_2O(l)}$

Moles of $\mathrm{CO_2}$ produced $= \left(\frac{3.38}{26} \times 2\right)$

$\mathrm{Mass\ of\ CO_2\ produced\ } = \left(\frac{3.38}{26} \times 2\right) \times 44 = 11.44$

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Q. 2 The first and second ionization constants of a weak dibasic acid $\mathrm{H_2\ A}$ are 8.1×10^{-8} and 1.0×10^{-13} respectively. 0.1 mol of $\mathrm{H_2\ A}$ was dissolved in 1L of 0.1 M HCl solution. The concentration of $\mathrm{HA^-}$ in the resultant solution is:

Option 1:
0.1 M

Option 2:
$9.53 \times 10^{-6}\mathrm{M}$

Option 3:
$8.1 \times 10^{-8}\mathrm{M}$

Option 4:
$1.0 \times 10^{-13}\mathrm{M}$

Correct Answer:
$8.1 \times 10^{-8}\mathrm{M}$

Solution:

$$\underset{0.1\mathrm{M}}{\mathrm{HCl}} \rightarrow \underset{0.1\mathrm{M}}{\mathrm{H}^{+}} + \underset{0.1\mathrm{M}}{\mathrm{Cl}^{-}}$$

$$\underset{0.1-\mathrm{x}}{\mathrm{H_2\ A}} \rightleftharpoons \underset{\mathrm{x}+0.1}{\mathrm{H}^{+}} + \underset{\mathrm{x}}{\mathrm{HA}^{-}}$$

(Due to common ion effect x is very less)

$$\mathrm{Ka_1} = \frac{\left[\mathrm{H}^{+}\right]\left[\mathrm{HA}^{-}\right]}{\left[\mathrm{H_2\ A}\right]}$$

$$\mathrm{Ka_1} = \frac{0.1 \times \left[\mathrm{HA}^{-}\right]}{0.1}$$

$$\left[\mathrm{HA}^{-}\right] = \mathrm{Ka_1} = 8.1 \times 10^{-8}$$

Q. 3 $\mathrm{SF_4}$ is isostructural with :

A. $\mathrm{BrF_4^{\ominus}}$

B. $\mathrm{CH_4}$

C. $\mathrm{IF_4^{\oplus}}$

D. $\mathrm{XeF_4}$

E. $\mathrm{XeO_2\ F_2}$

Choose the correct answer from the options given below:

Option 1:
C only

Option 2:
C and E only

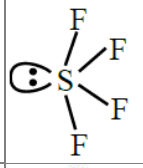
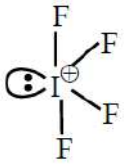
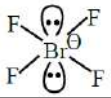
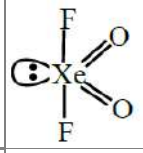
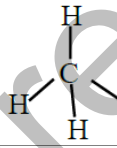
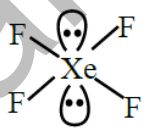
Option 3:
A and D only

Option 4:
B and E only

Correct Answer:
C and E only

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Solution:

Species	Structure / Shape
SF_4	 → see-saw
$IF_4^{\oplus}$	 → see-saw
$BrF_4^{\ominus}$	 → square planar
XeO_2F_2	 → see-saw
CH_4	 → Tetrahedral
XeF_4	 → square planar

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Q. 4 Gas 'A' undergoes change from state ' X ' to state ' Y '. In this process, the heat absorbed and work done by the gas is 10 J and 18 J respectively. Now gas is brought back to state ' X ' by another process during which 6 J of heat is evolved. In the reverse process of ' Y ' to ' X ',

Option 1:
18 J of the work is done by the gas ' A '.

Option 2:
2 J of the work is done by the gas ' A '.

Option 3:
12 J of the work is done on the gas 'A' by the surrounding.

Option 4:
14 J of the work is done on the gas ' A ' by the surrounding.

Correct Answer:
14 J of the work is done on the gas ' A ' by the surrounding.

Solution:
For process $\mathrm{X} \rightarrow \mathrm{Y}$

$\mathrm{w} = -18\ \mathrm{J}$
$\mathrm{q} = +10\ \mathrm{J}$
$\Delta \mathrm{U} = \mathrm{q} + \mathrm{w}$
$\Delta \mathrm{U}_{\mathrm{x} \rightarrow \mathrm{Y}} = 10 + (-18) = -8\ \mathrm{J}$

For reverse process :

$\Delta U_{Y \rightarrow X} = 8\ \mathrm{J}$
$q = -6\ \mathrm{J}$
$\Delta U = q + w$
$8 = -6 + w$
$w = 14\ \mathrm{J}$

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Q. 5 Solution A is prepared by dissolving 1 g of a protein (molar mass $= 50000\ \mathrm{g\ mol^{-1}}$) in 0.5 L of water at 300 K . Its osmotic pressure is x bar. Solution B is made by dissolving 2 g of same protein in 1 L of water at 300 K . Osmotic pressure of solution B is y bar. Entire solution of A is mixed with entire solution of B at same temperature. The osmotic pressure of resultant solution is z bar. x, y and z respectively are :

$\left(\mathrm{R} = 0.083\ \mathrm{L\ bar\ mol^{-1}\ K^{-1}}\right)$

Option 1:
$9.96 \times 10^{-4}; 9.96 \times 10^{-4}; 9.96 \times 10^{-4}$

Option 2:
$9.96 \times 10^{-4}; 9.96 \times 10^{-4}; 19.92 \times 10^{-4}$

Option 3:
$4.98 \times 10^{-4}; 4.98 \times 10^{-4}; 9.96 \times 10^{-4}$

Option 4:
$4.98 \times 10^{-4}; 4.98 \times 10^{-4}; 4.98 \times 10^{-4}$

Correct Answer:
$9.96 \times 10^{-4}; 9.96 \times 10^{-4}; 9.96 \times 10^{-4}$

Solution:

$$'x' = \frac{1}{50,000 \times 0.5} \times R \times 300 = 9.96 \times 10^{-4} \text{bar}$$

$$'y' = \frac{2}{50,000 \times 1} \times R \times 300 = 9.96 \times 10^{-4} \text{bar}$$

$$'z' = \frac{'x' \times \frac{1}{2} + 'y' \times 1}{\frac{3}{2}} = 9.96 \times 10^{-4} \text{bar}$$

Q. 6 At $25^{\circ}C, 20.0\ mL$ of 0.2 M weak monoprotic acid HX is titrated against 0.2 M NaOH . The pH of the solution (a) at the start of the titration (when NaOH has not been added) and (b) when 10 mL of NaOH is added respectively, are :

Given : $K_a = 5 \times 10^{-4}, pK_a = 3.3, \alpha \ll 1$

Option 1:
$0.7, 2.0$

Option 2:
$2.0, 3.3$

Option 3:
$1.1, 2.2$

Option 4:
$3.0, 2.2$

Correct Answer:
$2.0, 3.3$

Solution:
For $0.2M, 20mlHX$ solution,

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$$\text{pH of weak acid } = \frac{1}{2}[\text{pk}_a - \log c]$$
$$= \frac{1}{2}[4 - \log 5 - \log 0.2]$$
$$= \frac{1}{2}[4 - \log 5 + \log 5]$$
$$= 2$$
$$\text{HX} + \quad \text{NaOH} \longrightarrow \text{NaA} + \text{H}_2\text{O}$$
$$20\text{ml}, 0.2\text{M} \qquad 10\text{ml}, 0.2\text{M}$$
$$4 \text{ m mole} \quad - \quad 2 \text{ m mole}$$
$$2 \text{ m mole} \qquad 2 \text{ m mole}$$
$$[\text{HX} + \text{NaA}] \Rightarrow \text{acidic buffer}$$
$$\text{pH} = \text{pK}_a + \log \frac{\text{salt}}{\text{acid}}$$
$$\text{pH} = 3.3 + \log \frac{2}{2}$$

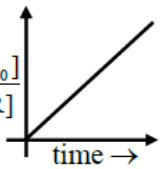
Q. 7 Consider the reaction $\text{aX} \rightarrow \text{bY}$, for which the rate constant at 30°C is $1 \times 10^{-3}\ \text{mol}^{-1}\text{Ls}^{-1}$. Which of the following statements are true ?

A. When concentration of ' X ' is increased to four times, the rate of reaction becomes 16 times.

B. The reaction is a second order reaction.

C. The half-life period is independent of the concentration of X .

D. Decomposition of N_2O_5 is an example of the above reaction.

E. 

is valid for the above reaction.

Choose the correct answer from the options given below:

Option 1:
A and B only

Option 2:
A, B and C only

Option 3:
A,B,D and E only

Option 4:
C and D only

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Correct Answer:
A and B only

Solution:
Rate of constant $(\mathrm{k}) = 10^{-3}\mathrm{M}^{-1}\mathrm{sec}^{-1}$

$\because 1 - \mathrm{n} = -1 \left[\text{ unit of } \mathrm{k} = \mathrm{M}^{1-\mathrm{n}}\mathrm{sec}^{-1}\right]$

$\mathrm{n} = 2 \Rightarrow 2^{\mathrm{nd}}$ order reaction

$\text{rate } = \mathrm{k}[\mathrm{A}]^2$

(A) If concentration of A is 4 times then reaction will become 16 times.
(B) Order of reaction is 2
(C) Half-life of the reaction is independent of the concentration of the reactant for 1st order, not for 2nd order reaction.
(D) Decomposition of $\mathrm{N_2O_5}$ is example of 1st order reaction

(E) Graph of $\ell \frac{[R]_0}{[R]_t}$ vs t is straight line for 1st order not for 2nd order reaction.

So statement (A) \& (B) are correct

Q. 8 The correct set that contains all kinds (basic, acidic, amphoteric and neutral) of oxides is :

Option 1:
$\mathrm{Na_2O}, \mathrm{K_2O}, \mathrm{Al_2O_3}$ and $\mathrm{As_2O_3}$

Option 2:
$\mathrm{Al_2O_3}, \mathrm{As_2O_3}, \mathrm{CO}$ and NO

Option 3:
$\mathrm{K_2O}, \mathrm{Cl_2O_7}, \mathrm{As_2O_3}$ and NO

Option 4:
$\mathrm{Na_2O}, \mathrm{N_2O}, \mathrm{Al_2O_3}$ and CO

Correct Answer:
$\mathrm{K_2O}, \mathrm{Cl_2O_7}, \mathrm{As_2O_3}$ and NO

Solution:

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Basic Oxides : Na_2O, K_2O

Acidic Oxides : Cl_2O_7

Neutral Oxides : N_2O, NO \& CO

Amphoteric Oxides : Al_2O_3, As_2O_3

Q. 9 Given below are two statements :
Statement-I : The second ionization enthalpy of B, Al and Ga is in the order of $B > Al > Ga$.

Statement-II : The correct order in terms of first ionization enthalpy is $Si < Ge < Pb < Sn$.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Both Statement-I and Statement-II are false.

Solution:
Order of II ionization energy for boron family elements is :

$$\underset{(2427KJ/mol)}{B} > \underset{(1979KJ/mol)}{Ga} > \underset{(1816KJ/mol)}{Al}$$

Order of I ionization energy for carbon family elements is :

$$\underset{(786KJ/mol)}{Si} > \underset{(761KJ/mol)}{Ge} > \underset{(715KJ/mol)}{Pb} > \underset{(708KJ/mol)}{Sn}$$

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Q. 10 Given below are two statements :
Statement-I : Among $\mathrm{Zn}, \mathrm{Mn}, \mathrm{Sc}$ and Cu , the energy required to remove the third valence electron is highest for Zn and lowest for Sc .

Statement-II : The correct order of the following complexes in terms of CFSE is

$[\mathrm{Co}(\mathrm{H_2O})_6]^{2+} < [\mathrm{Co}(\mathrm{H_2O})_6]^{3+} < [\mathrm{Co}(\mathrm{en})_3]^{3+}$.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Both Statement-I and Statement-II are true.

Solution:
Statement I

	$\mathrm{IE_3}(\mathrm{kJ/mol})$
Sc	2393
Mn	3260
Cu	3556
Zn	3837

$\mathrm{Sc} < \mathrm{Mn} < \mathrm{Cu} < \mathrm{Zn}\ (3^{\mathrm{rd}}\ \mathrm{IE})$

Statement II

$\mathrm{CFSE} \Rightarrow \mathrm{SFL} > \mathrm{WFL}$
$(\mathrm{en} > \mathrm{H_2O})$

CFSE $\propto$ Charge on metal

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$$Co^{+2} < Co^{+3}$$

$$[Co(H_2O)_6]^{2+} < [Co(H_2O)_6]^{3+} < [Co(en)_3]^{3+}$$

Q. 11 Which of the following complexes will show coordination isomerism ?

(A) $[Ag(NH_3)_2]\,[Ag(CN)_2]$

(B) $[Co(NH_3)_6]\,[Cr(CN)_6]$

(C) $[Co(NH_3)_6]\,[Co(CN)_6]$

(D) $[Fe(NH_3)_6]\,[Co(CN)_6]$

(E) $[Co(NH_3)_6]\,[Fe(CN)_6]$

Choose the correct answer from the options given below :

Option 1:
B, C and D Only

Option 2:
B, D and E Only

Option 3:
A, C and D Only

Option 4:
C, D and E Only

Correct Answer:
C, D and E Only

Solution:
Compound, $[Ag(NH_3)_2]\,[Ag(CN)_2]$ cannot show coordination isomerism. All other complexes (B, C, D and E) will show co-ordination isomerism.

Q. 12 Complete combustion of X g of an organic compound gave 0.25 g of CO_2 and 0.12 g of H_2O . If the \% of carbon is 25% and of hydrogen is 4.89%, then $X = ____ \times 10^{-3}$ g. (Nearest integer) (Molar mass of C, H and O are 12,1 and $16\ g\ mol^{-1}$ respectively.)

Option 1:
273

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Option 2:
27

Option 3:
2730

Option 4:
227

Correct Answer:
273

Solution:
Mass of carbon $= \frac{0.25}{44} \times 12$

$$\text{Mass \% of carbon} = \frac{\text{Mass of carbon}}{x} \times 100$$

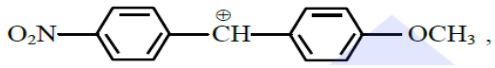
$$x = \frac{\frac{0.25 \times 12}{44} \times 100}{25}$$

$$x = \frac{12}{44}$$

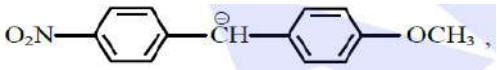
$$x = 0.2727 = 272.7 \times 10^{-3}$$

$$\simeq 273 \times 10^{-3}$$

Q. 13 Given below are two statements :
Statement I : In



The carbocation is stabilised by +R effect of $-\mathrm{OCH_3}$ group.

Statement II: In 

The carbanion is stabilised by -R effect of $-\mathrm{NO_2}$ group. In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

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Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:
Both statements are correct

Q. 14 The compound (X) on
(i) on heating in the presence of anhydrous $AlCl_3$ and HCl gas gives 2, 4-dimethyl pentane
(ii) aromatization gives toluene and
(iii) cyclisation gives methyl cyclohexane The correct name of compound (X) is :

Option 1:
Hept-2-ene

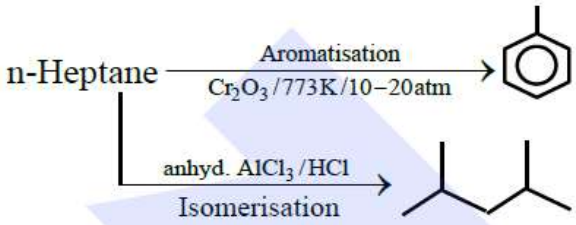
Option 2:
Hept-1,3,5-triene

Option 3:
Heptane

Option 4:
Hept-2,4,6-triene

Correct Answer:
Heptane

Solution:



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Q. 15 Correct statements regarding alkyl halides (R-X) among the following are:
(A) Alcohol being less polar solvent as compared to water, alcoholic KOH favours elimination reaction with $\mathrm{R}-\mathrm{X}$.
(B) Order of reactivity towards $\mathrm{S_N1}$ mechanism is
$\mathrm{C_6H_5-CH_2-Cl} > \mathrm{C_6H_5-CHCl-C_6H_5}$.
(C) Non substituted aryl halides exhibit properties similar to alkyl halides.
(D) Vinyl chloride is an example of haloalkene and allyl chloride is an example of haloalkyne.
(E) $\mathrm{R}-\mathrm{Cl}$ can be prepared by reacting $\mathrm{R}-\mathrm{OH}$ with $\mathrm{SOCl_2}$ but $\mathrm{Ar}-\mathrm{Cl}$ cannot be prepared by reacting $\mathrm{Ar}-\mathrm{OH}$ with $\mathrm{SOCl_2}$.

Choose the correct answer from the options given below :

Option 1:
A, B and C Only

Option 2:
B and D Only

Option 3:
A and E Only

Option 4:
D and E Only

Correct Answer:
A and E Only

Solution:
$\mathrm{Ar}-\mathrm{Cl}$ and $\mathrm{R}-\mathrm{Cl}$ shows different chemical properties

Rate of $\mathrm{S_N}^1 \propto$ stability of first formed carbocation
Water is more polar than alcohol
Vinyl alcohol, allyl alcohol, both are alkene unsaturated alcohols

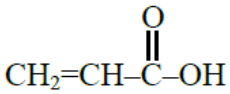
$$\mathrm{R-OH} \xrightarrow{\mathrm{SOCl_2}} \mathrm{R-Cl}$$

Phenol does not give $\mathrm{Ph}-\mathrm{Cl}$ with $\mathrm{SOCl_2}$.

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Q. 16 An organic compound " x " where molar ratio of C, O and H are equal, on treatment with $50\% \mathrm{KOH}$ under reflux followed by acidification produced " y ". The most likely structure of " y " is:
[Molar mass of ' x ' is $58\ \mathrm{g\ mol^{-1}}$]

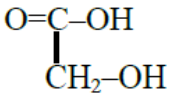
Option 1:



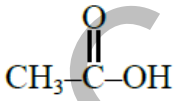
Option 2:

$\mathrm{CH_3-CH=CH-CH=O}$

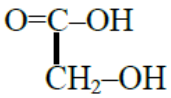
Option 3:



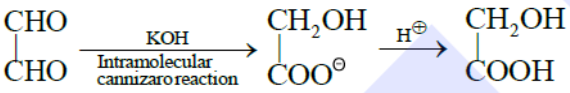
Option 4:



Correct Answer:

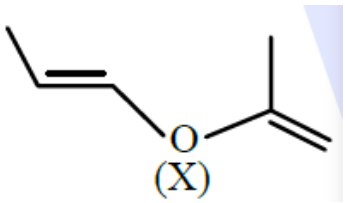


Solution:



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Q. 17 A molecule (X) with following structure under mild acidic condition is hydrolysed to produce (Y) and (Z). Identify the correct statements about (Y) and (Z).



(A)Both (Y) and (Z) have same molar mass

(B) (Y) and (Z) can be distinguished from each other by $\mathrm{NaHCO_3}$

(C) (Y) and (Z) react with HCN with same rates

(D) (Y) and (Z) undergo addition reaction with 2, 4-DNP

Choose the correct answer from the options given
below :

Option 1:
A, B and C Only

Option 2:
B and C Only

Option 3:
C and D Only

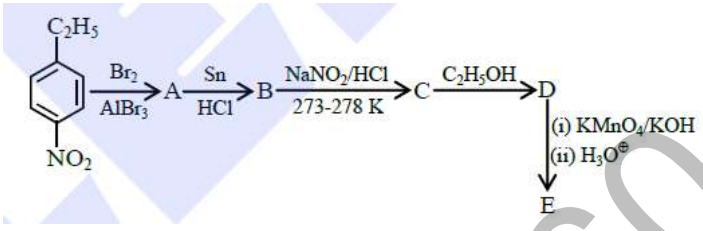
Option 4:
A and D Only

Correct Answer:
A and D Only

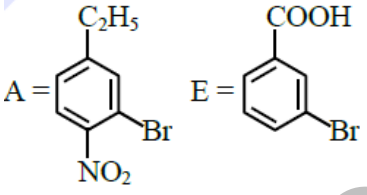
Solution:
Y will give positive fehling's test
Molar mass of $[\mathrm{Z}]\ (\mathrm{C_3H_6O}) = 58$
Molar mass of $[\mathrm{Y}]\ (\mathrm{C_3H_6O}) = 58$

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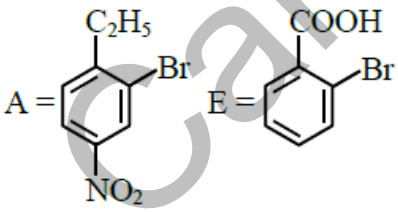
Q. 18 Identify compounds A and E in the following reaction sequence.



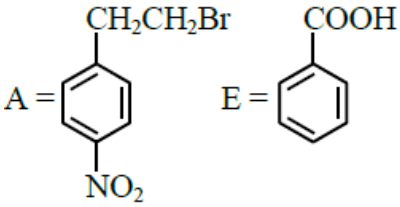
Option 1:



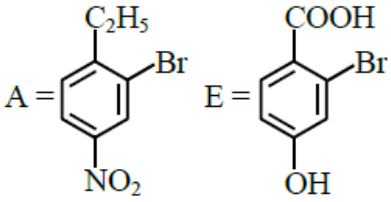
Option 2:



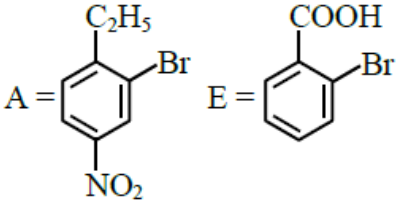
Option 3:



Option 4:

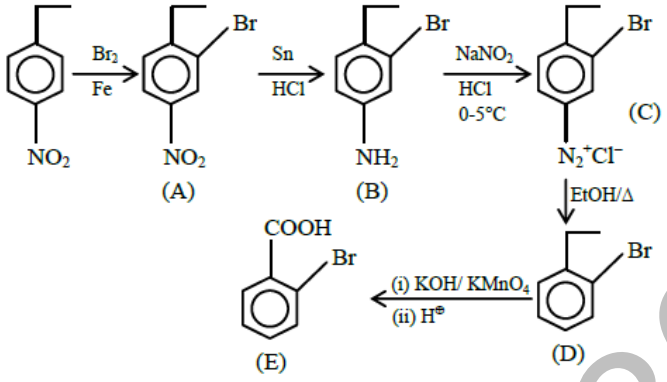


Correct Answer:



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Solution:



Q. 19 Identify the correct pair having amino acid (A) and the hormone (B) that is iodinated derivative of the
amino acid (A). (T and Y represent one letter code for amino acids)

Option 1:

Amino acid (A)	Hormone (B)
T	Insuline

Option 2:

Amino acid (A)	Hormone (B)
T	Thyroxine

Option 3:

Amino acid (A)	Hormone (B)
Y	Thyroxine

Option 4:

Amino acid (A)	Hormone (B)
Y	Insuline

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Correct Answer:

Amino acid (A)	Hormone (B)
Y	Thyroxine

Solution:
Thyroxine is derivative of Tyrosine and single letter symbol of Tyrosine amino acid is Y.

Q. 20 Among $Fe^{2+}, Fe^{3+}, Cr^{2+}$ and Zn^{2+}, the ion that shows positive borax bead test and with highest ionisation enthalpy is:

Option 1:
Fe^{2+}

Option 2:
Zn^{2+}

Option 3:
Cr^{2+}

Option 4:
Fe^{3+}

Correct Answer:
Fe^{3+}

Solution:
Zn^{2+} does not show Borax Bead Test.
Among remaining ions; Fe^{3+} has maximum ionization energy.

Q. 21 The surface of sodium metal is irradiated with radiation of wavelength x nm . The kinetic energy of ejected electrons is 2.8×10^{-20} J. The work function of sodium is 2.3 eV . The value of x is
____ $\times 10^2$ nm. (Nearest integer)
(Given : $h = 6.6 \times 10^{-34}$ J s; $1eV = 1.60 \times 10^{-19}$ J; $c = 3.0 \times 10^8$ ms^{-1})

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Correct Answer:
5

Solution:
for photoelectric effect :

$$\frac{hc}{\lambda} = W + K.E.$$
$$\frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{x} = 2.3 \times 1.6 \times 10^{-19} + 2.8 \times 10^{-20}$$
$$\frac{19.8 \times 10^{-26}}{x} = 3.96 \times 10^{-19}$$
$$x = 5 \times 10^{-7}$$
$$x = 500 \times 10^{-9}\ \mathrm{m}$$
$$x = 5 \times 10^{2}\ \mathrm{nm}$$

Q. 22 Consider the following gas phase reaction being carried out in a closed vessel at $25^{\circ}\mathrm{C}$.

$$2\ \mathrm{A}(\mathrm{g}) \rightarrow 4\ \mathrm{B}(\mathrm{g}) + \mathrm{C}(\mathrm{g})$$

$\frac{\text{time}}{(\text{min})}$	total pressure of the $\frac{\text{system}}{(\text{mm Hg})}$
30	300
∞	600

The pressure of $\mathrm{C}(\mathrm{g})$ at 30 minutes time interval would be _____ mm Hg. (Nearest integer)

Correct Answer:
20

Solution:
$2\ \mathrm{A}(\mathrm{g}) \rightarrow 4\ \mathrm{B}(\mathrm{g}) + \mathrm{C}(\mathrm{g})$
p°

$\mathrm{t} = 30\ \mathrm{min}\ \mathrm{p}^{\circ} - 2\mathrm{x} \quad 4\mathrm{x}$
$\mathrm{t} = \infty$
$\mathrm{p}_{\infty} = 2\mathrm{p}^{\circ} + \mathrm{p}^{\circ}/2 = 600$
$5\mathrm{p}^{\circ}/2 = 600$
$\mathrm{P}^{\circ} = 240\ \mathrm{mmHg}$
$\mathrm{Att} = 30\ \mathrm{min}, \mathrm{p} = 300 = \mathrm{p}^{\circ} - 2\mathrm{x} + 4\mathrm{x} + \mathrm{x}$
$300 = 240 + 3\mathrm{x}$
$3\mathrm{x} = 60$
$\mathrm{x} = 20$

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Pressure of $C(g) = 20\ mmHg$

Q. 23 Consider the following two half-cell reactions along with the standard reduction potential given :

$$CO_2 + 6H^+ + 6e^- \rightarrow CH_3OH + H_2O \quad E^\circ_{red} = 0.02\ V$$
$$\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O \quad E^\circ_{red} = 1.23\ V$$

The fuel cell was set up using the above two reactions such that the cell operates under the standard condition of 1 bar pressure and 298 K temperature. The fuel cell works with 80% efficiency. If the work derived from the cell using 1 mol of CH_3OH is used to compress an ideal gas isothermally against a constant pressure of 1 kPa , then the change in the volume of the gas, $\Delta V =$ ____ m^3. (Nearest integer)
Given : $F = 96500 C mol^{-1}$

Correct Answer:
560

Solution:
$E^\circ_{cell} = [1.23 - 0.02] = 1.21\ V$
$\Delta G^\circ = -6 \times 96500 \times 1.21\ J$
$W = \frac{80}{100} \times \Delta G^\circ = -P\Delta\ V$
$0.8 \times 6 \times 96500 \times 1.21 = 1 \times 10^3(\Delta\ V)$
$\Delta V = 560.472\ m^3$

Q. 24 Number of paramagnetic ions among the following d- and f-block metal ions is _____ .

$Mn^{2+}, Cu^{2+}, Zn^{2+}, Yb^{2+}, Sc^{3+}, La^{3+}, Gd^{3+}, Lu^{3+}, Ti^{4+}$
Ce^{4+}

(Atomic number of $Mn = 25, Cu = 29, Zn = 30,$

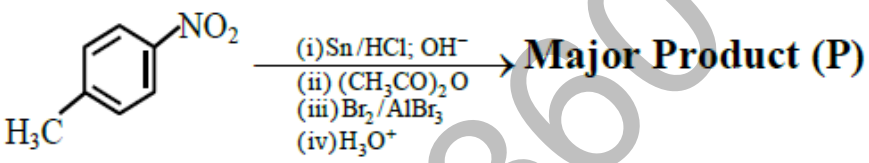
$Yb = 70, Sc = 21, La = 57, Gd = 64, Lu = 71$
$Ti = 22, Ce = 58$)

Correct Answer:
3

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Solution:
Paramagnetic : $\mathrm{Mn}^{+2}, \mathrm{Cu}^{+2}, \mathrm{Gd}^{+3}$
Diamagnetic : $\mathrm{Zn}^{+2}, \mathrm{Yb}^{+2}, \mathrm{Sc}^{+3}, \mathrm{La}^{+3}, \mathrm{Lu}^{+3}, \mathrm{Ti}^{+4}, \mathrm{Ce}^{+4}$

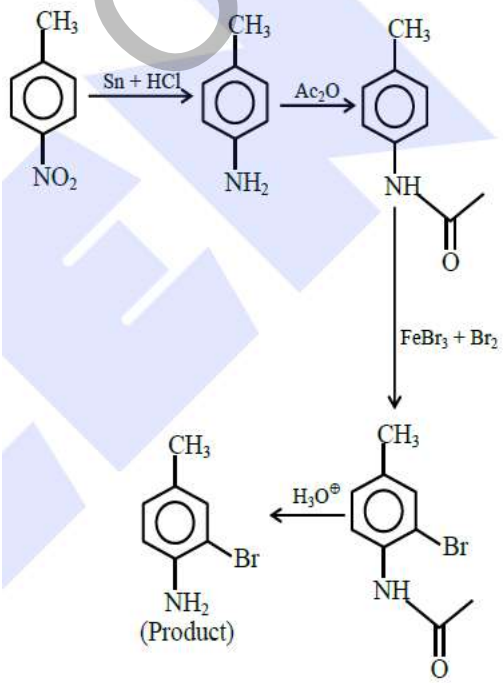
Q. 25 Consider the following reactions sequence



When the product (P) is subjected to Carius analysis using AgNO_3, 1.0 g of the product (P) will produce ____ g of the precipitate of AgBr .
(Nearest interger)
(Given : molar mass in gmol^{-1} $\mathrm{C}: 12, \mathrm{H}: 1, \mathrm{O}: 16,\ \mathrm{N}: 14, \mathrm{Br}: 80, \mathrm{Ag}: 108$)

Correct Answer:
1

Solution:



Molar mass of product $= 186$
Mass of $\mathrm{AgBr} = \frac{1}{186} \times 188 \approx 1$

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Maths

Q. 1 Let α, β be the roots of the equation $x^2 - 3x + r = 0$, and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 + 3x + r = 0$. If the roots of the equation $x^2 + 6x = m$ are $2\alpha + \beta + 2\mathrm{r}$ and $\alpha - 2\beta - \frac{\mathrm{r}}{2}$, then m is equal to:

Option 1:
-135

Option 2:
-567

Option 3:
135

Option 4:
567

Correct Answer:
567

Solution:
$\alpha + \beta = 3$
$\frac{\alpha}{2} + 2\beta = -3$

On solving, we get $\alpha = 6; \beta = -3$

Product of roots $= \alpha\beta = r \Rightarrow r = -18$

Now for $\mathrm{x}^2 + 6\mathrm{x} - \mathrm{m} = 0$

Product of roots $= -m$

$= (2\alpha + \beta + 2\mathrm{r})\left(\alpha - 2\beta - \frac{\mathrm{r}}{2}\right)$
$\Rightarrow -\mathrm{m} = (-27)(21)$
$\Rightarrow \mathrm{m} = 567$

Q. 2 Let the circle $C_1 : |z| = r$ and $C_2 : |z - 3 - 4i| = 5, z \in \mathbb{C}$, be such that C_2 lies within C_1. If z_1 moves on $\mathrm{C}_1, \mathrm{z}_2$ moves on C_2 and $\min |\mathrm{z}_1 - \mathrm{z}_2| = 2$, then max $|z_1 - z_2|$ is equal to :-

Option 1:
12

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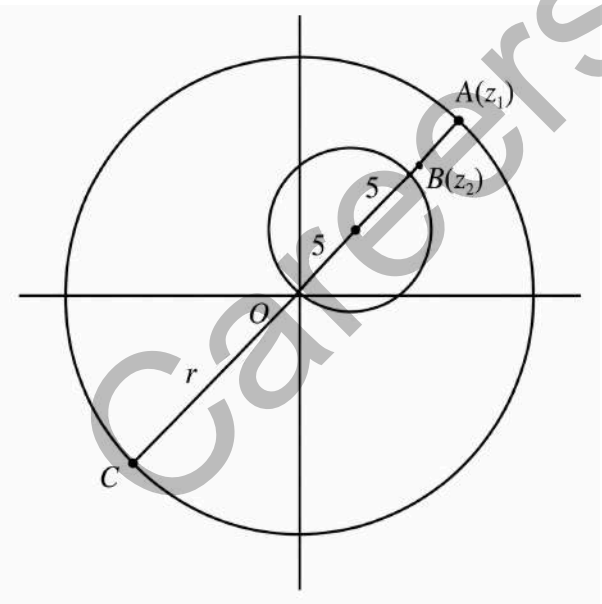
Option 2:
17

Option 3:
22

Option 4:
24

Correct Answer:
22

Solution:



$|z| = r$
$OA = r$
$\text{minimum}\,|z_1 - z_2| = 2$
$\Rightarrow r - 10 = 2 \Rightarrow r = 12$

maximum value of CB

$|z_1 - z_2| = r + 10 = 12 + 10 = 22$

Q. 3 If the system of equations

$$x + 5y + 6z = 4$$
$$2x + 3y + 4z = 7$$
$$x + 6y + az = b$$

has infinitely many solutions, then the point (a, b) lies on the line

Option 1:
$y - x = 3$

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Option 2:
$x - y = 3$

Option 3:
$x + y = 11$

Option 4:
$x + y = 12$

Correct Answer:
$x - y = 3$

Solution:

$$\mathrm{D} = \begin{vmatrix} 1 & 5 & 6 \\ 2 & 3 & 4 \\ 1 & 6 & \mathrm{a} \end{vmatrix} = 0$$

$$\Rightarrow \mathrm{a} = \frac{50}{7}$$

$$\mathrm{D}_2 = 0 \Rightarrow \begin{vmatrix} 1 & 5 & 4 \\ 2 & 3 & 7 \\ 1 & 6 & \mathrm{b} \end{vmatrix} = 0$$

$$\Rightarrow \mathrm{b} = \frac{29}{7}$$

$\therefore (\mathrm{a}, \mathrm{b})$ lies on $\mathrm{x} - \mathrm{y} = 3$

Q. 4 Let $a_1, a_2, a_3, \ldots\ldots$ be an A.P. and $g_1 = a_1, g_2, g_3, \ldots\ldots$ be an increasing G.P. If $\mathrm{a}_1 = \mathrm{a}_2 + \mathrm{g}_2 = 1$ and $\mathrm{a}_3 + \mathrm{g}_3 = 4$, then $\mathrm{a}_{10} + \mathrm{g}_5$ is equal to :

Option 1:
81

Option 2:
76

Option 3:
62

Option 4:
55

Correct Answer:
55

Solution:

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$\text{A.P. } : 1, a_2, a_3, \ldots$

$\text{G.P. } : 1, g_2, g_3, \ldots$

$a_2 + g_2 = 1 \Rightarrow 1 + d + r = 1 \Rightarrow d + r = 0$

$1 + 2d + r^2 = 4 \Rightarrow r^2 - 2r - 3 = 0$

$\Rightarrow r = 3, -1 \text{ (reject)}$

$\therefore d = -3$

$\therefore a_{10} = 1 + 9d = 1 + 9(-3) = -26$

$g_5 = 1.r^4 = 3^4 = 81$

$\therefore a_{10} + g_5 = 55$

Q. 5 The sum $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \ldots$ up to 8 terms, is :

Option 1:
70

Option 2:
71

Option 3:
72

Option 4:
73

Correct Answer:
71

Solution:

$$T_r = \frac{1^3 + 2^3 + 3^3 + \ldots\ldots r^3}{1 + 3 + 5 + \ldots\ldots + (2r - 1)} = \frac{\left(\frac{r(r+1)}{2}\right)^2}{r^2}$$

$$= \frac{r^2 + 2r + 1}{4}$$

$$S_n = \sum_{r=1}^{n} T_r$$

$$S_n = \frac{1}{4}\sum \left(r^2 + 2r + 1\right)$$

$$= \frac{1}{4}\left[\frac{n(n+1)(2n+1)}{6} + 2\frac{n(n+1)}{2} + n\right]$$

$$S_8 = \frac{1}{4}\left[\frac{8 \times 9 \times 17}{6} + 8 \times 9 + 8\right]$$

$$= \frac{1}{4}[204 + 72 + 8] = 71$$

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Q. 6 If for $3 \leq r \leq 30$,

$$\left({}^{30}C_{30-\tau}\right) + 3\left({}^{30}C_{31-\tau}\right) + 3\left({}^{30}C_{32-\tau}\right) + \left({}^{30}C_{33-\tau}\right) = {}^{m}C_{r},$$

then m equals :

Option 1:
31

Option 2:
32

Option 3:
33

Option 4:
34

Correct Answer:
33

Solution:
$${}^{3}C_{3} \cdot {}^{30}C_{30-r} + {}^{3}C_{2} \cdot {}^{30}C_{31-r} + {}^{3}C_{1} \cdot {}^{30}C_{32-r} + {}^{3}C_{0} \cdot {}^{30}C_{33-r}$$
$$= {}^{33}C_{33-r} \equiv {}^{m}C_{r} \equiv {}^{m}C_{m-r}$$
$$\Rightarrow m = 33$$

Q. 7 Let p_n denote the total number of triangles formed by joining the vertices of an n -side regular polygon. If $p_{n+1} - p_n = 66$, then the sum of all distinct prime divisors of n is :

Option 1:
7

Option 2:
8

Option 3:
5

Option 4:
6

Correct Answer:
5

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Solution:

$P_n = {}^nC_3$

$P_{n+1} - P_n = 66$

${}^{n+1}C_3 - {}^nC_3 = 66$

$\Rightarrow \frac{(n+1)n(n-1)}{6} - \frac{n(n-1)(n-2)}{6} = 66$

$\Rightarrow \frac{n(n-1)}{6}[n+1-n+2] = 66$

$\Rightarrow n(n-1) = 132$

$n = 12$

Prime divisors of 12 are 2 \& 3

$\text{sum } = 2+3 = 5$

Q. 8 A man throws a fair coin repeatedly. He gets 10 points for each head he throws and 5 points for each tail he throws. If the probability that he gets exactly 30 points is $\frac{m}{n}, \gcd(m,n) = 1$, then $m+n$ is equal to :

Option 1:
53

Option 2:
55

Option 3:
107

Option 4:
105

Correct Answer:
107

Solution:

$\mathrm{P} = (6 \text{ tail }) + (4 \text{ tail } + 1 \text{ head }) + (2 \text{ tail } + 2 \text{ head }) + (3 \text{ head })$

$\mathrm{P} = \left(\frac{1}{2}\right)^6 + \frac{5!}{1!4!}\left(\frac{1}{2}\right)^5 + \frac{4!}{2!2!}\left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^3$

$\mathrm{P} = \frac{1}{64} + \frac{5}{32} + \frac{3}{8} + \frac{1}{8}$

$\mathrm{P} = \frac{1+10+24+8}{64} = \frac{43}{64} = \frac{\mathrm{m}}{\mathrm{n}}$

$\mathrm{m}+\mathrm{n} = 107$

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Q. 9 The mean and variance of n observations are 8 and 16, respectively. If the sum of the first ($n-1$) observations is 48 and the sum of squares of the first ($n-1$) observations is 496 , then the value of n is :

Option 1:
21

Option 2:
16

Option 3:
13

Option 4:
7

Correct Answer:
7

Solution:

$x_1+x_2+\ldots+x_{n-1}+x_n=8n$

$48+x_n=8n$

$\Rightarrow x_n=8n-48$

$\Rightarrow 16=\dfrac{496+x_n^2}{n}-(8)^2$

$\Rightarrow 80n=496+x_n^2$

$\Rightarrow x_n^2=80n-496$

$\Rightarrow (8n-48)^2=80n-496$

$\Rightarrow 64(n-6)^2=8(10n-62)$

$\Rightarrow 8(n-6)^2=10n-62$

$\Rightarrow 4(n-6)^2=5n-31$

$\Rightarrow 4\left(n^2-36-12n\right)=5n-31$

$\Rightarrow 4n^2+144-48n=5n-31$

$\Rightarrow 4n^2-53n+175=0$

$\Rightarrow 4n^2-28n-25n+175=0$

$\Rightarrow 4n(n-7)-25(n-7)=0$

$n=7$

Q. 10 Let a circle pass through the origin and its centre be the point of intersection of two mutually perpendicular lines $x+(k-1)y+3=0$ and $2x+k^2y-4=0$. If the line $x-y+2=0$ intersects the circle at the points A and B , then $(\mathrm{AB})^2$ is equal to :

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Option 1:
10

Option 2:
27

Option 3:
18

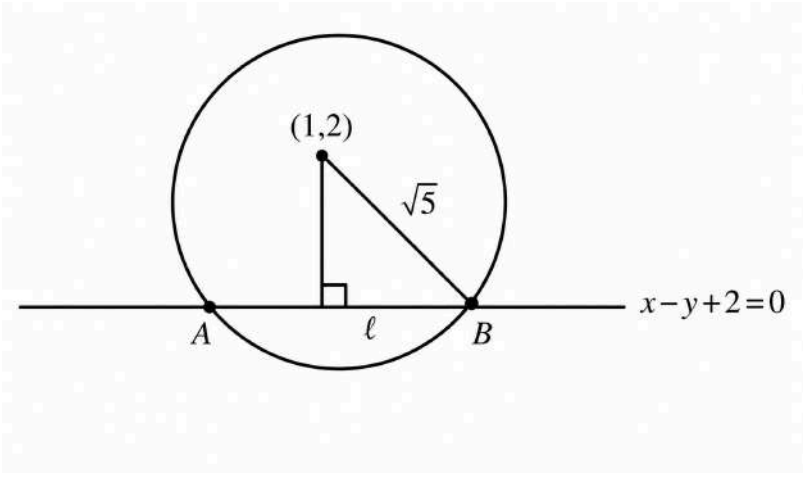
Option 4:
34

Correct Answer:
18

Solution:
$x+(k-1)y+3=0$
$2x+k^2y-4=0$
$\left(\frac{1}{1-k}\right)\left(\frac{2}{k^2}\right)=1$
$2=k^2-k^3$
$k^3-k^2+2=0$
$k=-1$

Solving :

$2(x-2y+3)=0$
$2x+y-4=0$
$-5y+10=0$
$y=2$



$x=1$

Centre $(1,2)$

$r=\sqrt{5}$

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So circle is $(x-1)^2+(y-2)^2=5$
Chord $x-y+2=0$

$$\mathrm{p}=\frac{1}{\sqrt{2}}$$

$$\ell=\sqrt{5-\frac{1}{2}}=\frac{3}{\sqrt{2}}$$

$$\mathrm{AB}=\frac{6}{\sqrt{2}} \quad \therefore \mathrm{AB}^2=18$$

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Q. 11 Let O be the origin, and P and Q be two points on the rectangular hyperbola $\mathrm{xy}=12$ such that the mid point of the line segment PQ is $\left(\frac{1}{2},-\frac{1}{2}\right)$. Then the area of the triangle OPQ equals :

Option 1:
$\frac{3}{2}$

Option 2:
$\frac{5}{2}$

Option 3:
$\frac{7}{2}$

Option 4:
$\frac{9}{2}$

Correct Answer:
$\frac{7}{2}$

Solution:
Equation of chord whose midpoint is $\left(\frac{1}{2},\frac{-1}{2}\right)$

$$\frac{x}{x_1}+\frac{y}{y_1}=2$$
$$2x-2y=2$$
$$x-y=1$$

solving with $\mathrm{xy}=12$

$P(-3,-4), Q(4,3)$

$\mathrm{Area}(\triangle OPQ)=\frac{1}{2}|-3\times 3-(-4)(4)|=\frac{7}{2}$

Q. 12 Let the parabola $y = x^2 + px + q$ passing through the point $(1, -1)$ be such that the distance between its vertex and the x-axis is minimum. Then the value of $p^2 + q^2$ is :

Option 1:
2

Option 2:
4

Option 3:
5

Option 4:
8

Correct Answer:
4

Solution:
parabola passes through $(1, -1)$ so

$-1 = 1 + p + q$
$p + q = -2$
$q = -2 - p$

Distance from x -axis

$$\frac{-D}{4a} = \frac{-\left(p^2 - 4q\right)}{4(1)} = \frac{4q - p^2}{4}$$
$$= \frac{4(-2-p) - p^2}{4} \quad \text{from } (1)$$
$$= \frac{-8 - 4p - p^2}{4} = \frac{-\left(p^2 + 4p + 8\right)}{4} = \frac{-\left((p+2)^2 + 4\right)}{4}$$

Minimum at $\mathrm{p} = -2 \Rightarrow \mathrm{q} = 0$

$p^2 + q^2 = 4 + 0 = 4$

Q. 13 Let $P = \left\{\theta \in [0, 4\pi] : \tan^2\theta \neq 1\right\}$ and $S = \left\{a \in Z : 2\left(\cos^8\theta - \sin^8\theta\right)\sec 2\theta = a^2, \theta \in P\right\}$. Then $\mathrm{n(S)}$ is :

Option 1:
0

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Option 2:
1

Option 3:
2

Option 4:
3

Correct Answer:
0

Solution:

$2\left(\cos^8\theta-\sin^8\theta\right)\sec 2\theta=\mathrm{a}^2$

$2\left(\cos^4\theta+\sin^4\theta\right)\left(\cos^2\theta+\sin^2\theta\right)\left(\cos^2\theta-\sin^2\theta\right)\sec 2\theta=\mathrm{a}^2$

$2\left(\cos^4\theta+\sin^4\theta\right)=\mathrm{a}^2$

$2\left(1-2\sin^2\theta\cos^2\theta\right)=\mathrm{a}^2$

$2\left(1-\frac{\sin^2 2\theta}{2}\right)=\mathrm{a}^2,\mathrm{a}\in\mathrm{Z}$

$\mathrm{a}^2=2-\sin^2 2\theta\in[1,2]$

$\mathrm{a}^2=1\text{ at }\sin^2 2\theta=1$

$2\theta=(2\mathrm{n}+1)\frac{\pi}{2}$

$\theta=(2\mathrm{n}+1)\frac{\pi}{4}\notin\mathrm{P}$

$\mathrm{n}(\mathrm{S})=0$

Q. 14 Let the vectors $\vec{a}=-\hat{i}+\hat{j}+3\hat{k}$ and $\vec{b}=\hat{i}+3\hat{j}+\hat{k}$. For some $\lambda,\mu\in\mathbf{R}$, let $\overrightarrow{\mathrm{c}}=\lambda\overrightarrow{\mathrm{a}}+\mu\overrightarrow{\mathrm{b}}$. If $\vec{c}\cdot(3\hat{i}-6\hat{j}+2\hat{k})=10$ and $\vec{c}\cdot(\hat{i}+\hat{j}+\hat{k})=-2$, then $|\overrightarrow{\mathrm{c}}|^2$ is equal to :

Option 1:
8

Option 2:
12

Option 3:
14

Option 4:
15

Correct Answer:
12

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Solution:

$\vec{c} = \lambda\vec{a} + \mu\vec{b}$

$\vec{c} = (\mu - \lambda)\hat{i} + (\lambda + 3\mu)\hat{j} + (3\lambda + \mu)\hat{k}$

$\vec{c} \cdot (3\hat{i} - 6\hat{j} + 2\hat{k}) = 10$

$3(\mu - \lambda) - 6(\lambda + 3\mu) + 2(3\lambda + \mu) = 10$

$-3\lambda - 13\mu = 10$

$\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = -2$

$5\mu + 3\lambda = -2$

Solving (1) & (2)

$\lambda = 1, \mu = -1$

$\vec{c} = -2\hat{i} - 2\hat{j} + 2\hat{k}$

$|\vec{c}|^2 = 12$

Q. 15 Let the point A be the foot of perpendicular drawn from the point $P(a, b, 0)$ on the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-\alpha}{3}$. If the midpoint of the line segment PA is $\left(0, \frac{3}{4}, \frac{-1}{4}\right)$, then the value of $a^2 + b^2 + \alpha^2$ is equal to:

Option 1:
1

Option 2:
2

Option 3:
6

Option 4:
9

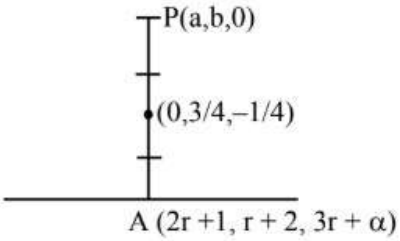
Correct Answer:
1

Solution:

$$\frac{2r + 1 + a}{2} = 0 \Rightarrow 2r + a = -1$$

$$\frac{r + 2 + b}{2} = \frac{3}{4} \Rightarrow r + b = -\frac{1}{2}$$

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$$\frac{3r+\alpha}{2}=\frac{-1}{4}\Rightarrow 3r+\alpha=\frac{-1}{2}$$

$$2\cdot a+1\cdot(b-3/4)+3\cdot\frac{1}{4}=0\Rightarrow a+b=0$$

$$\Rightarrow 2(-1-2r)+(-r-1/2)=0$$

$$\Rightarrow -3/2=3r=1\Rightarrow r=-1/2, a=0, b=0, \alpha=1$$

$$\therefore a^2+b^2+\alpha^2=1$$

Q. 16

Two adjacent sides of a parallelogram PQRS are given by $\overrightarrow{PQ}=\hat{j}+\hat{k}$ and $\overrightarrow{PS}=\hat{i}-\hat{j}$. If the side PS is rotated about the point P by an acute angle α in the plane of the parallelogram so that it becomes perpendicular to the side PQ , then $\sin^2\left(\frac{5\alpha}{2}\right)-\sin^2\left(\frac{\alpha}{2}\right)$ is equal to :

Option 1:

$\frac{1}{2}$

Option 2:

$\frac{\sqrt{3}}{2}$

Option 3:

$\frac{\sqrt{3}}{4}$

Option 4:

$\frac{2\sqrt{3}}{5}$

Correct Answer:

$\frac{\sqrt{3}}{2}$

Solution:

Let angle between $\overrightarrow{PQ}\&\overrightarrow{PS}$ is θ

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$$\cos\theta = \frac{\overrightarrow{PQ}\cdot\overrightarrow{PS}}{|\overrightarrow{PQ}||\overrightarrow{PS}|}$$

$$\cos\theta = \frac{0-1+0}{\sqrt{2}\sqrt{2}}$$

$$\cos\theta = \frac{-1}{2}$$

$$\theta = \frac{2\pi}{3}$$

$$\alpha = \frac{2\pi}{3} - \frac{\pi}{2} \text{ So } \alpha = \frac{\pi}{6}$$

$$\sin^2\frac{5\alpha}{2} - \sin^2\frac{\alpha}{2} = \sin\left(\frac{5\alpha}{2}+\frac{\alpha}{2}\right)\sin\left(\frac{5\alpha}{2}-\frac{\alpha}{2}\right)$$

$$= \sin\frac{6\alpha}{2} \times \sin\frac{4\alpha}{2}$$

$$= \sin 3\alpha \times \sin 2\alpha$$

$$= \sin\frac{\pi}{2} \times \sin\frac{\pi}{3}$$

$$= 1 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

Q. 17 The value of $\int_0^{20\pi}\left(\sin^4 x + \cos^4 x\right)dx$ is equal to:

Option 1:
$\frac{15\pi}{2}$

Option 2:
25π

Option 3:
15π

Option 4:
$\frac{25\pi}{2}$

Correct Answer:
15π

Solution:

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$$I = \int_0^{20\pi} \left(1 - 2\sin^2 x \cos^2 x\right) \cdot dx$$

$$= 20\pi - \frac{1}{2}\int_0^{20\pi} \sin^2 2x dx$$

$$20\pi - \frac{1}{2}(40)\int_0^{\frac{\pi}{2}} \sin^2 2x$$

$$20\pi - \frac{20}{2}\left(x - \frac{\sin 2x}{2}\right)_0^{\frac{\pi}{2}}$$

$$20\pi - 10\left(\frac{\pi}{2}\right) = 15\pi$$

Q. 18 Let $f(x)$ be a polynomial of degree 5 , and have extrema at $x = 1$ and $x = -1$. If $\lim_{x\to 0}\left(\frac{f(x)}{x^3}\right) = -5$, then $f(2) - f(-2)$ is equal to :

Option 1:
0

Option 2:
50

Option 3:
92

Option 4:
112

Correct Answer:
112

Solution:
Given $f(x)$ is a polynomial of degree 5
Also, $f'(1) = 0; f'(-1) = 0.$
Also $\because \lim_{x\to 0} \frac{f(x)}{x^3} = -5$ (fixed and finite)

$$\therefore f(0) = 0; f'(0) = 0; f'(0) = 0$$

$$\frac{f'''(0)}{6} = -5 \Rightarrow f'''(0) = -30$$

Hence, let $\mathrm{f'(x) = (ax + b)(x - 0)(x - 1)(x + 1)}$

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$\Rightarrow f'(x) = ax^4 + bx^3 - ax^2 - bx$
$f'(x) = 4ax^3 + 3bx^2 - 2ax - b$
$\because f''(0) = 0 \Rightarrow b = 0$
$f'''(x) = 12ax^2 + 6bx - 2a$
$\because f'''(0) = -2a = -30$
$\Rightarrow a = 15$
$\therefore f'(x) = 15x \cdot x \cdot (x-1)(x+1)$
$\Rightarrow f'(x) = 15x^4 - 15x^2$
$\Rightarrow f(x) = 3x^5 - 5x^3 + C$
$\because f(0) = 0 \Rightarrow C = 0$
$\therefore f(x) = 3x^5 - 5x^3$
$\therefore f(2) - f(-2) = 112$

Q. 19 Let $\mathrm{f(x)} = \int \left(\frac{16x+24}{x^2+2x-15}\right)\mathrm{dx}$. If $\mathrm{f(4)} = 14\log_e(3)$ and $f(7) = \log_c\left(2^\alpha \cdot 3^\beta\right), \alpha, \beta \in \mathbf{N}$, then $\alpha + \beta$ is equal to:

Option 1:
31

Option 2:
37

Option 3:
39

Option 4:
41

Correct Answer:
39

Solution:
$$\mathrm{f(x)} = \int \frac{8(2x+2)+8}{x^2+2x-15}\mathrm{dx}$$
$$\Rightarrow \mathrm{f(x)} = 8\ell\mathrm{n}\left|x^2+2x-15\right| + \ell\mathrm{n}\left|\frac{x-3}{x+5}\right| + \mathrm{C}$$
$\Rightarrow \mathrm{f(4)} = 14\ell\mathrm{n}3 + \mathrm{C}$
$\Rightarrow 14\ell\mathrm{n}3 = 14\ell\mathrm{n}3 + \mathrm{C}$
$\therefore \mathrm{C} = 0$

Now, $f(7) = 8\ell\mathrm{n}48 - \ell\mathrm{n}3$

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$= \ell n\left(\frac{(48)^8}{3}\right)$

$\Rightarrow f(7) = \ell n\left(2^{32} \times 3^7\right)$

$\alpha = 32, \beta = 7 \Rightarrow \alpha + \beta = 39$

Q. 20 Let $x = x(y)$ be the solution of the differential equation $2y^2\frac{dx}{dy} - 2xy + x^2 = 0, y > 1, x(e) = e$.
Then $x\left(e^2\right)$ is equal to:

Option 1:
$\frac{3}{2}e^2$

Option 2:
$\frac{2}{3}e^2$

Option 3:
e^2

Option 4:
$2e^2$

Correct Answer:
$\frac{2}{3}e^2$

Solution:
$2y(ydx - xdy) + x^2dy = 0$

$\Rightarrow -2yx^2d\left(\frac{y}{x}\right) + x^2dy = 0$

$\Rightarrow -2yd\left(\frac{y}{x}\right) + dy = 0$

$\Rightarrow -2d\left(\frac{y}{x}\right) + \frac{1}{y}dy = 0$

$\Rightarrow \frac{-2y}{x} + \log_e y = C$

Given $x(e) = e$

$\Rightarrow C = -1$

$\therefore \frac{-2y}{x} + \log_e y = -1$

$\Rightarrow \frac{2y}{x} - \log_e y = 1$

Put $y = e^2$, we get

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$$\Rightarrow \frac{2e^2}{x} - 2 = 1 \Rightarrow \frac{2e^2}{x} = 3 \Rightarrow x = \frac{2e^2}{3}$$

$$\Rightarrow x\left(e^2\right) = \frac{2e^2}{3}$$

Q. 21 Let $A = \{2, 3, 4, 5, 6\}$. Let R be a relation on the set $A \times A$ given by $(x, y)R(z, w)$ if and only if x divides z and $y \leq w$. Then the number of elements in R is _____ .

Correct Answer:
120

Solution:
For

$x = 2, \quad z = 2, 4, 6$
$x = 3 \quad z = 3, 6$
$x = 4 \quad z = 4$
$x = 5 \quad z = 5$
$x = 6 \quad z = 6$

Total case $= 8$
and no. of combination of y and w satisfying $y \leq w$

$= 1 + 2 + 3 + 4 + 5 = 15$

No. of relations satisfying $(x, y)R(z, w)$ is $8 \times 15 = 120$.

Q. 22 Consider the matrices $A = \begin{bmatrix} 2 & -2 \\ 4 & -2 \end{bmatrix}$ and $B \begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}$. If matrices P and Q are such that $PA = B$ and $AQ = B$, then the absolute value of the sum of the diagonal elements of $2(P + Q)$ is _____ .

Correct Answer:
34

Solution:

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$$A^{-1} = \frac{1}{4}\begin{bmatrix} -2 & 2 \\ -4 & 2 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}$$

$\because PA = B$

$\Rightarrow P = BA^{-1}$

$$= \frac{1}{2}\begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -21 & 12 \\ -7 & 4 \end{bmatrix}$$

$\because AQ = B$

$\Rightarrow Q = A^{-1}B$

$$= \frac{1}{2}\begin{bmatrix} -1 & 1 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -2 & -6 \\ -5 & -15 \end{bmatrix}$$

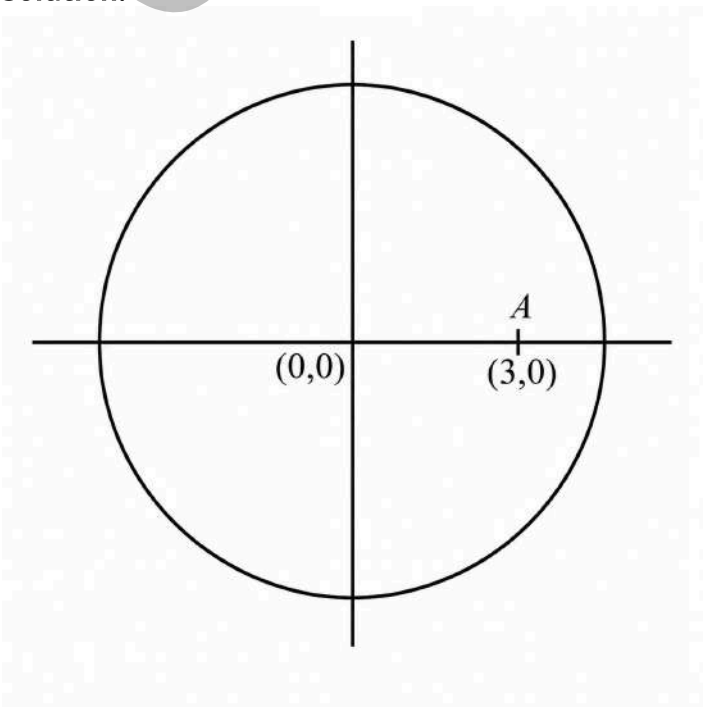
$\therefore |\text{tr}(2(P+Q))| = |(-21)+(4)+(-2)+(-15)|$

$= |-34| = 34$

Q. 23 Let A be the point $(3, 0)$ and circles with variable diameter AB touch the circle $x^2 + y^2 = 36$ internally. Let the curve C be the locus of the point B. If the eccentricity of C is e , then $72e^2$ is equal to _____ .

Correct Answer:
18

Solution:



Let $B(h, k)$

Equation of circle with AB as diameter

$$(x-h)(x-3)+(y-k)(y-0)=0$$

$$x^2+y^2-(h+3)x-ky+3h=0$$

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Centre $\left(\frac{h+3}{2}, \frac{k}{2}\right)$
Circle touches internally

$$\therefore C_1C_2 = |R - r|$$

$$\sqrt{\left(\frac{h+3}{2}\right)^2 + \left(\frac{k}{2}\right)^2} = \left|6 - \frac{1}{2}\left(\sqrt{(h-3)^2 + k^2}\right)\right|$$

$$\sqrt{(x+3)^2 + y^2} + \sqrt{(x-3)^2 + y^2} = 12$$

$2a = 12 \quad \therefore a = 6$

$(-3, 0)$ and $(3, 0)$ are foci

$2ae = 6$

$12e = 6$

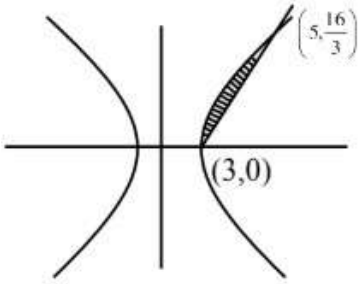
$e = \frac{1}{2}$

$\therefore 72e^2 = 72\left(\frac{1}{4}\right) = 18$

Q. 24 If the area of the region bounded by $16x^2 - 9y^2 = 144$ and $8x - 3y = 24$ is A, then $3\left(A + 6\log_e(3)\right)$ is equal to _____ .

Correct Answer:
24

Solution:



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$16x^2 - 9y^2 = 144$
$16x^2 - (8x - 24)^2 = 144$
$16x^2 - 64(x-3)^2 = 144 \Rightarrow x^2 - 4(x-3)^2 = 9$
$3x^2 - 24x + 45 \Rightarrow x^2 - 8x + 15 = 0$
$x = 3, 5$

$$\text{Area } = \int_3^5 \sqrt{\frac{16x^2 - 144}{9}} - \frac{1}{2} \cdot 2 \cdot \frac{16}{3}$$

$$= \frac{4}{3}\int_3^5 \sqrt{x^2 - 9} - \frac{16}{3}$$

$$= \frac{4}{3}\left(\frac{x}{2}\sqrt{x^2-9} - \frac{9}{2}\log_e\left(x + \sqrt{x^2-9}\right)\right)_3^5 - \frac{16}{3}$$

$$= \frac{4}{3}\left(\frac{5}{2}\cdot 4 - \frac{9}{2}\log_e 9 - \frac{3}{2}\cdot 0 + \frac{9}{2}\log_e 3\right) - \frac{16}{3}$$

$$\text{Area } = 8 - 6\log_e 3 = A$$

$\therefore A + 6\ln 3 = 8$
$\Rightarrow 3(A + 6\ln 3) = 24$

Q. 25 The number of points in the interval [2,4], at which the function $f(x) = \left[x^2 - x - \frac{1}{2}\right]$, where $[\cdot]$ denotes the greatest integer function, is discontinuous, is _____ .

Correct Answer:
10

Solution:
In $x \in [2, 4]$, range of $x^2 - x - \frac{1}{2}$ is $[1.5, 11.5]\mathrm{f(x)}$ is discontinuous at points where $\mathrm{x^2 - x - \frac{1}{2}}$ becomes integers.

$\therefore$ Integral values of $\mathrm{x^2 - x - \frac{1}{2}}$ are $2, 3, 4, 5, 6, 7, 8, 910, 11$

$\therefore$ Number of points where is discontinuous is 10

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JEE MAINS

2026

4th April

1st Shift

Question Paper

Physics

Q. 1 In a screw gauge when the circular scale is given five complete rotations it moves linearly by 2.5 mm . If the circular scale has 100 divisions, the least count of screw gauge is _____ mm.

Option 1:
1×10^{-2}

Option 2:
1×10^{-3}

Option 3:
5×10^{-2}

Option 4:
5×10^{-3}

Correct Answer:
5×10^{-3}

Solution:

$$\text{Pitch} \Rightarrow \frac{2.5\ \text{mm}}{5} = 0.5\ \text{mm} = 5 \times 10^{-4}\ \text{m}$$

$$\text{L.C.} \Rightarrow \frac{\text{Pitch}}{\text{No. of divisions}} = \frac{5 \times 10^{-4}}{100}\ \text{m}$$

$$= 5 \times 10^{-6}\ \text{m}$$

$$= 5 \times 10^{-3}\ \text{mm}$$

Q. 2 The increase in the pressure required to decrease the volume (ΔV) of water is $6.3 \times 10^{7}\ \text{N/m}^2$. The percentage decrease in the volume is _____ .
(Bulk modulus of water $= 2.1 \times 10^{9}\ \text{N/m}^2$.)

Option 1:
2%

Option 2:
3%

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Option 3:
6%

Option 4:
4%

Correct Answer:
3%

Solution:
Bulk modulus $\beta = \frac{-\Delta P}{(\frac{\Delta V}{V})}$

$100 \times \frac{\Delta V}{V} = \frac{-\Delta P}{\beta} \times 100 = -\frac{6.3\times10^7}{2.1\times10^9} \times 100 = -3\%$

Percentage decrease $= 3\%$

Q. 3 The time taken by a block of mass m to slide down from the highest point to the lowest point on a rough inclined plane is 50% more compared to the time taken by the same block on identical inclined smooth plane. Both inclined planes are at 45° with the horizontal. The coefficient of kinetic friction between the rough inclined surface and block is _____.

Option 1:
$3/4$

Option 2:
$2/3$

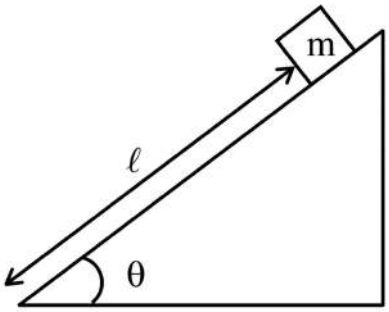
Option 3:
$5/9$

Option 4:
$4/9$

Correct Answer:
$5/9$

Solution:

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For Rough incline plane

$$a = g(\sin\theta - \mu\cos\theta)$$

$$t_1 = \sqrt{\frac{2\ell}{g(\sin\theta - \mu\cos\theta)}}$$

For smooth incline plane

$$t_2 = \sqrt{\frac{2\ell}{g\sin\theta}} \& t_1 = 1.5t_2$$

$$\sqrt{\frac{2\ell}{g(\sin\theta - \mu\cos\theta)}} = \frac{3}{2} \times \sqrt{\frac{2\ell}{g\sin\theta}}$$

$$(\sin\theta - \mu\cos\theta) = \frac{4}{9}\sin\theta$$

$$\frac{5}{9}\sin\theta = \mu\cos\theta \Rightarrow \frac{5}{9}\tan\theta$$

$$\mu = \frac{5}{9}\tan 45^\circ = \frac{5}{9}$$

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Q. 4 Two nuclei of mass number 3 combine with another nucleus of mass number 4 to yield a nucleus of mass number 10. If the binding energy per nucleon for the mass number 3, 4 and 10 are $5.6\mathrm{MeV}, 7.4\mathrm{MeV}$ and 6.1 MeV , respectively, then in the process, $\Delta \mathrm{Mc}^2 =$ ____ MeV .

Option 1:
6.9

Option 2:
7.9

Option 3:
2.2

Option 4:
4.3

Correct Answer:
2.2

Solution:
$\Delta mc^2 = [2 \times 3 \times 5.6 + 4 \times (7.4) - 6.1 \times 10]$
$= 2.2 \text{MeV}$

Q. 5 A solid sphere of mas M and radius R is divided into two unequal parts. The smaller part having mass M/8 is converted into a sphere of radius r and the larger part is converted into a circular disc of thickness t and radius $2R$. If I_1 is moment of inertia of a sphere having radius r about an axis through its centre and I_2 is the moment of inertia of a disc about its diameter, the ratio of their moment of inertia $I_2/I_1 =$ _____ .

Option 1:
35

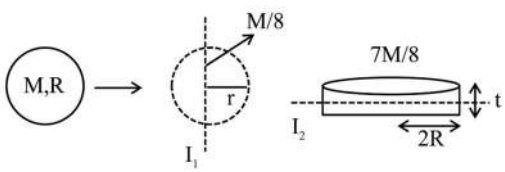
Option 2:
70

Option 3:
140

Option 4:
210

Correct Answer:
70

Solution:



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$$I_1 = \frac{2}{5}\left(\frac{M}{8}\right)r^2$$

$$I_2 = \frac{\frac{7M}{8} \times (2R)^2}{4}$$

$$M = \rho \times \frac{4\pi}{3} \times R^3$$

$$\frac{M}{8} = \rho \times \frac{4\pi}{3} \times |r|^3$$

$$r = \frac{R}{2}$$

$$\frac{I_1}{I_2} = \frac{\frac{2}{5}\left(\frac{M}{8}\right)\left(\frac{R}{2}\right)^2}{\frac{7M}{8\times 4} \times (2R)^2} = \frac{2}{5} \times \frac{1}{4 \times 7} = \frac{1}{70}$$

$$\frac{I_2}{I_1} = \frac{70}{1}$$

Q. 6 The two projectiles are projected with the same initial velocities at the 15° and 30° with respect to the horizontal. The ratio of their ranges is $1 : x$. The value of x is

Option 1:
$\sqrt{2}$

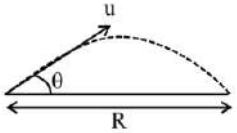
Option 2:
$\sqrt{3}$

Option 3:
$2\sqrt{3}$

Option 4:
$\frac{1}{\sqrt{2}}$

Correct Answer:
$\sqrt{3}$

Solution:



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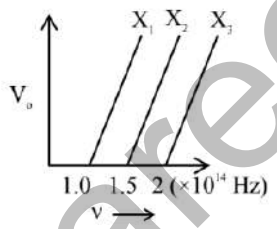
$$R_1 = \frac{u^2 \sin 2 \times 15^\circ}{g}$$

$$R_2 = \frac{u^2 \sin 2 \times 30^\circ}{g}$$

$$\frac{R_1}{R_2} = \frac{\sin 30^\circ}{\sin 60^\circ} = \frac{1}{\sqrt{3}} = \frac{1}{x}$$

$$x = \sqrt{3}$$

Q. 7 The graph shows variation of stopping potential $V_\circ$ with the frequency v of the incident radiation for three photosensitive metals $\mathrm{X}_1, \mathrm{X}_2$ and X_3. Which metal will give out electrons with greater kinetic energy, for the same wavelength of incident radiation?



Option 1:
(1) X_1

Option 2:
X_2

Option 3:
X_3

Option 4:
All the metals will give out photo electrons with same kinetic energies.

Correct Answer:
(1) X_1

Solution:

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$$\mathrm{hv} = \phi + \mathrm{ev}_0$$

$$\mathrm{v}_0 = \frac{\mathrm{h}v}{\mathrm{e}} - \frac{\phi}{\mathrm{e}} \Rightarrow 0 \Rightarrow \frac{\mathrm{h}v}{\mathrm{e}} = \frac{\phi}{\mathrm{e}}$$

$$\begin{pmatrix} \phi_1 : \phi_2 : \phi_3 \\ \mathrm{v}_1 : \mathrm{v}_2 : \mathrm{v}_3 \\ 1 : 1.5 : 2 \end{pmatrix}$$

As metal (1) having minimum value of work function

So x_1 will having maximum kinetic energy

Q. 8 A slit of width a is illuminated by light of wavelength λ. The linear separation between 1st and 3rd minima in the diffraction pattern produced on a screen placed at a distance D from the slit system is _____.

Option 1:
(1) $\frac{D\lambda}{a}$

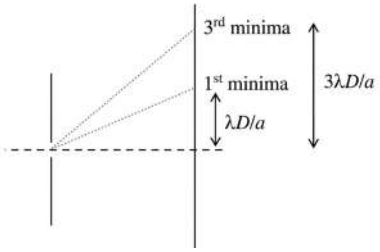
Option 2:
$1.5\frac{\mathrm{D}\lambda}{\mathrm{a}}$

Option 3:
$2\frac{\mathrm{D}\lambda}{\mathrm{a}}$

Option 4:
$3\frac{\mathrm{D}\lambda}{\mathrm{a}}$

Correct Answer:
$2\frac{\mathrm{D}\lambda}{\mathrm{a}}$

Solution:



Separation between 3rd &1st minima

$= \frac{2\mathrm{D}\lambda}{\mathrm{a}}$

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Q. 9 A string A of length 0.314 m of Young's modulus $2 \times 10^{10}\ \mathrm{N/m^2}$ is connected to another string B of length and Young's modulus both twice of those of A. This series combination of strings is then suspended from a rigid support and its free end is fixed to a load of mass 0.8 kg . The net change in length of the combination is _____ mm.
(radius of both the strings is 0.2 mm and acceleration due to gravity $= 10\ \mathrm{m/s^2}$)
(Mass of both strings is to be neglected as compared to the mass of load)

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Option 1:
3

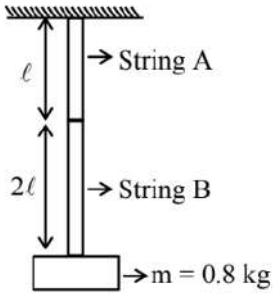
Option 2:
2

Option 3:
1.9

Option 4:
1

Correct Answer:
2

Solution:



$$\Delta\ell \Rightarrow \Delta\ell_1 + \Delta\ell_2$$

$$\Rightarrow \frac{\mathrm{mg}\ell}{\mathrm{AY}} + \frac{\mathrm{mg}(2\ell)}{\mathrm{A}(2\mathrm{Y})} = \frac{2\mathrm{mg}\ell}{\mathrm{AY}}$$

$$= \frac{2 \times 0.8 \times 10 \times (0.314)}{3.14 \times \left(2 \times 10^{-4}\right)^2 \times 2 \times 10^{10}} = 2\ \mathrm{mm}$$

Q. 10 One gas of n_1 mole of molecules at temperature T_1, volume V_1, and pressure P_1, and another gas of n_2 mole of molecules at temperature T_2, volume V_2, and pressure P_2, are mixed resulting in pressure P and volume V of the mixture. The temperature of the mixture is _____ .

Option 1:
$(T_1 + T_2)/2$

Option 2:
$\mathrm{T}_1\ \mathrm{T}_2\mathrm{PV}/\left(\mathrm{T}_2\mathrm{P}_1\ \mathrm{V}_1 + \mathrm{T}_1\mathrm{P}_2\ \mathrm{V}_2\right)$

Option 3:
$(T_2P_1V_1 + T_1P_2V_2)\,(T_1T_2PV)$

Option 4:
$|\mathrm{T}_1 - \mathrm{T}_2|/2$

Correct Answer:
$\mathrm{T}_1\ \mathrm{T}_2\mathrm{PV}/\left(\mathrm{T}_2\mathrm{P}_1\ \mathrm{V}_1 + \mathrm{T}_1\mathrm{P}_2\ \mathrm{V}_2\right)$

Solution:

$$\mathrm{n} = \mathrm{n}_1 + \mathrm{n}_2$$

$$\therefore \mathrm{PV} = \mathrm{nRT} \quad \Rightarrow \mathrm{n} = \frac{\mathrm{P}}{\mathrm{R}}$$

$$\frac{\mathrm{P(V)}}{\mathrm{T_{Final}}} = \frac{\mathrm{P_1\ V_1}}{\mathrm{T_1}} + \frac{\mathrm{P_2\ V_2}}{\mathrm{T_2}}$$

$$\frac{\mathrm{PV}}{\mathrm{RT_f}} = \frac{\mathrm{P_1\ V_1}}{\mathrm{RT_1}} + \frac{\mathrm{P_2\ V_2}}{\mathrm{RT_2}}$$

$$\mathrm{T_f} = \frac{\mathrm{PV}}{\mathrm{P_1\ V_1/T_1 + P_2\ V_2/T_2}}$$

Q. 11 An ideal gas undergoes a process maintaining relation between pressure (P) and volume (V) as $P = P_o\left(1 + \left(\frac{V_o}{V}\right)^2\right)^{-1}$, where P_o and V_o are constants. If two samples A and B (two moles each) with initial volume $\mathrm{V_o}$ and $3\ \mathrm{V_o}$ respectively undergo above mentioned process and attain same pressure, then the difference at the temperatures of these samples, $\mathrm{T_B - T_A}$ is _____
(R = gas constant)

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Option 1:
$\frac{9P_oV_o}{8R}$

Option 2:
$\frac{11P_oV_o}{10R}$

Option 3:
$\frac{7P_oV_o}{6R}$

Option 4:
$\frac{13P_oV_o}{11R}$

Correct Answer:
$\frac{11P_oV_o}{10R}$

Solution:
For

$V = V_0, P = \frac{P_0}{2}$

$\frac{P_0}{2} \times V_0 = 2 \times RT_A$

$T_A = \frac{P_0V_0}{4R}$

$\mathrm{V = 3\ V_0 \quad P = \frac{9P_0}{10}}$

$\mathrm{for\ \frac{9P_0}{10} \times 3\ V_0 = 2 \times RT_B}$

$\mathrm{T_B = \frac{27P_0\ V_0}{20R}}$

So,

$T_B - T_A$

$= \frac{11P_0V_0}{10R}$

Q. 12 A voltmeter with internal resistance of $\mathrm{x\Omega}$ can be used to measure upto 20 V . In order to increase its measuring range to 30 V , the required modification is to _____ .

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Option 1:
connect resistor of $\frac{X}{2}\Omega$, in series with voltmeter.

Option 2:
connect resistor of $\frac{x}{2}\Omega$, in parallel to voltmeter.

Option 3:
connect resistor of $x\Omega$, in series with voltmeter.

Option 4:
connect resistor of $2x\Omega$, in parallel to voltmeter.

Correct Answer:
connect resistor of $\frac{X}{2}\Omega$, in series with voltmeter.

Solution:
Since maximum current through voltmeter remains same

So,

$\frac{20}{x} = \frac{30}{x+R}$

$20\mathrm{x} + 20\mathrm{R} = 30\mathrm{x}$
$20\mathrm{R} = 10\mathrm{x}$
$\mathrm{R} = \frac{\mathrm{x}}{2}$

This extra resistance must be connected in series.

Q. 13 Two 4 bits binary numbers, $\mathrm{A} = 1101$ and $\mathrm{B} = 1010$ are given in the inputs of a logic circuit shown in figure below. The output (Y) will be :



Option 1:
$Y = 1101$

Option 2:
$\mathrm{Y} = 0010$

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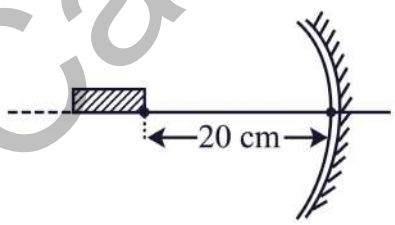
Option 3:
$\mathrm{Y} = 0111$

Option 4:
$Y = 1000$

Correct Answer:
$Y = 1101$

Solution:
$\mathrm{Y} = \overline{\overline{\mathrm{A}} \cdot \mathrm{B}} = \mathrm{A} + \overline{\mathrm{B}}$
$\mathrm{A} = 1101$
$\mathrm{B} = 1010$
$\overline{\mathrm{B}} = 0101$
$\mathrm{A} + \overline{\mathrm{B}} = 1101$

Q. 14 A rod of length 10 cm lies along the principle axis of a concave mirror of focal length 10 cm as shown in figure. The length of the image is _____ cm .



Option 1:
2.5

Option 2:
5

Option 3:
7.5

Option 4:
7

Correct Answer:
5

Solution:
For A

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$u = -20\ cm, f = -10\ cm$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \therefore v = \frac{u \times f}{u - f} = -20\ cm$

For B

$u = -30\ cm, f = -10\ cm$

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$v = \frac{uf}{u - f} = -15\ cm$

Length of image $= 20 - 15 = 5\ cm$

Q. 15 A parallel plate air capacitor is connected to a battery. The plates are pulled apart at uniform speed v. If x is the separation between the plates at any instant, then the time rate of change of electrostatic energy of the capacitor is proportional to x^{α}, where α is _____ .

Option 1:
-2

Option 2:
1

Option 3:
-1

Option 4:
2

Correct Answer:
-2

Solution:

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$$\text{Capacitance } = C = \frac{\varepsilon_0 A}{x}$$

$$\text{Energy stored } = U = \frac{1}{2}Cv^2$$

$$U = \frac{1}{2}\frac{\varepsilon_0 A}{x}v^2$$

$$\frac{dU}{dt} = \frac{1}{2}\varepsilon_0 A v^2\left(-\frac{1}{x^2}\right)\frac{dx}{dt}$$

$$\frac{dU}{dt} \propto \frac{1}{x^2}$$

Q. 16 An insulated wire is wound so that it forms a flat coil with $N = 200$ turns. The radius of the innermost turn is $r_1 = 3\ \text{cm}$, and of the outermost turn $r_2 = 6\ \text{cm}$. If 20 mA current flow in it then the magnetic moment will be $\alpha \times 10^{-2}\ \text{A.m}^2$. The value of α is _____ .

Option 1:
4.4

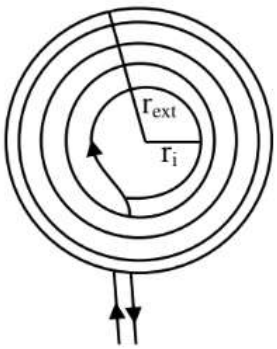
Option 2:
2.64

Option 3:
3.25

Option 4:
1.2

Correct Answer:
2.64

Solution:



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$$dM = (dI)\pi r^2 = (IdN)\pi r^2$$

$$M = \int_i^f I\left(\frac{N}{r_{ex} - r_i}\right) dr\pi r^2$$

$$M = \frac{I\pi N}{r_{ex} - r_i}\left[\frac{r^3}{3}\right]_{ri}^{r_{ex}}$$

$$M = \frac{I\pi N}{3\left(r_{ex} - r_{in}\right)}\left(r_{ex}^3 - r_i^3\right)$$

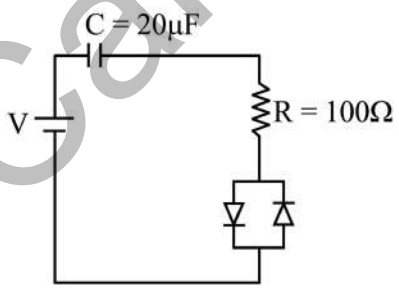
$$M = \frac{20 \times 10^{-3} \times 3.14 \times 200\left[6^3 - 3^3\right]10^{-6}}{3[3] \times 10^{-2}}$$

$$M = \frac{2373.84 \times 10^{-6}}{9 \times 10^{-2}}$$

$$M = 263.76 \times 10^{-4}$$

$$M = 2.64 \times 10^{-2}\mathrm{Am}^2$$

Q. 17 Consider a circuit consisting of a capacitor $(20\mu\ \mathrm{F})$, resistor (100Ω) and two identical diodes as shown in figure. The resistance of diode under forward biasing condition is 10Ω. The time constant of the circuit is $\alpha \times 10^{-3}$ s. The value of α is _____ .



Option 1:
2.2

Option 2:
2.0

Option 3:
2.1

Option 4:
2.4

Correct Answer:
2.2

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Solution:

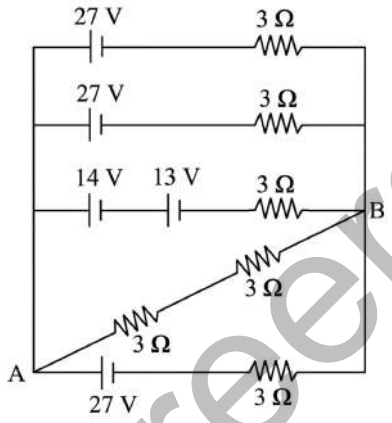
$\tau = \mathrm{RC}$

$\tau = (110) \times 20 \times 10^{-6} \mathrm{sec}$

$\tau = 2200 \times 10^{-6} \mathrm{sec}$

$\tau = 2.2 \times 10^{-3} \mathrm{sec}$

Q. 18 The voltage and the current between A and B points shown in the circuit are _____ .



Option 1:

$24 \mathrm{~V}, 12 \mathrm{~A}$

Option 2:

$24 \mathrm{~V}, 4 \mathrm{~A}$

Option 3:

$18 \mathrm{~V}, 12 \mathrm{~A}$

Option 4:

$27 \mathrm{~V}, 4 \mathrm{~A}$

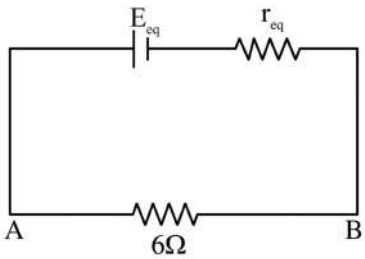
Correct Answer:

$24 \mathrm{~V}, 4 \mathrm{~A}$

Solution:

This circuit can be

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$$\text{Here : } \mathrm{E_{eq}} = \frac{\frac{27}{3}+\frac{27}{3}+\frac{27}{3}+\frac{27}{3}}{\frac{1}{3}+\frac{1}{3}+\frac{1}{3}+\frac{1}{3}}$$

$$\mathrm{E_{eq}} = 27\ \mathrm{V}$$

$$\mathrm{r_{eq}} = \frac{3}{4}\Omega$$

$$\mathrm{i}_{AB} = \frac{27}{6+\frac{3}{4}} = 4\ \mathrm{A}$$

$$\mathrm{V}_{AB} = 4 \times 6 = 24\ \mathrm{Volt}$$

Q. 19 A telescope with objective diameter R is used to observe a distant star emitting light of wavelength 500 nm , at a resolution of 5×10^{-7} radian. The value of R is _____ cm.

Option 1:
61

Option 2:
122

Option 3:
244

Option 4:
305

Correct Answer:
122

Solution:

$\theta = 5 \times 10^{-7}$ radian

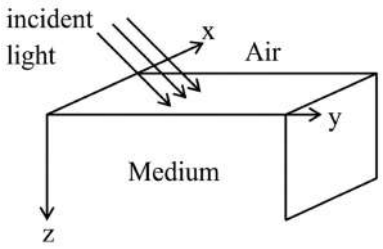
$\lambda = 500 \times 10^{-9}$ m

$$\theta = \frac{1.22\lambda}{\mathrm{R}}$$

$$\mathrm{R} = \frac{1.22\lambda}{\theta} = 1.22\ \mathrm{m} = 122\ \mathrm{cm}$$

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Q. 20 An unpolarized light is incident on the plane interface of air-dielectric medium shown in figure. If the incident angle is equal to Brewster angle, identify the expression representing reflected wave.



Option 1:
$\left(E_x\hat{i}+E_y\hat{j}\right)\sin(kx-kz-\omega t)$

Option 2:
$\left(E_x\hat{i}+E_z\hat{k}\right)\sin(kx+ky-\omega t)$

Option 3:
$\left(E_x\hat{j}+E_y\hat{k}\right)\sin(ky+kz-\omega t)$

Option 4:
$\left(E_x\hat{i}+E_y\hat{j}+E_z\hat{k}\right)\sin(kx+ky-kz-\omega t)$

Correct Answer:
$\left(E_x\hat{i}+E_y\hat{j}\right)\sin(kx-kz-\omega t)$

Solution:
Is the light is propagating along the positive z -axis and its is reflected from plane then correct phase for reflected light is $\mathrm{kx}-\mathrm{kz}-\omega \mathrm{t}$ and for electric field as the light incident on Brewster angle so the reflected light should be perfectly plane polarized that's why polarization direction should be y -axis so electric field of reflected light contain no z-

Q. 21 A 1 kg block subjected to two simultaneous forces $(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}+4\hat{\mathrm{k}})\mathrm{N}$ and $(3\hat{\mathrm{i}}-\hat{\mathrm{j}}-2\hat{\mathrm{k}})\mathrm{N}$ is moved a distance of 25 m along $(3\hat{\mathrm{i}}-4\hat{\mathrm{j}})$ direction. The work done in this process is _____ J.

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Correct Answer:
-

Solution:
$\overrightarrow{\mathrm{F}}_{\mathrm{Net}}=5\hat{\mathrm{i}}+2\hat{\mathrm{j}}+2\hat{\mathrm{k}}$
$\overrightarrow{\mathrm{S}}=15\hat{\mathrm{i}}-20\hat{\mathrm{j}}$
$\mathrm{W}=\overrightarrow{\mathrm{F}}\cdot\overrightarrow{\mathrm{S}}=75-40=35\ \mathrm{J}$

Q. 22 The surface tension of a soap solution is $3.5\times10^{-2}\ \mathrm{N/m}$. The work required to increase the radius of a soap bubble from 1 cm to 2 cm is $\alpha\times10^{-6}\ \mathrm{J}$. The value of α is _____ .
$(\pi=22/7)$

Correct Answer:
264

Solution:
$\mathrm{W.D.}\ =2\ \mathrm{T}\left[4\pi\mathrm{R}_2^2-4\pi\mathrm{R}_1^2\right]$
$=2\times3.5\times10^{-2}\times4\times\frac{22}{7}\left[2^2-1^2\right]\times10^{-4}$
$=2\times132\times10^{-6}$
$=264\times10^{-6}\ \mathrm{J}=\alpha\times10^{-6}\ \mathrm{J}$
$\alpha=264$

Q. 23 The velocity of a particle executing simple harmonic motion along x -axis is described as $v^2=50-x^2$, where x represents displacement. If the time period of motion is $\frac{x}{7}s$, the value of x is _____

Correct Answer:
44

Solution:
$V^2=\omega^2A^2-\omega^2x^2$

Comparing $\omega=1$

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$$T = \frac{2\pi}{\omega} = \frac{2\pi}{1} = 2\pi = \frac{x}{7}$$
$$x = 14\pi$$
$$x = 43.98 \approx 44$$

Q. 24 A body of mass 2 kg begins to move under the influence of time dependent force $\vec{F} = \left(2t\hat{i} + 6t^2\hat{j}\right)N$, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y-axis respectively. The power produced by the force at $t = 2$ s is ____ W.

Option 1:
200

Option 2:
-

Option 3:
-

Option 4:
-

Correct Answer:
200

Solution:

$$\vec{a} = \frac{\vec{F}}{m} = t\hat{i} + 3t^2\hat{j}$$
$$\vec{v} = \int_0^2 \vec{a} \cdot dt = \left[\frac{t^2}{2}\hat{i} + \frac{3t^3}{3}\hat{j}\right]_0^2$$
$$= (2\hat{i} + 8\hat{j}) m/s$$
$$\vec{F}_{t=2} = (4\hat{i} + 24\hat{j})N$$
$$P = \vec{F} \cdot \vec{v} = 8 + 192 = 200 \text{ W}$$

Q. 25 An inductor of 10 mH , capacitor of 0.1μ F and a resistor of 100Ω are connected in series aross an a.c power supply 220 V, 70 Hz. The power factor of the given circuit is 0.5 . The difference in the inductive reactance and capacitance reactance is $\sqrt{3}\alpha\Omega$. The value of α is ____ .

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Correct Answer:
100

Solution:

$\cos\phi = 0.5 = \frac{1}{2}$

$\phi = 60^\circ$

$\tan\phi = \frac{|x_L - x_C|}{R}$

$\tan 60^\circ = \sqrt{3} = \frac{x_L - x_C}{R}$

$|x_L - x_L| = R\sqrt{3} = 100\sqrt{3} = \sqrt{3}\alpha$

$\alpha = 100$

Chemistry

Q. 1 Number of moles and number of molecules in 1.4187 L of SO_2 at STP respectively are

Option 1:
$0.1266; 3.812 \times 10^{22}$

Option 2:
$0.0633; 3.812 \times 10^{22}$

Option 3:
$0.1266; 7.6238 \times 10^{22}$

Option 4:
$0.0633; 7.6238 \times 10^{22}$

Correct Answer:
$0.0633; 3.812 \times 10^{22}$

Solution:

$\text{moles} = \frac{\text{V lit}}{22.7 \text{ lit}} \Rightarrow \frac{1.4187 \text{ lit}}{22.7 \text{ lit}} \Rightarrow 0.0633$

no. of molecules = moles $\times 6.023 \times 10^{23}$

$\Rightarrow 0.0633 \times 6.032 \times 10^3$

$= 3.812 \times 10^{22}$

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Q. 2 What is the ratio of wave number of first line (lowest energy line) of Balmer series of H atomic spectrum to first line of its Brackett series?

Option 1:
$5:1$

Option 2:
$5:0.81$

Option 3:
$5:1.75$

Option 4:
$5:27$

Correct Answer:
$5:0.81$

Solution:
$\overline{\mathrm{V}}_1=\mathrm{R}_{\mathrm{H}}(\mathrm{z})^2\left[\frac{1}{2^2}-\frac{1}{3^2}\right] \Rightarrow \mathrm{I}^{\mathrm{st}}$ line of Balmer series
$\overline{\mathrm{V}}_2=\mathrm{R}_{\mathrm{H}}(\mathrm{z})^2\left[\frac{1}{4^2}-\frac{1}{5^2}\right] \Rightarrow \mathrm{I}^{\mathrm{x}}$ line of Brackett series

$\frac{\bar{V}_1}{\bar{V}_2}=\frac{500}{81} \Rightarrow 5:0.81$

Q. 3 Which of the following is correct set of 4 quantum number of 19th electron in Chromium (Atomic number $=24$) in accordance with Aufbau principle?

Option 1:
$\mathrm{n}=3, \ell=2, \mathrm{~m}=+2, \mathrm{~s}=+\frac{1}{2}$

Option 2:
$\mathrm{n}=3, \ell=2, \mathrm{~m}=-2, \mathrm{~s}=+\frac{1}{2}$

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Option 3:
$\mathrm{n}=4, \ell=1, \mathrm{~m}=0, \mathrm{~s}=+\frac{1}{2}$

Option 4:
$\mathrm{n}=4, \ell=0, \mathrm{~m}=0, \mathrm{~s}=+\frac{1}{2}$

Correct Answer:
$\mathrm{n}=4, \ell=0, \mathrm{~m}=0, \mathrm{~s}=+\frac{1}{2}$

Solution:
Cr $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
$\Rightarrow$ 19th e^- belongs to 4s subshell

$\mathrm{n}=4; \ell=0; \mathrm{m}=0; \mathrm{m_s}=+\frac{1}{2}$ or $-\frac{1}{2}$

Q. 4 Given below are two statements:
Statement I: For an ideal gas, heat capacity at constant volume is always greater than the heat capacity at constant pressure.

Statement II : In a constant volume process, no work is produced and all the heat withdrawn goes into the chaotic motion and is reflected by a temperature increase of the ideal gas.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

Solution:
From first law of thermodynamics

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$\Delta U = q + w$

constant volume $w = 0$

$\Delta U = q$

Also $C_P = C_V + R$
C_P = heat capacity at constant pressure
C_v = Heat capacity at constant volume.
R = constant

Q. 5 At $T(K)$, the equilibrium constant of $A_2(g) + B_2(g) \rightleftharpoons C(g)$ is 2.7×10^{-5}. What is the equilibrium constant for $\frac{1}{3} A_2(g) + \frac{1}{3} B_2(g) \rightleftharpoons \frac{1}{3} C(g)$ at the same temperature?

Option 1:
$\left(2.7 \times 10^{-5}\right)^3$

Option 2:
6×10^{-2}

Option 3:
$\sqrt{2.7 \times 10^{-5}}$

Option 4:
3×10^{-2}

Correct Answer:
3×10^{-2}

Solution:
$A_2(g) + B_2(g) \rightleftharpoons (g) K_{eq} = 2.7 \times 10^{-5}$
$\frac{1}{3} A_2(g) + \frac{1}{3} B_e(g) \rightleftharpoons \frac{1}{3} C(g) K_{eq}^1 = (K_{eq})^{1/3}$
$K_{eq}^1 = \left(2.7 \times 10^{-5}\right)^{1/3}$
$= 3 \times 10^{-2}$

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Q. 6 In order to oxidise a mixture of 1 mole each of $FeC_2O_4, Fe_2(C_2O_4)_3, FeSO_4$ and $Fe_2(SO_4)_3$ in acidic medium, the number of moles of $KMnO_4$ required is :

Option 1:
3

Option 2:
2

Option 3:
5

Option 4:
7

Correct Answer:
2

Solution:
Meq of KM_nO_4 = meq of FeC_2O_4 + meq of $Fe_2(C_2O_4)$ + meq of $FeSO_4$
moles $\times 5 = 1 \times 3 + 1 \times 6 + 1 \times 1$

$$\text{moles} = \frac{10}{5} = 2$$

Q. 7 Consider the first order reaction $R \rightarrow P$.
The fraction of molecules decomposed in the given first order reaction can be expressed as

Option 1:
$1 - e^{k_1 t}$

Option 2:
$1 + e^{k_1 t}$

Option 3:
$1 + e^{-k_1 t}$

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Option 4:
$1 - e^{-k_1 t}$

Correct Answer:
$1 - e^{-k_1 t}$

Solution:
It order of $R\mu^H$

$a - x = ae^{-K_1 t}$

$\frac{x}{a} = 1 - e^{-K_1 t}$

Q. 8 A monoatomic anion (A^-)has 45 neutrons and 36 electrons. Atomic mass, group in the periodic table and physical state at room temperature of the element (A) respectively are :

Option 1:
80,17 , liquid

Option 2:
81,16 , solid

Option 3:
80,16 , gas

Option 4:
81,15 , gas

Correct Answer:
80,17 , liquid

Solution:
$X = Br$; Atomic mass $= 80$
Period 4
Br_2 exist as liquid.

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Q. 9 Given below are two statements :

Statement I: The covalency of oxygen is generally two but it can exceed upto four. The oxidation state of oxygen in SO_2 is -2 and in OF_2 it is +2 .

Statement II : The anomalous behaviour of oxygen when compared to the other elements of group 16 is due to its small size and high electronegativity .

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:
Oxygen can have covalency 4
In SO_2; O.S. of $O = -2$
In OF_2; O.S. of $O = +2$
Anamolous behaviour of oxygen is due to high EN and small size.

Q. 10 The correct statements among the following are,
A. $Mo(VI)$ and $W(VI)$ are less stable than $Cr(VI)$.
B. Ce^{4+} and Tb^{4+} are oxidant while Eu^{2+} and Yb^{2+} are reductant.
C. Cm and Am have seven unpaired electrons.
D. Actinoid contraction is greater from element to element than lanthanoid contraction.

Choose the correct answer from the options given below:

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Option 1:
A and B Only

Option 2:
C and D Only

Option 3:
B and D Only

Option 4:
A and C Only

Correct Answer:
B and D Only

Solution:
(A) $\mathrm{Mo(VI)}$ \& $\mathrm{W(VI)}$ more stable than $\mathrm{Cr(VI)}$
(B) In lanthanoids +3 oxidation state is more stable hance +4 ions act as oxidising agents and +2 ions act as reducing agents.
(C) $\mathrm{Cm} - [\mathrm{Rn}]5\mathrm{f}^7 6\ \mathrm{d}^1 7\ \mathrm{S}^2 = 8$ unpaired electrons
$\mathrm{Am} - [\mathrm{Rn}]5\mathrm{f}^7 6\ \mathrm{S}^2 = 7$ unpaired electrons
(D) Actinoid contraction is more than lanthanoid contraction

Q. 11 Correct statements from the following are
A. Potassium dichromate is an oxidising agent and it oxidises $\mathrm{FeSO_4}$ to $\mathrm{Fe_2(SO_4)_3}$ in acidic medium.
B. Sodium dichromate can be used as primary standard in volumetric estimation.
C. $\mathrm{CrO_4^{2-}}$ and $\mathrm{Cr_2O_7^{2-}}$ are interconvertible in aqueous solution by varying the pH of the solution.
D. $\mathrm{Cr - O - Cr}$ bond angle in $\mathrm{Cr_2O_7^{2-}}$ is 126°.

Choose the correct answer from the options given below:

Option 1:
A, B and C only

Option 2:
A, C and D only

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Option 3:
A and C only

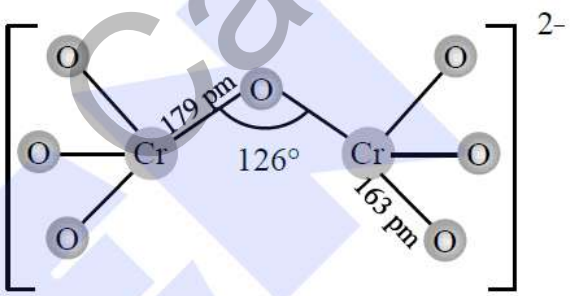
Option 4:
B and D only

Correct Answer:
A, C and D only

Solution:
(A) $MnO_4^- + Fe^{+2} + H^+ \longrightarrow Fe^{+3} + Mn^{+2} + H_2O$
(B) $Na_2Cr_2O_7$ generally not used as a primary standard in volumetric analysis because it is deliquescent.
(C) In acidic medium $Cr_2O_7^{2-}$ exists and it get converted into CrO_4^{2-} in basic medium.

$$2CrO_4^{2-} + 2H^+ \longrightarrow Cr_2O_7^{2-} + H_2O$$
$$Cr_2O_7^{2-} + 2OH^- \longrightarrow 2CrO_4^{2-} + H_2O$$

(D) $K_2Cr_2O_7$ is a good oxidising agent and bond angle of $Cr - O - Cr$ bond in $Cr_2O_7^{2-}$ is 126°



Dichromate ion

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Q. 12 Match the List-I with List-II

List-I Complex ion	List-II Calculated spin only magnetic moment (BM)
A. $[Cr(H_2O)_6]^{2+}$	I. 3.87
B. $[Co(H_2O)_6]^{2+}$	I. 5.95
C.$[Cu(H_2O)_6]^{2+}$	III. 4.90
D.$[Mn(H_2O)_6]^{2+}$	IV. 1.73

Choose the correct answer from the options given below:

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Option 1:
A-I, B-III, C-IV, D-II

Option 2:
A-II, B-I, C-III, D-IV

Option 3:
A-IV, B-II, C-I, D-III

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
$[Cr(H_2O)_6]^{2+} \rightarrow 4$ unpaired $e^-; 4.90BM(III)$
$[Co(H_2O)_6]^{2+} \rightarrow 0$ unpaired $e^-; 3.87BM(I)$
$[Cu(H_2O)_6]^{2+} \rightarrow 1$ unpaired $e^-; 1.73BM(IV)$
$[Mn(H_2O)_6]^{2+} \rightarrow 5$ unpaired $e^-; 5.92BM(II)$

Q. 13 Increasing order of electron withdrawing power of following functional groups is :

a. -CN

b. -COOH

c. $-NO_2$

d. -I

Option 1:

c < b < d < a

Option 2:

c < a < b < d

Option 3:

d < b < a < c

Option 4:

a < b < c < d

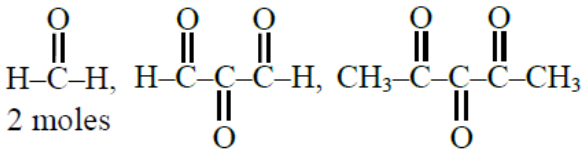
Correct Answer:

d < b < a < c

Solution:

Theory based

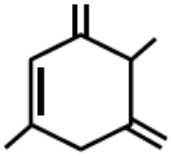
Q. 14 An alkene (X) on ozonolysis followed by reduction gives following products.



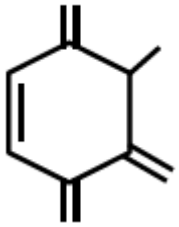
The alkene (X) is :

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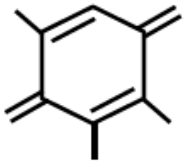
Option 1:



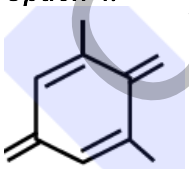
Option 2:



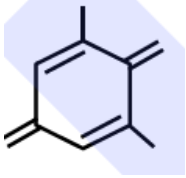
Option 3:



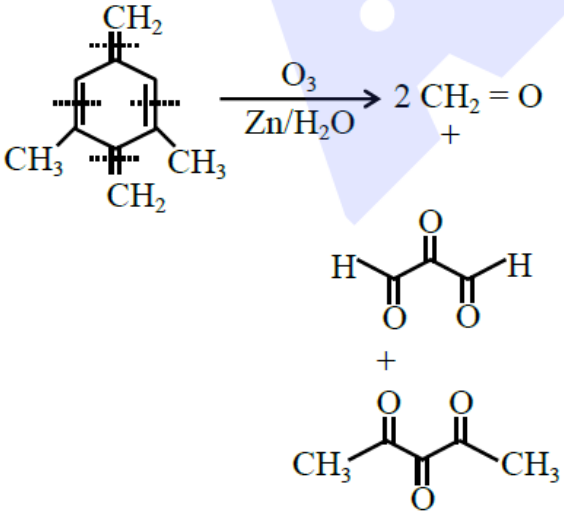
Option 4:



Correct Answer:



Solution:



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Q. 15 Match the List-I with List-II

List_I Name of reaction		List-II Reagent or catalyst used	
A.	Finkelstein reaction	I.	SbF_3
B.	Swarts reaction	II.	Na, dry ether
C.	Sandmeyer's reaction	III.	NaI
D.	Fitting reaction	IV	Cu_2Cl_2

Choose the correct answer from the options given below :

Option 1:
A-I, B-IV, C-III, D-III

Option 2:
A-III, B-I, C-IV, D-II

Option 3:
A-IV, B-II, C-I, D-III

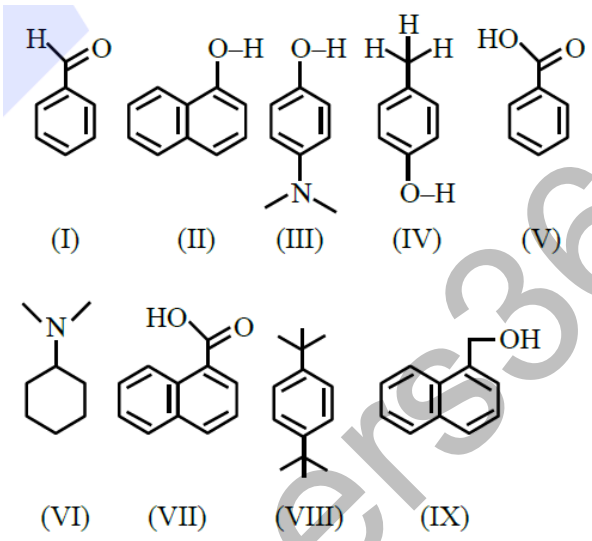
Option 4:
A-I, B-III, C-II D-IV

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
Theory based

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Q. 16 Amongst the following, the total number of compounds soluble in aqueous NaOH at room temperature is:



Option 1:
5

Option 2:
4

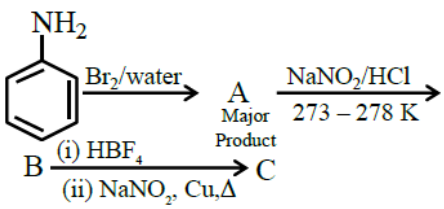
Option 3:
6

Option 4:
3

Correct Answer:
5

Solution:
II, III, IV, V, VII
Compounds which are more acidic than H_2O can soluble in NaOH .

Q. 17 Product C of the following reaction sequence will be



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Option 1:
1-Bromo-4-nitrobenzene

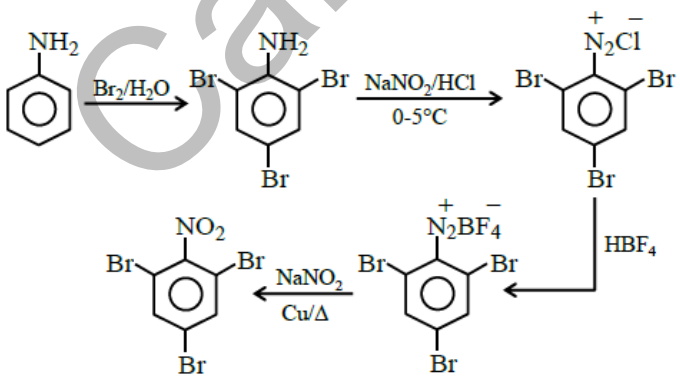
Option 2:
1,3,5-Tribromo-2-nitrobenzene

Option 3:
4-Bromo-1-nitrobenzene

Option 4:
1,3,5-Tribromobenzene

Correct Answer:
1,3,5-Tribromo-2-nitrobenzene

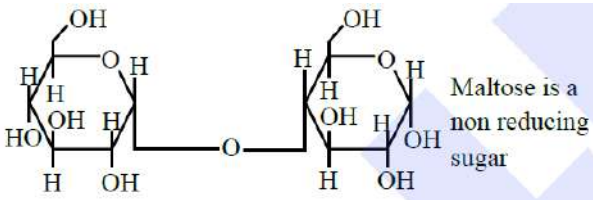
Solution:



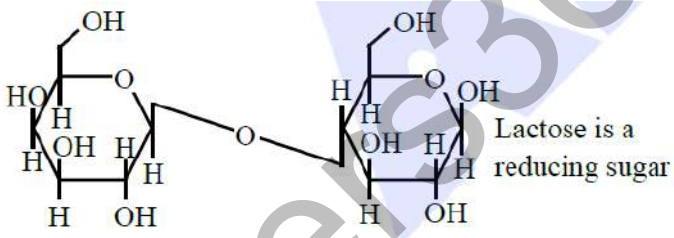
1,3,5-Tribromo
-2-nitrobenzene

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Q. 18 Statement-I : The structure of Maltose is given below :



Statement-II : The structure of Lactose is given below



In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

Solution:
Maltose and lactose both have hemiacetal linkage. So both are reducing sugars.

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Q. 19 Match the LIST -I with LIST-II

List-I Name of amino acid		List-II One letter symbol/type)	
A.	Arginine	I	D / Non-essential
B.	Aspartic acid	II	R / Essential
C.	Lysine	III	E / Non-essential
D.	Glutamic acid	IV	K/Essential

Choose the correct answer from the options given below :

Option 1:
A-II, B-I, C-IV, D-III

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-IV, C-I, D-II

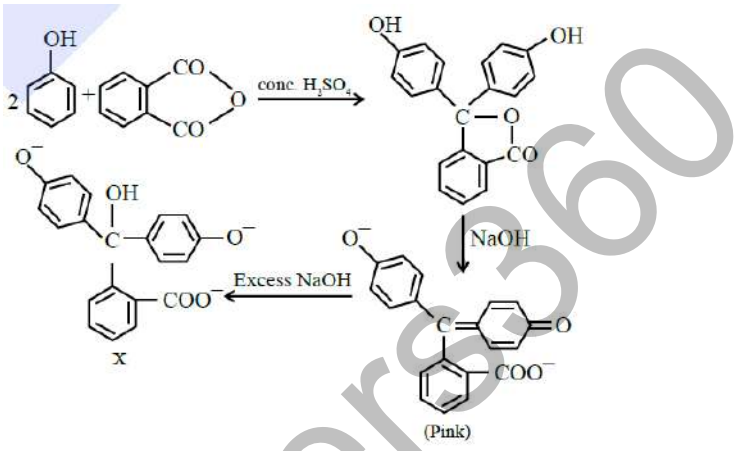
Option 4:
A-II, B-IV, C-I, D-III

Correct Answer:
A-II, B-I, C-IV, D-III

Solution:
Arginine (R) and Lysine (K) are essential amino acid. Aspartic acid (D) and Glutamic acid (E) are non-essential amino acid.

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Q. 20 Identify the colour of compound 'X' in the sequence of the reaction.



Option 1:
Violet

Option 2:
Green

Option 3:
Red

Option 4:
Colourless

Correct Answer:
Colourless

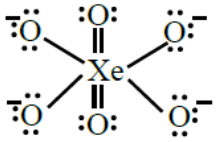
Solution:
Phenolphthalein indicator turns colourless in excess NaOH .

Q. 21 According to Lewis theory, the total number of bond-pairs and lone pair of electrons around the central atom of $\mathrm{XeO_6^{4-}}$ ion is _____ .

Correct Answer:
6

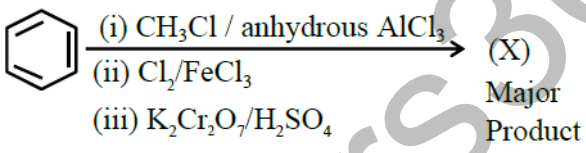
Solution:

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$\sigma = 6$
Lone pairs $= 0$

Q. 22 Consider the following sequence of reactions to give the major product (X)

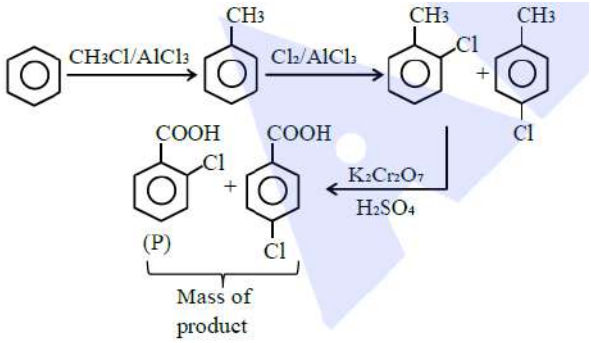


$\mathrm{P}g$ of the major product (X) formed is reacted with $\mathrm{NaHCO_3}$ soluion to liberate a gas which occupied $11.2\mathrm{dm}^3$ at STP.
$\mathrm{P} =$ ____ . g.
(Given molar mass in $\mathrm{gmol}^{-1}\mathrm{H}: 1, \mathrm{C}: 12, \mathrm{O}: 16, \mathrm{Cl}: 35.5$)

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Correct Answer:
78

Solution:



$$\text{Moles of P} = \frac{11.2}{22.4} = 0.5 \text{ mole}$$
$$\mathrm{M}_{\mathrm{C_7H_5O_2Cl}} = 156.5\mathrm{gm}$$
$$\mathrm{W}_{\mathrm{P}(\mathrm{C_7H_5O_2Cl})} = \frac{156.5}{2} = 78.25 \approx 78\mathrm{grams}$$

Q. 23 2.0 g of a bromo hydrocarbon (X) was subjected to Carius analysis, gave 3.36 g of AgBr . The percentage of carbon in the compound (X) is 26.7%. Total number of carbon atoms in the empirical formula for compound (X) is _____ .
(Given molar mass in $\mathrm{gmol}^{-1} \mathrm{H}: 1, \mathrm{C}: 12, \mathrm{Br}: 80, \mathrm{Ag}: 108$)

Correct Answer:
5

Solution:
$\mathrm{n_{Br}} = \frac{3.36}{(108+80)}$

$\mathrm{r} =$ no. of Br atoms in organic compound

$$\mathrm{n_{organic\ compound}} = \frac{3.36}{(108+80)\times \mathrm{x}} = \frac{2}{\mathrm{M}}$$

$$\mathrm{M} = \frac{2 \times 188 \times \mathrm{x}}{3.36}$$

(x is integer)

$$\mathrm{W_c} = \frac{112 \times \mathrm{x} \times 26.7}{100} = 30 \times \mathrm{x}$$

$$\mathrm{n_c} = \frac{30 \times \mathrm{x}}{12} = 2.5 \times \mathrm{x}$$

take $\mathrm{x} = 2, \mathrm{n_c} = 5$
(Empirical formula is $\mathrm{C_5H_4Br}$)

Q. 24 The pH a solution obtained by mixing 5 mL of $0.1 \mathrm{M} \mathrm{NH_4OH}$ solution with 250 mL of 0.1 M $\mathrm{NH_4Cl}$ solution is _____ $\times 10^{-2}$. (Nearest integer)
Given : $\mathrm{pK_b}\ (\mathrm{NH_4OH}) = 4.74$

$\log 2 = 0.30$
$\log 3 = 0.48$
$\log 5 = 0.70$

Correct Answer:
756

Solution:
On mixing final volume $= 255\mathrm{ml}$

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$$[NH_4OH] = \frac{5 \times 0.1}{255}, [NH_4Cl] = \frac{250 \times 0.1}{255}$$

$$p^{OH} = pk_b + \log \frac{[NH_4Cl]}{NH_4OH}$$

$$p^{OH} = 4.74 + \log \frac{250 \times 0.1}{5 \times 0.1}$$

$$P^H = 14 - P^{OH} = 14 - 6.44$$

Q. 25 A non-volatile, non-electrolyte solid solute when dissolved in 40 g of a solvent, the vapour pressure of the solvent decreased from 760 mm Hg to 750 mm Hg . If the same solution boils at 320 K , then the number of moles of the solvent present in the solution is _____ . (Nearest integer)
[Given : boiling point of the pure solvent $= 319.5\ \mathrm{K},\ K_b$ of the solvent $= 0.3\ \mathrm{K\ kg\ mol^{-1}}$]

Correct Answer:
5

Solution:

$$\frac{P^\circ - Ps}{Ps} = \text{i. molality} \times \frac{(\text{M. solvent})}{1000}$$

$$\Delta T_b = i.K_b.\ \text{molality} \Rightarrow \text{molality} = \frac{0.5}{0.3}$$

$$(\text{Molecular Mass}) = \frac{600}{75}\ \text{g}$$

$$\text{Moles} = \frac{40}{600/75} = 5$$

Maths

Q. 1 Let [·] denote the greatest integer function. If the domain of the function $f(x) = \cos^{-1}\left(\frac{4x+2[x]}{3}\right)$ is $[\alpha, \beta]$, then $12(\alpha + \beta)$ is equal to :

Option 1:
6

Option 2:
8

Option 3:
9

Option 4:
4

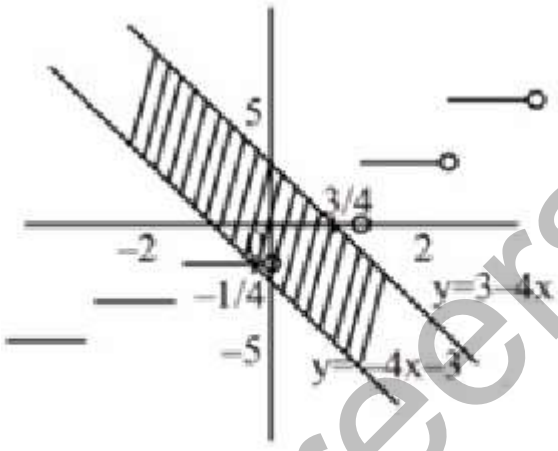
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Correct Answer:
6

Solution:

$$-1 \leq \frac{4x+2[x]}{3} \leq 1$$

$$-3-4x \leq 2[x] \leq 3-4x$$



$$x \in \left[\frac{-1}{4}, \frac{3}{4}\right]$$

$$12(\alpha+\beta)=6$$

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Q. 2 If the set of all solutions of $\left|x^2+x-9\right| = |x| + \left|x^2-9\right|$ is $[\alpha, \beta] \cup [\gamma, \infty)$, then $\left(\alpha^2+\beta^2+\gamma^2\right)$ is equal to :

Option 1:
9

Option 2:
18

Option 3:
36

Option 4:
72

Correct Answer:
18

Solution:

$|a|+|b|=|a+b|$
$\Rightarrow a \cdot b \geq 0$
$\Rightarrow x\left(x^2-9\right) \geq 0$
$x \in [-3,0] \cup [3,\infty]$
$\alpha^2+\beta^2+\gamma^2=18$

Q. 3 Let z be a complex number such that $|z+2|=|z-2|$ and $\arg\left(\frac{z+3}{z-i}\right)=\frac{\pi}{4}$. Then $|z|^2$ is equal to :

Option 1:
9

Option 2:
4

Option 3:
5

Option 4:
1

Correct Answer:
9

Solution:
$|z+2|=|z-2|$
$\Rightarrow z=0+iy \quad y \in R$
$\arg\left(\frac{3+iy}{iy-i}\right)=\frac{\pi}{4}$
$\Rightarrow \arg\left(\frac{y}{y-1}+\frac{3i}{1-y}\right)=\frac{\pi}{4}$
$\Rightarrow \frac{y}{y-1}=\frac{3}{1-y}>0$
$\Rightarrow z=-3i$
$|z|^2=9$

Q. 4 The number of functions $f:\{1,2,3,4\} \rightarrow \{a,b,c\}$, which are not onto, is :

Option 1:
48

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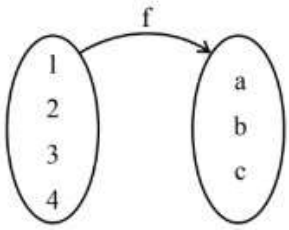
Option 2:
45

Option 3:
51

Option 4:
35

Correct Answer:
45

Solution:



No. of into function

$= {}^3\mathrm{C}_1 2^4 - {}^3\mathrm{C}_2 1 + 0$
$= 3 \times 16 - 3 \times 1$
$= 45$

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Q. 5 Let

$$S = \left\{ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \{0, 1, 2, 3, 4\} \text{ and } A^2 - 4A + 3I = 0 \right\}$$

be a set of 2×2 matrices. Then the number of matrices in S , for which the sum of the diagonal elements is equal to 4 , is:

Option 1:
20

Option 2:
17

Option 3:
21

Option 4:
19

Correct Answer:
19

Solution:

$a, b, c, d \in \{0, 1, 2, 3, 4\}$

$a + d = 4$ $\quad ad - bc = 3$

Case-I $\quad a = 0 \quad d = 4$

$0 - bc = 3$

No solution

Case-II $\quad a = 1 \quad d = 3$

$3 - bc = 3$

Case-III $\quad bc = 0 \quad \Rightarrow 9$ ways

$a = 2 \quad d = 2$

$4 - bc = 3$

$bc = 1 \quad \Rightarrow 1$ ways

Case-IV $\quad a = 3 \quad d = 1$

$bc = 0 \quad \Rightarrow 9$ way

Case-V $\quad a = 4 \quad d = 0 \Rightarrow$ No Solution

Total 19 matrix

Q. 6 Let $A = \begin{bmatrix} 1 & 1 & 2 \\ -2 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix}$. Then the sum of all elements of the matrix $\text{adj}\left(\text{adj}\left(2(\text{adj}\, A)^{-1}\right)\right)$ is equal to :

Option 1:

3

Option 2:

4

Option 3:

-4

Option 4:

-3

Correct Answer:

-3

Solution:

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$|A| = -4$

$As (\text{adj } A)^{-1} = \frac{A}{|A|} = \frac{-A}{4}$

$B = 2(\text{adj } A)^{-1} = \frac{-A}{2}$

$|B| = \left|\frac{-A}{2}\right|$

$= \frac{-1}{8}|A| = \frac{1}{2}$

$\text{adj}(\text{adj } B) = |B|^{n-2}B = |B|B$

$= \frac{B}{2} = \frac{-A}{4}$

$\text{Sum of element } = \frac{-12}{4} = -3$

Q. 7 The first term of an A.P. of 30 non-negative terms is $\frac{10}{3}$. If the sum of this A.P. is the cube of its last term, then its common difference is :

Option 1:
$\frac{5}{87}$

Option 2:
$\frac{25}{83}$

Option 3:
$\frac{15}{29}$

Option 4:
$\frac{5}{29}$

Correct Answer:
$\frac{5}{87}$

Solution:

$\text{a} = \frac{10}{3} : \text{S}_{30} = (\text{T}_{30})^3$

$\frac{30}{2}\left[2 \times \frac{10}{3} + 29\text{ d}\right] = \left[\frac{10}{3} + 29\text{ d}\right]^3$

$15\left(\frac{20}{3} + 29\text{ d}\right) = \left(\frac{10}{3} + 29\text{ d}\right)^3$

$15\left(\frac{10}{3} + \frac{10}{3} + 29\text{ d}\right) = \left(\frac{10}{3} + 29\text{ d}\right)^3$

By observation $\frac{10}{3} + 29\text{ d} = 5$

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$$29d = 5 - \frac{10}{3} = \frac{5}{3}$$

$$d = \frac{5}{87}$$

Q. 8 The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is:

Option 1:
5040

Option 2:
3050

Option 3:
3410

Option 4:
4320

Correct Answer:
4320

Solution:
No. of ways to arrange 7 students $= 7!$
Cases where all girls are together

$\mathrm{B_1\ B_2\ B_3\ B_4 G_1 G_2 G_3} = 5!3!$

$\text{Ans.} = 7! - 5!3!$

$= 7! - 6!$

$= 6 \times 6!$

$= 6 \times 720$

$= 4320$

Q. 9 Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \ldots + (1+x)^{99} + (1+kx)^{100}, x \neq 0$, is $\left(43n + \frac{101}{4}\right)\ (\ {}^{100}\mathrm{C}_3\ )$ for some $\mathrm{n} \in \mathrm{N}$, be p. Then the value of p +n is:

Option 1:
10

Option 2:
11

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Option 3:
12

Option 4:
13

Correct Answer:
11

Solution:

$$^{3}\mathrm{C}_{3}+{}^{4}\mathrm{C}_{3}+{}^{5}\mathrm{C}_{3}+\ldots+{}^{99}\mathrm{C}_{3}+{}^{100}\mathrm{C}_{3}\,\mathrm{K}^{3}=\left(43\mathrm{n}+\frac{101}{4}\right)^{100}\mathrm{C}_{3}$$

$$\Rightarrow{}^{4}\mathrm{C}_{4}+{}^{4}\mathrm{C}_{3}+{}^{5}\mathrm{C}_{3}+\ldots+{}^{99}\mathrm{C}_{3}+{}^{100}\mathrm{C}_{3}\,\mathrm{K}^{3}=\left(43\mathrm{n}+\frac{101}{4}\right){}^{100}\mathrm{C}_{3}$$

$$={}^{100}\mathrm{C}_{4}+{}^{100}\mathrm{C}_{3}\,\mathrm{K}^{3}=\left(43\mathrm{n}+\frac{101}{4}\right){}^{100}\mathrm{C}_{3}$$

$$\frac{100}{1^{100}\mathrm{C}_{4}}+\mathrm{K}^{3}=43\mathrm{n}+\frac{101}{4}$$

$$\frac{97}{4}+\mathrm{K}^{3}=43\mathrm{n}+\frac{101}{4}$$

$$\mathrm{K}^{3}=43\mathrm{n}+1$$

$$n=5$$

$$\mathrm{K}^{3}=216$$

$$\mathrm{K}=6=\mathrm{P}$$

$$\mathrm{P}+\mathrm{n}=11$$

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Q. 10 Suppose that the mean and median of the nonnegative numbers $21, 8, 17, a, 51, 103, b, 13, 67, (\mathrm{a}>\mathrm{b})$, are 40 and 21, respectively. If the mean deviation about the median is 26 , then 2 a is equal to :

Option 1:
109

Option 2:
117

Option 3:
161

Option 4:
131

Correct Answer:
131

Solution:

$$8, 13, b, 17, 21, 51, a, 67, 103$$

$$\text{mean } = 40 = \frac{280 + a + b}{9}$$

$$a + b = 80$$

mean deviation about median

$$= \frac{13 + 8 + 4 + 30 + 46 + 82 + (a - 21) + (21 - b)}{9} = 26$$

$$a - b = 51$$

$$\Rightarrow 2a = 131$$

Q. 11 Let the line $\mathrm{L_1 : x + 3 = 0}$ intersect the lines $\mathrm{L_2 : x - y = 0}$ and $\mathrm{L_3 : 3x + y = 0}$ at the points A and B , respectively. Let the bisector of the obtuse angle between the lines $\mathrm{L_2}$ and $\mathrm{L_3}$ intersect the line $\mathrm{L_1}$ at the point C . Then $\mathrm{BC^2 : AC^2}$ is equal to :

Option 1:
5:1

Option 2:
1:5

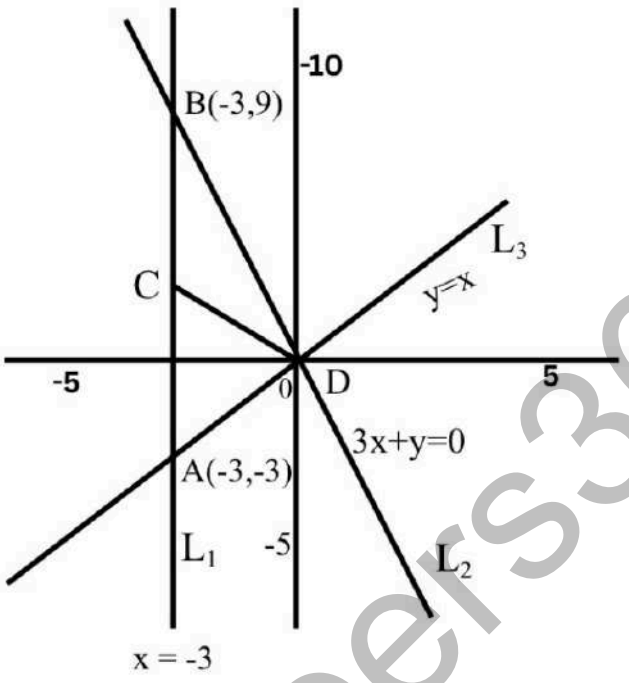
Option 3:
2:3

Option 4:
3:2

Correct Answer:
5:1

Solution:

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$$\frac{BC^2}{AC^2} = \left(\frac{OB}{OA}\right)^2 = \frac{9+81}{9+9}$$
$$= 5$$

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Q. 12 Let the vertex A of a triangle ABC be $(1, 2)$, and the mid-point of the side AB be $(5, -1)$. If the centroid of this triangle is $(3, 4)$ and its circumcenter is (α, β), then $21(\alpha + \beta)$ is equal to:

Option 1:
309

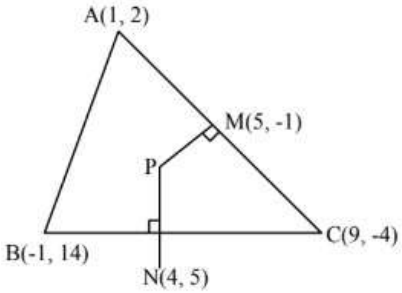
Option 2:
403

Option 3:
497

Option 4:
524

Correct Answer:
497

Solution:



$\text{equation of PM} : 4x - 3y = 23$

$\text{equation of PM} : 5x - 9y = -25$

$P\left(\frac{94}{7}, \frac{215}{21}\right)$

$21(\alpha + \beta) = 497$

Q. 13 Suppose the two chords, drawn from the point (1, 2) on the circle $x^2 + y^2 + x - 3y = 0$ are bisected by the y -axis. If the other ends of these chords are R and S , and the mid point of the line segment RS is (α, β), then $6(\alpha + \beta)$ is equal to :

Option 1:
1

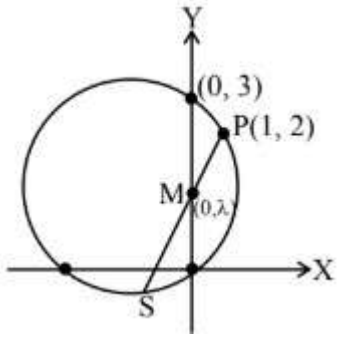
Option 2:
3

Option 3:
4

Option 4:
6

Correct Answer:
3

Solution:



chord PS bisected by y -axis
at point $\mathrm{M}(0, \lambda)$

$\therefore \mathrm{S}(-1, 2\lambda - 2)$

Which satisfy equation of circle

$\Rightarrow (-1)^2 + (2\lambda - 2)^2 - 1 - 3(2\lambda - 2) = 0$

$\Rightarrow \lambda = 1 \quad \& \quad \lambda = \frac{5}{2}$

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Therefore two such chord PS \& PR will exist
When $\lambda = 1 \Rightarrow \ \mathrm{S}(-1, 0)$
When $\lambda = \frac{5}{2} \quad \mathrm{R}(-1, 3)$
$\therefore$ Mid point of $\mathrm{R\&\ S}$ will be $\left(-1, \frac{3}{2}\right)$

$$\therefore \alpha = -1, \beta = \frac{3}{2}$$
$$\therefore 6(\alpha + \beta) = 3$$

Q. 14 A line with direction ratios $1, -1, 2$ intersects the lines $\frac{x}{2} = \frac{y}{3} = \frac{z+1}{3}$ and $\frac{x+1}{-1} = \frac{y-2}{1} = \frac{z}{4}$ at the points P and Q , respectively. If the length of the line segment PQ is α, then $225\alpha^2$ is equal to:

Option 1:
1024

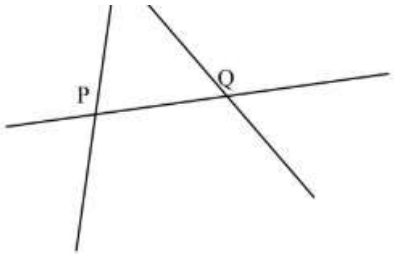
Option 2:
1014

Option 3:
1104

Option 4:
1204

Correct Answer:
1014

Solution:



$\mathrm{P}(2\lambda, 3\lambda, 3\lambda - 1)$ and $\mathrm{Q}(-\mu - 1, \mu + 2, 4\mu)$
$\mathrm{D.R\ of\ PQ}\ 2\lambda + \mu + 1, 3\lambda - \mu - 2, 3\lambda - 4\mu - 1,$
$$\frac{2\lambda + \mu + 1}{1} = \frac{3\lambda - \mu - 2}{-1} = \frac{3\lambda - 4\mu - 1}{2}$$

Solving $\lambda = \frac{1}{5}$ and $\mu = \frac{-8}{15}$

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$P\left(\frac{2}{5}, \frac{3}{5}, \frac{-2}{5}\right) \& Q\left(\frac{-7}{15}, \frac{22}{15}, \frac{-32}{15}\right)$

$PQ^2 = \left(\frac{13}{15}\right)^2 + \left(\frac{13}{15}\right)^2 + \left(\frac{26}{15}\right)^2$

$\alpha^2 = \frac{1014}{225} \therefore 225\alpha^2 = 1014$

Q. 15 The square of the distance of the point $(-2, -8, 6)$ from the line $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z}{-1}$ along the line $\frac{x+5}{1} = \frac{y+5}{-1} = \frac{z}{2}$ is equal to:

Option 1:
3

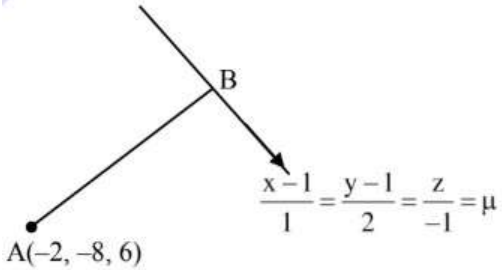
Option 2:
6

Option 3:
8

Option 4:
12

Correct Answer:
6

Solution:



DR or AB is $1, -1, 2$

$\therefore$ Line AB is $\frac{x+2}{1} = \frac{y+8}{-1} = \frac{z-6}{2} = \lambda$

B is ($\lambda - 2, -\lambda - 8, 2\lambda + 6$)

B is $(\mu + 1, 2\mu + 1, -\mu)$

Solving point B is $(-3, -7, 4)$

$AB = \sqrt{(-1)^2 + (1)^2 + (2)^2} = \sqrt{6}$

$AB^2 = 6$

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Q. 16 If $y=\tan^{-1}\left(\frac{3\cos x-4\sin x}{4\cos x+3\sin x}\right)+2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$, then $\frac{dy}{dx}$ at $x=\frac{\sqrt{3}}{2}$ is equal to :

Option 1:
3

Option 2:
-1

Option 3:
1

Option 4:
2

Correct Answer:
1

Solution:

$$y=\tan^{-1}\left(\frac{\frac{3}{4}-\tan x}{1+\frac{3}{4}\tan x}\right)+2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$$

$$\text{For}\quad 2\tan^{-1}\left(\frac{x}{1+\sqrt{1-x^2}}\right)$$

$$\text{put}\quad x=\sin\theta$$

$$\therefore 2\tan^{-1}\left(\frac{\sin\theta}{1+\cos\theta}\right)=\theta=\sin^{-1}x$$

Hence

$$y=\tan^{-1}\left(\frac{3}{4}\right)-\tan^{-1}(\tan x)+\sin^{-1}x$$

$$\frac{dy}{dx}=-1+\frac{1}{\sqrt{1-x^2}}$$

$$x=\frac{\sqrt{3}}{2}$$

$$\frac{dy}{dx}=-1+2=1$$

Q. 17 Let f be a real polynomial of degree n such that $\mathrm{f}(\mathrm{x})=\mathrm{f}'(\mathrm{x})\mathrm{f}''(\mathrm{x})$, for all $\mathrm{x}\in\mathbb{R}$. If $\mathrm{f}(0)=0$, then $36\left(\mathrm{f}'(2)+\mathrm{f}''(2)+\int_0^2\mathrm{f}(\mathrm{x})\mathrm{dx}\right)$ is equal to :

Option 1:
42

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Option 2:
46

Option 3:
56

Option 4:
66

Correct Answer:
56

Solution:
Let degree of polynomial be ' n '

$n = n - 1 + n - 2 \Rightarrow n = 3$
$f(x) = ax^3 + bx^2 + cx$
$\left(ax^3 + bx^2 + cx\right) = \left(3ax^2 + 2bx + c\right)(6ax + 2b)$
$x^3 : a = 18a^2 \Rightarrow a = \frac{1}{18}$
$x^2 : b = 6ab + 12ab$
$x : c = 4b^2 + 6ac \Rightarrow \frac{2C}{3} = 4b^2 \Rightarrow b^2 = \frac{c}{6}$

const : $bc = 0 \Rightarrow b = c = 0$

$\int_0^2 f(x)dx + f'(2) + f''(2)$
$\int_0^2 ax^3 dx + 12a + 12a = 4a + 24a = 28a = \frac{28}{18} = \frac{14}{9}$
$36\left(\frac{14}{9}\right) = 56$

Q. 18 The area of the region $\{(x, y) : y \leq \pi - |x|, y \leq |\, x \sin x\,|, y \geq 0\}$ is :

Option 1:
$1 + \frac{\pi^2}{8}$

Option 2:
$2 + \frac{\pi^2}{4}$

Option 3:
$\frac{\pi^2}{8} - 1$

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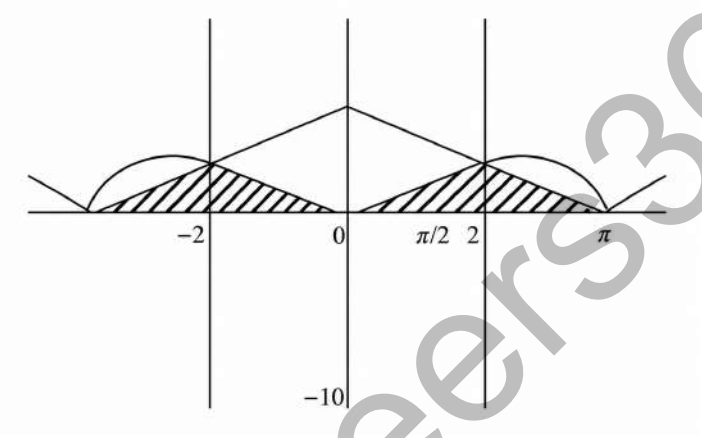
Option 4:

$4+\frac{\pi^2}{2}$

Correct Answer:

$2+\frac{\pi^2}{4}$

Solution:



$$\text{Area };2\left(\int_0^{\pi/2} x\sin x dx+\int_{\pi/2}^{\pi}(\pi-x)dx\right)$$

$$=2(-x\cos x+\sin x)_0^{\pi/2}+2\left(\frac{(\pi-x)^2}{2}\Big|_{\pi}^{\pi/2}\right)$$

$$=2+\frac{\pi^2}{4}$$

Q. 19 Let $\int_{-2}^{2}(|\sin x|+[x\sin x])dx=2(3-\cos 2)+\beta$, where [•] is the greatest integer function. Then $\beta\sin\left(\frac{\beta}{2}\right)$ equals.

Option 1:

1

Option 2:

2

Option 3:

4

Option 4:

8

Correct Answer:

2

Solution:

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$$\int_{-2}^{2}(|\sin x|+[x\sin x])dx$$

$$\Rightarrow 2\int_{0}^{2}(|\sin x|+[x\sin x])dx$$

$$\Rightarrow 2\left[\int_{0}^{2}\sin x dx+\int_{0}^{2}[x\sin x]dx\right]$$

$$2\left([-\cos x]_0^2+\int_0^{\alpha}0dx+\int_{\alpha}^{2}2dx\right)$$

$$2(3-\cos 2-2-\alpha)$$

$$2(3-\cos 2)-2\alpha$$

where α is Root of $x\sin x=1$

Now comparing $\beta=-2\alpha$

$$\alpha=-\frac{\beta}{2}$$

$$-\frac{\beta}{2}\sin\left(\frac{-\beta}{2}\right)=1$$

$$\beta\sin\left(\frac{\beta}{2}\right)=2$$

Q. 20 Let $y=y(x)$ be the solution of the differential equation $\frac{dy}{dx}=\left(1+x+x^2\right)\left(1-y+y^2\right), y(0)=\frac{1}{2}$. Then $(2y(1)-1)$ is equal to

Option 1:

$\sqrt{3}\tan\left(\frac{11\sqrt{3}}{6}\right)$

Option 2:

$\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{12}\right)$

Option 3:

$\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$

Option 4:

$\frac{\sqrt{3}}{2}\tan\left(\frac{11\sqrt{3}}{6}\right)$

Correct Answer:

$\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$

Solution:

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$$\frac{dy}{y^2-y+1}=\left(x^2+x+1\right)dx$$

$$\frac{dy}{\left(y-\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2}=\left(x^2+x+1\right)dx$$

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{y-\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)=\frac{x^3}{3}+\frac{x^3}{2}+x+C$$

$$\frac{2}{\sqrt{3\sqrt{3}}}=\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=\frac{x^3}{3}+\frac{x^3}{2}+x+C$$

$$y(0)=\frac{1}{2}\Rightarrow 0=0+0+0+C\Rightarrow C=0$$

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=\frac{x^3}{3}+\frac{x^2}{2}+x$$

$$\text{put } x=1$$

$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=\frac{1}{3}+\frac{1}{2}+1=\frac{2+3+6}{6}=\frac{11}{6}$$

$$\tan^{-1}\left(\frac{2y-1}{\sqrt{3}}\right)=\frac{11\sqrt{3}}{12}$$

$$\frac{2y-1}{\sqrt{3}}=\tan\left(\frac{11\sqrt{3}}{12}\right)$$

$$\Rightarrow 2y(1)-1=\sqrt{3}\tan\left(\frac{11\sqrt{3}}{12}\right)$$

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Q. 21 A coin is tossed 8 times. If the probability that exactly 4 heads appear in the first six tosses and exactly 3 heads appear in the last five tosses is p, then 96 p is equal to _____ .

Correct Answer:
9

Solution:
$x_1x_2x_3x_4x_5x_6x_7x_8$

Exactly 4 heads in first six Tosses.

$(x_1x_2x_3x_4x_5x_6)$

Exactly 3 heads in last 5 Tosses.
i.e. $(x_4x_5x_6x_7x_8)$
Common tosses $(x_4x_5x_6)$
Let k be number of heads on $(x_4x_5x_6)$
$\mathrm{k}=1\therefore$ ways to overlap $={}^3\mathrm{C}_1$
${}^3\mathrm{C}_3\times{}^3\mathrm{C}_1\times{}^2\mathrm{C}_2=3$ ways

$k = 2$
${}^3C_2 \times {}^3C_2 \times {}^2C_1 = 18$
$k = 3$
${}^3C_1 \times 1 = 3$

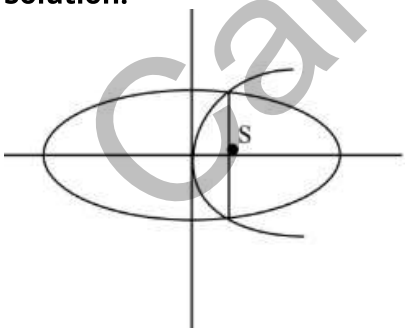
Total ways $24\left(\frac{1}{2}\right)^8$

$96 \times \frac{24}{28} = \frac{6\times24}{16} = 9$

Q. 22 Consider the parabola $P : y^2 = 4kx$ and the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Let the line segment joining the points of intersection of P and E , be their latus rectums. If the eccentricity of E is e , then $e^2 + 2\sqrt{2}$ is equal to ____ .

Correct Answer:
3

Solution:



Focus of parabola ($k, 0$)
Focus of Ellipse (ae, 0) $k = ae$

Also $4k = 2\frac{b^2}{a}$

$$4ae = \frac{2\,b^2}{a}$$
$$\Rightarrow 4a^2e = 2\,b^2$$
$$4a^2e = 2a^2\left(1 - e^2\right)$$
$$2e = 1 - e^2$$
$$e^2 + 2e + 1 = 2$$
$$(e+1)^2 = 2$$
$$e + 1 = \sqrt{2} \quad \therefore e = \sqrt{2} - 1$$
$$e^2 + 2\sqrt{2} = 3$$

Q. 23 If $A = \frac{\sin 3^\circ}{\cos 9^\circ} + \frac{\sin 9^\circ}{\cos 27^\circ} + \frac{\sin 27^\circ}{\cos 81^\circ}$ and $B = \tan 81^\circ - \tan 3^\circ$, then $\frac{B}{A}$ is equal to_____.

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Correct Answer:
2

Solution:

$$\text{Consider } \mathrm{E} = \frac{\sin\theta}{\cos 3\theta} \Rightarrow \mathrm{E} = \frac{2\sin\theta\cos\theta}{2\cos 3\theta\cos\theta}$$

$$\Rightarrow \mathrm{E} = \frac{\sin 2\theta}{2\cos 3\theta\cos\theta} \Rightarrow \mathrm{E} = \frac{\sin(30-\theta)}{2\cos 3\theta\cos\theta}$$

$$\Rightarrow \mathrm{E} = \frac{1}{2}[\tan 3\theta - \tan\theta]$$

$$\mathrm{A} = \frac{1}{2}[\tan 9^\circ - \tan 3^\circ + \tan 27^\circ - \tan 9^\circ + \tan 81^\circ - \tan 27^\circ]$$

$$\therefore \mathrm{A} = \frac{1}{2}[\tan 81^\circ - \tan 3^\circ]$$

$$\therefore \frac{\mathrm{B}}{\mathrm{A}} = 2$$

Q. 24

Let $\overrightarrow{\mathrm{a_k}} = (\tan\theta_\mathrm{k})\hat{\mathrm{i}} + \hat{\mathrm{j}}$ and $\overrightarrow{\mathrm{b_k}} = \hat{\mathrm{i}} - (\cos\theta_\mathrm{k})\hat{\mathrm{j}}$, where $\theta_k = \frac{2^{k-1}\pi}{2^n+1}$, for some $n \in N, n > 5$. Then the value of $\frac{\sum_{k=1}^{n}|\vec{a}_k|^2}{\sum_{k=1}^{n}|\vec{b}_k|^2}$ is _____ .

Correct Answer:
3

Solution:

$$|a_k|^2 = 1 + \tan^2\theta_k = \sec^2\theta_k$$

$$|b_k|^2 = 1 + \cot^2\theta_k = \operatorname{cosec}^2\theta_k$$

$$\text{Since, } \cot\theta - \tan\theta = 2\cot 2\theta$$

$$\therefore -\operatorname{cosec}^2\theta - \sec^2\theta = -4\operatorname{cosec}^2 2\theta$$

$$\sec^2\theta = 4\operatorname{cosec}^2 2\theta - \operatorname{cosec}^2\theta$$

$$\Sigma\sec^2\theta_k = 4\Sigma\operatorname{cosec}^2 2\theta_k - \Sigma\operatorname{cosec}^2\theta_k$$

$$\text{Now}$$

$$\Sigma\operatorname{cosec} 2\theta_k = \Sigma\operatorname{cosec}^2\theta_k$$

$$\text{since}$$

$$\sum_{k=1}^{n}\operatorname{cosec}^2\frac{2^k\pi}{2^n+1} = \sum_{k=1}^{n}\operatorname{cosec}^2\frac{2^{k-1}\pi}{2^n+1}$$

$$\text{because } \operatorname{cosec}^n\frac{2^n\pi}{2^n+1} = \operatorname{cosec}\frac{\pi}{2^n+1}$$

$$\therefore \Sigma\sec^2\theta_k = 3\Sigma\operatorname{cosec}^2\theta_k$$

$$\frac{\sum\sec^2\theta_k}{\sum\operatorname{cosec}^2\theta_k} = 3$$

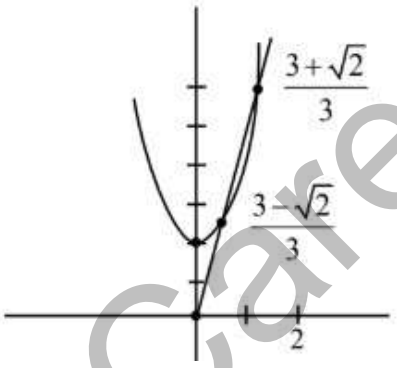
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Q. 25 The number of points, at which the fucntion

$$f(x) = \max\left\{6x, 2+3x^2\right\} + |x-1|\cos\left|x^2 - \frac{1}{4}\right|, x \in (-\pi, \pi), \text{ is not differentiable, is}_____$$

Correct Answer:
3

Solution:
$2 + 3x^2 = 6x \Rightarrow 3x^2 - 6x + 2 = 0$



$x = \frac{6 \pm \sqrt{12}}{6} = \frac{3 \pm \sqrt{2}}{3}$

$|x-1|\cos\left|x^2 - \frac{1}{4}\right|$ is differentiable at $x = 1$
Total number of points where $f(x)$ is nonDifferentiable is 3

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JEE MAINS

2026

4th April

2nd Shift

Question Paper

Physics

Q. 1 Match the LIST-I with LIST-II

LIST-I		LIST-II	
A	Planck's constant	I	$\mathrm{ML^2\ T^{-2}}$
B	Stopping potential	II	$\mathrm{T^{-1}}$
C	Work function	III	$\mathrm{ML^2\ T^{-1}}$
D	Threshold frequency	IV	$\mathrm{ML^2\ T^{-3}\ A^{-1}}$

Choose the correct answer from the options given below:

Option 1:
A-III, B-IV, C-I, D-II

Option 2:
A-I, B-II, C-III, D-IV

Option 3:
A-IV, B-III, C-I, D-II

Option 4:
A-I, B-IV, C-III, D-II

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
(i) $[\mathrm{h}] = \frac{[\mathrm{E}]}{[\mathrm{v}]} = \frac{\mathrm{ML^2\ T^{-2}}}{\mathrm{T^{-1}}} = \mathrm{ML^2\ T^{-1}}$
(ii) Stopping potential : $\mathrm{E = q \cdot V_0}$

$$[\mathrm{V_0}] = [\mathrm{v_0}] = \frac{[\mathrm{E}]}{[\mathrm{q}]} = \frac{\mathrm{ML^2\ T^{-2}}}{\mathrm{AT}} = \mathrm{ML^2\ T^{-3}\ A^{-1}}$$

(iii) Work function $(\phi) \rightarrow$ Energy form

$[\phi] = \mathrm{ML^2\ T^{-2}}\$$

(iv) Threshold freq. $[\mathrm{v_0}] = \mathrm{T^{-1}}$

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Q. 2 Two cars A and B are moving in the same direction along a straight line with speeds $100\ \mathrm{km/h}$ and $80\ \mathrm{km/h}$, respectively such that car A is moving ahead of car B. A person in car B throws a stone with a speed v so that it hits the car A with a speed of $5\ \mathrm{m/s}$. The value of v is
_____ km/h.

Option 1:
18

Option 2:
28

Option 3:
38

Option 4:
48

Correct Answer:
38

Solution:



$$\mathrm{V} = \overrightarrow{\mathrm{V}}_{\mathrm{SB}}$$
$$\mathrm{V} = \overrightarrow{\mathrm{V}}_{\mathrm{S}} - \overrightarrow{\mathrm{V}}_{\mathrm{B}}$$
$$\mathrm{V}_{\mathrm{B}} + \mathrm{V} = \overrightarrow{\mathrm{V}}_{\mathrm{S}}$$
$$\overrightarrow{\mathrm{V}}_{\mathrm{S}} = \mathrm{V} + 80$$
$$\overrightarrow{\mathrm{V}}_{\mathrm{SA}} = 5\ \mathrm{m/s} = 18\ \mathrm{km/hr}$$
$$\overrightarrow{\mathrm{V}}_{\mathrm{S}} - \overrightarrow{\mathrm{V}}_{\mathrm{A}} = \overrightarrow{\mathrm{V}}_{\mathrm{SA}}$$
$$\mathrm{V} + 80 - 100 = 18\ \mathrm{km/hr}$$
$$\mathrm{V} - 20 = 18$$
$$\mathrm{V} = 38\ \mathrm{km/hr}$$

Q. 3 At $t = 0$, a body of mass 100 g starts moving under the influence of a force $(5\hat{\mathrm{i}} + 10\hat{\mathrm{j}})\mathrm{N}$. After 2 s its position is $(2x\hat{\mathrm{i}} + 5y\hat{\mathrm{j}})\mathrm{m}$. The ratio $\mathrm{x} : \mathrm{y}$ is _____ .

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Option 1:
$1:2$

Option 2:
$2:5$

Option 3:
$5:2$

Option 4:
$5:4$

Correct Answer:
$5:4$

Solution:

$\vec{a}=\frac{\vec{F}}{m}=\frac{5\hat{i}+10\hat{j}}{0.1}=(50\hat{i}+100\hat{j})$

$\vec{s}=\vec{u}t+\frac{1}{2}\vec{a}t^2$

$\vec{s}=0+\frac{1}{2}[50\hat{i}+100\hat{j}][2]^2$

$\vec{s}=100\hat{i}+200\hat{j}=\vec{r}=2x\hat{i}+5y\hat{j}$

$2x=100 \quad 5y=200$

$x=50$

$\frac{x}{y}=\frac{5}{4}$

Q. 4 If x and y coordinates of a projectile as a function of time (t) are given as $24t$ and $43.6t-4.9t^2$, respectively, then the angle (in degrees) made by the projectile with horizontal when $\mathrm{t}=2$ s is _____ .

Option 1:
60

Option 2:
45

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Option 3:
30

Option 4:
75

Correct Answer:
45

Solution:

$x = 24t \quad y = 43.6t - 4.9t^2$

$v_x = \dfrac{dx}{dt} = 24$

$v_y = \dfrac{dy}{dt} = 43.6 - 9.8t$

$\tan\theta = \dfrac{v_y}{v_x} = \dfrac{43.6 - 9.8t}{24}$

$\text{At } t = 2 \quad \dfrac{43.6 - 19.6}{24}$

$\tan\theta = 1$

$\theta = 45^\circ$

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Q. 5 The height in terms of radius of the earth (R), at which the acceleration due to gravity becomes $\frac{g}{9}$, where g is acceleration due to gravity on earth's surface, is _____ .

Option 1:
$\sqrt{3}R$

Option 2:
$2\sqrt{2}R$

Option 3:
$2R$

Option 4:
$\frac{4}{9}\mathrm{R}$

Correct Answer:
$2R$

Solution:

$$g_h = \frac{g}{\left(1+\frac{h}{R}\right)^2}$$
$$\frac{g}{9} = \frac{g}{\left(1+\frac{h}{R}\right)^2}$$
$$1+\frac{h}{R} = 3$$
$$h = 2R$$

Q. 6 A metal string A is suspended from a rigid support and its free end is attached to a block of mass M . Second block having mass 2 M is suspended at the bottom of the first block using a string B . The area of cross sections of strings A and B are same. The ratio of lengths of strings of A to B is 2 and the ratio of their Young's moduli (Y_A/Y_B) is 0.5 . The ratio of elongations in A to B is ____

Option 1:
1

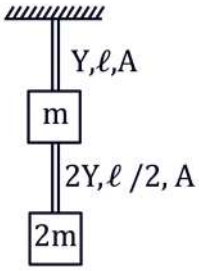
Option 2:
4

Option 3:
8

Option 4:
6

Correct Answer:
6

Solution:



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$$\Delta \ell = \frac{T \ell}{YA}$$

$$\Delta \ell_1 = \frac{3mg\ell}{YA}$$

$$\Delta \ell_2 = \frac{2mg(\ell/2)}{2YA} = \frac{mg\ell}{2YA}$$

$$\frac{\Delta \ell_1}{\Delta \ell_2} = 6$$

Q. 7 A water spray gun is attached to a hose of cross sectional area $30\ \mathrm{cm}^2$. The gun comprises of 10 perforations each of cross sectional area of $15\ \mathrm{mm}^2$. If the water flows in the hose with the speed of $50\ \mathrm{cm/s}$, calculate the speed at which the water flows out from each perforation. (Neglect any edge effects)

Option 1:
$100\ \mathrm{m/s}$

Option 2:
$10\ \mathrm{m/s}$

Option 3:
$1000\ \mathrm{m/s}$

Option 4:
$15 \times 10^2\ \mathrm{m/s}$

Correct Answer:
$10\ \mathrm{m/s}$

Solution:
Using equation of continuity

$$AV = 10av$$

$$v = \frac{AV}{10a} = \frac{30 \times 50}{10 \times .15} = 1000\ \mathrm{cm/s} = 10\ \mathrm{m/s}$$

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Q. 8 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: If the average kinetic energy of $\mathrm{H_2}$ and $\mathrm{O_2}$ molecules, kept in two different sized containers are same, then their temperatures will be same.
Reason R: The r.m.s. speed of $\mathrm{H_2}$ and $\mathrm{O_2}$ molecules are same at same temperature.
Choose the correct answer from the options given below

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

Option 2:
Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Option 3:
$\mathbf{A}$ is true but $\mathbf{R}$ is false

Option 4:
$\mathbf{A}$ is false but $\mathbf{R}$ is true

Correct Answer:
$\mathbf{A}$ is true but $\mathbf{R}$ is false

Solution:

Avg. kinetic energy of molecule will be $\frac{f}{2}\mathrm{KT}$ But $\mathrm{v_{rms}} = \sqrt{\frac{3\mathrm{RT}}{\mathrm{M}}}$

Molar mass m different so v_{rms} is different

Q. 9 The temperature of a metal strip having coefficier of linear expansion α is increased from T_1 to T resulting in increase of its length by $\Delta \mathrm{L}_1$. Th temperature is further increased from T_2 to T_3 suc that the increase in its length is $\Delta \mathrm{L}_2$.
Given $\mathrm{T}_3 + \mathrm{T}_1 = 2\ \mathrm{T}_2$ and $\mathrm{T}_2 - \mathrm{T}_1 = \Delta \mathrm{T}$, the value o $\Delta \mathrm{L}_2$ is _____ .

Option 1:
$\Delta \mathrm{L}_1 \left[1 + 2\alpha^2 (\Delta\ \mathrm{T})^2\right]$

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Option 2:
$\Delta L_1 \left[1+\alpha^2(\Delta\ T)^2\right]$

Option 3:
$\Delta L_1[1+2\alpha\Delta\ T]$

Option 4:
$\Delta L_1[1+\alpha\Delta T]$

Correct Answer:
$\Delta L_1[1+\alpha\Delta T]$

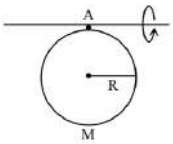
Solution:
$\Delta L_1 = L_0\alpha\left(\ T_2 - T_1\right)$

So final length at $T_2 = L_0 + \Delta L_1$

$\Delta L_2 = (L_0 + \Delta L_1)\alpha\,(T_3 - T_2)$ and $T_3 - T_2 = T_2 - T_1 = \Delta T$

$$\begin{aligned}\Delta\ L_2 &= L_0\alpha\Delta\ T + \Delta L_1\alpha\Delta\ T \\ &= \Delta L_1 + \Delta L_1\alpha\Delta\ T \\ &= \Delta L_1(1+\alpha\Delta T)\end{aligned}$$

Q. 10 A uniform disc of radius R and mass M is free to oscillate about the axis A as shown in the figure. For small oscillations the time period is _____ . (g is acceleration due to gravity)



Option 1:
$2\pi\sqrt{\frac{5R}{4g}}$

Option 2:
$2\pi\sqrt{\frac{2R}{3g}}$

Option 3:
$2\pi\sqrt{\frac{3R}{2g}}$

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Option 4:
$2\pi\sqrt{\frac{3R}{g}}$

Correct Answer:
$2\pi\sqrt{\frac{5R}{4g}}$

Solution:
$I_{\text{disc}} = \frac{mR^2}{4} + mR^2 = \frac{5mR^2}{4}$

$\therefore$ For small displacement

$\tau = -\mathrm{mgR}\theta$

$\therefore \alpha = -\frac{\mathrm{mgR}}{\frac{5\mathrm{mR}^2}{4\,\mathrm{g}}}\theta$

$\therefore \omega = \sqrt{\frac{4\,\mathrm{g}}{5\mathrm{R}}}$

$\therefore\ \mathrm{T} = 2\pi\sqrt{\frac{5\mathrm{R}}{4\,\mathrm{g}}}$

Q. 11 A rigid dipole undergoes a simple harmonic motion about its centre in the presence of an electric field $\overrightarrow{\mathrm{E}}_1 = \mathrm{E}_0\hat{\mathrm{x}}$. If another electric field $\overrightarrow{\mathrm{E}}_2 = 2\mathrm{E}_0(\hat{\mathrm{y}} + \hat{\mathrm{z}})$ is introduced to the system, what will be the percentage change in the frequency of the oscillation (approximate) ?

Option 1:
73%

Option 2:
63%

Option 3:
83%

Option 4:
53%

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Correct Answer:
73%

Solution:

Initial freq. $f_1 = \frac{1}{2\pi}\sqrt{\frac{P(E_0)}{I}}$

Final $EF = E_0\hat{i} + 2E_0(\hat{j} + \hat{k})$

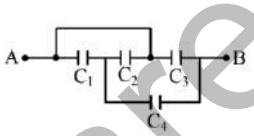
Magnitude $= 3\mathrm{E}_0$

Final Freq. $\mathrm{f}_2 = \frac{1}{2\pi}\sqrt{\frac{\mathrm{P}(3\mathrm{E}_0)}{\mathrm{I}}}$

% change $= \frac{\Delta f}{f_1} = (\sqrt{3} - 1) \times 180 \simeq 73$

Q. 12 From the circuit given below, the capacitance between terminals A and B shown in the circuit is

_____ μF. (take $\mathrm{C}_1 = \mathrm{C}_2 = \mathrm{C}_3 = 1\mu\ \mathrm{F}$ and $\mathrm{C}_4 = 2\mu\ \mathrm{F}$.)



Option 1:
2

Option 2:
7/2

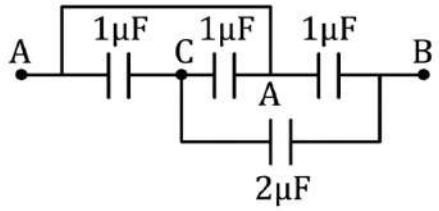
Option 3:
7/3

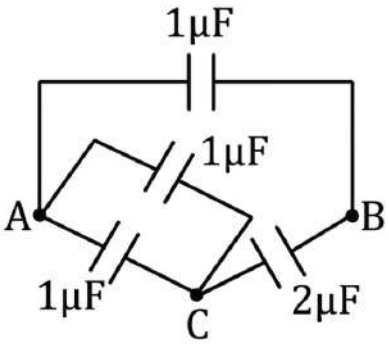
Option 4:
5/2

Correct Answer:
7/2

Solution:

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$C_{AB} = 2\mu\,F$

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Q. 13 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A : In electrostatics, a conductor does not store any net charge inside.

Reason R : Inside the capacitor (with no dielectric medium), the free charge carriers, if placed between the plates of capacitor, experience force and drift.

Choose the correct answer from the options given below.

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

Option 2:
Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of A.

Option 3:
$\mathbf{A}$ is true but $\mathbf{R}$ is false

Option 4:
$\mathbf{A}$ is false but $\mathbf{R}$ is true

Correct Answer:
Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of A.

Solution:
Theoretical

Q. 14 A solenoid has a core made of material with relative permeability 400 . The magnetic field produced in the interior of solenoid is 1.0 T . The magnetic intensity is SI units is $\alpha \times 10^5$. The value of α is _____ . (Free space permeability $\mu_0 = 4\pi \times 10^{-7}$ SI units).

Option 1:
$\frac{25}{\pi}$

Option 2:
$\frac{1}{16\pi}$

Option 3:
$\frac{1}{\pi}$

Option 4:
$\frac{1}{4\pi}$

Correct Answer:
$\frac{1}{16\pi}$

Solution:

$$H = \frac{B}{\mu} = \frac{B}{\mu_0 \mu_r} = \frac{1}{400 \times 4\pi \times 10^{-7}}$$

$$H = \frac{1}{16\pi} \times 10^5$$

Q. 15 A magnetic field vector in an electromagnetic wave is represented by, $\vec{B} = B_0 \sin\left(2\pi v t - \frac{2\pi x}{\lambda}\right)\hat{j}$ Its associated electric field vector is _____ .

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Option 1:

$\overrightarrow{\mathrm{E}}=-v\lambda\,\mathrm{B}_0\sin\left(2\pi v\mathrm{t}-\frac{2\pi \mathrm{x}}{\lambda}\right)\hat{\mathrm{k}}$

Option 2:

$\overrightarrow{\mathrm{E}}=-v\lambda\,\mathrm{B}_0\sin\left(2\pi v\mathrm{t}-\frac{2\pi \mathrm{x}}{\lambda}\right)\hat{\mathrm{i}}$

Option 3:

$\overrightarrow{\mathrm{E}}=v\lambda\,\mathrm{B}_0\sin\left(2\pi v\mathrm{t}-\frac{2\pi \mathrm{x}}{\lambda}\right)\hat{\mathrm{k}}$

Option 4:

$\overrightarrow{\mathrm{E}}=v\lambda\,\mathrm{B}_0\sin\left(2\pi v\mathrm{t}-\frac{2\pi \mathrm{x}}{\lambda}\right)\hat{\mathrm{i}}$

Correct Answer:

$\overrightarrow{\mathrm{E}}=-v\lambda\,\mathrm{B}_0\sin\left(2\pi v\mathrm{t}-\frac{2\pi \mathrm{x}}{\lambda}\right)\hat{\mathrm{k}}$

Solution:

$\hat{E}\times\hat{B}=\hat{C}$

$\hat{B}=\hat{j}$

$\hat{C}=\hat{i}$

$\therefore \hat{E}=-\hat{k}$

$\frac{E_0}{B_0}=C=v\lambda$

$E_0=B\times v\lambda$

Q. 16 A convex lens is made from glass material having refractive index of 1.4 with same radius of curvature on both sides. The ratio of its focal length and radius of curvature is _____ .

Option 1:

0.5

Option 2:

2.5

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Option 3:
0.8

Option 4:
1.25

Correct Answer:
1.25

Solution:

$$\frac{1}{\mathrm{f}}=(1.4-1)\left(\frac{1}{\mathrm{R}}-\frac{1}{-\mathrm{R}}\right)$$

$$\frac{1}{\mathrm{f}}=\frac{0.8}{\mathrm{R}}$$

$$\frac{\mathrm{f}}{\mathrm{R}}=1.25$$

Q. 17 An unpolarized light of certain intensity passes through a combination of two polarizers whose transmission axes are at 30° and 90°, respectively, with respect to the horizontal axis. A third polarizer with its transmission axis at 60° with the horizontal axis is placed between the two existing polarizers. The ratio of the output intensities with and without the third polarizer is _____ .

Option 1:
$3/4$

Option 2:
$4/3$

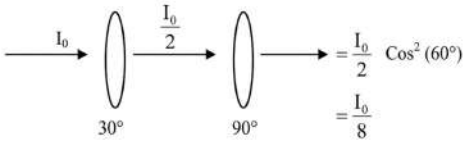
Option 3:
$9/4$

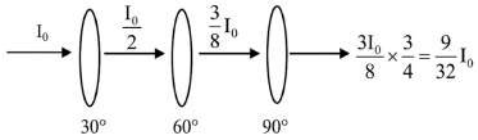
Option 4:
$4/9$

Correct Answer:
$9/4$

Solution:

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$\frac{9I_0}{32} \times \frac{8}{I_0} = \frac{9}{4}$

Q. 18 In Rutherford's alpha-particle scattering experiment, only a few alpha particles rebound back because
(A)The size of gold nucleus is very small as compared to the size of gold atom.
(B) Alpha particle and gold nucleus have equal charge.
(C) The impact parameter is minimum for a few alpha particles.
(D) A few alpha particles have very high kinetic energy.
(E) Only a few alpha particles undergo head-on collision with the nuclei.
Choose the correct answer from the options given below :

Option 1:
A, B Only

Option 2:
B, E Only

Option 3:
C,D Only

Option 4:
A,C,E Only

Correct Answer:
A,C,E Only

Solution:
A,C, E only
Theoretical

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Q. 19 The de Broglie wavelength associated with an electron accelerated through a potential difference V is λ_c and the de Broglie wavelength associated with a proton accelerated through the same potential difference is λ_p. If their corresponding masses are m_e and m_p, respectively, then the ratio of their de Broglie wavelengths $\left(\frac{\lambda_e}{\lambda_p}\right)$ is _____ .

Option 1:
$\sqrt{\frac{m_p}{m_e}}$

Option 2:
$\sqrt{\frac{m_e}{m_p}}$

Option 3:
$\frac{m_p}{m_e}$

Option 4:
$\left(\frac{m_p}{m_e}\right)^2$

Correct Answer:
$\sqrt{\frac{m_p}{m_e}}$

Solution:
$\lambda = \frac{h}{\sqrt{2mqV}} \propto \frac{1}{\sqrt{m}}$

So $\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}}$

$\sqrt{\frac{m_e}{m_p}}$

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Q. 20 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: A diode under reverse- biased condition provides very small current which is nearly independent of voltage until a critical limit at which the current increases drastically.
Reason R : Below the critical voltage limit, only majority charge carriers flow which increases drastically above critical voltage.
Choose the correct answer from the options given below.

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

Option 2:
Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

Option 3:
$\mathbf{A}$ is true but $\mathbf{R}$ is false

Option 4:
$\mathbf{A}$ is false but $\mathbf{R}$ is true

Correct Answer:
$\mathbf{A}$ is true but $\mathbf{R}$ is false

Solution:
Theoretical

Q. 21 A diode has Zener voltage of 10 V and maximum power dissipation of 0.5 W , then the minimum resistance to be used in series with this diode for safety when it is connected to a 25 V power supply is _____ Ω.

Correct Answer:
300

Solution:
Maximum current from diode

$i = \frac{P}{V} = \frac{0.5}{10} = 0.05\ \text{A}$

Now let resistance is R

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$$V_B = i(R + R_{diode})$$
$$25 = 0.05 \times R + 10$$
$$R = \frac{15}{0.05} = 300\Omega$$

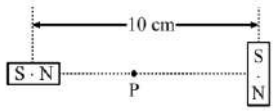
Q. 22 A gun mounted on the ground fires bullets in all directions with same speed. The farthest distance the bullets could reach is 6.4 m . The speed of the bullets from the gun is _____ m/s . (take $g = 10\ m/s^2$)

Correct Answer:
8

Solution:
$$R_{max} = \frac{v^2}{g} = 6.4$$
$$v = \sqrt{64} = 8\ m/s$$

Q. 23 Two identical small bar magnets each of dipole moment $3\sqrt{5}\ J/T$ are placed at a center to center separation of 10 cm , with their axes perpendicular to each other as shown in figure. The value of magnetic field at the point P midway between the magnets is $\alpha \times 10^{-3}\ T$. The value of α is _____ .

$(\mu_0 = 4\pi \times 10^{-7} Tm/A)$



Correct Answer:
12

Solution:
$$B_{net} = \sqrt{B_1^2 + B_2^2} = \frac{\mu_0}{4\pi}\frac{M}{(r)^3}\sqrt{5}$$

$$= 10^{-7} \times 3\sqrt{5} \times \sqrt{5} \times \frac{8}{10^{-3}}$$
$$= 120 \times 10^{-4} = 12mT$$

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Q. 24 A circular coil of radius 2 cm and 125 turns carries a current of 1 A . The coil is placed in a uniform magnetic field of magnitude 0.4 T. The axis of the coil makes an angle of 30° with the direction of the magnetic field. The torque acting on the coil is $\alpha \times 10^{-4}\ \mathrm{N.m}$. The value of α is _____ .

$(\pi = 3.14)$

Correct Answer:
314

Solution:

$$\tau = |\overrightarrow{\mathrm{M}} \times \overrightarrow{\mathrm{B}}|$$

$$= |\mathrm{NI}\,\overrightarrow{\mathrm{A}} \times \overrightarrow{\mathrm{B}}|$$

$$\tau = 125 \times 1 \times \pi\left(\frac{2}{100}\right)^2 \times 0.4 \times \sin 30^{\circ}$$

$$= 100\pi \times 10^{-4}\ \mathrm{N-m}$$

So $\alpha = 100\pi = 314$

Q. 25 In a double slit experiment, when one of the slits is covered by a transparent mica sheet of refractive index 1.56 , the central fringe shifts to the position of 7th bright fringe, obtained with both slits uncovered. If the light source wavelength is 450 nm , the thickness of mica sheet is $\alpha \times 10^{-9}\ \mathrm{m}$. The value of α is _____ .

Correct Answer:
5625

Solution:

$$\Delta \mathrm{y} = (\mu - 1)\mathrm{t}\frac{\mathrm{D}}{\mathrm{d}} = 7\frac{\lambda \mathrm{D}}{\mathrm{d}}$$

$$(\mu - 1)\mathrm{t} = 7\lambda$$

$$(1.56 - 1)\mathrm{t} = 7 \times 450\ \mathrm{m}$$

$$\mathrm{t} = 5625\ \mathrm{nm}$$

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Chemistry

Q. 1 The correct order of total number of atoms of present in
(A) 2 moles of cyclohexane
(B) 684 g of sucrose
(C) 90.8 L of dihydrogen at STP

Option 1:
$\mathrm{C} > \mathrm{A} > \mathrm{B}$

Option 2:
$\mathrm{C} > \mathrm{B} > \mathrm{A}$

Option 3:
$\mathrm{B} > \mathrm{C} > \mathrm{A}$

Option 4:
$\mathrm{B} > \mathrm{AC}$

Correct Answer:
$\mathrm{B} > \mathrm{C} > \mathrm{A}$

Solution:
(A) Mole of atoms in $\mathrm{C_6H_{12}}$

$= 2 \times 18$
$= 36$

(B) Moles of atoms of sucrose $= \frac{604}{342} \times 45$
(C) $\mathrm{n_{H_2}} = \frac{90.8}{22.7} = 4$ moles

Mole fo atoms $= 4 \times 2 = 8$
$(\mathrm{B}) > (\mathrm{A}) > (\mathrm{C})$

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Q. 2 The species having identical radii according to the Bohr's theory are :

A. H (first orbit)

B. He^{+}(first orbit)

C. He^{+}(second orbit)

D. Li^{2+} (first orbit)

E. Be^{3+} (Second orbit)

Choose the correct answer from the options given below:

Option 1:
A and C Only

Option 2:
A and E Only

Option 3:
B and E Only

Option 4:
C and D Only

Correct Answer:
A and E Only

Solution:
(A) $\rightarrow$ same
(B) $\rightarrow$ same
(C) $\rightarrow$ same
(D) $\rightarrow \mathrm{r}=\frac{90\times(1)^2}{3}=\frac{90}{3}$
(E) $\rightarrow$ same

Q. 3 Which of the following pictorial diagram most correctly represents the π^* (π - antibonding) molecular orbital between two atoms if the internuclear axis is taken to be in the z -direction $(\xrightarrow{\text{z-axis}})$?

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Option 1:



Option 2:



Option 3:



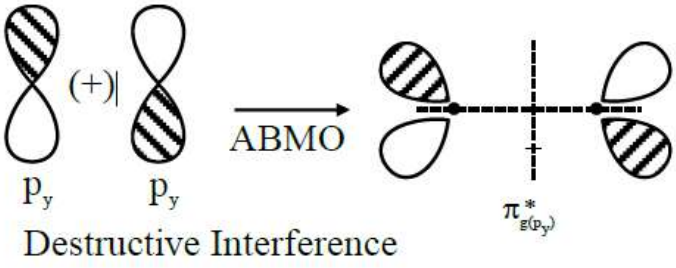
Option 4:



Correct Answer:



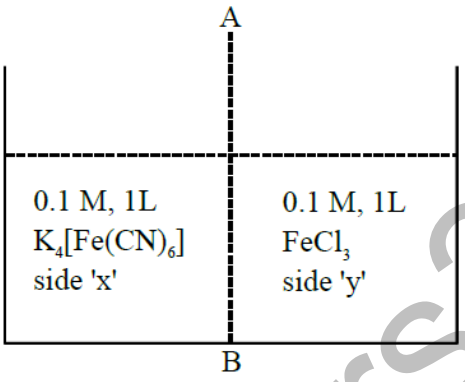
Solution:



Destructive Interference

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Q. 4 At $27^{\circ}C, 0.1M, 1\ L\ K_4\left[Fe(CN)_6\right]$ aqueous solution and $0.1M, 1\ LFeCl_3$ aqueous solution are placed in a container separated by a semi permeable membrane AB . Assume complete dissociation of both the solutes. Which of the following statement is correct?



Option 1:
Blue color is formed on both sides.

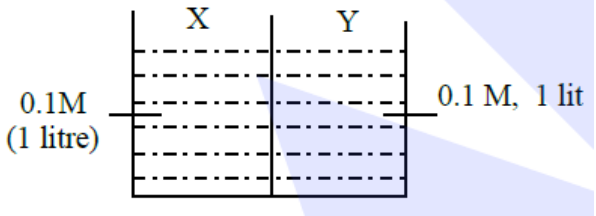
Option 2:
Ionic solutes in aqueous solution can pass through semi-permeable membrane.

Option 3:
Solution on side ' y ' is hypotonic.

Option 4:
To cause the reverse flow of solvent during osmosis, external pressure (any value) should be applied to side ' x '.

Correct Answer:
Solution on side ' y ' is hypotonic.

Solution:



Q. 5 20 mL of a solution of acetic acid required 28.4 mL of 0.1 M NaOH for its neutralization. A solution (X) was prepared by mixing 20 mL of the above acetic acid and 14.2 mL of 0.1 M NaOH solution. What is the pH of the solution (X) ? $\left(pK_a\right.$ value of acetic acid is 4.75).

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Option 1:
7.0

Option 2:
4.45

Option 3:
3.5

Option 4:
4.82

Correct Answer:
4.45

Solution:
In experiment $1, 28.4\text{ml}$ of NaOH is required for complete neutralisation, therefore for 14.2 ml , half equivalence point will be achieved to form acidic buffer.

Q. 6 Match the LIST-I with LIST -II

List-I Reaction		**List-II** **Mechanism**	
A.	Williamson Synthesis	I.	Electrophilic addition
B.	Friedel Craft Reaction	II.	Free radical substitution
C.	Bromination of vinyl benzene	III.	Nucleophilic substitution
D.	Chlorination of toluene in light	IV	Electrophilic substitution

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-II, D-IV

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Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-I, B-III, C-IV, D-II

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
Theory based

Q. 7 The 1st ionization enthalpy for Mg is $+737\ \mathrm{kJ/mol}$. The most probable estimated value of the 2nd ionization enthalpy of Mg is _____ .

Option 1:
$-906\ \mathrm{kJ/mol}$

Option 2:
$-856\ \mathrm{kJ/mol}$

Option 3:
$+1450\ \mathrm{kJ/mol}$

Option 4:
$+590\ \mathrm{kJ/mol}$

Correct Answer:
$+1450\ \mathrm{kJ/mol}$

Solution:
Since successive ionization energy increases thus $\mathrm{IE}_2(\mathrm{Mg}) > \mathrm{IE}_1(\mathrm{Mg})$

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Q. 8 The electronegativity of a group 13 element ' E ' is same as that of Ge (on Pauling scale and upto one decimal point). The CORRECT statements about E^{3+} are

A. It can act as a reducing agent.

B. It can act as an oxidizing agent.

C. E^{3+} is more stable than E^{+}.

D. The standard electrode potential value for $\mathrm{E}^{3+}/\mathrm{E}$ is positive.

Choose the correct answer from the options given below

Option 1:
A and C Only

Option 2:
B and C Only

Option 3:
B and D Only

Option 4:
A and D Only

Correct Answer:
B and D Only

Solution:
EN of ' $\mathrm{T}\ell$ ' is 1.8 that is same as ' Ge '

$$\text{' E '} = \mathrm{T}\ell$$

$\mathrm{T}\ell^{+3}$ acts as oxidizing agent
Stability order : $\mathrm{T}\ell^{+3} < \mathrm{T}\ell^{+1}$ (due to inert pair effect)
The standard electrode potential value $\mathrm{E}^{\circ}\left(\mathrm{T}\ell^{3+}/\mathrm{T}\ell\right)$

$$= +1.26\ \mathrm{V}$$

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Q. 9 Pairs of elements with the same number of electrons in their respective 4 f orbital are
[Atomic number, Eu-63, Gd-64, Dy-66, Ho-67, Tm-69, Yb-70, Lu-71, Hf-72]
A. (Eu and Gd)
B. (Dy and Ho)
C. (Yb and Hf)
D. (Lu and Tm)

Choose the correct answer from the options given below:

Option 1:
B and C Only

Option 2:
A and B Only

Option 3:
A and D Only

Option 4:
A and C Only

Correct Answer:
A and C Only

Solution:
$\mathrm{Gd} - [\mathrm{Xe}]4\mathrm{f}^{7}5\ \mathrm{d}^{1}6\ \mathrm{s}^{2}$
$\mathrm{Eu} - [\mathrm{Xe}]4\mathrm{f}^{7}6\ \mathrm{s}^{2}$
Dy $-[\mathrm{Xe}]4\mathrm{f}^{10}6\ \mathrm{s}^{2}$
$\mathrm{Ho} - [\mathrm{Xe}]4\mathrm{f}^{11}6\ \mathrm{s}^{2}$
$\mathrm{Hf} - [\mathrm{Xe}]4\mathrm{f}^{14}6\ \mathrm{s}^{2}5\ \mathrm{d}^{2}$
$\mathrm{Yb} - [\mathrm{Xe}]4\mathrm{f}^{14}6\ \mathrm{s}^{2}$
$\mathrm{Lu} - [\mathrm{Xe}]4\mathrm{f}^{14}5\ \mathrm{d}^{1}6\ \mathrm{s}^{2}$
$\mathrm{Tm} - [\mathrm{Xe}]4\mathrm{f}^{13}6\ \mathrm{s}^{2}$

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Q. 10 Consider the metal complexes $[Ni(en)_3]^{2+}(A)$, $[NiCl_4]^{2-}(B)$ and $[Ni(NH_3)_6]^{2+}(C)$. Choose the CORRECT option by considering the number of unpaired electron present in (A), (B) and (C) respectively and the order of frequency of absorption.

Option 1:
$2, 2, 2$ and $(A) > (C) >$ (B)

Option 2:
$0, 2, 0$ and $(A) > (C) >$ (B)

Option 3:
$2, 2, 0$ and $(B) > (C) >$ (A)

Option 4:
$2, 2, 2$ and $(C) >$ (A) $>$ (B)

Correct Answer:
$2, 2, 2$ and $(A) > (C) >$ (B)

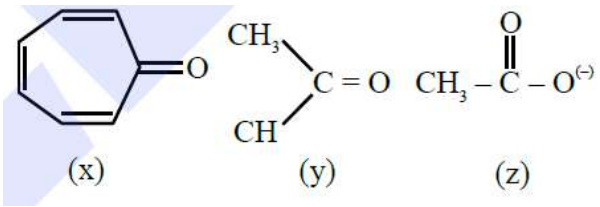
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Solution:
(A) $Ni^{2+} \Rightarrow 3\ d^8, SFL, t_{2g}^{2,2,2}e_g^{1,1} \Rightarrow$ unpaired $e^- = 2$
(B) $Ni^{2+} \Rightarrow 3\ d^8$, WFL, tetrahedral, $e^{2,2}t_2^{2,1,1}$

$\Rightarrow$ unpaired $e^- = 2$
(C) $Ni^{2+} \Rightarrow 3\ d^8$, SFL, $t_{2g}^{2,2,2}e_g^{1,1} \Rightarrow$ unpaired $e^- = 2$

As $\Delta_o > \Delta_t$ and Ligand strength : en $> NH_3 > Cl^-$
Strength of ligand increases, Δ increases, hence more is the frequency required of absorption.

Q. 11 Consider the following molecules/species:



(x) (y) (z)

The correct order of carbon - oxygen double bond length is:

Option 1:
$x > y > z$

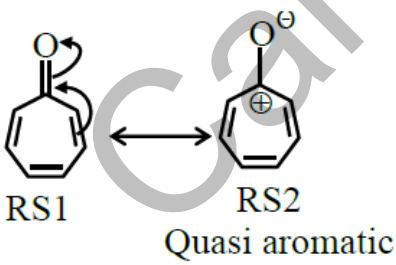
Option 2:
$y > z > x$

Option 3:
$z > x > y$

Option 4:
$x > z > y$

Correct Answer:
$x > z > y$

Solution:



$C - O$ bond length is maximum in compound (b) because RS2 of this compound has aromatic character.

Q. 12 Consider $|x|$ is the difference in oxidation states of Mn in highest manganese fluoride and highest manganese oxide. The ions with $|x|$ number of unpaired electrons from the following are:

A. Sc^{3+}
B. Zn^{2+}
C. V^{2+}
D. Fe^{2+}
E. Co^{2+}

Choose the correct answer from the options given below:

Option 1:
A and B Only

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Option 2:
C, D and E Only

Option 3:
C and E Only

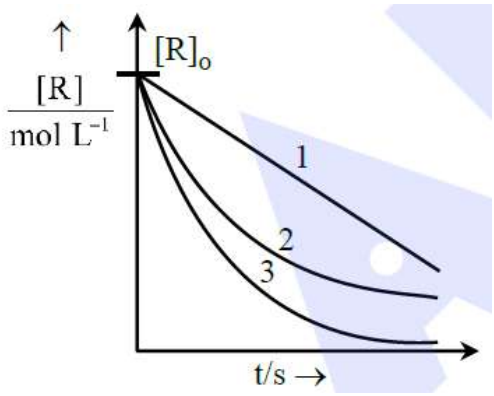
Option 4:
B and E Only

Correct Answer:
C and E Only

Solution:
Oxide of Mn in its highest O.S. $= \mathrm{Mn_2O_7}\ [\mathrm{Mn^{+7}}]$
Fluoride of Mn in its highest O.S. $= \mathrm{MnF_4}\ [\mathrm{Mn^{+4}}]$
⇒ Difference in O.S. $= |\mathrm{x}| = 3$
$\mathrm{Sc^{+3}} \Rightarrow [\mathrm{Ar}]; 0$ unpaired $\mathrm{e^-}$
$\mathrm{Zn^{2+}} \Rightarrow [\mathrm{Ar}]3\ \mathrm{d^{10}}; 0$ unpaired electrons
$\mathrm{Fe^{+2}} \Rightarrow [\mathrm{Ar}]3\ \mathrm{d^6}; 4$ unpaired electrons
$\mathrm{Co^{+2}} \Rightarrow [\mathrm{Ar}]3\ \mathrm{d^7}; 3$ unpaired electrons
$\mathrm{V^{+2}} \Rightarrow [\mathrm{Ar}]3\ \mathrm{d^3}; 3$ unpaired electrons

---

Q. 13 Consider the given graph showing variation of reactant concentration with time.
Three different reactions were started with identical initial concentration of reactants. Which of the following statement is correct?



Option 1:
The order of all the three reactions is same.

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Option 2:
The rate constant of reaction 3 is larger than the rate constant of reaction 2 if the order of reaction is same for both.

Option 3:
The SI unit of rate constant of reaction 1 is s^{-1}.

Option 4:
Thermal decomposition of HI on gold surface is an example of reaction 2 .

Correct Answer:
The rate constant of reaction 3 is larger than the rate constant of reaction 2 if the order of reaction is same for both.

Solution:
⇒ Curve I^{st} represent zero order kinetics

$$[R_t] = [R]_0 - Kt$$

$$\Rightarrow K_3 > K_2$$

⇒ Reaction-1 is zero order having unit of rate constant same as rate of reaction i.e., molarity / second.

Q. 14 Compound (X) is subjected to the sequence of reactions as shown above. Molar mass of the major product (Y) formed is ____ $gmol^{-1}$.
(Given molar mass in $gmol^{-1} C: 12, H: 1, O: 16$)

Option 1:
90

Option 2:
118

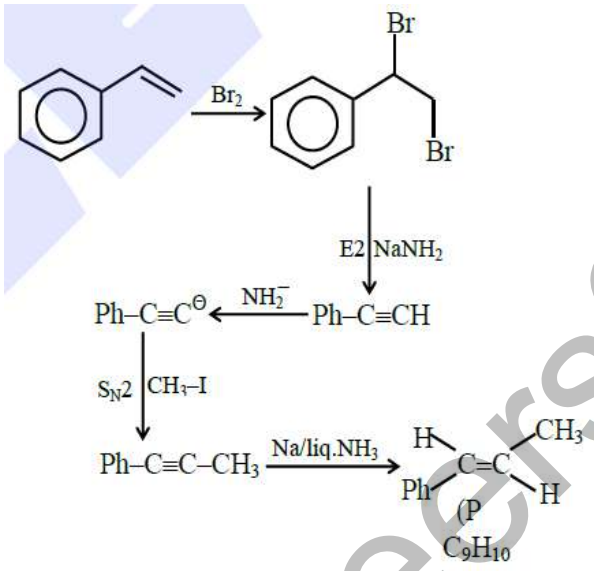
Option 3:
160

Option 4:
125

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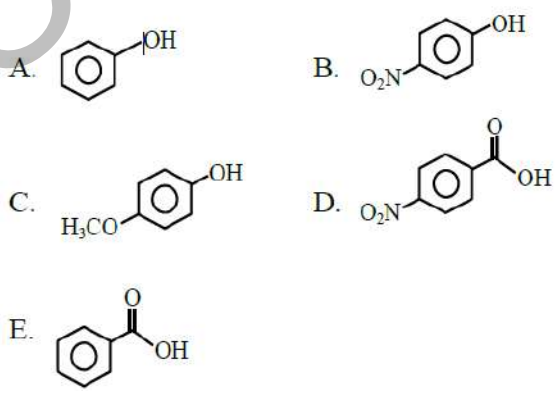
Correct Answer:
118

Solution:



Molar mass of Y = 118 gm

Q. 15 The descending order of acidity among the following compounds is



Choose the correct answer from the option given below options

Option 1:
B > D > E > A >C

Option 2:
D > B > E > A > C

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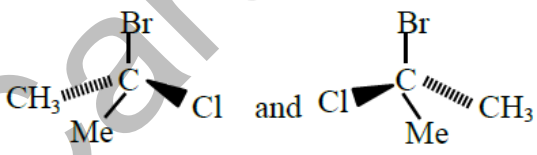
Option 3:
C > A > B > D > E

Option 4:
D > E > B > A > C

Correct Answer:
D > E > B > A > C

Solution:
Theory based.
Benzoic acid is more acidic than phenol.
Electron withdrawing group in increases acidicity.

Q. 16 The following structures are



Option 1:
enantiomer

Option 2:
Identical molecules

Option 3:
Diastereomers

Option 4:
meso compound

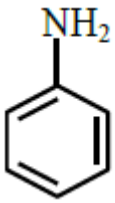
Correct Answer:
Identical molecules

Solution:
No chiral center (2 methyl groups)

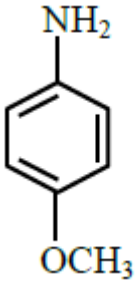
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Q. 17 The strongest conjugate acid will result from

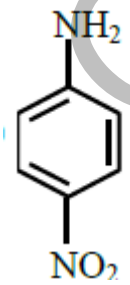
Option 1:



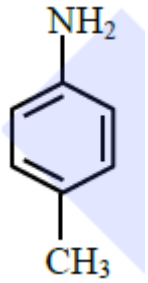
Option 2:



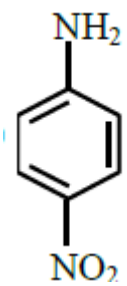
Option 3:



Option 4:

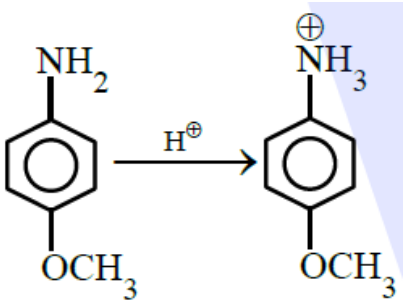


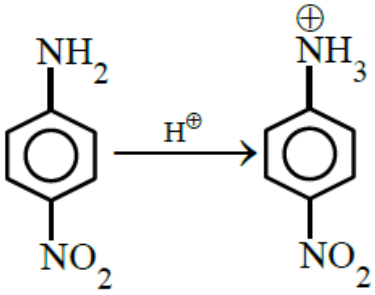
Correct Answer:

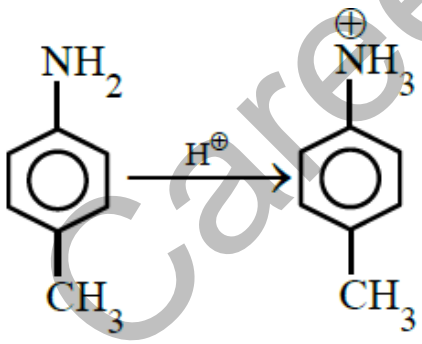


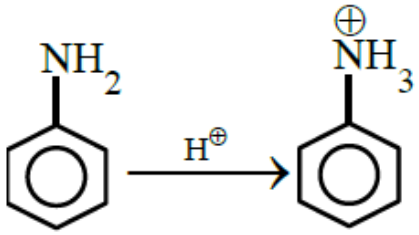
Solution:

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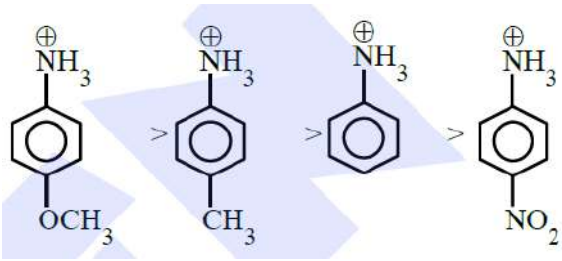








Stability order of conjugated acid :

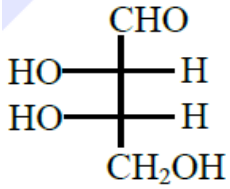


For stronger conjugate acid, we need weaker base.

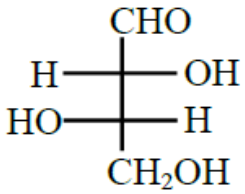
Q. 18 A D-aldotetrose on oxidation with concentrated HNO_3 resulted in optically inactive dicarboxylic acid. The structure of the D-aldotetrose is :

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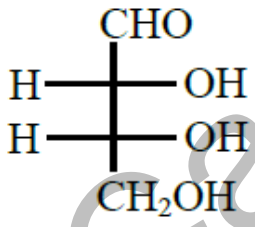
Option 1:



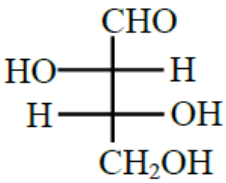
Option 2:



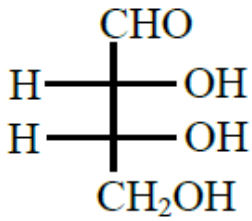
Option 3:



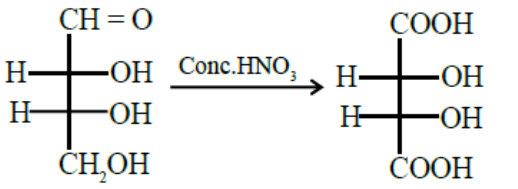
Option 4:



Correct Answer:



Solution:



D-Aldotetrose

Optically inactive
Dicarboxylic acid

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Q. 19 Among $Fe^{3+}, Pb^{2+}, Cu^{2+}$ and Mn^{2+}, identify the one that gets precipitated out while passing $H_2\ S$ in presence of NH_4OH as group reagent. The highest possible oxidation state of the corresponding metal is

Option 1:
+3

Option 2:
+4

Option 3:
+2

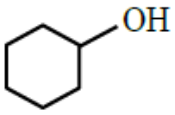
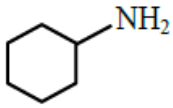
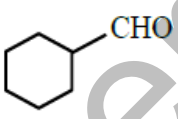
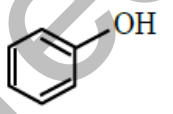
Option 4:
+7

Correct Answer:
+7

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Solution:
Group IV reagent : $NH_4OH + H_2\ S$
Mn^{+2} will be precipitated out and hence highest possible oxidation state of Mn is +7

Q. 20 Match the LIST-I with LIST-II

List-I Compound		List-I Test	
A.		**I.**	**Hinsberg's reagent test**
B.		**II.**	**Phthalein dye test**
C.		**III.**	**Lucas test**
D		**IV**	**Tollen's test**

Choose the correct answer from the options given below :

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Option 1:
A-III, B-I, C-IV , D-II

Option 2:
A-III, B-IV, C-I , D-II

Option 3:
A-I, B-III, C-II , D-IV

Option 4:
A-I, B-II, C-III , D-IV

Correct Answer:
A-III, B-I, C-IV , D-II

Solution:

Theory based.
Tollen's test is used for test for aldehydes.
Heinsberg's reagent is used to test for amines.
Lucas test is used to test for alcohols.
Phthalein dye test is used for test of phenols.

Q. 21 If 3.365 g of ethanol (l) burnt completely in a bomb calorimeter at 298.15 K , the heat produced is 99.472 kJ . The $\left|\Delta \mathrm{H_f}^\circ\right|$ of ethanol at 298.15 K is ____ $\times 10^2$ $\mathrm{kJ\ mol^{-1}}$. (Nearest integer)
Given :
Standard enthalpy of combustion of graphite $= -393.5\ \mathrm{kJ\ mol^{-1}}$

Standard enthalpy of formation of water (l) $= -285.8\ \mathrm{kJ\ mol^{-1}}$
Molar mass in gmol $^{-1}$ of $\mathrm{C, H, O}$ are 12,1 and 16 respectively.

Correct Answer:
282

Solution:
$\Delta \mathrm{U}^\circ_{\text{combustion}} = \frac{-\mathrm{q}}{\text{moles}} = \frac{-99.472}{3.365/46} \Rightarrow 1359.8\mathrm{KJ/mole}$
$\mathrm{C_2H_5OH}(\ell) + 3\mathrm{O_2}(\mathrm{\ g}) \rightarrow 2\mathrm{CO_2}(\mathrm{\ g}) + 3\mathrm{H_2O}(\ell); \Delta \mathrm{H_r}^\circ = \Delta \mathrm{H_C}^\circ$
$\Delta \mathrm{H_C}^\circ = \Delta \mathrm{U}^\circ + \Delta \mathrm{n_g RT}$
$= -1359.8 + \frac{(-1) \times 8.314 \times 298.15}{1000}$
$= -1362.27\mathrm{KJ/mole}$
$-1362.27 = [2 \times (-393.5) + 3 \times (-285.8)] -$
$\Delta \mathrm{H_f}^\circ \mathrm{C_2H_5OH}(\ell)$
$-1362.27 = -787 - 857.4 - \Delta \mathrm{H_f}^\circ\left[\mathrm{C_2H_5OH}(\ell)\right]$
$\Delta \mathrm{H_f}^\circ\left[\mathrm{C_2H_5OH}(\ell)\right]$
$\simeq -3 \times 10^2$

Q. 22 For the following reaction at $50^\circ \mathrm{C}$ and 2 atm pressure,

$$2\ \mathrm{N_2O_5}(\mathrm{\ g}) \rightleftharpoons 2\ \mathrm{N_2O_4}(\mathrm{\ g}) + \mathrm{O_2}(\mathrm{\ g})$$

$\mathrm{N_2O_5}$ is 50% dissociated
The magnitude of standard free energy change at this temperature is x .
$\mathrm{x} =$ ____ $\mathrm{Jmol^{-1}}$ [Nearest integer]
Given : $\mathrm{R} = 8.314\ \mathrm{mol^{-1}\ K^{-1}}, \log 2 = 0.30,$

$\log 3 = 0.48, \ln 10 = 2.303, {}^\circ\mathrm{C} + 273 = \mathrm{K}$

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Correct Answer:
2474

Solution:

$$
\begin{array}{lccc}
 & 2\,N_2O_5(q) \rightleftharpoons 2\,N_2O_4(\,g) + O_2(\,g) & & \\
t=0 & 1 & - & - \\
t_{eqm.} & 1-2x & 2x & x \\
 & 1-0.5 & 0.5 & 0.25 \\
 & 0.5 & 0.5 & 0.25 \\
 & \frac{0.5}{1.25}\times 2 & \frac{0.5}{1.25}\times 2 & \frac{0.25}{1.25}\times 2
\end{array}
$$

$$
\begin{aligned}
K_P &= \frac{P^2_{N_2O_4}\cdot P_{O_2}}{P^2_{N_2O_5}} = 0.4 \\
\Delta G^\circ &= -2.303RT\log\,K_p \\
&= -2.303\times 8.314\times 323\log(4/10) \\
&= 2473.81\ J/mole
\end{aligned}
$$

Q. 23 An electrochemical cell. consist of the following two redox couples,

$M^{x+}(aq)/M(\,s)\left[E^{\ominus}_{red} = +0.15\ V\right]$ and
$Fe^{3+}(aq)/Fe(\,s)\left[E^{\ominus}_{red} = -0.036\ V\right]$

The cell EMF (E_{cell}) is recorded to be 0.2057 V .
If the reaction quotient of the electrochemical reaction is found to be 10^{-2}, then the value of x is
_____ (Nearest integer)
[Given : M is a p-block metal and

$\frac{2.303RT}{F} = 0.059\ V\Big]$

Correct Answer:
2

Solution:

$$E_{cell} = E^\circ_{cell} - \frac{0.059}{3x}\log 10^{-2}$$

$$0.2057 = 0.186 - \frac{0.059}{3x}(-2)$$

$$3x = \frac{0.059\times 2}{0.0197}$$

$$x = 1.99 \simeq 2$$

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Q. 24 For a first order reaction $A \rightarrow B$

t/min	[A]/M
0	0.6500
X	0.0650
20	0.00065

$x =$ _____ min. (Nearest integer)

Correct Answer:
7

Solution:

$$K = \frac{1}{t} \ln \left[\frac{A_0}{A_t} \right]$$

$$K = \frac{1}{20} \ln \left[\frac{0.6500}{0.00065} \right] \Rightarrow \frac{3 \ln 10}{20} \text{ min}^{-1}$$

$$\text{Also, } x = \frac{1}{K} \ln \left[\frac{A_0}{A_x} \right] \Rightarrow \frac{20}{3 \ln 10} \ln \left(\frac{0.6500}{0.0650} \right)$$

$$= \frac{20}{3} \text{ min} \Rightarrow 6.7 \text{ min}$$

Q. 25 In sulphur estimation, 2.0×10^{-3} mol of an organic compound (X) (molar mass 76 g mol^{-1}) gave 0.4813 g of barium sulphate (molar mass 233gmol^{-1}). The percentage of sulphur in the compound (X) is
_____ $\times 10^{-1}\%$
(Nearest integer)

Correct Answer:
435

Solution:

$$\% \text{ of S} = \frac{\text{Mass of S}}{\text{Mass of} BaSO_4} \times \frac{m}{w} \times 100$$

$m = \text{gm of } BaSO_4 \text{ppt}$

$w = \text{gm of Organic compound}$

$$w = 2 \times 10^{-3} \times 76 = 0.152 \text{gm}$$

$$\% \text{ of S} = \frac{32}{233} \times \frac{0.4813}{0.152} \times 100 = 43.487$$

In form of 10^{-1}

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$= 43.487 \times 10^{-1} = 434.87 \approx 435$

Maths

Q. 1 For the function $f : [1, \infty) \to [1, \infty)$ defined by $f(x) = (x - 1)^4 + 1$, among the two statements :

(I) The set $S = \{x \in [1, \infty) : f(x) = f^1(x)\}$ contains exactly two elements, and

(II) The set $S = \{x \in [1, \infty) : f(x) = f^1(x + 1)\}$ is an empty set,

Option 1:
only (I) is TRUE

Option 2:
only (II) is TRUE

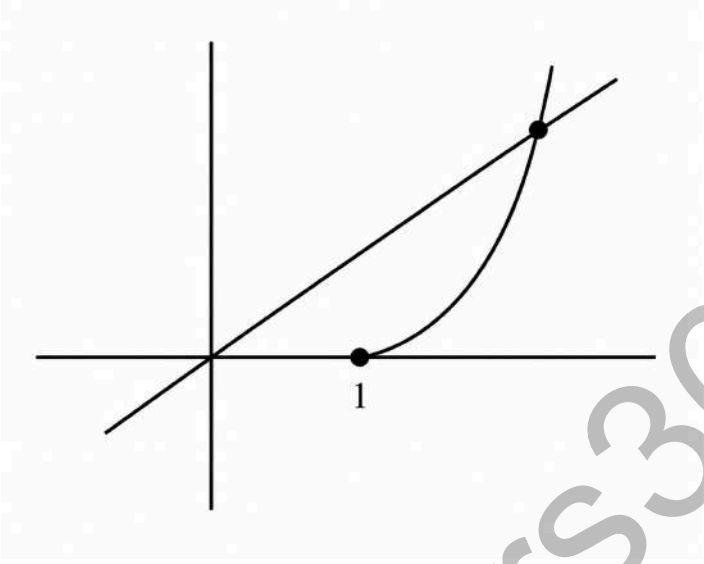
Option 3:
both (I) and (II) are TRUE

Option 4:
neither (I) nor (II) is TRUE

Correct Answer:
only (I) is TRUE

Solution:
$f(x) = (x - 1)^4 + 1$
$f'(x) = 4(x - 1)^3; f'(x) \geq 0$
$\therefore f(x)$ is increasing
$\therefore (x - 1)^4 + 1 = x$
$(x - 1)\left[(x - 1)^3 - 1\right] = 0$
$x = 1, x = 2$ are two solutions

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Now

$f^{-1}(x) = (x-1)^{1/4} + 1$
$f^{-1}(x+1) = x^{1/4} + 1$
$f^{-1}(x+1) = f(x)$
$x^{1/4} + 1 = (x-1)^4 + 1$
$x = (x-1)^{16}$

Above equation has one solution using graph.

Q. 2 Let $S = \{z \in \mathbb{C} : z^2 + 4z + 16 = 0\}$

Then $\sum_{z \in S} |z + \sqrt{3}i|^2$ is equal to :

Option 1:
42

Option 2:
23

Option 3:
27

Option 4:
38

Correct Answer:
38

Solution:

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$z^2+4z+16=0$
$\Rightarrow (z+2)^2=-12$
$\Rightarrow z=-2\pm 2\sqrt{3}i$
$\Rightarrow S=\{-2+2\sqrt{3}i,-2-2\sqrt{3}i\}$
$\therefore \sum_{z\in S}|z+\sqrt{3}i|^2$
$=|-2+2\sqrt{3}i+\sqrt{3}i|^2+|-2-2\sqrt{3}i+\sqrt{3}i|^2$
$=|-2+3\sqrt{3}i|^2+|-2-\sqrt{3}i|^2$
$=38$

Q. 3 If the system of equations :

$x+y+z=5$
$x+2y+3z=9$
$x+3y+\lambda z=\mu$

has infinitely many solutions, then the value of $\lambda+\mu$ is :

Option 1:
16

Option 2:
18

Option 3:
19

Option 4:
21

Correct Answer:
18

Solution:
$D=\lambda-5$
$D_1=\lambda+\mu-18$
$D_2=4\lambda-2\mu+6$
$D_3=\mu-13$
$\text{Here } x=\frac{D_1}{D};y=\frac{D_2}{D};z=\frac{D_3}{D}$

For infinitely many solutions

$D=D_1=D_2=D_3=0$
$\Rightarrow \lambda=5;\mu=13$
$\therefore \lambda+\mu=18$

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Q. 4 If $\alpha = 1$ and $\beta = 1 + \mathrm{i}\sqrt{2}$, where $i = \sqrt{-1}$ are two roots of the equation $\mathrm{x}^3 + \mathrm{ax}^2 + \mathrm{bx} + \mathrm{c} = 0, \mathrm{a}, \mathrm{b}, \mathrm{c} \in \mathbb{R}$, then $\int_{-1}^{1} \left(x^3 + ax^2 + bx + c\right)dx$ is equal to :

Option 1:
-2

Option 2:
-4

Option 3:
-8

Option 4:
-10

Correct Answer:
-8

Solution:

$\text{Roots are } 1 \pm i\sqrt{2} \& 1$

$-a = \text{ sum of roots}$

$a = -3$

$-c = 1(1 + \sqrt{2}i)(1 - i\sqrt{2})$

$c = -3$

$I = \int_{-1}^{1} \left(x^3 - 3x^2 + bx - 3\right)dx$

$= 2\int_{0}^{1} \left(-3x^2 - 3\right)dx$

$= 2\left(-x^3 - 3x\right)_0^1 = -8$

Q. 5 If the quadratic equation $(\lambda + 2)\mathrm{x}^2 - 3\lambda\mathrm{x} + 4\lambda = 0, \lambda \neq -2$, has two positive roots, then the number of possible integral values of λ is :

Option 1:
1

Option 2:
2

Option 3:
3

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Option 4:
4

Correct Answer:
2

Solution:
$f(x) = (\lambda + 2)x^2 - 3\lambda x + 4\lambda$
$C - 1 \quad af\,(0) > 0$
$(\lambda + 2)4\lambda > 0$
$\Rightarrow \lambda < -2 \text{ or } \lambda > 0$
$C - 2 \quad -\frac{b}{2a} > 0$
$\frac{3\lambda}{2(\lambda + 2)} > 0$
$\Rightarrow \lambda < -2 \text{ or } \lambda > 0$
$C - 3 \quad D \geq 0$
$(-3\lambda)^2 - 4(\lambda + 2) \times 4\lambda \geq 0$
$\lambda(7\lambda + 32) \leq 0$
$\Rightarrow \lambda \in \left[\frac{-32}{7}, 0\right]$

Intersection of C-1, C-2 and C-3

$\Rightarrow \lambda \in \left[\frac{-32}{7}, -2\right)$
$\lambda \in [-4.57, -2)$
$\Rightarrow \lambda = -4, -3$

$\therefore$ Number of values of $\lambda = 2$

Q. 6 Let $A = \begin{bmatrix} 1 & 2 & 7 \\ 4 & -2 & 8 \\ 3 & 8 & -7 \end{bmatrix}$ and $\det(A - \alpha I) = 0$, where α is a real number. If the largest possible value of α is p, then the circle $(x - p)^2 + (y - 2p)^2 = 320$, intersects the co-ordinate axes at

Option 1:
1 point

Option 2:
2 points

Option 3:
3 points

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Option 4:
4 points

Correct Answer:
3 points

Solution:

$$\begin{vmatrix} 1-\alpha & 2 & 7 \\ 4 & -2-\alpha & 8 \\ 3 & 8 & -7-\alpha \end{vmatrix} = 0$$

$\Rightarrow (1-\alpha)[(\alpha+2)(\alpha+7)-64]-2[-28-4\alpha-24]+7[32+6+3\alpha]=0$

$\Rightarrow \alpha^3+8\alpha^2-88\alpha-320=0$

$(\alpha-8)\left(\alpha^2+16\alpha+40\right)=0$

$\alpha=8, \alpha=-8\pm 2\sqrt{6}$

So $p=-8$

$\Rightarrow (x-8)^2+(y-16)^2=320=8^2+16^2$

put $\quad y=0 \Rightarrow x=16,0$

put $\quad x=0 \Rightarrow y=32,0$

$\Rightarrow (16,0),(0,32),(0,0)$

$\Rightarrow 3$ points of intersection

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Q. 7 Let $\alpha=\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\infty$ and $\beta=\frac{1}{3}+\frac{1}{9}+\frac{1}{27}+\ldots\infty$. Then the value of $(0.2)^{\log_{\sqrt{5}}(\alpha)}+(0.04)^{\log_5(\beta)}$ is equal to :

Option 1:
4

Option 2:
5

Option 3:
8

Option 4:
25

Correct Answer:
8

Solution:

$$\alpha = \frac{1/4}{1-\frac{1}{2}} = \frac{1}{2}$$

$$\beta = \frac{1/3}{1-\frac{1}{3}} = \frac{1}{2}$$

$$(0.2)^{\log_{\sqrt{5}}\alpha} = \alpha^{\log_{\sqrt{5}}\frac{1}{5}} = \left(\frac{1}{2}\right)^{-2} = 4$$

$$(0.04)^{\log_5\beta} = 5^{-2\log_5\beta} = \left(\frac{1}{2}\right)^{-2} = 4$$

$$\Rightarrow 4+4=8$$

Q. 8 For 10 observations $x_1, x_2, \ldots, x_{10}$, if $\sum_{i=1}^{10}(x_i+2)^2 = 180$ and $\sum_{i=1}^{10}(x_i-1)^2 = 90$, then their standard deviation is :

Option 1:
2

Option 2:
$\sqrt{3}$

Option 3:
$2\sqrt{2}$

Option 4:
3

Correct Answer:
3

Solution:

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$$\sum_{i=1}^{10}(x_i+2)^2=180$$

$$\sum_{i=1}^{10}x_i^2+4\sum_{i=1}^{10}x_i+\sum_{i=1}^{10}4=180$$

$$\sum_{i=1}^{10}x_i^2+4\sum_{i=1}^{10}x_i=180-40$$

Also $\sum_{i=1}^{10}(x_i-1)^2=90$

$$\sum_{i=1}^{10}x_i^2-2\sum_{i=1}^{10}x_i+\sum_{i=1}^{10}1=90$$

$$\sum_{i=1}^{10}x_i^2-2\sum_{i=1}^{10}x_i=90-10$$

$$\sum_{i=1}^{10}x_i^2+4\sum_{i=1}^{10}x_i=140$$

$$\sum_{i=1}^{10}x_i^2-2\sum_{i=1}^{10}x_i=80$$

$\sum_{i=1}^{10}x_i^2=100$ and $\sum_{i=1}^{10}x_i=10$

$$\sigma^2=\frac{\sum_{i=1}^{10}x_i^2}{N}-\left(\frac{\sum_{i=1}^{10}x_i}{N}\right)^2$$

$$=\frac{100}{10}-\left(\frac{10}{10}\right)^2$$

$$\sigma^2=10-1=9$$

$\Rightarrow \sigma=3$

Q. 9 In the expansion of $\left(9x-\frac{1}{3\sqrt{x}}\right)^{18}, x>0$, if the term independent of x is $(221)\mathrm{k}$. then k is equal to:

Option 1:
84

Option 2:
78

Option 3:
168

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Option 4:
198

Correct Answer:
84

Solution:
General term in above expression is

$$T_{r+1} = {}^{18}C_r (9x)^{18-r} \left(-\frac{1}{3\sqrt{x}}\right)^r$$
$$= \left(-\frac{1}{3}\right)^r {}^{18}C_r 9^{18-r} \cdot x^{18-\frac{3r}{2}}$$

For term independent of x, $18 - \frac{3r}{2} = 0 \Rightarrow r = 12$
$\therefore$ Coefficient of term independent of x

$$= \left(-\frac{1}{3}\right)^{12} {}^{18}C_{12} 9^{18-12}$$
$$= 18564$$
$$= 221k \text{ (given)}$$
$$\Rightarrow k = 84$$

Q. 10 Let $P(3\cos\alpha, 2\sin\alpha), \alpha \neq 0$, be a point on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$, Q be a point on the circle $x^2 + y^2 - 14x - 14y + 82 = 0$ and R be a point on the line $x + y = 5$ such that the centroid of the triangle PQR is $\left(2 + \cos\alpha, 3 + \frac{2}{3}\sin\alpha\right)$. Then the sum of the ordinates of all possible points R is:

Option 1:
6

Option 2:
2

Option 3:
4

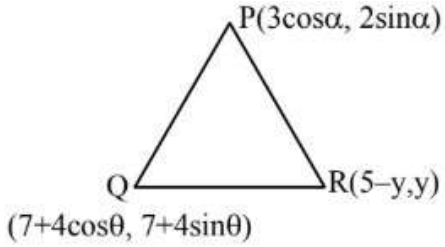
Option 4:
8

Correct Answer:
8

Solution:
Centroid :

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$$\left(\frac{3\cos\alpha+7+4\cos\theta+5-y}{3}, \frac{2\sin\alpha+7+4\sin\theta+y}{3}\right)$$
$$=\left(\cos\alpha+2, \frac{2}{3}\sin\alpha+3\right)$$



On comparison

$$\cos\alpha=\frac{y-6}{4} \& \sin\alpha=\frac{2-y}{4}$$
$$\sin^2\alpha+\cos^2\alpha=1 \Rightarrow (y-6)^2+(y-2)^2=16$$
$$y^2-8y+12=0$$
$$\Rightarrow \text{ Sum of ordinates } =8$$

Q. 11 Let $H: \frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be a hyperbola such that the distance between its foci is 6 and the distance between its directrices is $\frac{8}{3}$. If the line $x=\alpha$ intersects the hyperbola H at the points A and B such that the area of the triangle AOB is $4\sqrt{15}$, where O is the origin, then α^2 equals

Option 1:
12

Option 2:
16

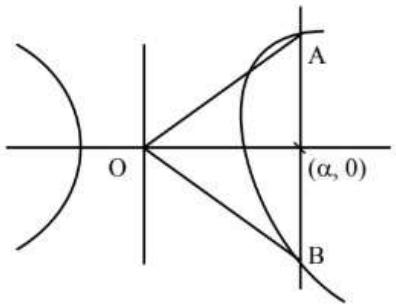
Option 3:
24

Option 4:
25

Correct Answer:
16

Solution:
$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

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Given:

$2ae = 6 \Rightarrow ae = 3$

$\frac{2a}{e} = \frac{8}{3} \Rightarrow \frac{a}{e} = \frac{4}{3}$

$a^2 = 4 \quad e = \frac{3}{2} \quad b^2 = 5$

$\frac{x^2}{4} - \frac{y^2}{5} = 1 \quad$ solve with $x = \alpha$

$y = \pm\sqrt{\frac{5(\alpha^2 - 4)}{4}}$

$A\left(\alpha, \sqrt{\frac{5(\alpha^2 - 4)}{4}}\right) \quad B\left(\alpha, -\sqrt{\frac{5(\alpha^2 - 4)}{4}}\right)$

$\text{Ar. of } \Delta OAB = \frac{1}{2}|\alpha|\sqrt{5(\alpha^2 - 4)} = 4\sqrt{15}$

$\Rightarrow 16 \times 15 = \frac{1}{4}\alpha^2(5)\left(\alpha^2 - 4\right)$

$\Rightarrow \alpha^2 = 16$

Q. 12 $\max_{0 \le x \le \pi}\left(16\sin\left(\frac{x}{2}\right)\cos^3\left(\frac{x}{2}\right)\right)$ is equal to :

Option 1:
$\frac{3\sqrt{3}}{2}$

Option 2:
$3\sqrt{3}$

Option 3:
$4\sqrt{3}$

Option 4:
$6\sqrt{3}$

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Correct Answer:
$3\sqrt{3}$

Solution:

$E = 16 \sin \frac{x}{2} \cos^3 \frac{x}{2}$

$E = 4 \sin x[1 + \cos x]$

$\frac{dE}{dx} = 4[\cos x + \cos 2x]$

$= 8 \cos \frac{3x}{2} \cos \frac{x}{2} = 0$

$\Rightarrow \cos \frac{3x}{2} = 0 \text{ or } \cos \frac{x}{2} = 0$

$\Rightarrow x = \left\{\frac{\pi}{3}, \pi\right\}$ are critical points of the function

$E(0) = 0$

$E(\pi) = 0$

$E\left(\frac{\pi}{3}\right) = 3\sqrt{3}$

$\therefore$ maximum value of $E = 3\sqrt{3}$

Q. 13 The shortest distance between the lines

$\vec{r} = \left(\frac{1}{3}\hat{i} + 2\hat{j} + \frac{8}{3}\hat{k}\right) + \lambda(2\hat{i} - 5\hat{j} + 6\hat{k})$

and $\vec{r} = \left(-\frac{2}{3}\hat{i} - \frac{1}{3}\hat{k}\right) + \mu(\hat{j} - \hat{k}), \lambda, \mu \in \mathbb{R}$, is :

Option 1:
$\sqrt{5}$

Option 2:
3

Option 3:
$2\sqrt{3}$

Option 4:
$\sqrt{15}$

Correct Answer:
3

Solution:

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$\vec{a}_1 = \frac{1}{3}\hat{i} + 2\hat{j} + \frac{8}{3}\hat{k}$

$\vec{a}_2 = \frac{-2}{3}\hat{i} - \frac{1}{3}\hat{k}$

$\vec{b}_1 = 2\hat{i} - 5\hat{j} + 6\hat{k}, \quad \vec{b}_2 = \hat{j} - \hat{k}$

$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -5 & 6 \\ 0 & 1 & -1 \end{vmatrix} = -\hat{i} + 2\hat{j} + 2\hat{k}$

$\vec{a}_1 - \vec{a}_2 = \hat{i} + 2\hat{j} + 3\hat{k}$

$\text{S.D} = \left| \frac{(\vec{a}_1 - \vec{a}_2) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$

$\text{S.D} = \left| \frac{(\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (-\hat{i} + 2\hat{j} + 2\hat{k})}{3} \right|$

$= \frac{9}{3} = 3$

Q. 14 If $\left(2\alpha + 1, \alpha^2 - 3\alpha, \frac{\alpha - 1}{2}\right)$ is the image of $(\alpha, 2\alpha, 1)$ in the line $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z}{1}$, then the possible value(s) of α is(are) :

Option 1:
Only 3

Option 2:
Only 3 and -1

Option 3:
Only 3, $\frac{1}{4}$ and -1

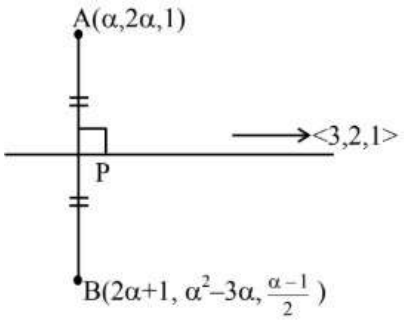
Option 4:
Only 3 and $\frac{1}{4}$

Correct Answer:
Only 3

Solution:
$\therefore$ P is mid point of AB and lies on the given line

$\Rightarrow \text{P}\left(\frac{3\alpha+1}{2}, \frac{\alpha^2-\alpha}{2}, \frac{\alpha+1}{4}\right)$

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Now put coordinates of P in the line

$$\Rightarrow \frac{\frac{3\alpha+1}{2}-2}{3}=\frac{\frac{\alpha^2-\alpha}{2}-1}{2}=\frac{\alpha+1}{4}$$
$$\Rightarrow \frac{\alpha-1}{2}=\frac{\alpha+1}{4}=\frac{\alpha^2-\alpha-2}{4}$$
$$\Rightarrow \alpha=3$$

Now for $\alpha=3$, clearly AB is perpendicular to

$3\hat{i}+2\hat{j}+k$

for $\alpha=3$ point B is image of point A

Q. 15 Let $\hat{u}$ and $\hat{v}$ be unit vectors inclined at an acute angle such that $|\hat{u}\times\hat{v}|=\frac{\sqrt{3}}{2}$. If $\vec{A}=\lambda\hat{u}+\hat{v}+(\hat{u}\times\hat{v})$. Then λ is equal to :

Option 1:
$\frac{4}{3}(\vec{A}\cdot\hat{u})-\frac{2}{3}(\vec{A}\cdot\hat{v})$

Option 2:
$\frac{2}{3}(\vec{A}\cdot\hat{u})-\frac{1}{3}(\vec{A}\cdot\hat{v})$

Option 3:
$\frac{4}{3}(\vec{A}\cdot\hat{u})+\frac{2}{3}(\vec{A}\cdot\hat{v})$

Option 4:
$(\vec{A}\cdot\hat{u})-\frac{1}{2}(\vec{A}\cdot\hat{v})$

Correct Answer:
$\frac{4}{3}(\vec{A}\cdot\hat{u})-\frac{2}{3}(\vec{A}\cdot\hat{v})$

Solution:

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$$|\hat{u} \times \hat{v}| = \frac{\sqrt{3}}{2}$$

$$|\hat{u}||\hat{v}| \sin\theta = \frac{\sqrt{3}}{2} \quad \therefore \theta = \frac{\pi}{3}$$

$$\text{and } \hat{u} \cdot \hat{v} = |\hat{u}||\hat{v}| \cos\frac{\pi}{3} = \frac{1}{2}$$

$$\vec{A} = \lambda\hat{u} + \hat{v} + \hat{u} \times \hat{v}$$

Dot with $\hat{u}$

$$\vec{A} \cdot \hat{u} = \lambda(1) + \hat{u} \cdot \hat{v} + \hat{u} \cdot (\hat{u} \times \hat{v})$$

$$\vec{A} \cdot \hat{u} = \lambda + \frac{1}{2}$$

$$\Rightarrow 2\vec{A} \cdot \hat{u} = 2\lambda + 1$$

Dot equation (1) with $\hat{v}$

$$\vec{A} \cdot \hat{v} = \lambda(\hat{u} \cdot \hat{v}) + \hat{v} \cdot \hat{v} + \hat{v} \cdot (\hat{u} \times \hat{v})$$

$$\vec{A} \cdot \hat{v} = \frac{\lambda}{2} + 1$$

$$\vec{A} \cdot \hat{v} - \frac{\lambda}{2} = 1$$

From (2) and (3)

$$2\vec{A} \cdot \hat{u} - 2\lambda = \vec{A} \cdot \hat{v} - \frac{\lambda}{2}$$

$$\therefore \lambda = \frac{4}{3}\vec{A} \cdot \hat{u} - \frac{2}{3}\vec{A} \cdot \hat{v}$$

Q. 16 Let for some $\alpha \in \mathbb{R}$, $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying $f(x+y) = f(x) + 2y^2 + y + \alpha xy$ for all $x, y \in \mathbb{R}$. If $f(0) = -1$ and $f(1) = 2$, then the value of $\sum_{n=1}^{5}(\alpha + f(n))$ is :

Option 1:
110

Option 2:
140

Option 3:
150

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Option 4:
170

Correct Answer:
140

Solution:
$f(x+y)=f(x)+2y^2+y+\alpha xy$

Put $\mathrm{x}=0$ in eq.(1)

$\Rightarrow \mathrm{f(y)}=-1+2\mathrm{y}^2+\mathrm{y}$

Now put $\mathrm{x}=\mathrm{y}=1$ in eq. (1)

$\Rightarrow f(2)=f(1)+3+\alpha$
$\Rightarrow 9=2+3+\alpha \Rightarrow \alpha=4$

$\text{Now } \sum_{n=1}^{5}(\alpha+f(n))=\sum_{n=1}^{5}\left(2y^2+y+3\right)$
$=\frac{2\times5\times6\times11}{6}+\frac{5\times6}{2}+3\times5=140$

Q. 17 Let $\mathrm{A}=\{(\mathrm{a},\mathrm{b},\mathrm{c}):\mathrm{a},\mathrm{b},\mathrm{c} \text{ are non-negative integers and } \mathrm{a}+\mathrm{b}+2\mathrm{c}=22\}$.
The $\mathrm{n(A)}$ is equal to :

Option 1:
121

Option 2:
124

Option 3:
144

Option 4:
169

Correct Answer:
144

Solution:

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$$c = 0 \Rightarrow a + b = 22 \rightarrow {}^{22+2-1}C_{2-1} = {}^{23}C_1$$
$$c = 1 \Rightarrow a + b = 20 \rightarrow {}^{20+2-1}C_{2-1} = {}^{21}C_1$$
$$\vdots$$
$$\vdots$$
$$c = 10 \Rightarrow a + b = 2 \rightarrow {}^{2+2-1}C_{2-1} = {}^{3}C_1$$
$$c = 11 \Rightarrow a + b = 0 \rightarrow {}^{0+2-1}C_{2-1} = {}^{1}C_1$$

Number of non-negative integral solutions

$${}^{1}C_1 + {}^{3}C_1 + {}^{5}C_1 + \ldots + {}^{23}C_1$$
$$1 + 3 + 5 \ldots\ldots\ldots + 23$$
$$= 144$$

Q. 18 The area of the region bounded by the curves $x + 3y^2 = 0$ and $x + 4y^2 = 1$ is equal to :

Option 1:
$\frac{1}{3}$

Option 2:
$\frac{2}{3}$

Option 3:
$\frac{4}{3}$

Option 4:
$\frac{5}{3}$

Correct Answer:
$\frac{4}{3}$

Solution:
$$x = -3y^2 \text{ and } x = 1 - 4y^2$$
$$A = \int_{-1}^{1} \left((1 - 4y^2) + 3y^2 \right) dy$$
$$A = \int_{-1}^{1} (1 - y^2) dy = 2\int_{0}^{1} (1 - y^2) dy$$
$$A = \frac{4}{3}$$

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Q. 19 Let $y = y(x)$ be the solution of the differential equation :

$$\frac{dy}{dx} + \left(\frac{6x^2+(3x^2+2x^3+4)e^{-2x}}{(x^3+2)(2+e^{-2x})}\right)y = 2 + e^{-2x}$$

$x \in (-1, 2)$, satisfying $y(0) = \frac{3}{2}$.
If $y(1) = \alpha\left(2 + e^{-2}\right)$, than α is equal to :

Option 1:
$\frac{13}{8}$

Option 2:
$\frac{6}{13}$

Option 3:
$\frac{12}{13}$

Option 4:
$\frac{13}{12}$

Correct Answer:
$\frac{13}{12}$

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Solution:

$$\text{I.F.} = e^{\int \frac{6x^2+e^{-2x}\left(3x^2+2x^3+4\right)}{\left(x^3+2\right)\left(2+e^{-2x}\right)}dx}$$

$$= e^{\int \frac{6x^2+e^{-2x}\left(3x^2-2x^3-4\right)+\left(4x^3+8\right)e^{-2x}}{\left(x^3+2\right)\left(2+e^{-2x}\right)}dx}$$

Let $\left(x^3+2\right)\left(2+e^{-2x}\right) = t$

$$\left(6x^2 + e^{-2x}\left(3x^2 - 2x^3 - 4\right)\right)dx = dt$$

$$\text{I.F.} = e^{\ln\left(x^3+2\right)\left(2+e^{-2x}\right)\left| - \int \frac{4e^{-2x}dx}{2+e^{-2x}}\right.}$$

$$= e^{\ln\left|\left(x^3+2\right)\left(2+e^{-2x}\right)\right| - 2\ln\left|2+e^{-2x}\right|}$$

$$= e^{\ln\left|\frac{x^3+2}{2+e^{-2x}}\right|} = \frac{x^3+2}{2+e^{-2x}}$$

$$\Rightarrow y \cdot \left(\frac{x^3+2}{2+e^{-2x}}\right) = \int \left(\frac{x^3+2}{2+e^{-2x}}\right)\left(e^{-2x}+2\right)dx + C$$

$$y \cdot \left(\frac{x^3+2}{2+e^{-2x}}\right) = \frac{x^4}{4} + 2x + C \text{ at } y(0) = \frac{3}{2} \Rightarrow C = 1$$

So $y \cdot \left(\frac{x^3+2}{2+e^{-2x}}\right) = \frac{x^4}{4} + 2x + 1$
at $x = 1$

$$y \cdot \left(\frac{3}{2+e^{-2}}\right) = \frac{1}{4} + 2 + 1$$

$$\Rightarrow y = \frac{13}{12}\left(e^{-2}+2\right)$$

So $\alpha = \frac{13}{12}$

Q. 20 The integral $\int_0^1 \cot^{-1}\left(1+x+x^2\right)dx$ is equal to :

Option 1:
$2\tan^{-1}2 + \frac{1}{2}\log_e\left(\frac{5}{4}\right) + \frac{\pi}{2}$

Option 2:
$2\tan^{-1}2 + \frac{1}{2}\log_e\left(\frac{5}{4}\right) - \frac{\pi}{2}$

Option 3:
$2\tan^{-1}2 - \frac{1}{2}\log_e\left(\frac{5}{4}\right) + \frac{\pi}{2}$

Option 4:
$2\tan^{-1}2 - \frac{1}{2}\log_e\left(\frac{5}{4}\right) - \frac{\pi}{2}$

Correct Answer:
$2\tan^{-1}2 - \frac{1}{2}\log_e\left(\frac{5}{4}\right) - \frac{\pi}{2}$

Solution:

$$= \int_0^1 \tan^{-1}\left(\frac{1}{1+x+x^2}\right)dx$$

$$= \int_0^1 \tan^{-1}\left(\frac{(x+1)-x}{1+x(x+1)}\right)dx$$

$$= \int_0^1 \left(\tan^{-1}(x+1) - \tan^{-1}x\right)dx$$

$$= \left(x\tan^{-1}(x+1) - \frac{1}{2}\ell n\left|1+(1+x)^2\right| + \tan^{-1}(x+1)\right)_0^1$$

$$- \left(x\tan^{-1}x - \frac{1}{2}\ell n\left|1+x^2\right|\right)_0^1$$

$$= \left(2\tan^{-1}2 - \frac{\pi}{4} - \frac{1}{2}\ell n\frac{5}{2}\right) - \left(\frac{\pi}{4} - \frac{1}{2}\ell n2\right)$$

$$= 2\tan^{-1}2 - \frac{1}{2}\ell n\left(\frac{5}{4}\right) - \frac{\pi}{2}$$

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Q. 21 From a month of 31 days, 3 different dates are selected at random. If the probability that these dates are in an increasing A.P. is equal to $\frac{a}{b}$, where $a, b \in N$ and $\gcd(a, b) = 1$, then $a + b$ is equal to _____ .

Correct Answer:
944

Solution:
Total numbers of ways of selecting

$3 \text{ numbers } = {}^{31}C_3 = \frac{31 \times 30 \times 29}{3 \times 2 \times 1} = 4495$

a, b, c are in A.P.
⇒ either a & c both odd numbers or both even numbers

$= {}^{16}C_2 + {}^{15}C_2$
$= 120 + 105$
$= 225$

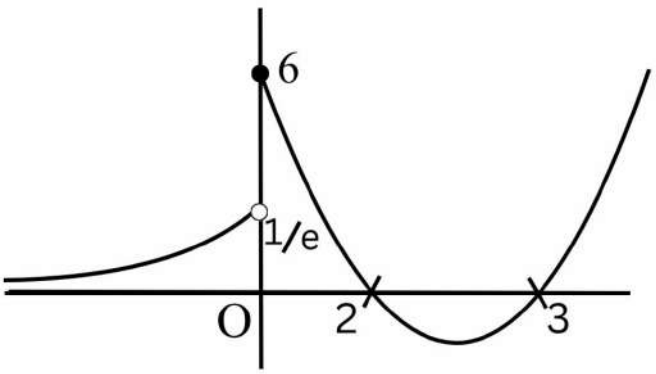
Probability $= \frac{225}{4495} = \frac{45}{899}$

$= \frac{a}{b} \Rightarrow a + b = 944$

Q. 22 Let $f(x) = \begin{cases} e^{x-1} & , x < 0 \\ x^2 - 5x + 6 & , x \geq 0 \end{cases}$ and $g(x) = f(|x|) + |f(x)|$. If the number of points where g is not continuous and is not differentiable are α and β respectively, then $\alpha + \beta$ is equal to _____

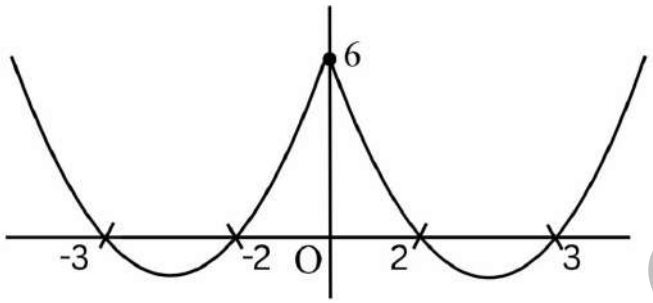
Correct Answer:
4

Solution:
Graph of f(x)

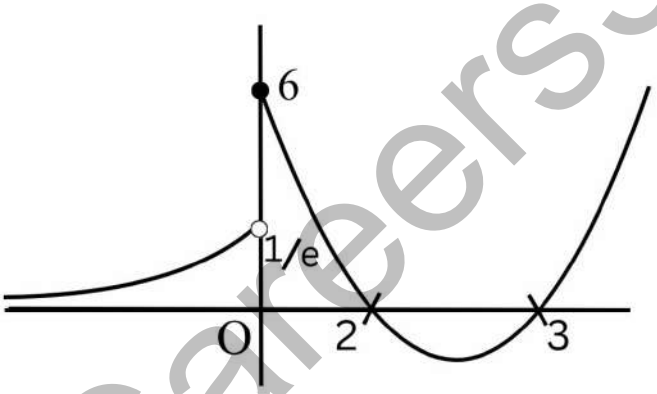


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Now graph of $f(|x|)$



$f(|x|)$ is continuous function and it is non diff. at $x = 0$ Graph of $|f(x)|$



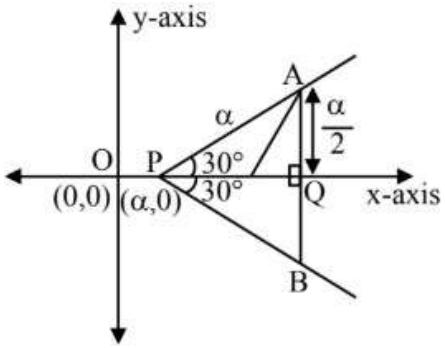
$|f(x)|$ is discontinuous at $x = 0$
$|f(x)|$ is non-diff. at $x = 0, 2, 3$
$g(x) = f(|x|) + |f(x)|$
$g(x)$ will be discontinuous at $x = 0$
$g(x)$ will be non diff at $x = 0, 2, 3$
$\alpha = 1, \beta = 3$
$\alpha + \beta = 1 + 3 = 4$

Q. 23 Let A, B be points on the two half-lines $x - \sqrt{3}|y| = \alpha, \alpha > 0$ at a distance of α from their point of intersection P . The line segment AB meets the angle bisector of the given half-lines at the point Q . If $PQ = \frac{9}{2}$ and R is the radius of the circumcircle of $\triangle PAB$, then $\frac{\alpha^2}{R}$ is equal to _____ .

Correct Answer:
9

Solution:
$\triangle PAB$ is equilateral

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$$PQ = \frac{\sqrt{3}\alpha}{2} = \frac{9}{2} \quad \therefore \quad \alpha = \frac{9}{\sqrt{3}}$$

$$\alpha = 3\sqrt{3}$$

$$\frac{\alpha}{2r} = \cos 30^\circ$$

$$\alpha = \frac{\sqrt{3}}{2} \cdot 2R$$

$$\alpha = \sqrt{3}R$$

$$R = 3$$

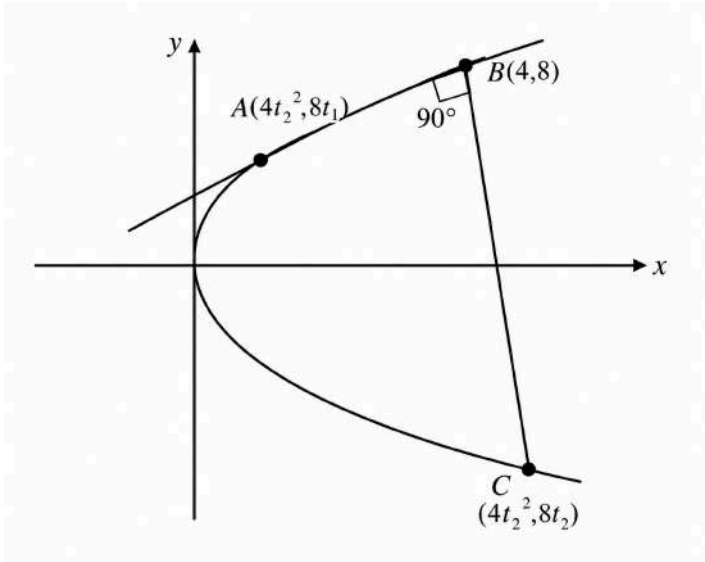
$$\frac{\alpha^2}{R} = \frac{(3\sqrt{3})^2}{3} = 9$$

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Q. 24 Let A, B and C be the vertices of a variable right angled triangle inscribed in the parabola $y^2 = 16x$. Let the vertex B containing the right angle be $(4, 8)$ and the locus of the centroid of $\triangle ABC$ be a conic C_0. Then three times the length of latus rectum of C_0 is _____ .

Correct Answer:
16

Solution:



Variable point A \& C are shown in the figure

$\because \mathrm{m_{AB}} \cdot \mathrm{m_{BC}} = -1$
$\Rightarrow \mathrm{t_1 + t_2 + t_1 t_2} = -5 \ldots . . (1)$

Suppose locus of centroid of $\triangle \mathrm{ABC}$ is (h, k)

$\therefore 3\,\mathrm{h} = 4 + 4\mathrm{t}_1^2 + 4\mathrm{t}_2^2 \& 3\mathrm{k} = 8 + 8\mathrm{t}_1 + 8\mathrm{t}_2$

by eliminating $t_1 \& t_2$ using equation (1) also we get $h = \frac{9}{48}k^2 + \frac{40}{3}$
$\therefore$ locus of centroid of parabola is

$$\mathrm{x} = \frac{9}{48}\mathrm{y}^2 + \frac{40}{3}$$
$$\therefore \ell(\text{ L.R. }) = \frac{48}{9}$$
$$\therefore 3\ell(\text{ L.R. }) = \frac{48}{3} = 16$$

Q. 25 Let f be a twice differentiable function such that

$f(\mathrm{x}) = \int_0^{\mathrm{x}} \tan(\mathrm{t} - \mathrm{x})\mathrm{dt} - \int_0^{\mathrm{x}} f(\mathrm{t}) \tan \mathrm{t}\mathrm{dt}, \mathrm{x} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Then $f''\left(\frac{\pi}{6}\right) + 12f'\left(-\frac{\pi}{6}\right) + f\left(\frac{\pi}{6}\right)$ is equal to _____ .

Correct Answer:
5

Solution:

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$$f(x)=\int_0^x \tan(x-t-x)dt-\int_0^x f(t)\tan t dt$$

$$f(x)=-\int_0^x \tan t dt-\int_0^x f(t)\tan t dt$$

$$f'(x)=-\tan x-f(x)\tan x$$

$$\frac{dy}{y+1}=-\tan x dx$$

$$y+1=\cos x\cdot c$$

$$f(0)=0\Rightarrow 1=c$$

$$y=\cos x-1=f(x)$$

$$f\left(\frac{\pi}{6}\right)=\frac{\sqrt{3}}{2}-1$$

$$f'(x)=-\sin x$$

$$f'\left(\frac{-\pi}{6}\right)=\frac{1}{2}$$

$$f''(x)=-\cos x$$

$$f''\left(\frac{\pi}{6}\right)=-\frac{\sqrt{3}}{2}$$

$$f''\left(\frac{\pi}{6}\right)+12f'\left(\frac{-\pi}{6}\right)+f\left(\frac{\pi}{6}\right)=5$$

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JEE MAINS

2026

5th April

1st Shift

Question Paper

Physics

Q. 1 In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and 7th Vernier division coincides with a main scale reading. If the value of 1 main scale division is 1 mm and there are 10 Vernier scale divisions, then the Vernier caliper has

Option 1:
0.07 cm negative zero error

Option 2:
0.7 cm negative zero error

Option 3:
0.07 cm positive zero error

Option 4:
0.7 cm positive zero error

Correct Answer:
0.07 cm positive zero error

Solution:
Scale reading = main scale + Least count × V.S.R.

$$= 0 + \frac{1}{10} \times 7 = 0.7 \text{ mm}$$
$$= 0.07 \text{ cm}$$

Q. 2 L, C and R represents physical quantitie inductance, capacitance and resistanc respectively.

The dimensional formula $\mathrm{ML}^2\ \mathrm{T}^{-4}\ \mathrm{A}^{-2}$ correspond to _____ .

Option 1:
$\frac{\mathrm{R}}{\sqrt{\mathrm{LC}}}$

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Option 2:
$\frac{\mathrm{R}}{\mathrm{LC}}$

Option 3:
$\frac{\mathrm{C}}{\sqrt{\mathrm{LR}}}$

Option 4:
$\frac{1}{\mathrm{R}}\sqrt{\frac{\mathrm{L}}{\mathrm{C}}}$

Correct Answer:
$\frac{\mathrm{R}}{\sqrt{\mathrm{LC}}}$

Solution:

$$\mathrm{R}=\frac{\mathrm{V}}{\mathrm{I}}=\frac{\mathrm{w}}{\mathrm{qI}}=\frac{\mathrm{ML}^2\,\mathrm{T}^{-2}}{\mathrm{I}^2\,\mathrm{T}}=\mathrm{ML}^2\,\mathrm{T}^{-3}\mathrm{I}^{-2}$$

$$\mathrm{U}=\frac{1}{2}\mathrm{LI}^2\rightarrow\mathrm{ML}^2\,\mathrm{T}^{-2}=\mathrm{LA}^2$$

$$\mathrm{L}=\mathrm{ML}^2\,\mathrm{T}^{-2}\,\mathrm{A}^{-2}$$

$$\mathrm{Q}=\mathrm{CV}$$

$$\mathrm{C}=\frac{\mathrm{IT}\times\mathrm{q}}{\mathrm{ML}^2\,\mathrm{T}^{-2}}=\frac{\mathrm{A}^2\,\mathrm{T}^2}{\mathrm{ML}^2\,\mathrm{T}^{-2}}$$

$$\mathrm{C}=\mathrm{M}^{-1}\,\mathrm{L}^{-2}\,\mathrm{A}^{-2}\,\mathrm{T}^4$$

$$\mathrm{LC}=\mathrm{ML}^2\,\mathrm{T}^{-2}\,\mathrm{A}^{-2}\times\mathrm{M}^{-1}\,\mathrm{L}^{-1}\,\mathrm{A}^2\,\mathrm{T}^4$$

$$\mathrm{LC}=\mathrm{T}^2$$

$$\frac{\mathrm{R}}{\sqrt{\mathrm{LC}}}=\frac{\mathrm{ML}^2\,\mathrm{T}^{-3}\,\mathrm{A}^{-2}}{\mathrm{T}}=\left(\mathrm{ML}^2\,\mathrm{T}^{-4}\,\mathrm{A}^{-2}\right)$$

Q. 3 When one moves from a point 16 km below the earth's surface to a point 16 km above the earth's surface. The change in g is approximately $\alpha\%$. The value of α is _____ .
(Take radius of the earth $=6400\ \mathrm{km}$.)

Option 1:
0.12

Option 2:
0.25

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Option 3:
0.50

Option 4:
0.75

Correct Answer:
0.25

Solution:

$$g_{above} = g - \frac{2hg}{R} = 10 - \frac{2\,h}{R}$$

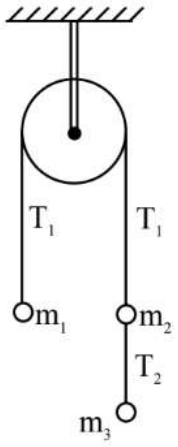
$$g_{below} = g - \frac{hg}{R}$$

$$\Delta g = g_{above} - g_{below} = g - \frac{2hg}{R} - g +$$

$$\Delta g = \frac{hg}{R}$$

$$\Delta g = \frac{h}{16 \times 100} = 0.25\%$$

Q. 4 Three masses $m_1 = 4\ kg,\ m_2 = 4\ kg$ and $m_3 = 6\ kg$ are suspended from a fixed smooth frictionless pully as shown in the figure below. The value of T_1/T_2 is _____ (take $g = 10\ m/s^2$)



Option 1:
$\frac{5}{3}$

Option 2:
$\frac{2}{3}$

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Option 3:

$\frac{3}{5}$

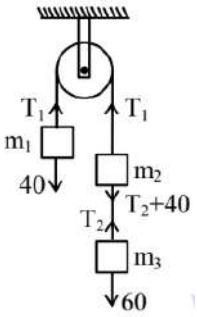
Option 4:

$\frac{2}{5}$

Correct Answer:

$\frac{5}{3}$

Solution:



$60 - T_2 = 6a$

$T_2 + 40 - T_1 = 4a$

$60 = 14a$

$a = \frac{60}{14} = \frac{30}{7}$

$T_1 = 40 + 4 \times \frac{30}{7} = \frac{280 + 120}{7}$

$T_1 = \frac{400}{7}$

$60 - T_2 = 66 \times \frac{30}{7}$

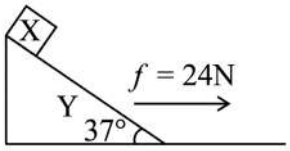
$60 - \frac{180}{7} = T_2$

$\frac{420 - 180}{7} = T_2 = \frac{240}{7}$

$\frac{T_1}{T_2} = \frac{5}{3}$

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Q. 5 A wedge Y with mass of 10 kg and all frictionless surfaces and the inclined surface making 37° with horizontal. A block X with mass 2 kg is placed at the highest point of the wedge as shown in figure is at rest. At $\mathrm{t}=0$ wedge (Y) is pulled toward right with constant force (f) of 24 N . Taking the block X at rest at $\mathrm{t}=0$, the time taken by it to slide down 8.8 m on the slope, while Y is on the move, is _____ s. (take $\tan\left(37^{\circ}\right)=3/4$ and $g=10\ \mathrm{m/s^2}$)



Option 1:
2

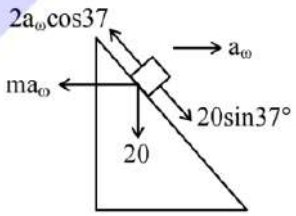
Option 2:
4

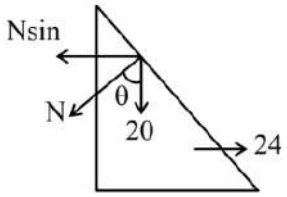
Option 3:
$\sqrt{2}$

Option 4:
$2\sqrt{2}$

Correct Answer:
2

Solution:





$20\sin 37^{\circ}-2\mathrm{a}_{\omega}\cos 37^{\circ}=2\mathrm{a}_{\mathrm{rel}}$
$20\cos 37^{\circ}-\mathrm{N}+2\mathrm{a}_{\omega}\sin 37^{\circ}=0$
$24-\mathrm{N}\sin 37^{\circ}=10\times\mathrm{a}_{\omega}$

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$24-(20\cos 37^{\circ}+2a_{\omega}\sin 37^{\circ})\times \sin 37$

$24-\left(16+2a_{\omega}\times \frac{3}{5}\right)\frac{3}{5}=10a_{\omega}$

$24-\frac{48}{5}-\frac{18}{25}a_{\omega}=10a_{\omega}$

$\frac{72}{5}=\left(10-\frac{18}{25}\right)a_{\omega}$

$\frac{72}{5}=a_{\omega}=\frac{180}{116}=\frac{90}{58}$

$a_{\omega}=\frac{45}{29}$

$20\sin 37^{\circ}-2\times \frac{45}{29}\times \frac{4}{5}\times \cos 37^{\circ}=2$

$12-\frac{72}{29}=2a_{rel}$

$\frac{348-72}{29}=2a_{rel}$

$\frac{276}{29\times 2}=a_{rel}\,;\frac{138}{29}=a_{rel}$

$t=\sqrt{\frac{2\times 8.8\times 29}{138}}$

$t=\sqrt{\frac{2\times 8.8\times 29}{10\times 138}}$

$t=\sqrt{3.69}$

$t\approx 2sec$

After rounding off

Q. 6 The Young's modulus of steel wire of radius r and length L is Y . If the radius r and length L of the wire are doubled then the value of Y

Option 1:
increases by two times

Option 2:
reduces by half

Option 3:
remains unchanged

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Option 4:
becomes one fourth

Correct Answer:
remains unchanged

Solution:
Young's modulus is the property of material not depends on dimensions of material.

Q. 7 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R Statement I**: Change in internal energy of a system containing n mole of ideal gas can be written as $\Delta \mathrm{U}=\mathrm{nC}_{\mathrm{v}}\left(\mathrm{T}_{\mathrm{f}}-\mathrm{T}_{\mathrm{i}}\right)=\frac{\mathrm{nR}}{\gamma-1}\left(\mathrm{T}_{\mathrm{f}}-\mathrm{T}_{\mathrm{i}}\right)$, where $\gamma=\frac{\mathrm{C}_{\mathrm{p}}}{\mathrm{C}_{\mathrm{v}}}, \mathrm{T}_{\mathrm{i}}=$ initial temperature, $\mathrm{T}_{\mathrm{f}}=$ final temperature.

Statement II : Relation between degree of freedom f and $\gamma\left(=\mathrm{C}_{\mathrm{p}} / \mathrm{C}_{\mathrm{v}}\right)$ is $\left(\gamma=1+\frac{2}{f}\right)$

Choose the correct answer from the options given

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

Option 2:
Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Option 3:
$\mathbf{A}$ is true but $\mathbf{R}$ is false

Option 4:
$\mathbf{A}$ is false but $\mathbf{R}$ is ttue

Correct Answer:
Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Solution:
Both statement are correct but between does not explain the statement-1

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Q. 8 Consider the following statements:

A. Zeroth law of thermodynamics gives concept of temperature

B. First law of thermodynamics gives concept of internal energy

C. In isothermal expansion of ideal gas, $\Delta Q \neq \Delta W$

D. Product of intensive and extensive variables is extensive

E. The ratio of any extensive variable to mass will be an extensive variable

Choose the correct combination of statements from the options given below:

Option 1:
C, D and E Only

Option 2:
A, B and C Only

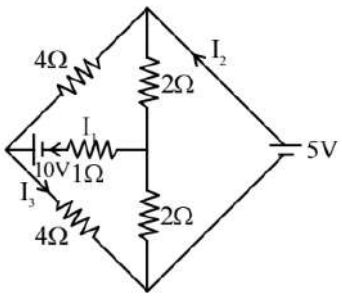
Option 3:
A, B and D Only

Option 4:
B, C and D Only

Correct Answer:
A, B and D Only

Solution:
Statement A, B and D are correct

Q. 9 Refer to the figure given below. The values of I_1, I_2 and I_3 are _____ .



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Option 1:
$I_1 = 2.5\ A, I_2 = 1.875\ A, I_3 = 1.875\ A$

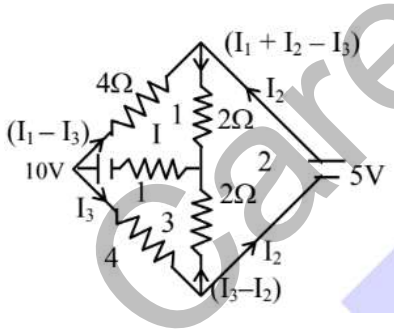
Option 2:
$I_1 = 1.875\ A, I_2 = 2.5\ A, I_3 = 1.875\ A$

Option 3:
$I_1 = 1.875\ A, I_2 = 1.875\ A, I_3 = 2.5\ A$

Option 4:
$I_1 = 2.5\ A, I_2 = 2.5\ A, I_3 = 1.875\ A$

Correct Answer:
$I_1 = 2.5\ A, I_2 = 1.875\ A, I_3 = 1.875\ A$

Solution:



Loop-1 :

$-4\left(i_1 - i_3\right) - 2\left(i_1 + i_2 - i_3\right) - i_1 + 10 = 0$

$-7i_1 - 2i_2 + 6i_3 + 10 = 0$.....(1)

Loop-2

$-5 - 2\left(i_3 - i_2\right) + 2\left(i_1 + i_2 - i_3\right) = 0$

$2i_1 + 4i_2 - 4i_3 - 5 = 0$.....(2)

Loop-3

$-i_1 + 10 - 4i_3 - 2\left(i_3 - i_2\right) = 0$

$2i_2 - 6i_3 - i_1 + 10 = 0$.....(3)

On solving equation- (1) & (2) & 3)
$i_1 = 2.5\ A$
$i_2 = 1.875\ A$
$i_3 = 1.875\ A$

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Q. 10 An electron of mass m is moving in an electric field $\vec{E} = -2E_0\hat{i}\,(E_0 = \text{constant} > 0)$, with an initial velocity $\overrightarrow{V} = \mathrm{v_o}\hat{\mathrm{i}} \quad (\mathrm{v_0} = \text{constant} > 0)$. If $\lambda_0 = \frac{\mathrm{h}}{4\mathrm{mv_0}}$, its de Broglie wavelength at time t is

_____ . (e = charge of electron)

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Option 1:

$\frac{4\lambda_0}{\left[1-\frac{E_0 e}{2m}\frac{t}{v_0}\right]}$

Option 2:

$\frac{4\lambda_0}{\left[1+\frac{E_0 e}{2m}\frac{t}{v_0}\right]}$

Option 3:

$\frac{4\lambda_0}{\left[1+\frac{2\mathrm{E_0 e}}{\mathrm{m}}\frac{\mathrm{t}}{\mathrm{v_0}}\right]}$

Option 4:

$\frac{4\lambda_0}{\left[1-\frac{2E_0 e}{m}\frac{t}{v_0}\right]}$

Correct Answer:

$\frac{4\lambda_0}{\left[1+\frac{2\mathrm{E_0 e}}{\mathrm{m}}\frac{\mathrm{t}}{\mathrm{v_0}}\right]}$

Solution:

$$\mathrm{V = U + at}$$

$$\mathrm{V = V_0 + \left(\frac{2eE_0}{m}\right)t}$$

$$\because \mathrm{a = \frac{(2E_0)e}{m}}$$

$$\lambda = \frac{\mathrm{h}}{\mathrm{mV}}$$

$$\lambda = \frac{\mathrm{h}}{\mathrm{m}\left[\mathrm{V_0} + \frac{2\mathrm{eE_0 t}}{\mathrm{m}}\right]}$$

Put $\lambda_0 = \frac{\mathrm{h}}{4\mathrm{mV_0}} \Rightarrow \mathrm{h} = 4\lambda_0\mathrm{mV_0}$

$$\lambda = \frac{4\,m\lambda_0\,V_0}{m\left[V_0 + \frac{2eE_0}{m}t\right]}$$

$$\lambda = \frac{4\lambda_0}{1 + \frac{2eE_0}{mV_0}t}$$

Q. 11 In the hydrogen atom, the electron makes a transition from the higher orbit (i) to a lower orbit (f). The ratio of the radius of the orbits in given by $r_i : r_f = 16 : 4$. The wavelength of photon emitted due to this transition is _____ nm. (Given Rydberg constant $= 1.0973 \times 10^7/m$)

Option 1:
121

Option 2:
242

Option 3:
486

Option 4:
974

Correct Answer:
486

Solution:
Given : $\frac{r_i}{r_f} = \frac{16}{4} = \frac{n_i^2}{n_f^2}$

$$\frac{n_i}{n_f} = \frac{4}{2}$$

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) = 1.0973 \times 10^7\left(\frac{1}{(2)^2} - \frac{1}{(4)^2}\right)$$

$$\lambda = 4.86 \times 10^{-7}\ m \approx 486\ nm$$

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Q. 12 A displacement current of 4.0 A can be set up in the space between two parallel plates of $6\mu\ \mathrm{F}$ capacitor. The rate of change of potential difference across the plates of the capacitor is nearly $\alpha \times 10^6\ \mathrm{V/s}$. The value of α is _____ .

Option 1:
0.58

Option 2:
0.67

Option 3:
0.82

Option 4:
0.75

Correct Answer:
0.67

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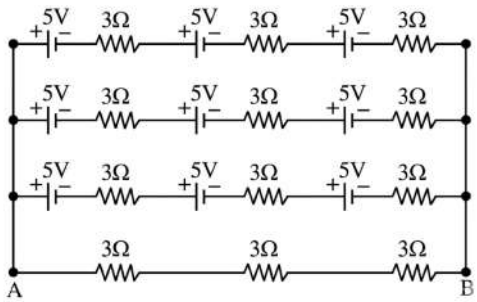
Solution:

$$I_d = C\frac{dV}{dt}$$

$$4 = \left(6 \times 10^{-6}\right)\left(\alpha \times 10^6\right)$$

$$\alpha = \frac{2}{3} \approx 0.67$$

Q. 13 Refer to the figure given below, current between terminals A and B is _____ A.



Option 1:
12.5

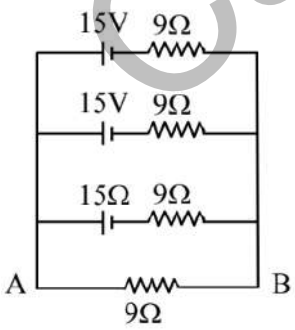
Option 2:
1.25

Option 3:
7.5

Option 4:
5

Correct Answer:
1.25

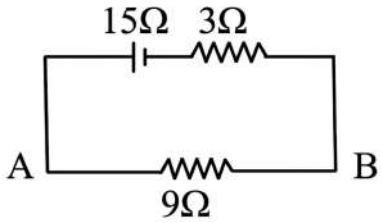
Solution:
Solving each branch as series



$$\varepsilon_{eq} = \frac{\frac{15}{9} + \frac{15}{9} + \frac{15}{9}}{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}} = 15\text{volt}$$

$$\text{and } \frac{1}{r_{eq}} = \frac{1}{9} + \frac{1}{9} + \frac{1}{9}$$

$$r_{eq} = 3\Omega$$



$$I_{AB} = \frac{15}{12} = 1.25\text{ A}$$

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Q. 14 In Young's double slit experiment, the frings width of the interferences pattern produced on the screen is $2.4\mu\ \mathrm{m}$. If the experiment is carried out in another medium having refractive index 1.2, the fringe width will be _____ $\mu\mathrm{m}$.

Option 1:
1.2

Option 2:
2

Option 3:
2.4

Option 4:
2.88

Correct Answer:
2

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Solution:
Since Fringe width in a medium becomes

$\beta' = \frac{\beta}{\mu} \qquad \left(\text{Here } \beta = \frac{\lambda D}{d}\right)$

$\beta' = \frac{\beta}{\mu} = \frac{2.4}{1.2} = 2\mu\ \mathrm{m}$

Q. 15 A ray of light passing through an equilateral prism is having velocity $2.12 \times 10^8\ \mathrm{m/s}$ in the prism material, then the minimum angle of deviation is
_____ degrees.

Option 1:
45

Option 2:
30

Option 3:
28

Option 4:
58

Correct Answer:
30

Solution:

$$\mu = \frac{\sin\left[\frac{\delta_{\min}+A}{2}\right]}{\sin\frac{A}{2}}$$

$$\mu = \frac{C}{V} = \frac{3\times10^8}{2.12\times10^8} = \sqrt{2}$$

$$\sqrt{2} = \frac{\sin\left[\frac{\delta_{\min}+60^\circ}{2}\right]}{\sin 30^\circ}$$

$$\sin\left[\frac{\delta_{\min}+60^\circ}{2}\right] = \frac{1}{\sqrt{2}}$$

$$\delta_{\min}+60^\circ = 90^\circ \Rightarrow \delta_{\min} = 30^\circ$$

Q. 16 Light source having wavelength 331 nm is used to generate photo-electrons whose stopping potential is 0.2 V . The work function of the used metal in the experiment is $\alpha \times 10^{-19}$ J. The value of α is

_____.

($h = 6.62\times10^{-34}$ J s, $e = 1.6\times10^{-19}$C and $c = 3\times10^8$ m/s)

Option 1:
3.68

Option 2:
4.68

Option 3:
5.68

Option 4:
2.68

Correct Answer:
5.68

Solution:

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$$V_s = \frac{1}{e}\left(\frac{hc}{\lambda} - \phi\right)$$

$$\phi = \frac{hc}{\lambda} - eV_s$$

$$= \left(\frac{6.62 \times 10^{-34} \times 3 \times 10^{8}}{331 \times 10^{-9}}\right) - \left(1.6 \times 10^{-19} \times 0.2\right)$$

$$= 5.68 \times 10^{-19}\ J$$

Q. 17 A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-convex lenses. One of them is used in place of original objective lens. To retain same magnification keeping the object distance unchanged, the tube length has to be :-

Option 1:
increased two times

Option 2:
increased $\frac{3}{2}$ times

Option 3:
decreased two times

Option 4:
decreased $\frac{3}{2}$ times

Correct Answer:
increased two times

Solution:

$$M = \frac{L}{F_0}\frac{D}{F_e}$$

$$\frac{1}{F_0} = (n-1)\left(\frac{2}{R}\right)$$

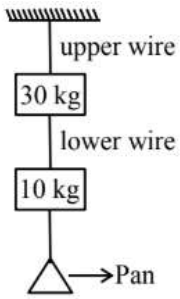
But when cut

$\frac{1}{F_0^1} = (n-1)\left(\frac{1}{R}\right) = \frac{1}{2F_0}$

So, $\frac{L}{F_0} = \frac{L'}{F_0^1} = 2\ L$

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Q. 18 Two wires as shown in the figure below, made of steel and have breaking stress of $12 \times 10^8\ \mathrm{N/m^2}$. Area of cross-section of upper wire is $0.008\ \mathrm{cm^2}$ and of lower wire is $0.004\ \mathrm{cm^2}$. The maximum mass that can be added to pan without breaking any wire is _____ kg . (take $\mathrm{g} = 10\ \mathrm{m/s^2}$)



Option 1:
56

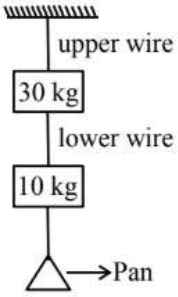
Option 2:
38

Option 3:
96

Option 4:
5.6

Correct Answer:
38

Solution:



$\mathrm{T_1 = mg}$
$\mathrm{T_2 = (10 + m)g}$
$\mathrm{T_3 = (30 + 10 + m)g = g(40 + m)g}$

Stress in upper wire

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$$= \frac{(40+\mathrm{m})\mathrm{g}}{0.008\ \mathrm{cm}^2}$$
$$= \frac{(40+\mathrm{m})\times 10}{8\times 10^{-7}}$$

Stress in lower wire $=$

$$= \frac{(10+\mathrm{m})\mathrm{g}}{0.004\ \mathrm{cm}^2}$$
$$= \frac{(10+\mathrm{m})\times 10}{4\times 10^{-7}}$$

Upper wire breakes :

$\frac{(40+m)\times 10}{8\times 10^{-7}} = 12\times 10^{8}$

So, maximum mass, without breaking any wire is 38 kg

Q. 19 An a.c. source of angular frequency ω is connected across a resistor R and a capacitor C in series. The current is observed as I . Now the frequency of the source is changed to $\omega/4$, (keeping the voltage unchanged) the current is found to be $I/3$. The ratio of resistance to reactance at frequency ω is

Option 1:
$\sqrt{\frac{6}{7}}$

Option 2:
$\sqrt{\frac{3}{5}}$

Option 3:
$\sqrt{\frac{7}{8}}$

Option 4:
$\sqrt{\frac{3}{4}}$

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Correct Answer:

$\sqrt{\frac{7}{8}}$

Solution:

As we know that current in RLC circuit is given by

$$I = \frac{V}{Z} = \frac{V}{\left[R^2 + (X_L - X_C)^2\right]^{1/2}}$$

$$I = \frac{V_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

As circuit is RC

$$I_0 = \frac{V_0}{\left(R^2 + X_C^2\right)^{1/2}}$$

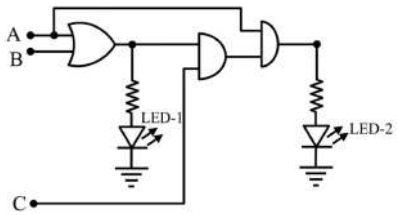
$$\frac{I_0}{3} = \frac{V_0}{\left[R^2 + (4X_C)^2\right]^{1/2}}$$

$$R^2 + (X_C)^2 = \frac{\left(R^2 + 16X_C^2\right)}{9}$$

$$\frac{8R^2}{9} = \frac{7X_C^2}{9}$$

$$\frac{R}{X_C} = \sqrt{\frac{7}{8}}$$

Q. 20 For the given logic circuit, which of the following inputs combination will make both LED-1 and LED-2 to glow ?



Option 1:

A = 0, B = 1, C = 1

Option 2:

A = 1, B = 0, C = 0

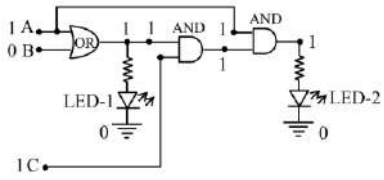
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Option 3:
A = 1, B = 0, C = 1

Option 4:
A = 1, B = 1, C = 0

Correct Answer:
A = 1, B = 0, C = 1

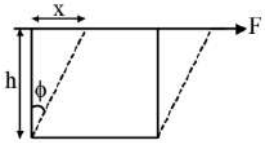
Solution:



Q. 21 A cube has side length 5 cm and modulus of rigidity $10^5\ \mathrm{N/m^2}$. The displacement produced by a force of 10 N in the upper face of cube is _____ mm.

Correct Answer:
2

Solution:



$$\eta = \frac{F/A}{\phi} = \frac{Fh}{Ax}$$

$$x = \frac{Fh}{A\eta} = \frac{10 \times 5 \times 10^{-2}}{25 \times 10^{-4} \times 10^5} = \frac{1}{500}\ \mathrm{m}$$

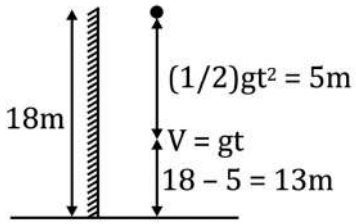
$$\therefore x = 2\ \mathrm{mm}$$

Q. 22 From 18 m height above the ground a ball is dropped from rest. The height above the ground at which the magnitude of velocity equal to the magnitude of acceleration (in the same set of units) due to gravity is _____ m.
(Take $g = 10\ \mathrm{m/s^2}$ and neglect the air resistance)

Correct Answer:
13

Solution:

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At any time t after it is released velocity $\mathrm{V}=\mathrm{gt}$
Acceleration of the ball remains constant $=g$
$\therefore$ Both will be equal at $\mathrm{t}=1\mathrm{sec}$
$\therefore$ Position of ball from ground will be $(18-5)\mathrm{m}$

Q. 23 A transverse wave on a string is described by $\mathrm{y}=3\sin(36\mathrm{t}+0.018\mathrm{x}+\pi/4)$. Where x, y are in cm and t in seconds. The least distance between the two successive crests in the wave is _____ cm. (Nearest integer) $(\pi=3.14)$

Correct Answer:
349

Solution:


Least distance between successive crest $=\lambda$

$\mathrm{y}=\mathrm{A}\sin(\omega\mathrm{t}+\mathrm{kx}+\phi), \omega=2\pi\mathrm{f} \& \mathrm{k}=\dfrac{2\pi}{\lambda}$
$\&\mathrm{y}=\mathrm{A}\sin(36\mathrm{t}+0.018\mathrm{x}+\pi/4)$
$\Rightarrow \mathrm{k}=0.018\ \mathrm{cm}^{-1} \quad \Rightarrow \dfrac{2\pi}{\lambda}=0.018$
$\lambda=\dfrac{2\pi}{0.018}=\dfrac{2\times 3.14}{18\times 10^{-3}}\ \mathrm{cm}=348.88\ \mathrm{cm}$
$\approx 349\ \mathrm{cm}\ (\text{nearest integer})$

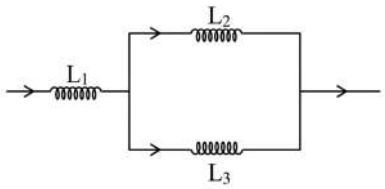
Q. 24 The charged particle moving in a uniform magnetic field of $(3\hat{i}+2\hat{j})\mathrm{T}$ has an acceleration $\left(4\hat{\mathrm{i}}-\frac{\mathrm{x}}{2}\hat{\mathrm{j}}\right)\mathrm{m/s}^2$. The value of x is

Correct Answer:
12

Solution:
$\overrightarrow{\mathrm{F}_\mathrm{B}} \propto (\overrightarrow{\mathrm{V}}\times\overrightarrow{\mathrm{B}}) \quad \Rightarrow \quad \overrightarrow{\mathrm{F}_\mathrm{B}} \perp \overrightarrow{\mathrm{B}}$
$\overrightarrow{\mathrm{a}} \perp \overrightarrow{\mathrm{B}} \rightarrow \overrightarrow{\mathrm{a}}\cdot\overrightarrow{\mathrm{B}}=0 \quad \Rightarrow$
$12-\mathrm{x}=0 \qquad x=12$

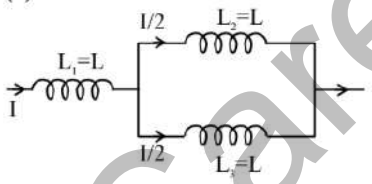
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Q. 25 In the given circuit below inductance values of L_1, L_2 and L_3 are same. The magnetic energy stored in the entire circuit is (U_1) and that stored in the L_2 inductor is (U_l). $(\mathrm{U}_l/\mathrm{U}_l)$ is _____ .
(Ignore the mutual inductance if any)



Correct Answer:
6

Solution:



Magnetic energy of inductor is $\mathrm{U} = \frac{1}{2}\mathrm{Li}^2$
So $\mathrm{U_t} = \frac{1}{2}\mathrm{LI}^2 + \frac{1}{2}\,\mathrm{L}\left(\frac{\mathrm{I}}{2}\right)^2 + \frac{1}{2}\,\mathrm{L}\left(\frac{\mathrm{I}}{2}\right)^2 = \frac{3}{4}\mathrm{LI}^2$

$\mathrm{U}_\ell = \frac{1}{2}\,\mathrm{L}\left(\frac{\mathrm{I}}{2}\right)^2 \Rightarrow \frac{\mathrm{U_t}}{\mathrm{U}_\ell} = 6$

Chemistry

Q. 1 How many grams of residue is obtained by heating 2.76 g of silver carbonate ?
(Given : Molar mass of C, O and Ag are 12, 16 and $108\ \mathrm{g\ mol^{-1}}$ respectively)

Option 1:
1.08 g

Option 2:
2.16 g

Option 3:
3.24 g

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Option 4:
4.32 g

Correct Answer:
2.16 g

Solution:

$$Ag_2CO_3(s) \xrightarrow{\Delta} 2Ag(s) + CO_2(g) + \frac{1}{2}O_2(g)$$

$2.76g$

$\frac{1}{100}$ mol

$\text{Moles of Ag(s)} = \frac{2}{100}$ mol

$\text{Mass of residue } = 2.16$ g

Q. 2 Arrange the following atomic orbitals of multi electron atoms in order of increasing energy.

(A) $n = 3, \ell = 2, m = +1$
(B) $n = 4, \ell = 0, m = 0$
(C) $n = 6, \ell = 1, m = 0$
(D) $n = 5, \ell = 1, m = +1$
(E) $n = 2, \ell = 1, m = +1$

Choose the correct answer from the options given below :

Option 1:
$C < D < B < A < E$

Option 2:
$B < A < E < C < D$

Option 3:
$E < C < D < B < A$

Option 4:
$E < B < A < D < C$

Correct Answer:
$E < B < A < D < C$

Solution:
(A) $n = 3, l = 2, m = +1; (n + l) = 5$
(B) $n = 4, l = 0, m = 0; (n + l) = 4$
(C) $n = 6, l = 1, m = 0; (n + l) = 7$
(D) $n = 5, l = 1, m = +1; (n + l) = 6$
(E) $n = 2, l = 1, m = +1; (n + l) = 3$

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More the value of ($\mathrm{n}+l$), more will be the energy of that particular orbital.
Ans. $\mathrm{E}<\mathrm{B}<\mathrm{A}<\mathrm{D}<\mathrm{C}$

Q. 3 Identify the correct statements from the following :
A. Heisenberg uncertainty principle is applicable to electron.
B. The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.
C. The energy of 2 s orbital of H atom is equal to the energy of 2s orbital of Li.
D. The electronic configuration of Cr is $[\mathrm{Ar}]3\ \mathrm{d}^5 4\ \mathrm{s}^1$

Choose the correct answer from the options given below:

Option 1:
A, B and C only

Option 2:
A, B and D only

Option 3:
B, C, and D only

Option 4:
A, C and D only

Correct Answer:
A, B and D only

Solution:
(A) Heisenberg uncertainty principle is applicable to all the moving microscopic particles.
(B) $2p_x$ has less value of n thus has lesser size than $3\mathrm{p_x}$.
(C) $\mathrm{E_{2\,s}(H)} \neq \mathrm{E_{2\,s}(Li)}$
(D) $\mathrm{Cr}: -[\mathrm{Ar}]3\ \mathrm{d}^5 4\ \mathrm{s}^1$

Correct statement are A, B and D.

Q. 4 What is the mole fraction of water in 10% by weight ($\mathrm{w/w}$) of aqueous urea solution ?
[Given : Molar mass of H, O, C and N are 1, 16, 12 and $14\ \mathrm{g\ mol^{-1}}$ respectively]

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Option 1:
0.825

Option 2:
0.032

Option 3:
0.867

Option 4:
0.967

Correct Answer:
0.967

Solution:

$$\text{Mass ratio } = \text{ mol ratio } \times \text{GMM ratio}$$

$$\frac{10}{90} = \frac{\mathrm{n}}{\mathrm{N}} \times \frac{60}{18}$$

$$\frac{\mathrm{n}}{\mathrm{N}} = \frac{1}{30}$$

$$\mathrm{x}_{\mathrm{H_2O}} = \frac{\mathrm{W}}{\mathrm{n}+\mathrm{N}} = \frac{30}{31} = 0.967$$

Q. 5 $\mathrm{M_3\ A_2}$ is a sparingly soluble salt of molar mass $\mathrm{ygmol^{-1}}$ and solubility $\times \mathrm{gL^{-1}}$. The ratio of the molar concentration of the anion ($\mathrm{A^{3-}}$) to the solubility product of the salt is

Option 1:
$\frac{1}{54} \cdot \frac{y^4}{x^4}$

Option 2:
$\frac{y^5}{108x^4}$

Option 3:
$108 \cdot \frac{x^5}{y^5}$

Option 4:
$\frac{1}{108} \frac{y^4}{x^4}$

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Correct Answer:
$\frac{1}{54} \cdot \frac{y^4}{x^4}$

Solution:

$$\mathrm{M_3\ A_2(\ s) \rightleftharpoons 3M^{+2}\ (aq.) + 2\ A^{-3}\ (aq.)}$$

Molarity

$\frac{x}{y}$	–	–
–	$3x/y$	$2x/y$

$K_{sp} = \left[M^{+2}\right]^3 \left[A^{-3}\right]^2$

$K_{sp} = [3x/y]^3 [2x/y]^2$

$$\text{Ratio of } \frac{\left[A^{-3}\right]}{K_{sp}} = \frac{2x/y}{\left[\frac{3x}{y}\right]^3 \left[\frac{2x}{y}\right]^2}$$

$$= \frac{1}{54} \frac{y^4}{x^4}$$

Q. 6 Arrange the following resultant mixtures in increasing order of their pH values

A. $10\ \mathrm{mL0.2MCa(OH)_2} + 25\ \mathrm{mL0.1MHCl}$

B. $10\ \mathrm{mL0.01MH_2SO_4} + 10\ \mathrm{mL0.01MCa(OH)_2}$

C. $10\ \mathrm{mL0.1MH_2SO_4} + 10\ \mathrm{mL0.1MKOH}$

Choose the correct answer from the options given below:

Option 1:
B < C < A

Option 2:
C < A < B

Option 3:
C < B < A

Option 4:
A < C < B

Correct Answer:
C < B < A

Solution:

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$$(\mathrm{A}) \quad \mathrm{H}^{+}_{(\mathrm{aq})} + \mathrm{OH}^{-}_{(\mathrm{aq})} \rightarrow \mathrm{H_2O}(\ell)$$

$\mathrm{t}=0 \quad 2.5\mathrm{mmol} \quad 4\mathrm{mmol}$
$\mathrm{t}=\infty \quad 0 \quad 1.5\mathrm{mmol}$

an final solution, $\left[\mathrm{OH}^{-}\right]=\dfrac{1.5\mathrm{mmol}}{35\mathrm{ml}}=\dfrac{1.5}{35}$

$\mathrm{pOH}=-\log \dfrac{1.5}{35}$

$\mathrm{pH}=14+\log\left(\dfrac{1.5}{35}\right)$

$$(\mathrm{B}) \quad \mathrm{H}^{+}_{(\mathrm{aq})} + \mathrm{OH}^{-}_{(\mathrm{aq})} \rightarrow \mathrm{H_2O}(\ell)$$

$\mathrm{t}=0 \quad 0.2\mathrm{mmol} \quad 0.2\mathrm{mmol}$ final $\mathrm{pH}=7$

$$(\mathrm{C}) \quad \mathrm{H}^{+}_{(\mathrm{aq})} + \mathrm{OH}^{-}_{(\mathrm{aq})} \rightarrow \mathrm{H_2O}(\ell)$$

$t=0 \quad 2\mathrm{mmol} \quad 1\mathrm{mmol}$
$t=\infty \quad 1\mathrm{mmol} \quad 0$

Final $\mathrm{pH}<7$
So, order of pH values :

$(\mathrm{C})<(\mathrm{B})<(\mathrm{A})$

Q. 7 First order gas phase reaction
$\mathrm{A} \rightarrow \mathrm{B}+\mathrm{C}$
p_i = initial pressure of gas A, p_t = total pressure of the reaction mixture at time t

Expression of rate constant (k) is

Option 1:
$\frac{1}{t}\ln\frac{p_i}{2p_i-p_t}$

Option 2:
$\frac{1}{t}\ln\frac{2p_i}{p_i-p_t}$

Option 3:
$\frac{1}{t}\ln\frac{p_i}{3p_i-2p_t}$

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Option 4:
$\frac{1}{t}\ln\frac{3p_i}{4p_i-p_t}$

Correct Answer:
$\frac{1}{t}\ln\frac{p_i}{2p_i-p_t}$

Solution:
$\mathrm{A(g)} \rightarrow \mathrm{B(g)} + \mathrm{C(g)}$

$t=0 \quad P_i \quad - \quad -$
$t=t \quad P_i - x \quad x \quad x$

As given $(P_i - x) + x + x = P_t$

$\mathrm{x} = (\mathrm{P_t} - \mathrm{P_i})$

$$\mathrm{K} = \frac{1}{\mathrm{t}}\ell\mathrm{n}\frac{\mathrm{P_i}}{\mathrm{P_i}-\mathrm{x}} = \frac{1}{\mathrm{t}}\ell\mathrm{n}\frac{\mathrm{P_i}}{\mathrm{P_i}-(\mathrm{P_t}-\mathrm{P_i})}$$

$$\mathrm{K} = \frac{1}{\mathrm{t}}\ell\mathrm{n}\frac{\mathrm{P_i}}{2\mathrm{P_i}-\mathrm{P_t}}$$

Q. 8

Statement I: The correct order of electronegativity of fluorine, oxygen nitrogen is $\mathrm{F} > \mathrm{O} > \mathrm{N}$.
Statement II : The oxidation state of oxygen in $\mathrm{OF_2}$ is +2 and in $\mathrm{Na_2O}$ is -2 .
In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

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Solution:
Statement-I

$$\left.\begin{array}{l}(\mathrm{EN})_{\mathrm{F}}=4.0 \\ (\mathrm{EN})_{\mathrm{O}}=3.5 \\ (\mathrm{EN})_{\mathrm{N}}=3.0\end{array}\right\} \text{ on pauling scale}$$

Statement-II
Oxidation state of oxygen in $\mathrm{OF_2}$

$$\mathrm{OF_2} \Rightarrow \mathrm{x}-1\times 2=0$$
$$\mathrm{x}=+2$$

Oxidation state of oxygen in $\mathrm{Na_2O}$

$$\mathrm{Na_2O} \Rightarrow +1\times 2+\mathrm{x}=0$$
$$\mathrm{x}=-2$$

Both the statement I and statement II are correct.

Q. 9 Correct statements from the following are :
A. Nitrogen in oxidation states from +1 to +4 disproportionates in acid medium.
B. Nitrogen has the ability of form $\mathrm{d}\pi-\mathrm{p}\pi$ multiple bonds with itself and other elements with small size and high electronegativity.
C. N-N single bond is stronger than P-P single bond.
D. Nitrogen has highest density in its group due to small size.
E. The maximum covalency of nitrogen is four since it has only four valence orbitals for bonding.
Choose the correct answer from the options given below :

Option 1:
B, C, and D only

Option 2:
C, D, and E only

Option 3:
A, C and E only

Option 4:
A and E only

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Correct Answer:
A and E only

Solution:
(A) In the case of nitrogen, all oxidation states from +1 to +4 tend to disproportionate in acid solution. For example.

$$3HNO_2 \rightarrow HNO_3 + H_2O + 2NO$$

(B) Nitrogen form $p_\pi - p_\pi$ bonds with itself and other elements with small size and high electronegativity, $d_\pi - p_\pi$ is not possible with itself as it has no d-orbital.
(C) The single $N - N$ bond is weaker than the single $P - P$ bond because of high intereelecronic repulsion of the non-bonding electrons (lone pair), owing to the small bond length.
(D) Nitrogen has lowest density in its group.
(E) The maximum covalency of nitrogen is four.

Correct answer are (A) and (E)

Q. 10 Which of the following is NOT a physical or chemical characteristics of interstitial compounds?

Option 1:
They have high melting points, higher than those of pure metals

Option 2:
They are very soft ionic in nature

Option 3:
They retain metallic conductivity

Option 4:
They are chemically inert and usually nonstoichiometric

Correct Answer:
They are very soft ionic in nature

Solution:
The physical and chemical characteristies of interstitial compounds are :
(1) They have high melting points, higher than those of pure metals.
(2) They are very hard, some borides approach diamond in hardness.
(3) They retain metallic conductivity
(4) They are chemically inert and usually nonstoichiometric.

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Ans. (2)

Q. 11 The correct statements about metal carbonyls are

A. The metal-carbon bonds in metals carbonyls possess both σ and π-character.

B. Due to synergic bonding interactions between metals and CO ligand the metal-carbon bond becomes weak.

C. The metal-carbon σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of metal.

D. The metal-carbon π bond is formed by the donation of electrons from filled d-orbital of metal into vacant π^* orbital of CO.

Choose the correct answer from the options given below:

Option 1:
A and B only

Option 2:
A, C and D only

Option 3:
B and C only

Option 4:
A and D only

Correct Answer:
A, C and D only

Solution:
(A) The metal-carbon bonds in metal carbonyls possess both σ and π-character.
(B) The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal i.e. metal carbon bond becomes strong.
(C) The metal-carbon σ is formed by the donation of lone pair of electrons on the carbonyl carbon into a vacant orbital of metal.
(D) The metal-carbon π bond is formed by the donation of electrons from filled d-orbial of metal into vacant π^* orbital of CO.
Correct answer are A, C, D

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Q. 12 Given below are two statements :

Statement I : Each electron in e_g orbitals destabilizes the orbitals by $+0.6\Delta_0$ and each electron in the t_{2g} orbitals stabilizes the orbitals by $-0.4\Delta_0$ in an octahedral field on the basis of crystal field theory.

Statement II : All the d-orbitals of the transition metals have the same energy in their free atomic state but when a complex is formed the ligands destroy the degeneracy of these orbitals on the basis of crystal field theory.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II are correct

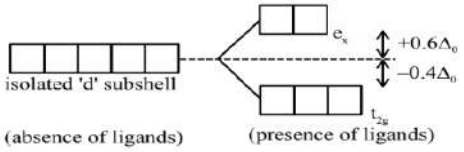
Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is correct but Statement II is incorrect

Option 4:
Statement I is incorrect but Statement II is correct

Correct Answer:
Both Statement I and Statement II are correct

Solution:
Statement-I



Statement-II
The degeneracy of the d-orbitals is removed due to ligand electron-metal electron repulsions according to crystal field theory.
Both the statement I and II are correct.

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Q. 13 Given below are two statements :
Statement I: On the basis of inductive effect, the order of stability of alkyl carbanions is $CH_3^- > CH_3 - CH_2^- > (CH_3)_2CH^- > (CH_3)_3C^-$.
Statement II : Allyl and benzyl carbanions are more stabilised by inductive effect and not by resonance effect.
In the light of the above statements choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are correct

Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is correct and Statement II is incorrect

Option 4:
Statement I is incorrect and Statement II is correct

Correct Answer:
Statement I is correct and Statement II is incorrect

Solution:
Statement-I

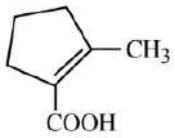
$\overset{\ominus}{C}H_3 > CH_3 - \overset{\ominus}{C}H_2 > (CH_3)_2\overset{\ominus}{C}H > (CH_3)_3\overset{\ominus}{C}$

It is correct order of anion stability and it is based on inductive effect.
Statement-II

Both are stabilised by resonance
Statement-I : Correct
Statement-II : Incorrect

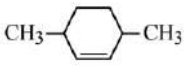
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Q. 14 "P" is hydrocarbon of molecular formula : C_8H_{14}. On ozonolysis, "P" forms "Q". "Q" on treatment with alkali under reflux condition produces. " R ", which on treatment with $I_2/NaOH$ gives a yellow precipitate. Acidification of the solution gives " S ". The structure of " S " given below :

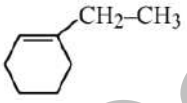


The correct structure of " P " is

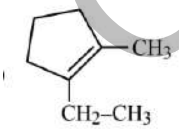
Option 1:



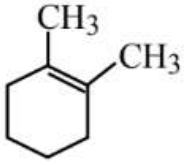
Option 2:



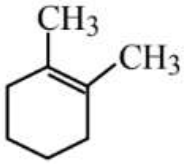
Option 3:



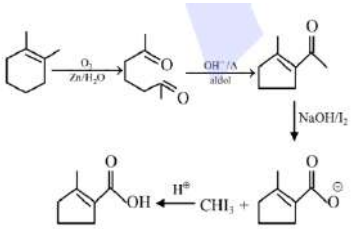
Option 4:



Correct Answer:

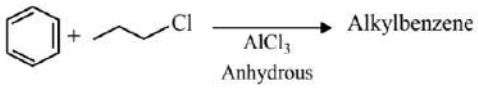


Solution:



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Q. 15 For the following Friedel Craft's alkylation reaction, which of the statements are correct?



A. Major product is n-propyl benzene.
B. iso-propyl carbocation intermediate is also generated
C. Multiple substitution is inevitable.
D. Introducing electron-donating substituent on benzene will not produce any alkylbenzene.
Choose the correct answer from the options given below :

Option 1:
A and D only

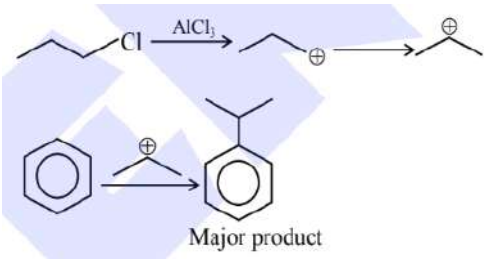
Option 2:
B and C only

Option 3:
A and C only

Option 4:
B and D only

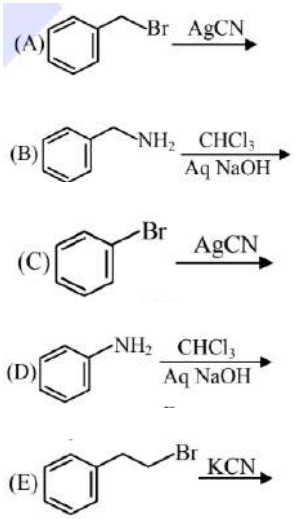
Correct Answer:
B and C only

Solution:



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Q. 16 Benzyl isocyanide can be obtained from



Choose the correct answer from the options given below :

Option 1:
A and B only

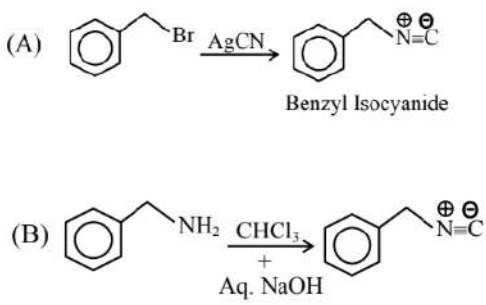
Option 2:
A and C only

Option 3:
B and D only

Option 4:
D and E only

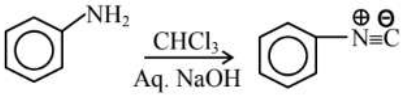
Correct Answer:
A and B only

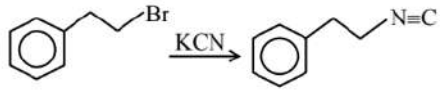
Solution:



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(C) 

(D) 

(E) 

(A) & (B) gives correct product.
Ans. (1) (A) & (B) only

Q. 17 Consider compounds A, B and C with following structural formulae

$$\mathrm{A} = \mathrm{CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - OH}$$
$$\mathrm{B} = \mathrm{CH_2 = CH - CH_2 - CH_2 - CH_3}$$
$$\mathrm{C} = \mathrm{HO - CH_2 - CH_2 - CH(OH) - CH_3}$$

For the conversion of B from A , reagent (D) required is _____ and structural formula of product (E) obtained when C undergoes same reaction using excess reagent (D) is _____

Option 1:

D	E
Conc. $\mathrm{H_2SO_4}$	$\mathrm{CH_2 = CH - CH(OH)/CH_3}$

Option 2:

D	E
PCC	$\mathrm{HO - CH_2 - CH_2 - CH = CH_2}$

Option 3:

D	E
PCC	$\mathrm{CH_2 = CH - CH = CH_2}$

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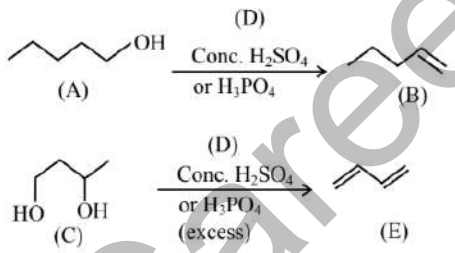
Option 4:

D	E
Conc. H_2SO_4 or H_3PO_4	$CH_2 = CH - CH = CH_2$

Correct Answer:

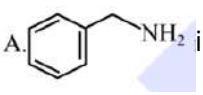
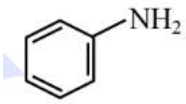
D	E
Conc. H_2SO_4 or H_3PO_4	$CH_2 = CH - CH = CH_2$

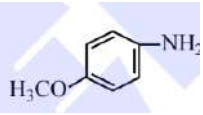
Solution:

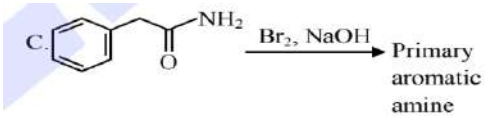


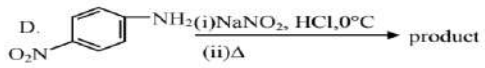
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Q. 18 Identify the incorrect statements :

A.  is a stronger base than 

B.  can be synthesized by Gabriel phthalimide synthesis.

C. 

D. 

will dissolve in NaOH

Choose the correct answer from the options given below :

Option 1:

A and D only

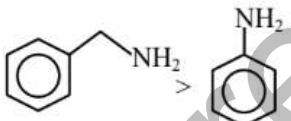
Option 2:
A and C only

Option 3:
B and C only

Option 4:
A and B only

Correct Answer:
B and C only

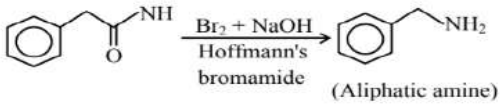
Solution:

(A) 

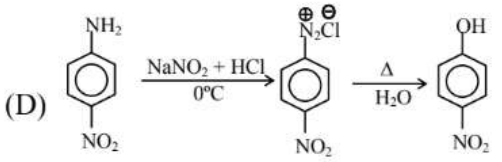
(Basic strength order)

(B) 

Cannot be synthesized by Gabriel pathalamide synthesis

(C) 

(Aliphatic amine)

(D) 

Soluble in NaOH

Only (B) and (C) are incorrect hence Ans. is (3)

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Q. 19 Identify the correct statements

A. Glucose exist in two anomeric forms.

B. Anomers of glucose differ in configuration at C-1 in cyclic hemiacetal structure.

C. Melting point of α-anomer of glucose is greater than β-anomer.

D. Specific rotation of α-anomer is $+19^{\circ}$ while for β-anomer is $+112^{\circ}$

E. α and β-anomers of glucose are prepared by crystallization of saturated glucose solution at 303 K and 371 K respectively.

Choose the correct answer from the options given below:

Option 1:
A and B only

Option 2:
B and C only

Option 3:
A, B and D only

Option 4:
A, B and E only

Correct Answer:
A, B and E only

Solution:
It is based on theory

Q. 20 **Statement I** : Sodium dichromate and potassium dichromate are classified as primary standard in titrimetric analysis .

Statement II : Phenolphthalein is a weak base, therefore it dissociates in acidic medium.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

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Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are false

Solution:
$Na_2Cr_2O_7$ is hygroscopic in nature and is not used as primary standard.

Phenolphthalein is weak acid.

Q. 21 Consider the following species

$BrF_5, XeF_5^-, BF_4^-, ICl_4^-, XeF_4, SF_4, NH_4^+, ClF_3, XeF_2, ICl_2^-$

Number of species having $sp^3\,d$ hybridized central atom is _____

Correct Answer:
4

Solution:

Species	Hybridisation
BrF_5	$sp^3\,d^2$
$XeF_5^{\ominus}$	$sp^3\,d^3$
$BF_4^{\ominus}$	sp^3
$ICl_4^{\ominus}$	$sp^3\,d^2$
XeF_4	$sp^3\,d^2$
SF_4	$sp^3\,d$
NH_4^+	sp^3
ClF_3	$sp^3\,d$
XeF_2	$sp^3\,d$
$ICl_2^{\ominus}$	$sp^3\,d$

Number of species having $sp^3\,d$ hybridisation are 4.

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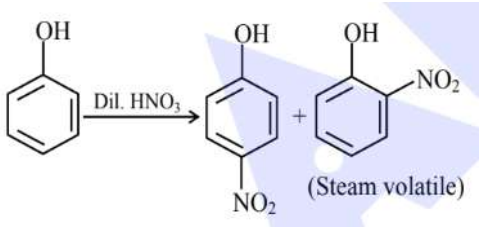
Q. 22 In an estimation of sulphur by Carius method 0.2 of the substance gave 0.6 g of $\mathrm{BaSO_4}$. Th percentage of sulphur in the substance is _____ \%.
(Given molar mass in $\mathrm{gmol^{-1}}$ $\mathrm{S:32, BaSO_4:231}$)

Correct Answer:
42

Solution:
Moles of $\mathrm{BaSO_4}$ formed $\xrightarrow[231\mathrm{gm}]{0.6\mathrm{gm}} = 0.0026$
Moles of $\mathrm{BaSO_4}$ = Moles of S
Mass of sulphur $(\mathrm{W_s}) \xrightarrow[231]{0.6} \times 32$
% of sulphur $(S) = \frac{0.6\times32}{231\times0.2} \times 100$
$= \frac{1920}{462} = 41.558 \approx 42$

Q. 23 One mole of phenol is treated with dilute $\mathrm{HNO_3}$ at 298 K to give a mixture of products. The mixture is separated by steam distillation. The seam volatile compound (X) is separated. The increase in percentage of oxygen in (X) with respect to phenol is _____ $10^{-1}\%$
(Given molar mass in $\mathrm{gmol^{-1}H:1, C:12,\ N:14}$, O:16)

Correct Answer:
175

Solution:


$\%$ oxygen in phenol $= \frac{16}{94} \times 100 = 17.02\%$
$\%$ oxygen in o-Nitrophenol $(\mathrm{C_6H_5NO_3})$
Molecular mass $= \mathrm{C_6H_5NO_3} = 139\ \mathrm{g/m}$
$= \frac{48}{139} \times 100 = 34.53\%$
$\%$ increase $= (34.53 - 17.02) = 17.5$
$17.5 \times 10^{-1} = 175$ Answer

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Q. 24 Match the List-I with List-II

List-I Complex ion	List-II Calculated spin only magnetic moment (BM)
A. $[Cr(H_2O)_6]^{2+}$	I. 3.87
B. $[Co(H_2O)_6]^{2+}$	I. 5.95
C.$[Cu(H_2O)_6]^{2+}$	III. 4.90
D.$[Mn(H_2O)_6]^{2+}$	IV. 1.73

Choose the correct answer from the options given below:

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Option 1:
A-I, B-III, C-IV, D-II

Option 2:
A-II, B-I, C-III, D-IV

Option 3:
A-IV, B-II, C-I, D-III

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
$[Cr(H_2O)_6]^{2+} \rightarrow 4$ unpaired $e^-; 4.90BM(III)$
$[Co(H_2O)_6]^{2+} \rightarrow 0$ unpaired $e^-; 3.87BM(I)$
$[Cu(H_2O)_6]^{2+} \rightarrow 1$ unpaired $e^-; 1.73BM(IV)$
$[Mn(H_2O)_6]^{2+} \rightarrow 5$ unpaired $e^-; 5.92BM(II)$

Q. 25 If the half life of a first order reaction is 6.93 minutes then the time required for completion of 99% of the reaction will be _____ minutes.
(Given : $\log 2 = 0.3010$)

Correct Answer:
46

Solution:
$t_{1/2} = \frac{\ell n 2}{K} \Rightarrow K_1 = \frac{0.693}{6.93} \Rightarrow 0.1\ \text{min}^{-1}$
$t_{99\%} = \frac{1}{K} \ell n \left[\frac{A_0}{A_t}\right]$
$= \frac{1}{0.1} \ln \left[\frac{100}{1}\right] \Rightarrow \frac{2\ell n 10}{0.1}$
$= \frac{2 \times 2.303}{0.1}$
$\Rightarrow 46.06\ \text{min} \approx 46$

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Maths

Q. 1 Let $a, b \in C$. Let α, β be the roots of the equation $x^2 + ax + b = 0$. If $\beta - \alpha = \sqrt{11}$ and $\beta^2 - \alpha^2 = 3i\sqrt{11}$, then $\left(\beta^3 - \alpha^3\right)^2$ is equal to :

Option 1:
160

Option 2:
176

Option 3:
194

Option 4:
187

Correct Answer:
176

Solution:
$\beta - \alpha = \sqrt{11} \ldots . (1)$

and $\beta^2-\alpha^2=3\mathrm{i}\sqrt{11}$

$\Rightarrow(\beta+\alpha)(\beta-\alpha)=3\mathrm{i}\sqrt{11}$
$\Rightarrow\beta+\alpha=3\mathrm{i}\ldots.($

from (1) and (2)

$\beta=\dfrac{\sqrt{11}+3\mathrm{i}}{2},\alpha=\dfrac{3\mathrm{i}-\sqrt{11}}{2}$
$\Rightarrow\alpha\beta=\dfrac{-9-11}{4}\Rightarrow\alpha\beta=-5$

$\text{Now }\left(\beta^3-\alpha^3\right)^2=(\beta-\alpha)^2\left(\beta^2+\alpha^2+\alpha\beta\right)^2$

$=(\beta-\alpha)^2\left((\beta+\alpha)^2-\alpha\beta\right)^2$
$=(11)(-9+5)^2=176$

Q. 2 Let the sum of the first n terms of an A.P. be $3n^2+5n$. Then the sum of squares of the first 10 terms of the A.P. is :

Option 1:
10220

Option 2:
12860

Option 3:
15220

Option 4:
19780

Correct Answer:
15220

Solution:
$S_n=3n^2+5n$
$T_n=S_n-S_{n-1}$
$\Rightarrow T_n=\left(3n^2+5n\right)-\left(3(n-1)^2\right)+5(n-1))$
$\Rightarrow T_n=6n+2$
$\sum_{n=1}^{10}T_n^2=\sum_{n=1}^{10}(6n+2)^2$
$=36\sum_{n=1}^{10}n^2+24\sum_{n=1}^{10}n+\sum_{n=1}^{10}4$
$=15220$

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Q. 3 Let A be a 3×3 matrix such that

$$A^T \begin{bmatrix}1\\0\\1\end{bmatrix} = \begin{bmatrix}5\\2\\2\end{bmatrix}, \quad A^T \begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}3\\1\\1\end{bmatrix}, \quad A \begin{bmatrix}1\\0\\1\end{bmatrix} = \begin{bmatrix}3\\4\\4\end{bmatrix} \quad \text{and } A \begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}1\\3\\1\end{bmatrix}$$

If $\det(A) = 1$, then $\det\left(\text{adj}\left(A^2 + A\right)\right)$ is equal to :

Option 1:
16

Option 2:
25

Option 3:
49

Option 4:
64

Correct Answer:
64

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Solution:

$$\text{Let } A = \begin{pmatrix}\alpha_1 & \alpha_2 & \alpha_3\\ \beta_1 & \beta_2 & \beta_3\\ \gamma_1 & \gamma_2 & \gamma_3\end{pmatrix}$$

$$\text{Now } A = \begin{pmatrix}\alpha_1 & \beta_1 & \gamma_1\\ \alpha_2 & \beta_2 & \gamma_2\\ \alpha_3 & \beta_3 & \gamma_3\end{pmatrix} \begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}3\\1\\1\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}\gamma_1\\ \gamma_2\\ \gamma_3\end{pmatrix} = \begin{pmatrix}3\\1\\1\end{pmatrix}$$

$$\text{Now } \begin{pmatrix}\alpha_1 & \alpha_2 & \alpha_3\\ \beta_1 & \beta_2 & \beta_3\\ \gamma_1 & \gamma_2 & \gamma_3\end{pmatrix} \begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}1\\3\\1\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}\alpha_3\\ \beta_3\\ \gamma_3\end{pmatrix} = \begin{pmatrix}1\\3\\1\end{pmatrix}$$

$$A^T \begin{pmatrix}1\\0\\1\end{pmatrix} - A^T \begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}5\\2\\2\end{pmatrix} - \begin{pmatrix}3\\1\\1\end{pmatrix}$$

$$\Rightarrow A^T \begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}2\\1\\1\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\end{pmatrix} = \begin{pmatrix}2\\ 1\\ 1\end{pmatrix}$$

$$\text{Now A(l} \begin{pmatrix}1\\ 0\\ 1\end{pmatrix} - \text{A} \begin{pmatrix}0\\ 0\\ 1\end{pmatrix} = \begin{pmatrix}3\\ 4\\ 4\end{pmatrix} - \begin{pmatrix}1\\ 3\\ 1\end{pmatrix}$$

$$\Rightarrow \text{A} \begin{pmatrix}1\\ 0\\ 0\end{pmatrix} = \begin{pmatrix}2\\ 1\\ 3\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}\alpha_1\\ \beta_1\\ \gamma_1\end{pmatrix} = \begin{pmatrix}2\\ 1\\ 3\end{pmatrix}$$

$$\text{A} = \begin{pmatrix}2 & 1 & 1\\ 1 & \beta_2 & 3\\ 3 & 1 & 1\end{pmatrix}$$

$$|\text{A}| = 2\left(\beta_2 - 3\right) - 1(1-9) + 1\left(1 - 3\beta_2\right) = 1$$

$$\Rightarrow 2\beta_2 - 6 + 8 + 1 - 3\beta_2 = 1$$

$$\Rightarrow \beta_2 = 2$$

$$\text{So A} = \begin{pmatrix}2 & 1 & 1\\ 1 & 2 & 3\\ 3 & 1 & 1\end{pmatrix}$$

$$\text{A}^2 = \begin{pmatrix}2 & 1 & 1\\ 1 & 2 & 3\\ 3 & 1 & 1\end{pmatrix}\begin{pmatrix}2 & 1 & 1\\ 1 & 2 & 3\\ 3 & 1 & 1\end{pmatrix} = \begin{pmatrix}8 & 5 & 6\\ 13 & 8 & 10\\ 10 & 6 & 7\end{pmatrix}$$

$$\text{A}^2 + \text{A} = \begin{pmatrix}10 & 6 & 7\\ 14 & 10 & 13\\ 13 & 7 & 8\end{pmatrix} \Rightarrow \det\left(\text{A}^2 + \text{A}\right) = 8$$

$$\Rightarrow \text{Det}\left(\text{adj}\left(\text{A}^2 + \text{A}\right)\right) = 8^2 = 64$$

Q. 4 Consider the system of linear equations in x, y, z :

$$x + 2y + tz = 0$$
$$6x + y + 5tz = 0$$
$$3x + t^2y + f(t)z = 0$$

Where $f : \text{R} \to \text{R}$ is a differentiable function. If this system has infinitely many solutions for all $t \in R$, then f

Option 1:
is a constant function

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Option 2:
is strictly increasing on R

Option 3:
is strictly decreasing on R

Option 4:
has two critical points

Correct Answer:
is strictly increasing on R

Solution:

$$D=\begin{vmatrix}1 & 2 & t\\ 6 & 1 & 5t\\ 3 & t^2 & f(t)\end{vmatrix}=0$$

$$\Rightarrow 1\left(f(t)-5t^3\right)-2(6f(t)-15t)+t\left(6t^2-3\right)=0$$

$$f(t)=\frac{t^3+12t}{11}$$

$$f'(t)=\frac{1}{11}\left(3t^2+12\right)>0 \quad \forall t\in R$$

$\Rightarrow \mathrm{f(t)}$ is strictly increasing on R

Q. 5 $\sum_{n=1}^{10}\left(\frac{528}{n(n+1)(n+2)}\right)$ is equal to :

Option 1:
65

Option 2:
130

Option 3:
220

Option 4:
440

Correct Answer:
130

Solution:

Let $\sum_{n=1}^{10}\frac{528}{n(n+1)(n+2)}=\lambda$

Let $\mathrm{T_n}=\frac{1}{\mathrm{n(n+1)(n+2)}}$

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$$T_n = \frac{1}{2}\left\{\frac{(n+2)-n}{n(n+1)(n+2)}\right\}$$

$$T_n = \frac{1}{2}\left\{\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right\}$$

$$T_1 = \frac{1}{2}\left\{\frac{1}{1.2} - \frac{1}{2.3}\right\}$$

$$T_2 = \frac{1}{2}\left\{\frac{1}{2.3} - \frac{1}{3.4}\right\}$$

$$\vdots$$

$$T_{10} = \frac{1}{2}\left\{\frac{1}{10.11} - \frac{1}{11.12}\right\}$$

$$S_n = \frac{1}{2}\left[\frac{1}{2} - \frac{1}{132}\right]$$

$$S_n = \frac{1}{2}\left[\frac{66-1}{132}\right] = \frac{65}{264}$$

So, $\lambda = \frac{65}{264} \times 528$

$\lambda = 130$

Q. 6 Let $\tan A, \tan B$, where $A, B \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$.
Then $20\sin^2\left(\frac{A+B}{2}\right)$ is equal to :

Option 1:
$10 + \sqrt{10}$

Option 2:
$10 - 2\sqrt{10}$

Option 3:
$10 - 3\sqrt{10}$

Option 4:
$10 - \sqrt{10}$

Correct Answer:
$10 - 3\sqrt{10}$

Solution:

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$x^2 - 2x - 5 = 0$

$\tan A + \tan B = 2; \tan A \tan B = -5$

$\therefore \tan(A + B) = \dfrac{2}{1 - (-5)} = \dfrac{1}{3}$

$\Rightarrow \cos(A + B) = \dfrac{3}{\sqrt{10}}$

$\therefore 20\left(\sin^2\left(\dfrac{A + B}{2}\right)\right) = \dfrac{10}{2}(1 - \cos(A + B))$

$= 10\left(1 - \dfrac{3}{\sqrt{10}}\right) = (10 - 3\sqrt{10})$

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Q. 7 A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the the envelope just two consecutive letters AN are visible. The probability, that the letter came from ANANTPUR, is :

Option 1:
$\frac{7}{10}$

Option 2:
$\frac{10}{17}$

Option 3:
$\frac{12}{19}$

Option 4:
$\frac{7}{19}$

Correct Answer:
$\frac{10}{17}$

Solution:

$P(I) : \dfrac{1}{2} P(A/I) \rightarrow \dfrac{1}{5}$

$P(II) : \dfrac{1}{2} P(A/II) \rightarrow \dfrac{2}{7}$

$\therefore \text{ Reqd. Prob } = \dfrac{\frac{1}{2} \times \frac{2}{7}}{\frac{1}{2} \times \frac{1}{5} + \frac{1}{2} \times \frac{2}{7}} = \dfrac{\frac{2}{7} \times 5}{\frac{17}{7}} = \dfrac{10}{17}$

Q. 8 The mean deviation about the mean for the data

x_i	5	7	9	10	12	15
f_i	8	6	2	2	2	6

is equal to :

Option 1:
$\frac{40}{13}$

Option 2:
$\frac{42}{13}$

Option 3:
$\frac{44}{13}$

Option 4:
$\frac{46}{13}$

Correct Answer:
$\frac{44}{13}$

Solution:

$$\text{M.D.} = \frac{\sum f_i |x_i - \bar{x}|}{\sum f_i}$$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{40 + 42 + 18 + 20 + 24 + 90}{26}$$

$$= \frac{234}{26} = 9$$

$$\text{M.D.} = \frac{8(4) + 6(2) + 2(0) + 2(1) + 2(3) + 6(6)}{26}$$

$$= \frac{84}{26} = \frac{44}{13}$$

Q. 9 Let a focus of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be $S(4, 0)$ and its eccentricity be $\frac{4}{5}$. If the point $\mathrm{P}(3, \alpha)$ lies on E and O is the origin, then the area of $\triangle \mathrm{POS}$ is equal to :

Option 1:
$12/5$

Option 2:
$14/5$

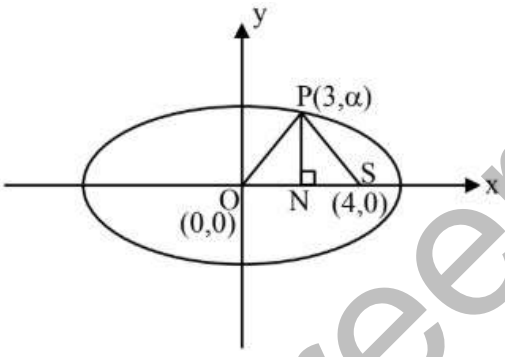
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Option 3:
$24/5$

Option 4:
$48/5$

Correct Answer:
$24/5$

Solution:



$a > b$

$\text{Focus } (ae, 0) = (4, 0)$

$\therefore ae = 4$

$a\left(\frac{4}{5}\right) = 4 \Rightarrow a = 5$

$\because b^2 = a^2\left(1 - e^2\right)$

$b^2 = 25\left(1 - \frac{16}{25}\right)$

$\therefore E : \frac{x^2}{25} + \frac{y^2}{9} = 1$

$\because$ P lies on the ellipse E

$\therefore \frac{9}{25} + \frac{\alpha^2}{9} = 1$

$\therefore \alpha = \pm\frac{12}{5}$

$\therefore \text{ Area of } \Delta_{\text{POS}} = \frac{1}{2}(\text{OS})(\text{PN})$

$= \frac{1}{2} \times 4 \times \frac{12}{5} = \frac{24}{5}$

Q. 10 Let P be a moving point on the circle

$x^2 + y^2 - 6x - 8y + 21 = 0$

Then, the maximum distance of P from the vertex of the parabola $x^2 + 6x + y + 13 = 0$ is equal to :

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Option 1:
8

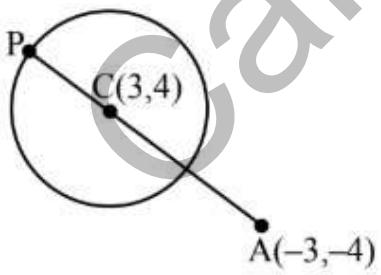
Option 2:
10

Option 3:
12

Option 4:
9

Correct Answer:
12

Solution:
Centre : $C(3,4) \& r=2$
Parabola $(x+3)^2=-(y+4)$
Vertex is $\mathrm{A}(-3,-4)$



$\mathrm{AP}_{\max }=\mathrm{AC}+\mathrm{r}=\sqrt{36+64}+2=12$

Q. 11 In an equilateral triangle PQR , let the vertex P be at $(3,5)$ and the side QR be along the line $x+y=4$. If the orthocentre of the triangle PQR is (α, β), then $9(\alpha+\beta)$ is equal to :

Option 1:
16

Option 2:
27

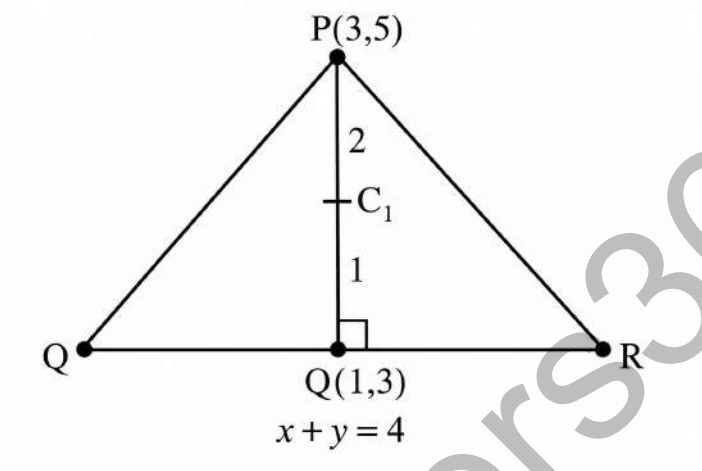
Option 3:
36

Option 4:
48

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Correct Answer:
48

Solution:



$Q: \frac{x-3}{1} = \frac{y-5}{1} = -\frac{(3+5-4)}{1+1}$

$Q(1,3)$

So orthocentre is $\left(\frac{2+3}{3}, \frac{6+5}{3}\right)$

$\left(\frac{5}{3}, \frac{11}{3}\right)$

So $9(\alpha+\beta) = 3(16) = 48$

Q. 12 The sum of all the integral values of p such that the equation $3\sin^2 x + 12\cos x - 3 = p, x \in R$, has at least one solution, is :

Option 1:
-54

Option 2:
-60

Option 3:
-75

Option 4:
-84

Correct Answer:
-75

Solution:

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$P = 12\cos x - 3\cos^2 x$
$P = -3\left(\cos^2 x - 4\cos x\right)$
$P = -3\left((\cos x - 2)^2 - 2\right)$
put $\cos x = -1 \Rightarrow P = -15$
put $\cos x = 1 \Rightarrow P = 9$
$-15 \le P \le 9$
Sum of all integers $= -(10 + 11 + \ldots + 15) = -75$

Q. 13 The square of the distance of the point $\mathrm{P}(5, 6, 7)$ from the line $\frac{x-2}{2} = \frac{y-5}{3} = \frac{z-2}{4}$ is equal to :

Option 1:
3

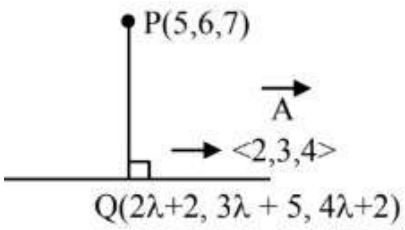
Option 2:
5

Option 3:
6

Option 4:
8

Correct Answer:
6

Solution:
$\overrightarrow{\mathrm{PQ}} \cdot \overrightarrow{\mathrm{A}} = 0$
$\Rightarrow (2\lambda - 3)2 + (3\lambda - 1)3 + (4\lambda - 5)4 = 0$
$\Rightarrow \lambda = 1$



Point $Q = (4, 8, 6)$
$PQ^2 = 6$

Q. 14 Let $\vec{a} = \sqrt{7}\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + 2\hat{k}$. If $\vec{r}$ is a vector such that $\overrightarrow{\mathrm{r}} \times \overrightarrow{\mathrm{a}} + \overrightarrow{\mathrm{a}} \times \overrightarrow{\mathrm{b}} = \overrightarrow{0}$ and $\overrightarrow{\mathrm{r}} \cdot \overrightarrow{\mathrm{a}} = 0$, then $|\overrightarrow{3\mathrm{r}}|^2$ is equal to :

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Option 1:
44

Option 2:
54

Option 3:
86

Option 4:
132

Correct Answer:
44

Solution:

$\vec{r} \times \vec{a} - \vec{b} \times \vec{a} = \overrightarrow{0}$

$(\vec{r} - \vec{b}) \times \vec{a} = \overrightarrow{0}$

$\vec{r} - \vec{b} = \lambda\vec{a}$

$\vec{r} = \vec{b} + \lambda\vec{a}$

$\vec{r}.a = 0 \Rightarrow \vec{a}\vec{b} + \lambda|\vec{a}|^2 = 0$

$\lambda = -\frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} = -\frac{(1-2)}{9} = \frac{1}{9}$

$\vec{r} = \vec{b} + \frac{\vec{a}}{9}$

$|3\vec{r}|^2 = 9|\vec{r}|^2 = 9\left(b^2 + \frac{a^2}{81} + \frac{2(\vec{a} \cdot \vec{b})}{9}\right) = 44$

Q. 15 The square of the distance of the point of intersection of the line $\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(a\hat{i} - \hat{j}), a \neq 0$ and $\vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + a\hat{k})$ from the origin is :

Option 1:
5

Option 2:
10

Option 3:
17

Option 4:
26

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Correct Answer:
17

Solution:
Equating both the equations

$(\hat{i}+\hat{j}-\hat{k})+\lambda(a\hat{i}-\hat{j})=(4\hat{i}-\hat{k})+\mu(2\hat{i}+a\hat{k})$
$1+a\lambda=4+2\mu$
$1-\lambda=0\Rightarrow\lambda=1$
$-1=-1+a\mu$
$a\mu=0\Rightarrow\mu=0$
$a=3$
point of intersection $=\hat{i}+\hat{j}-\hat{k}+3\hat{i}-\hat{j}=4\hat{i}-\hat{k}$
Ans. 17

Q. 16 The area of the region $R=\left\{(x,y):xy\leq 27,1\leq y\leq x^2\right\}$ is equal to :

Option 1:
$78\log_e 3-\frac{52}{3}$

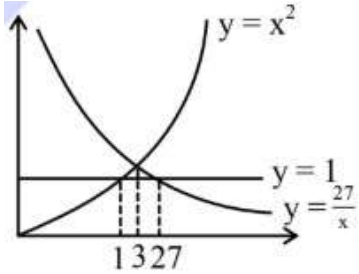
Option 2:
$54\log_e 3-\frac{52}{3}$

Option 3:
$54\log_e 3-\frac{26}{3}$

Option 4:
$54\log_e 3+\frac{26}{3}$

Correct Answer:
$54\log_e 3-\frac{52}{3}$

Solution:



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$$\text{Area } = \int_1^3 (x^2-1)dx + \int_3^{27}\left(\frac{27}{x}-1\right)dx$$
$$= \left[\frac{x^3}{3}-x\right]_1^3 + [27\ell nx - x]_3^{27}$$
$$= \frac{26}{3} - 2 + 27\ln 9 - 24$$
$$= 27\ln 9 - \frac{26\times 2}{3}$$
$$= 54\ln 3 - \frac{52}{3}$$

Q. 17 The product of all possible values of α, for which

$\lim_{x\to 0}\left(\frac{1-\cos(\alpha x)\cos((\alpha+1)x)\cos((\alpha+2)x)}{\sin^2((\alpha+1)x)}\right) = 2,$

is :

Option 1:
-2

Option 2:
1

Option 3:
-1

Option 4:
$\frac{5}{4}$

Correct Answer:
-1

Solution:
$\lim_{x\to 0}\frac{1-\cos\alpha x\cdot\cos((\alpha+1)x)\cos((\alpha+2)x)}{(\alpha+1)^2x^2}$

By using L.H. Rule we get

$$\Rightarrow \frac{1}{2}(\alpha+1)^2\left[\alpha^2+(\alpha+1)^2+(\alpha+2)^2\right] = 2$$
$$\alpha^2+(\alpha+1)^2+(\alpha+2)^2 = 4(\alpha+1)^2$$
$$\alpha^2+(\alpha+2)^2 = 3(\alpha+1)^2$$
$$\Rightarrow \alpha^2+2\alpha-1=0$$
$$\text{Product } = -1$$

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Q. 18 The value of the integral $\int_0^{\infty} \frac{\log_e(x)}{x^2+4}dx$ is:

Option 1:
$\frac{\pi \log_e(2)}{2}$

Option 2:
$\frac{\pi \log_e(2)}{4}$

Option 3:
$1+\pi \log_e(2)$

Option 4:
$2+\pi \log_e(2)$

Correct Answer:
$\frac{\pi \log_e(2)}{4}$

Solution:
Put $x=2t \Rightarrow dx=2dt$

$$I=\int_0^{\infty} \frac{\ell n2t}{4t^2+4}(2dt)=\frac{1}{2}\int_0^{\infty} \frac{\ell n2+\ell nt}{t^2+1}dt$$

$$=\frac{1}{2}\int_0^{\infty} \frac{\ell n2}{t^2+1}dt+\frac{1}{2}\int_0^{\infty} \frac{\ell nt}{t^2+1}dt$$

$$\left[\frac{\ln 2}{2}\tan^{-1}t\right]_0^{\infty}+I_1$$

$$=\frac{\ell n2}{2}\cdot\frac{\pi}{2}+I_1$$

$$I_2=\frac{1}{2}\int_0^{\infty} \frac{\ell nt}{t^2+1}dt \quad t=\frac{1}{u}\Rightarrow dt=-\frac{du}{u^2}$$

$$=\frac{1}{2}\int_{\infty}^{0} \frac{\ell n(1/u)}{\frac{1}{u^2}+1}\left(-\frac{du}{u^2}\right)$$

Add $\Rightarrow I_1=0$

$I=\frac{\pi \ell n2}{4}$

Q. 19 Let $f:R\to R$ be a differentiable function such that $f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)}{3}$ for all $x,y\in R$ and $f'(0)=3$. Then the minimum value of the function $g(x)=3+e^x f(x)$, is

Option 1:
$3\left(\frac{e+1}{e}\right)$

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Option 2:
$3\left(\frac{e-1}{e}\right)$

Option 3:
$\frac{3-e}{e}$

Option 4:
3 e

Correct Answer:
$3\left(\frac{e-1}{e}\right)$

Solution:
$f\left(\frac{x+y}{3}\right)=\frac{f(x)+f(y)}{3},\ \text{put } x=y=0$

$f(0)=\frac{2f(0)}{3}$

$\Rightarrow f(0)=0\ldots\ldots(1)$

$f'\left(\frac{x+y}{3}\right)\cdot\frac{1}{3}=\frac{1}{3}f'(x)$

$\text{put } x=0$

$f'\left(\frac{y}{3}\right)\frac{1}{3}=\frac{1}{3}\times 3$

$f'\left(\frac{y}{3}\right)=3$

$\text{put } y=3x$

$f'(x)=3$

$\text{integrate both sides}$

$f(x)=3x+c$

from eq. (1)

$f(0)=0$

$\Rightarrow f(x)=3x$

Now

$g(x)=3+e^{x}\cdot 3x$

$g'(x)=3\left[e^{x}+x\cdot e^{x}\right]$

$=3e^{x}(x+1)$

$g'(x)=0,\ \text{at } x=-1$

$(g(x))_{\min}=3+e^{-1}(-3)$

$=3\left[1-\frac{1}{e}\right]=\frac{3(e-1)}{e}$

Q. 20 The value of the integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\left(\frac{4-\operatorname{cosec}^2 x}{\cos^4 x}\right)dx$ is :

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Option 1:
$\frac{11}{\sqrt{3}}$

Option 2:
$\frac{16}{\sqrt{3}}$

Option 3:
$\frac{32}{3\sqrt{3}}$

Option 4:
$\frac{64}{3\sqrt{3}}$

Correct Answer:
$\frac{32}{3\sqrt{3}}$

Solution:

$$= \int_{\pi/6}^{\pi/3} \frac{4}{\cos^4 x} dx - \int_{\pi/6}^{\pi/3} \frac{\operatorname{cosec}^2 x}{\cos^4 x} dx$$

$$= \int_{\pi/6}^{\pi/3} \frac{4}{\cos^4 x} dx - \left[-\frac{\cot x}{\cos^4 x} - \int_{\pi/6}^{\pi/3} (-\cot x) \frac{-4}{\cos^5 x} (-\sin x) dx \right]$$

$$= \frac{\cot x}{\cos^4 x} \Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= \frac{1/\sqrt{3}}{(1/2)^4} - \frac{\sqrt{3}}{\left(\frac{\sqrt{3}}{2}\right)^4}$$

$$= \frac{16}{\sqrt{3}} - \frac{\sqrt{3} \cdot 16}{9} = \frac{16}{\sqrt{3}} - \frac{16}{3\sqrt{3}} = \frac{16}{\sqrt{3}} \left(1 - \frac{1}{3}\right)$$

$$\frac{32}{3\sqrt{3}} = \frac{32\sqrt{3}}{9}$$

Q. 21 Let $A = \{1, 2, 3, 4, 5, 6\}$. The number of one-one functions $\mathrm{f} : \mathrm{A} \to \mathrm{A}$ such that $\mathrm{f}(1) \geq 3, \mathrm{f}(3) \leq 4$ and $f(2) + f(3) = 5$, is ____ .

Correct Answer:
72

Solution:
$f(2) + f(3) = 5$
C-I: $f(2) = 4, f(3) = 1$
so, $f(1) = 3, 5, 6 \to 3$ choices
$\mathrm{f}(4) \to 3$ choices
$\mathrm{f}(5) \to 2$ choices

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$f(6) \to 1$ choices
so total function for C-I : $3 \times 3 \times 2 \times 1 = 18$
Similarly
C-II : $\mathrm{f}(2) = 1, \mathrm{f}(3) = 4 \to 18$ function
C-III : $\mathrm{f}(2) = 3, \mathrm{f}(3) = 2 \to 18$ function
C-IV : $\mathrm{f}(2) = 2, \mathrm{f}(3) = 2 \to 18$ function

Answer total 72 function

Q. 22 Two players A and B play a series of games of badminton. The player, who wins 5 games first, wins the series. Assuming that no game ends in a draw, the number of ways, in which player A wins the series is _____ .

Correct Answer:
126

Solution:
Maximum number of games will be 9
A must win 5 gamese so,
Number of ways $= {}^{9}\mathrm{C}_{5} = 126$
Answer $= 126$

Q. 23 If the sum of the coefficients of x^7 and x^{14} in the expansion of $\left(\frac{1}{x^3} - x^4\right)^n, x \neq 0$, is zero, then the value of n is _____ .

Correct Answer:
21

Solution:
$T_{r+1} = {}^{n}C_r \left(\frac{1}{x^3}\right)^{n-r} \left(-x^4\right)^r$

$T_{r+1} = {}^{n}C_r x^{7r-3n}(-1)^r$

for coeff. of $x^7 \Rightarrow 7r_1 - 3n = 7$

$\Rightarrow \mathrm{r}_1 = \frac{7+3\mathrm{n}}{7}$

coeff. of $x^7 \Rightarrow {}^{n}C_{r_1}(-1)^{r_1}$
for coeff. of $x^{14} \Rightarrow 7r_2 - 3n = 14$

$\Rightarrow r_2 = \frac{14+3n}{7} = r_1 + 1$

coeff. of $x^{14} \Rightarrow {}^{n}C_{r_1+1}(-1)^{r_1+1}$

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$\Rightarrow {}^{n}C_{r_1}(-1)^{r_1} + {}^{n}C_{r_1+1}(-1)^{r_1+1} = 0$
$\Rightarrow {}^{n}C_{r_1} = {}^{n}C_{r_1+1} \Rightarrow r_1 + r_1 + 1 = n$
$\Rightarrow 2r_1 + 1 = n \Rightarrow 2\left(\frac{7+3n}{7}\right) + 1 = n$

Ans. $\Rightarrow \mathrm{n} = 21$

Q. 24 If $\frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1}\left(\frac{2^{p-1}}{1+2^{2p-1}}\right) = \alpha$, then $\tan\alpha$ is equal to _____ .

Correct Answer:
2048

Solution:
$$\alpha = \frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1}\left(\frac{2^{p-1}}{1+2^{2p-1}}\right)$$
$$= \frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1}\left(\frac{2^{p}-2^{p-1}}{1+2^{p}2^{p-1}}\right)$$
$$= \frac{\pi}{4} + \sum_{p=1}^{11} \left(\tan^{-1}(2^{p}) - \tan^{-1}\left(2^{p-1}\right)\right)$$
$$= \frac{\pi}{4} + \tan^{-1}\left(2^{11}\right) - \tan^{-1}\left(2^{0}\right) = 2^{11}$$

Ans. 2048

Q. 25 Let $y = y(x)$ be the solution of the differential equation $x\sin\left(\frac{y}{x}\right)dy = \left(y\sin\left(\frac{y}{x}\right) - x\right)dx, y(1) = \frac{\pi}{2}$ and let $\alpha = \cos\left(\frac{\mathrm{y}\left(\mathrm{e}^{12}\right)}{\mathrm{e}^{12}}\right)$. Then the number of integral value of p, for which the equation $x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$ represents a circle of radius $\mathrm{r} \leq 6$, is _____ .

Option 1:
11

Option 2:
-

Option 3:
-

Option 4:
-

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Correct Answer:
11

Solution:

$$\sin\frac{y}{x}\frac{dy}{dx}=\frac{y}{x}\sin\frac{y}{x}-1$$

$$\text{put } y=tx \Rightarrow \frac{dy}{dx}=t+x\frac{dt}{dx}$$

$$\sin t\left(t+x\frac{dt}{dx}\right)=t\sin t-1$$

$$x\sin t\frac{dt}{dx}+1=0$$

$$\sin t dt+\frac{dx}{x}=0$$

$$\Rightarrow -\cos t+\ell nx=C$$

$$\Rightarrow -\cos\left(\frac{y}{x}\right)+\ell nx=C$$

$$y(1)=\frac{\pi}{2}\Rightarrow C=0$$

$$\Rightarrow \cos\left(\frac{y}{x}\right)=\ell nx$$

$$\text{cosec}\left(\frac{y\left(e^{12}\right)}{e^{12}}\right)=12\Rightarrow\alpha=12$$

$$x^2+y^2-2px+2py+14=0$$

$$r=\sqrt{p^2+p^2-14}$$

$$\Rightarrow r\le 6\Rightarrow r^2\le 36$$

$$2p^2-14\le 36$$

$$p^2\le 25$$

$$p\in[-5,5]$$

$$\text{number of integral value of } p=11$$

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JEE MAINS

2026

5th April

2nd Shift

Question Paper

Physics

Q. 1 Match List-I with List-II.

List-I		List-II	
A	Meter (L)	I	$\sqrt{\frac{hc}{G}}$
B	Second (S)	II	$\sqrt{\frac{Gh}{c^5}}$
C	Kilogram (M)	III	$\sqrt{\frac{K^2 L^2 c^3}{Gh}}$
D	Kelvin (K)	IV	$\sqrt{\frac{Gh}{c^3}}$

where h (Planck's constant), G (gravitational constant) and c (speed of light in vacuum) as fundamental units.

Choose the correct answer from the options given below:

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Option 1:
A-II, B-IV, C-I, D-III

Option 2:
A-IV, B-II, C-I, D-III

Option 3:
A-IV, B-I, C-II, D-III

Option 4:
A-III, B-I, C-II, D-IV

Correct Answer:
A-IV, B-II, C-I, D-III

Solution:

$[h] = \left[ML^2\ T^{-1}\right]$

$[c] = \left[LT^{-1}\right]$

and $[G] = \left[M^{-1}\ L^3\ T^{-2}\right]$

then $\left[\sqrt{\frac{hc}{G}}\right] = \left[\frac{ML^2\ T^{-1}LT^{-1}}{M^{-1}\ L^3\ T^{-2}}\right]^{1/2} = [M]$

$\left[\sqrt{\frac{Gh}{c^5}}\right] = \left[\frac{M^{-1}\ L^3\ T^{-2}ML^2\ T^{-1}}{L^5\ T^{-5}}\right]^{1/2} = [T]$

$\left[\sqrt{\frac{k^2\ L^2c^3}{Gh}}\right] = \left[\frac{k^2\ L^2\ L^3\ T^{-3}}{M^{-1}\ L^3\ T^{-2}ML^2\ T^{-1}}\right]^{1/2} = [k]$

$\left[\sqrt{\frac{Gh}{c^3}}\right] = \left[\frac{M^{-1}\ L^3\ T^{-2}ML^2\ T^{-1}}{L^3\ T^{-3}}\right]^{1/2} = [L]$

$A \rightarrow IV, B \rightarrow II, C \rightarrow I, D \rightarrow III$

Q. 2 In an experiment to determine the resistance of a given wire using Ohm's law, the voltmeter and ammeter readings are noted as 10 V and 5 A , respectively. The least counts of voltmeter and ammeter are 500 mV and 200 mA , respectively. The estimated error in the resistance measurement is _____ Ω

Option 1:
0.25

Option 2:
2

Option 3:
2.5

Option 4:
0.18

Correct Answer:
0.18

Solution:

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$$V = IR$$
$$R = \frac{10}{5} = 2\Omega$$
$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$
$$\frac{\Delta R}{R} = \frac{500 \times 10^{-3}}{10} + \frac{200 \times 10^{-3}}{5}$$
$$\frac{\Delta R}{2} = 0.05 + 0.04$$
$$\Delta R = 0.09 \times 2 = 0.18\Omega$$

Q. 3 A mass of 1 kg kept in a inclined plane with 30° inclination with respect to horizontal plane and it is at rest initially. Then the whole assembly is moved up with constant velocity of $4\ \mathrm{m/s}$. The work done by the frictional force in time 2 s is _____ J.
(Take $g = 10\ \mathrm{m/s^2}$)

Option 1:
20

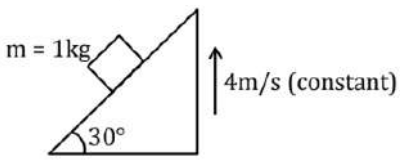
Option 2:
25

Option 3:
30

Option 4:
10

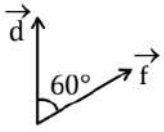
Correct Answer:
20

Solution:



$$f = mg \sin 30 = 5\ \mathrm{N}$$

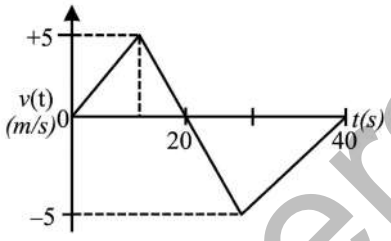
work done in 2 sec will be



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$d = 4 \times 2 = 8\ \text{m}$
$W = fd\cos 60^\circ$
$W = (5)(8)\cos 60^\circ$
$W = 20\ \text{joule}$

Q. 4 The velocity (v) versus time (t) plot of a particle is shown in the figure, for a time interval of 40s. The total distance travelled by the particle and the average velocity during this period are, respectively _____



Option 1:
25 m and zero

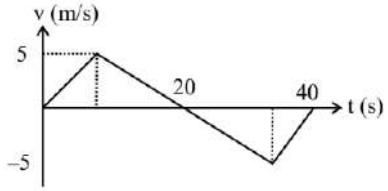
Option 2:
50 m and zero

Option 3:
100 m and zero

Option 4:
100 m and 2.5 m/s

Correct Answer:
100 m and zero

Solution:



Total distance travel

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$$d = \frac{1}{2}(20)(5) + \frac{1}{2}(20) \times 5$$
$$d = 100 \text{ m}$$
$$\text{Average velocity } = \frac{\text{Total displacement}}{\text{Total time}}$$
$$\langle \text{v} \rangle = \frac{0}{40} = 0$$

Q. 5 A wheel initially at rest is subjected to a uniform angular acceleration about its axis. In the first 2 s it rotates through an angle θ_1 and in the next 2 s it rotates through an angle θ_2. The ratio $\frac{\theta_2}{\theta_1}$ is _____ .

Option 1:
6

Option 2:
3

Option 3:
4

Option 4:
$\frac{1}{3}$

Correct Answer:
3

Solution:
Let α be angular acceleration
In first 2 sec

$\theta_1 = \frac{1}{2} \cdot \alpha(2)^2 = 2\alpha$

Angular covered in 4 sec

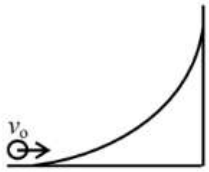
$\theta' = \frac{1}{2}(\alpha)(4)^2 = 8\alpha$

Angle in next 2 sec

$\theta_2 = \theta' - \theta_1 = 8\alpha - 2\alpha$
$\theta_2 = 6\alpha$
$\frac{\theta_2}{\theta_1} = \frac{6\alpha}{2\alpha} = 3$

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Q. 6 An object of uniform density rolls up the curved path with the initial velocity v_0 as shown in the figure. If the maximum height attained by an object is $\frac{7v_0^2}{10\ g}$ ($g =$ acceleration due to gravity), the object is a _____ .



Option 1:
solid cylinder

Option 2:
ring

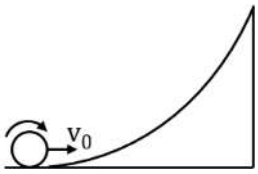
Option 3:
disc

Option 4:
solid sphere

Correct Answer:
solid sphere

Solution:
$h = \frac{7v_0^2}{10\ g}$

Applying work energy theorem



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$$-\mathrm{mgh}=0-\frac{1}{2}\mathrm{mv}^2-\frac{1}{2}\mathrm{I}\omega^2$$

$$-\mathrm{mg}\left(\frac{7\mathrm{v}_0^2}{10\,\mathrm{g}}\right)=-\frac{1}{2}\mathrm{mv}_0^2-\frac{1}{2}\frac{\mathrm{Iv}_0^2}{\mathrm{R}^2}$$

$$\frac{7\mathrm{mv}_0^2}{10}=\frac{\mathrm{mv}_0^2}{2}+\frac{\mathrm{Iv}_0^2}{2\mathrm{R}^2}$$

$$\frac{\mathrm{mv}_0^2}{5}=\frac{\mathrm{Iv}_0^2}{2\mathrm{R}^2}$$

$$\mathrm{I}=\frac{2\mathrm{mR}^2}{5}$$

Q. 7 A body of mass m is taken from the surface of earth to a height equal to twice the radius of earth ($\mathrm{R_e}$). The increase in potential energy will be _____ .
(g is acceleration due to gravity at the surface of earth)

Option 1:
$\frac{1}{2}mg\mathrm{R_e}$

Option 2:
$\frac{3}{4}mgR_e$

Option 3:
$\frac{1}{4}mg\mathrm{R_c}$

Option 4:
$\frac{2}{3}mg\mathrm{R_c}$

Correct Answer:
$\frac{2}{3}mg\mathrm{R_c}$

Solution:

$$\mathrm{U}_1=\frac{-\mathrm{GMm}}{\mathrm{R_e}}$$

$$\mathrm{U}_2=\frac{-\mathrm{GMm}}{\mathrm{R_e}+2\mathrm{R_e}}=-\frac{\mathrm{GMm}}{3\mathrm{R_e}}$$

$$\Delta\mathrm{U}=\mathrm{U}_2-\mathrm{U}_1=\frac{-\mathrm{GMm}}{3\mathrm{R_e}}+\frac{\mathrm{GMm}}{\mathrm{R_e}}$$

$$\Delta\mathrm{U}=\frac{2}{3}\frac{\mathrm{GMm}}{\mathrm{R_e}}=\frac{2}{3}\left[\frac{\mathrm{GM}}{\mathrm{R_e^2}}\right](\mathrm{mR_e})$$

$$\Delta\mathrm{U}=\frac{2}{3}\mathrm{mgR_e}$$

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Q. 8 Eight mercury drops, each of radius r, coalesce to form a bigger drop. The surface energy released in this process is _____ . (S is the surface tension of mercury).

Option 1:
$8\pi r^2 S$

Option 2:
$16\pi r^2 S$

Option 3:
$64\pi r^2 S$

Option 4:
$4\pi r^2 S$

Correct Answer:
$16\pi r^2 S$

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Solution:
Volume conservation

$$(8)\left[\frac{4}{3}\pi r^3\right] = \frac{4}{3}\pi R^3$$
$$R = 2r$$
$$U_i = (8)\left(4\pi r^2\right)(s); s = \text{ surface tension}$$
$$U_i = 32\pi r^2 s$$
$$U_f = \left(4\pi R^2\right)(s)$$
$$U_f = 4\pi(2r)^2(s) = 16\pi r^2 s$$
$$\text{Energy released } = U_i - U_f$$
$$\text{Energy released } = 16\pi r^2 s$$

Q. 9 An ideal gas at pressure P and temperature T is expanding such that $PT^3 =$ constant. The coefficient of volume expansion of the gas is _____ .

Option 1:
$\frac{2}{T}$

Option 2:
$\frac{1}{T}$

Option 3:
$\frac{4}{T}$

Option 4:
$\frac{3}{T}$

Correct Answer:
$\frac{4}{T}$

Solution:

$$\mathrm{PT^3 = \ constant\ (C)}$$

$$\mathrm{\frac{nRT}{V} \cdot T^3 = C}$$

$$\mathrm{V = \frac{nR}{C} \cdot \ T^4}$$

Volume expansion coeff.

$$\gamma = \frac{1}{\mathrm{V}} \frac{\mathrm{dV}}{\mathrm{dT}}$$

$$\gamma = \frac{\mathrm{C}}{\mathrm{nRT^4}} \cdot \frac{\mathrm{nR}}{\mathrm{C}} \cdot 4\ \mathrm{T^3} = \frac{4}{\mathrm{T}}$$

Ans. $\gamma = \frac{4}{T}$

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Q. 10 Match List-I with List-II.

List-I		List-II	
A	$\sin^2 \omega t$	I	Periodic with time period $\mathrm{T} = \frac{\pi}{\omega}$ but not simple harmonic motion (SHM)
B	$\sin^3(2\omega t)$	II	Periodic with time period $\mathrm{T} = \frac{2\pi}{\omega}$ but Not SHM
C	$\sin(\omega t) + \cos(\pi\omega t)$	III	periodic with time period $\mathrm{T} = \frac{\pi}{\omega}$ and SHM
D	$\cos \omega t + \cos 2\omega t$	IV	Non-periodic

Choose the correct answer from the options given below.

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Option 1:
A-III, B-I, C-IV, D-II

Option 2:
A-II, B-I, C-III, D-IV

Option 3:
A-III, B-II, C-IV, D-I

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
(A) $\sin^2 \omega t = \frac{1}{2} - \frac{1}{2}\cos 2\omega t$
it represents SHM motion having period $= \frac{\pi}{\omega}$
(B) $\sin^3 2\omega t = \frac{3}{4}\sin 2\omega t - \frac{1}{4}\sin 6\omega t$
common period $\frac{\pi}{\omega}$ but not SHM.
(C) $\sin \omega t + \cos \pi\omega t \rightarrow$ no common period hence it is a nonperiodic function
(D) $\cos \omega t + \cos 2\omega t$
common period $\frac{2\pi}{\omega}$ but it do not represent SHM.
$\mathrm{A} \rightarrow \mathrm{III}, \mathrm{B} \rightarrow \mathrm{I}, \mathrm{C} \rightarrow \mathrm{IV}, \mathrm{D} \rightarrow \mathrm{II}$

Q. 11 A metal rod of length L rotates about on end at origin with a uniform angular velocity ω. The magnetic field radially falls off as $B(\mathrm{r}) = \mathrm{B_o e^{-\lambda r}}$; λ being a positive constant. The emf induced (neglecting the centripetal force on electrons in the rod) is:

Option 1:
$B_o\omega\left[\frac{1}{\lambda^2} - \mathrm{e}^{-\lambda \mathrm{L}}\left(\frac{1}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

Option 2:
$B_0\omega\left[\frac{1}{\lambda^2} + \mathrm{e}^{-\lambda \mathrm{L}}\left(\frac{1}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

Option 3:
$B_o\omega\left[\frac{4}{\lambda^2} - \mathrm{e}^{-2\lambda \mathrm{L}}\left(\frac{1}{\lambda^2} + \frac{2L}{\lambda}\right)\right]$

Option 4:
$B_o\omega\left[\frac{3}{\lambda^2} - \mathrm{e}^{-3\lambda \mathrm{L}}\left(\frac{3}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

Correct Answer:
$B_o\omega\left[\frac{1}{\lambda^2} - \mathrm{e}^{-\lambda \mathrm{L}}\left(\frac{1}{\lambda^2} + \frac{L}{\lambda}\right)\right]$

Solution:

$$e = \int_0^L B(\omega r)dr = B_0\omega\int_0^L e^{-\lambda r}\cdot r dr$$

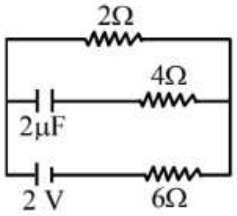
$$= B_0\omega\left[r\int e^{-\lambda r}dr - \int 1\cdot\frac{e^{-\lambda r}}{-\lambda}dr\right]_0^L$$

$$= B_0\omega\left[\frac{re^{-\lambda r}}{-\lambda} - \frac{e^{-\lambda r}}{\lambda^2}\right]_0^L$$

$$= B_0\omega\left[\left(0 + \frac{1}{\lambda^2}\right) - \left(\frac{Le^{-\lambda L}}{\lambda} + \frac{e^{-\lambda L}}{\lambda^2}\right)\right]$$

$$e = B_0\omega\left[\frac{1}{\lambda^2} - e^{-\lambda L}\left(\frac{1}{\lambda^2} + \frac{L}{\lambda}\right)\right]$$

Q. 12 Under steady state condition the potential difference across the capacitor in the circuit is _____ V.



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Option 1:
0.5

Option 2:
1.5

Option 3:
0

Option 4:
2

Correct Answer:
0.5

Solution:

$$\mathrm{i}=\frac{2}{6+2}=\frac{1}{4}\ \mathrm{A}$$

$$\mathrm{v_C}=\mathrm{v_{2\Omega}}=\mathrm{i}\times 2=\frac{1}{2}\mathrm{v}=0.5\ \mathrm{V}$$

Q. 13 A particle of charge q and mass m is projected from origin with an initial velocity $\vec{v}=\left(\frac{v_0}{\sqrt{2}}\hat{\mathrm{x}}+\frac{v_0}{\sqrt{2}}\hat{\mathrm{y}}\right)$. There exists a uniform magnetic field $\vec{B}=B_0\hat{z}$ and a space varying electric field $\vec{E}=E_0\mathrm{e}^{-\lambda x}\hat{\mathrm{x}}$ within the region $0\leq \mathrm{x}\leq L$. After travelling a distance such that x -coordinate has changed from $x=0$ to $x=L$, the change in the kinetic energy is ____ .

Option 1:
$\frac{qE_0}{\lambda}\left[1-\mathrm{e}^{-\lambda \mathrm{L}}\right]$

Option 2:
$\left(\frac{v_0\mathrm{qB}_0}{2\lambda}\right)\left[2-\mathrm{e}^{-2\lambda\,\mathrm{L}}\right]$

Option 3:
$\frac{qE_0}{\lambda}\left[1+\mathrm{e}^{-\lambda \mathrm{L}}\right]$

Option 4:
$q\left[\frac{E_0+v_0B_0}{\lambda}\right]\left[1-\mathrm{e}^{-\lambda \mathrm{L}/2}\right]$

Correct Answer:
$\frac{qE_0}{\lambda}\left[1-\mathrm{e}^{-\lambda \mathrm{L}}\right]$

Solution:

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$$\Delta K = W_e + W_m = W_e + 0$$
$$= \int_0^L qEdx = qE_0 \int_0^L e^{-\lambda x} dx$$
$$= \frac{qE_0}{\lambda}\left(1 - e^{-\lambda L}\right)$$

Q. 14 Given below are two statements : one is labelled as **Assertion (A)** : and the other is labelled as **Reason (R).**

Assertion (A) : The electromagnetic wave exerts pressure on the surface on which they are allowed to fall.

Reason (R) : There is no mass associated with the electromagnetic waves.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both (A) and (R) are true and (R) is the correct explanation of (A)

Option 2:
Both (A) and (R) are true but (R) is not the correct explanation of (A)

Option 3:
(A) is true but (R) is false

Option 4:
(A) is false but (R) is true

Correct Answer:
Both (A) and (R) are true but (R) is not the correct explanation of (A)

Solution:
Assertion and reason both are correct but not related.

Q. 15 A thin convex lens and a thin concave lens are kept in contact and are co-axial. Which of the following statements is correct for this combination of two lenses?

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Option 1:
behaves as concave lens if $|f_{\text{convex}}| > |f_{\text{concave}}|$

Option 2:
behaves as concave lens if $|\text{f}_{\text{convex}}| < |\text{f}_{\text{concave}}|$

Option 3:
behaves as convex lens if $|f_{\text{convex}}| > |f_{\text{concave}}|$

Option 4:
Focal length of the lens system will change if the position of two lenses are interchanged

Correct Answer:
behaves as concave lens if $|f_{\text{convex}}| > |f_{\text{concave}}|$

Solution:
$1 \rightarrow$ convex lens
$2 \rightarrow$ concave lens

$\frac{1}{\text{f}} = \frac{1}{\text{f}_1} + \frac{1}{-\text{f}_2} : \frac{1}{\text{f}_1} - \frac{1}{\text{f}_2}$

for $\text{f}_1 > \text{f}_2 \therefore \text{f} < 0$
$\therefore$ concave lens.

Q. 16 An object AB is placed 15 cm on the left of a convex lens P of focal length 10 cm . Another convex lens Q is now placed 15 cm right of lens P . If the focal length of lens Q is 15 cm , the final image is _____

Option 1:
virtual, formed at 7.5 cm right of lens Q , with a size bigger than that of AB

Option 2:
real, formed at 7.5 cm right of lens Q , with a size same than that of AB

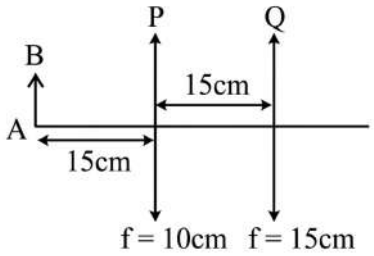
Option 3:
formed at infinity.

Option 4:
real, formed at 7 cm right of lens Q , with a size smaller than that of AB

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Correct Answer:
real, formed at 7.5 cm right of lens Q , with a size same than that of AB

Solution:



for $\mathrm{P} \frac{1}{\mathrm{V}}-\frac{1}{-15}=\frac{1}{10}$

$\therefore \mathrm{v}=+30 \mathrm{~cm}=+15 \mathrm{~cm}$ right of Q.

for $Q \frac{1}{\mathrm{V}'}-\frac{1}{15}=\frac{1}{15}$

$\mathrm{V}'=7.5 \mathrm{~cm}$

$\mathrm{m}=\mathrm{m}_1 \mathrm{~m}_2=\frac{30}{-15} \times \frac{7.5}{15}=-1$

Q. 17 The maximum intensity in a Young's double slit experiment is I_0. Distance between the slits (d) is 5λ, where λ is the wavelength of light used. The intensity of the fringe, exactly opposite to one of the slits on the screen, placed at $\mathrm{D}=10 \mathrm{~d}$ is _____

Option 1:
$\frac{I_0}{4}$

Option 2:
$\frac{I_0}{2}$

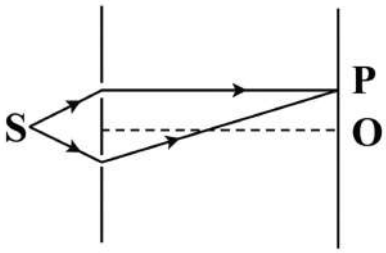
Option 3:
I_0

Option 4:
$\frac{3I_0}{4}$

Correct Answer:
$\frac{I_0}{2}$

Solution:

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$$\text{At } Py = \frac{d}{2}$$

$$\Delta Q = \frac{2\pi}{\lambda}\frac{yd}{D} = \frac{2\pi}{\lambda}\frac{d}{2}\frac{d}{D} = \frac{\pi d^2}{\lambda D}$$

$$\Delta Q = \frac{\pi \times 25\lambda^2}{\lambda \times 10d} = \frac{5\pi\lambda}{2 \times 5\lambda} = \frac{\pi}{2}$$

$$I = I_0 \cos^2\left(\frac{\Delta Q}{2}\right) = I_0 \cos^2\left(\frac{\pi}{2 \times 2}\right) = \frac{I_0}{2}$$

Q. 18 An electron is travelling with a velocity v in free space and when it enters a medium, its velocity is reduced by 20%. The de Broglie wavelength of electron in the medium is $\alpha\lambda_0$, where λ_0 is its de Broglie wavelength in free space. The value of α is _____

Option 1:
1.20

Option 2:
1.0

Option 3:
1.25

Option 4:
0.75

Correct Answer:
1.25

Solution:

$$\lambda_0 = \frac{\mathrm{h}}{\mathrm{mv}_0}$$

$$\lambda = \frac{\mathrm{h}}{\mathrm{mv}} = \frac{\mathrm{h}}{\mathrm{m}\left(0.8\mathrm{v}_0\right)} = \frac{1}{0.8}\frac{\mathrm{h}}{\mathrm{mv}_0} = 1.25\lambda_0$$

$$\therefore \alpha = 1.25$$

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Q. 19 Assuming the experimental mass of $^{12}_{6}\mathrm{C}$ as 12 u , the mass defect of $^{12}_{6}\mathrm{C}$ atom is _____ $\mathrm{MeV/c^2}$. (Mass of proton $= 1.00727\mathrm{u}$. mass of neutron $= 1.00866\mathrm{u}, 1\mathrm{u} = 931.5\mathrm{MeV/c^2}$ and c is the speed of the light in vacuum.)

Option 1:
127.5

Option 2:
89.03

Option 3:
272.0

Option 4:
92.0

Correct Answer:
89.03

Solution:

$$\Delta \mathrm{m} = (6\ \mathrm{m_p} + 6\ \mathrm{m_n}) - \mathrm{m_c}$$
$$= 6(1.00727 + 1.00866) - 12$$
$$= 0.09558\mathrm{amu} = 89.03\mathrm{MeV/C^2}$$

Q. 20 In a semiconductor p-n diode, the doping concentrations on p -side and n -side are 10^{15} atoms/ $\mathrm{cm^3}$ and 10^{18} atoms $/\mathrm{cm^3}$, respectively. Which one of the following statement is true?

Option 1:
Widths of depletion region on either side of the interface are equal

Option 2:
The depletion region width is more on p-side compared to that in n -side

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Option 3:
The depletion region width is more on n -side compared to that in p -side

Option 4:
No depletion region forms because of unequal doping concentrations on p and n-sides

Correct Answer:
The depletion region width is more on p-side compared to that in n -side

Solution:
$N_P x_P = N_n x_n$

$\mathrm{N_p} < \mathrm{N_n} \therefore \mathrm{x_p} > \mathrm{x_n}$ in depletion region

---

Q. 21 A copper wire of length 3 m is stretched by 3 mm by applying an external force. The volume of the wire is $600 \times 10^{-6}\ \mathrm{m}^3$. The elastic potential energy stored in the wire in stretched condition would be _____ J.
$\left(\text{ Given Young modulus of copper } = 1.1 \times 10^{11}\ \mathrm{N/m^2}\right)$

Correct Answer:
33

Solution:
$$\mathrm{U} = \frac{1}{2}\mathrm{y}(\text{ strain })^2 \times \text{ volume}$$
$$= \frac{1}{2} \times 1.1 \times 10^{11} \times \left(\frac{3 \times 10^{-3}}{3}\right)^2 \times 600 \times 10^{-6}$$
$$\mathrm{U} = 33\ \mathrm{J}$$

Q. 22 The heat extracted out of x gram of water initially at $50^{\circ}\mathrm{C}$ to cool it down to $0^{\circ}\mathrm{C}$ is sufficient to evaporate $(1000 - \mathrm{x})$ gram of water also initially at $50^{\circ}\mathrm{C}$. The value of x (closest integer) is _____ .
(Take latent heat of water $2256\ \mathrm{kJ/kg.K}$, specific heat capacity of water $4200\ \mathrm{J/kg.K}$)

Option 1:
922

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Option 2:
-

Option 3:
-

Option 4:
-

Correct Answer:
922

Solution:
$x \times 4200 \times 50 = (1000 - x) \times 4200 \times 50 + (1000 - x) \times 2256 \times 10^3$
$x = 1000 - x + (1000 - x) \times 10.74$
$x = 921.52$

Q. 23 A series LCR circuit with R $= 20\Omega,\ \mathrm{L} = 1.6\mathrm{H}$ and $\mathrm{C} = 40\mu\ \mathrm{F}$ is connected to a variable frequency a.c. source. The inductive reactance at resonant frequency is _____ Ω.

Correct Answer:
200

Solution:
For resonance

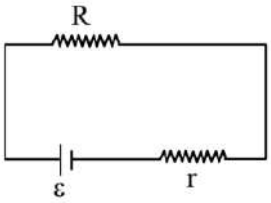
$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1.6 \times 40 \times 10^{-6}}} = \frac{10^3}{8} = 125$$
$$X_L = \omega L = 125 \times 1.6 = 200\Omega$$

Q. 24 When an external resistance of 5Ω is connected across terminals of a cell, a current of 0.25 A flows through it. When the 5Ω resistor is replaced by a 2Ω resistor, a current of 0.5 A flows through it. The internal resistance of the cell is _____ Ω.

Correct Answer:
1

Solution:

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$$\mathrm{i}=\frac{\varepsilon}{\mathrm{R}+\mathrm{r}}$$
$$0.25=\frac{\varepsilon}{5+\mathrm{r}}$$
$$0.5=\frac{\varepsilon}{2+\mathrm{r}}$$

on solving, $\mathrm{r}=1\Omega$

Q. 25 A circular loop of radius 20 cm and resistance 2Ω is placed in a time varying magnetic field $\vec{B}=\left(2t^2+2t+3\right)T$. At $t=0$, for the plane of the loop being perpendicular to the magnetic field and, the induced current in the loop at $\mathrm{t}=3$ s is $\frac{\alpha}{50}$ A. The value of α is _____ (Take $\pi=22/7$)

Correct Answer:
44

Solution:
$$\phi=\mathrm{B}\cdot\pi\mathrm{r}^2$$
$$\varepsilon=\frac{\mathrm{d}\phi}{\mathrm{dt}}=\pi\mathrm{r}^2\frac{\mathrm{dB}}{\mathrm{dt}}$$
$$\mathrm{i}=\frac{\varepsilon}{\mathrm{R}}=\frac{\pi\mathrm{r}^2}{\mathrm{R}}\cdot\left(\frac{\mathrm{dB}}{\mathrm{dt}}\right)=\frac{\pi}{2}\left(\frac{1}{5}\right)^2(4\mathrm{t}+2)$$

for $t=3$

$$i=\frac{11}{7}\times\frac{1}{25}\times14$$
$$i=\frac{44}{50}$$

Chemistry

Q. 1 What volume of hydrogen gas at STP would be liberated by action of 50 mL of $\mathrm{H_2SO_4}$ of 50% purity (density $=1.3\ \mathrm{g\ mL^{-1}}$) on 20 g of zinc?

Given : Molar mass of $\mathrm{H, O, S, Zn}$ are $1, 16, 32, 65\mathrm{gmol^{-1}}$ respectively.

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Option 1:
5.824 L

Option 2:
7.428 L

Option 3:
6.892 L

Option 4:
8.375 L

Correct Answer:
6.892 L

Solution:

$$\text{Mass of } H_2SO_4 = \frac{50 \times 1.3}{98} \times \frac{50}{100} = \frac{32.5}{98}$$

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq.) + H_2(g)$$

$$\frac{20}{65} \quad \frac{32.5}{98}$$

$$\text{Moles of } H_2 \text{ formed } = \frac{20}{65}$$

$$\text{Volume of } H_2 = \frac{20}{65} \times 22.7 = 6.9 \text{ L}.$$

Q. 2 Which of the following statement(s) is/are true ?
(A) If two orbitals have the same value of ($n + l$), the orbital with lower value of n will have lower energy.
(B) Energies of the orbitals in the same subshell increase with increase in atomic number.
(C) The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.
(D) Among 5f,6s, 4d, 5p and 5d orbitals, none of the orbitals have 2 radial nodes.

Choose the correct answer from the options given below :

Option 1:
A, B and C only

Option 2:
A and C only

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Option 3:
C and D only

Option 4:
A only

Correct Answer:
A and C only

Solution:
(a) If two orbitals are having same value of " $n+\ell$ " then the orbital having lower value of ' n ' will have lower energy.
(b) Energies of the orbitals in the same subshell decreases with increase in atomic number.
(c) Size of $2p_x < 3p_x$

Orbital	Radial node ($\mathrm{n}-\ell-1$)
5f	1
6s	5
4d	1
5p	3
5d	2

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Q. 3 The covalent radii of atoms A and B are $\mathrm{r_A}$ and $\mathrm{r_B}$, respectively. The covalent bond length and total length of AB molecule are respectively :

Option 1:
$(\mathrm{r_A}+\mathrm{r_B}), 2\,(\mathrm{r_A}+\mathrm{r_B})$

Option 2:
$\frac{1}{2}(\mathrm{r_A}+\mathrm{r_B}), (\mathrm{r_A}+\mathrm{r_B})$

Option 3:
$(\mathrm{r_A}+\mathrm{r_B}), (\mathrm{r_A}+\mathrm{r_B})$

Option 4:
$2\left(\mathrm{r_A}+\mathrm{r_B}\right), \frac{1}{2}\left(\mathrm{r_A}+\mathrm{r_B}\right)$

Correct Answer:
$\left(\mathrm{r_A}+\mathrm{r_B}\right), 2\left(\mathrm{r_A}+\mathrm{r_B}\right)$

Solution:
According to NCERT
Bond length $= r_A + r_B$
Total length of molecule $= 2\left(r_A + r_B\right)$

Q. 4 Consider the following data for the reaction

$\mathrm{X_2(\ g) + Y_2(\ g) \rightleftharpoons 2XY(\ g)}$

at $600\ \mathrm{K}^{\circ}$. The $\Delta_{\mathrm{r}} \mathrm{G}^{\ominus}$ (in $\mathrm{kJmol^{-1}}$) for the reaction is:

Compound	$\Delta_f \mathrm{H}^{\ominus}_{600\ \mathrm{K}}$ $\left(\mathrm{kJmol^{-1}}\right)$	$S^{\ominus}_{600K}$ $\left(J\mathrm{mol^{-1}\ K^{-1}}\right)$
XY(g)	42	200
$\mathrm{X_2(\ g)}$	8	140
$\mathrm{Y_2(\ g)}$	80	250

Option 1:
-21000

Option 2:
-10

Option 3:
-1000

Option 4:
-9.012

Correct Answer:
-10

Solution:

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$$\Delta_r G = \Delta_r H - T \cdot \Delta_r S$$
$$\Delta_r H = (2 \times 42 - 80 - 8) = -4\ \text{kJ/mol}$$
$$\Delta_r S = (400 - 250 - 140) = +10\ \text{J/K} \cdot \text{mol}$$
$$\Delta_r G = -4000 - 600 \times (10)$$
$$= -10,000\ \text{J/mol} = -10\ \text{kJ/mol}$$

Q. 5 The correct order of molar heat capacities measured at 298 K and 1 bar is :

Option 1:
Copper (s) > Bromine (l) > Helium (g)

Option 2:
Bromine (l) > Copper (s) > Helium (g)

Option 3:
Helium (g) > Bromine (l) > Copper (s)

Option 4:
Helium (g) > Bromine (l) = Copper (s)

Correct Answer:
Bromine (l) > Copper (s) > Helium (g)

Solution:
Generally molar heat capacity of liquids is greater than solids because liquids have greater degrees of freedom \& thus can store more energy with lesser temperature rise.

Also for monoatomic gases molar heat capacity is low as only transalational degrees of freedom are present.
∴ order is :

$$\mathrm{Br_2}(\ell) > \mathrm{Cu}(\mathrm{s}) > \mathrm{He}(\mathrm{g})$$

Q. 6 The reaction $\mathrm{A(g)} \rightleftharpoons \mathrm{B(g)} + \mathrm{C(g)}$ was initiated with the amount ' a ' of $\mathrm{A(g)}$. At equilibrium it is found that the amount of $\mathrm{A(g)}$ remaining is ($\mathrm{a - x}$) at a total pressure of p .
The equilibrium constant K_p of the reaction can be calculated from the expression :

Option 1:
$\frac{x^2}{a^2+x^2} \times p$

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Option 2:
$\frac{x^2}{a^2-x^2} \times p$

Option 3:
$\frac{a+x^2}{x^2} \times p$

Option 4:
$\frac{a-x^2}{x^2} \times p$

Correct Answer:
$\frac{x^2}{a^2-x^2} \times p$

Solution:

$$A(g) \rightleftharpoons B(g) + C(g)$$

$t = 0 \quad a \quad 0$

$t_{cq} \quad a-x \quad x \quad x$

$$K_p = \frac{(x)(x)}{(a-x)} \times \left(\frac{P}{a+x}\right)$$

$$K_p = \frac{x^2 P}{a^2 - x^2}$$

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Q. 7 One half cell in a voltaic cell is constructed by dipping silver rod in $\mathrm{AgNO_3}$ solution of unknown concentration, other half cell is Zn rod dipped in 1 molar solution of $\mathrm{ZnSO_4}$. A voltage of 1.60 V is measured at 298 K for this cell. What is the concentration of $\mathrm{Ag^+}$ ions used in terms of $\log \mathrm{x}$ $\left(\mathrm{x} = \left[\mathrm{Ag^+}\right]\right)$?

$$\mathrm{E^{\Theta}_{Zn^{2+}/Zn}} = -0.76\ \mathrm{V}, \mathrm{E^{\Theta}_{Ag^{+}/Ag}} = +0.80\ \mathrm{V},$$

$$\frac{2.303\mathrm{RT}}{\mathrm{F}} = 0.059\ \mathrm{V}$$

Option 1:
$\frac{2}{3.9}$

Option 2:
$\frac{4}{5.9}$

Option 3:
$\frac{2.9}{2}$

Option 4:
$\frac{5.9}{4}$

Correct Answer:
$\frac{4}{5.9}$

Solution:
Cell Reaction :

$$\mathrm{Zn(\ s) + 2Ag^{\oplus}(aq) \longrightarrow 2Ag(\ s) + Zn^{+2}(aq)}$$

$$\mathrm{E_{cell}\ = E^{\circ}_{cell} - \frac{0.059}{n}\log(Q)}$$

$$1.6 = 1.56 - \frac{0.059}{2}\log\frac{1}{\left(\mathrm{Ag^{+}}\right)^{2}}$$

$$0.04 = +0.059\log\left[\mathrm{Ag^{+}}\right]$$

$$\log\left[\mathrm{Ag^{+}}\right] = \frac{4}{50}$$

Q. 8 Given below are two statements :
Statement-I : The number of pairs among $[\mathrm{Al_2O_3}, \mathrm{Cr_2O_3}]$, $[\mathrm{Cl_2O_7}, \mathrm{Mn_2O_7}]$, $[\mathrm{Na_2O}, \mathrm{V_2O_3}]$ and $[\mathrm{CO}, \mathrm{N_2O}\ ]$ that **contain oxides of same nature (acidic, basic, neutral or amphoteric) is 4 .**
Statement-II : Among $\mathrm{Na_2O}$, $\mathrm{Al_2O_3}$, CO and $\mathrm{Cl_2O_7}$, the most basic and acidic oxides are $\mathrm{Na_2O}$ and $\mathrm{Cl_2O_7}$, respectively.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Both Statement-I and Statement-II are true.

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Solution:
Statement I :
Al_2O_3, Cr_2O_3 : Amphoteric
CO, N_2O : Neutral
Na_2O, V_2O_3 : Basic
Cl_2O_7, Mn_2O_7 : Acidic
Statement II :
Most basic oxide : Na_2O
Most acidic oxide : Cl_2O_7

Q. 9 Given below are two statements :
Statement-I : Aluminium upon reaction with NaOH forms $[Al(OH)_6]^{3-}$ ion.
Statement-II : The geometry of ICl_4^-, ClO_3^- and IBr_2^- is square planar, pyramidal and linear respectively.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Statement-I is false but Statement-II is true.

Solution:
Statement I :

$$Al + NaOH(\text{ excess }) \rightarrow Na\,[Al(OH)_4] + H_2(\text{ g})$$

Statement II :
$ClO_3^{\ominus}$ is sp^3 and pyramidal
$ICl_4^{\ominus}$ is $sp^3\,d^2$ and square planar
$IBr_2^{\ominus}$ is $sp^3\,d$ and linear.

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Q. 10 Given below are two statements :

Statement-I : Presence of large number of unpaired electrons in transition metal atoms results in higher enthalpies of their atomisation.

Statement-II : $\mathrm{d_{xy} = d_{xz} = d_{yz} < d_{x^2-y^2} = d_{z^2}}$ and $d_{x^2-y^2} = d_{z^2} < d_{xy} = d_{xz} = d_{yz}$ are the d-orbital splittings in $[\mathrm{Fe(H_2O)_6}]^{3+}$ and $[\mathrm{Ni(Cl)_4}]^{2-}$ complex ions respectively.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are correct.

Option 2:
Both Statement-I and Statement-II are incorrect.

Option 3:
Statement-I is correct but Statement-II is incorrect.

Option 4:
Statement-I is incorrect but Statement-II is correct.

Correct Answer:
Both Statement-I and Statement-II are correct.

Solution:
Statemet I :
More are the number of unpaired electrons , stronger is metal-metal boding thus higher enthalpy of atomisation.

Statement II :
$[\mathrm{Fe(H_2O)_6}]^{3+}$ is octahedral complex.
Thus, $\left[\mathrm{t_{2\,g}\left(d_{xy} = d_{yz} = d_{zx}\right) < e_g\left(d_{x^2-y^2} = d_{z^2}\right)}\right]$
$[\mathrm{NiCl_4}]^{2-}$ is tetrahedral complex.
Thus, $\left[\mathrm{e\left(d_{x^2-y^2} = d_{z^2}\right) < t_2\left(d_{xy} = d_{yz} = d_{zx}\right)}\right]$

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Q. 11 Identify the correct statements from the following:

(A) $[Fe(C_2O_4)_3]^{3-}$ is the most stable complex among $[Fe(OH)_6]^{3-}$, $[Fe(C_2O_4)_3]^{3-}$ and $[Fe(SCN)_6]^{3-}$

(B) The stability of $[Cu(NH_3)_4]^{2+}$ is greater than that of $[Cu(en)_2]^{2+}$

(C) The hybridization of Fe in $K_4[Fe(CN)_6]$ is d^2sp^3

(D) $[Fe(NO_2)_3Cl_3]^{3-}$ exhibits linkage isomerism

(E) NO_2^- and SCN^- ligands are NOT ambidentate ligands

Choose the correct answer from the options given below:

Option 1:
A, B, C, D and E

Option 2:
B, C and D only

Option 3:
A, C and D only

Option 4:
A, C and E only

Correct Answer:
A, C and D only

Solution:

(A) $[Fe(C_2O_4)_3]^{3-}$ has oxalate ion as ligand which leads to chelation thus maximum stability among given complexes.

(B) Stability : $[Cu(NH_3)_4]^{2+} < [Cu(en)_2]^{2+}$; due to higher CFSE and chelation.

(C) $K_4[Fe(CN)_6]$

$\Rightarrow Fe^{2+} = [Ar]3d^6$

In presence of SFL the electronic configuration of Fe^{+2} $\left[t_{2g}^6, e_g^0\right]$, thus d^2sp^3 hybridized.

(D) $[Fe(NO_2)_3Cl_3]^3-$ has NO_2^- as ambidentate ligand thus can show linakge isomerism.

(E) NO_2^- and SCN^- both are ambidentate ligands.

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Q. 12 Match List-I with List-II.

List-I (Purification technique)		List-II (Used to separate)	
A	Simple distillation	I	Steam volatile compound
B	Fractional distillation	II	Two liquids with large difference in boiling points
C	Steam distillation	III	Liquid decomposing at its boiling point
D	Distillation under reduced pressure	IV	Two liquids with close boiling points

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-I, D-IV

Option 2:
A-II, B-IV, C-I, D-III

Option 3:
A-II, B-IV, C-III, D-I

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-II, B-IV, C-III, D-I

Solution:
It is theory based question.

Q. 13 IUPAC name of the some alkenes are given below :
Find out the correct stability order.
A. 2-Methylbut-2-ene
B. cis-But-2-ene
C. 2,3-Dimethylbut-2-ene
D. Prop-1-ene

Choose the correct answer from the options given below:

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Option 1:
$\mathrm{C} > \mathrm{A} > \mathrm{B} > \mathrm{D}$

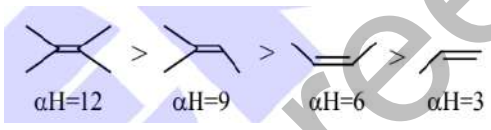
Option 2:
$\mathrm{C} > \mathrm{A} > \mathrm{D} > \mathrm{B}$

Option 3:
$\mathrm{B} > \mathrm{D} > \mathrm{A} > \mathrm{C}$

Option 4:
$\mathrm{A} > \mathrm{B} > \mathrm{C} > \mathrm{D}$

Correct Answer:
$\mathrm{C} > \mathrm{A} > \mathrm{B} > \mathrm{D}$

Solution:



Stability of alkene $\propto$ Number of α hydrogen.

Q. 14 Identify the correct IUPAC name of hydrocarbon (x) containing three primary carbon atoms and with molar mass $72\ \mathrm{g\ mol^{-1}}$.

Option 1:
1,1-Dimethylcyclopropane

Option 2:
2,2-Dimethylpropane

Option 3:
2-Methylbutane

Option 4:
n-pentane

Correct Answer:
2-Methylbutane

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Solution:

(A) 

(1°C = 2)

(B) 

(1°C = 4)

(C) 

(1°C = 3)

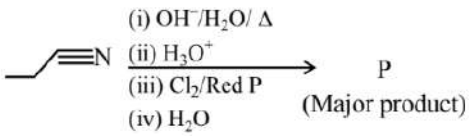
(B) 

(1°C = 2)

Mol. mass $= 72$

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Q. 15 Complete the following reaction sequence and give the name of major product ' P '.



Option 1:
2-Chloropropanoic acid

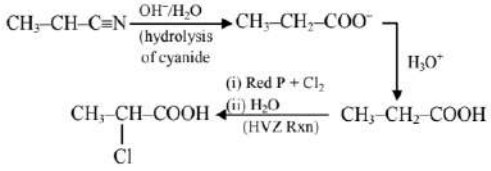
Option 2:
3-Chloropropanoic acid

Option 3:
1-Chloropropane

Option 4:
2-Chloropropane

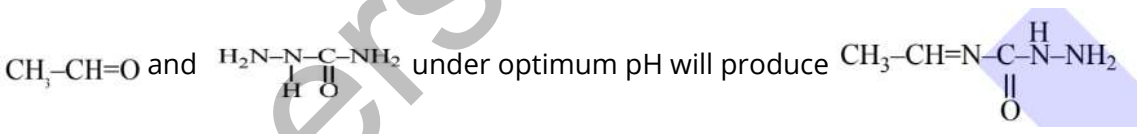
Correct Answer:
2-Chloropropanoic acid

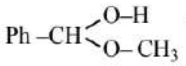
Solution:



Q. 16 Given below are two statements :

Statement-I : The condensation reaction between



Statement-II : The molecule,  will generate $\mathrm{Ph-CH=O}$ in the presence of dilute acid.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

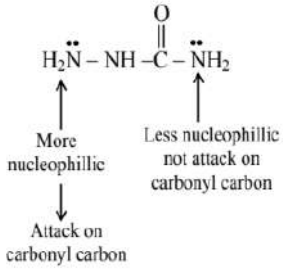
Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Statement-I is false but Statement-II is true.

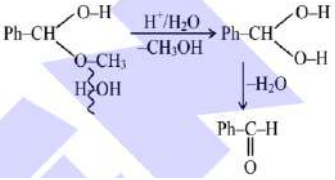
Solution:
Statement-I

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Hence statement-I is incorrect.

Statement-II



Statement-II is correct.

Q. 17 Given below are two statements :

Statement-I : Heating benzamide with bromine in an ethanolic solution of sodium hydroxide will give benzylamine.

Statement-II : Nitration of aniline with $\mathrm{HNO_3/H_2SO_4}$ at 288 K produces *m*-nitroaniline in higher amount than *o*-nitroaniline (pH adjusted).

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

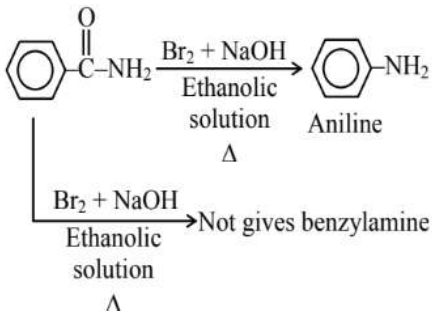
Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Statement-I is false but Statement-II is true.

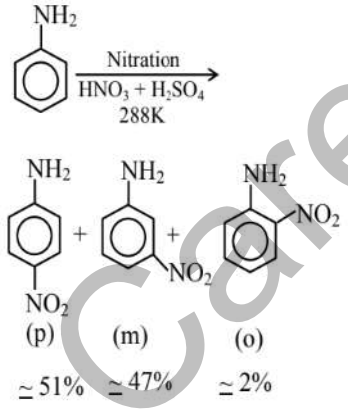
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Solution:
Statement-I



Hence Statement-I is incorrect

Statement-II



Hence Statement-II is correct

Q. 18 Identify the incorrect statement about tertiary structure of proteins.

Option 1:
They can be fibrous or globular in structure.

Option 2:
The main forces that stabilize the structure are hydrogen bonding, disulphide links, van der Waals and electrostatic forces of attraction.

Option 3:
The structure remains intact when exposed to pH changes.

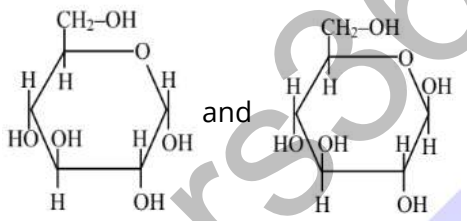
Option 4:
A linear polypeptide chain will convert to a secondary structure and then further folding of the secondary structure will convert to tertiary structure.

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Correct Answer:
The structure remains intact when exposed to pH changes.

Solution:
On changing pH tertiary strutcure of protein disrupted and amino acids intact only by peptide linkage.

Q. 19 Given below are two statements :

statement I  and are two anomers of D-(+)-glucose.

Statement-II : The open chain forms of D-glucose and D-fructose contain three similar chiral carbons at C_3, C_4 and C_5.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Both Statement-I and Statement-II are true.

Solution:
α-D-Glucose and β-D-Glucose are anomers.
At C-3, C-4 and C-5 are identical configuration in glucose and fructose.

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Q. 20 A paper dipped in a dil. H_2SO_4 solution of ' X ' upon treatment with SO_2 gas turns into green. The compound ' X ' is :

Option 1:
KI-starch

Option 2:
$KMnO_4$

Option 3:
$Pb(CH_3COO)_2$

Option 4:
$K_2Cr_2O_7$

Correct Answer:
$K_2Cr_2O_7$

Solution:
SO_2 gas acts as reducing agent.

$$K_2Cr_2O_7 + SO_2 \xrightarrow{H_2SO_4} \underset{\text{Green solution}}{Cr_2(SO_4)_3} + SO_4^{2-}$$

Q. 21 The total number of unpaired electrons present in the d^3, d^4 (low spin) d^5 (high spin), d^6 (high spin) and d^7 (low spin) octahedral complex systems is
____ .

Correct Answer:
15

Solution:
$d^3 \rightarrow \left[t_{2g}^3, e_g^0\right]$, 3 unpaired electrons

d^4(low spin) $\rightarrow \left[t_{2g}^4, e_g^0\right]$, 2 unpaired electrons

d^5 (high spin) $\rightarrow \left[t_{2g}^3, e_g^2\right]$, 5 unpaired electrons

d^6 (high spin) $\rightarrow \left[t_{2g}^4, e_g^2\right]$, 4 unpaired electron

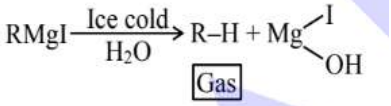
d^7(low spin) $\rightarrow \left[t_{2g}^6, e_g^1\right]$, 1 unpaired electrons

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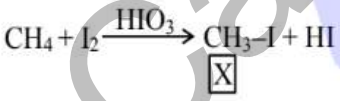
Q. 22 RMgI when treated with ice cold water liberated a gas which occupied $1.4\mathrm{dm}^3/\mathrm{g}$ at STP. The gas produced is further reacted with iodine in presence of HIO_3 to give compound (X). Compound (X) in presence of Na and dry ether produced compound (Y). Molar mass of compound (Y) is _____ gmol^{-1}.
(Nearest integer)

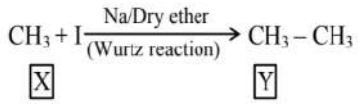
Correct Answer:
30

Solution:



Vol. $= 1.4\mathrm{dm}^3/\mathrm{gm}$ at STP
Hence according to this mol. Mass of gas is $= 16$
Hence liberated gas is $= \mathrm{CH}_4$ (Methane)





Molar mass = 30

Q. 23 20 g hemoglobin in a 1 L aqueous solution (A) at 300 K is separated from pure water by semi permeable membrane. At equilibrium the height of solution in a tube dipped in a solution (A) is found to be 80.0 mm higher than the tube dipped in water.

The molar mass of hemoglobin is _____ kgmol^{-1}. (Nearest integer)
(Given : $\mathrm{g} = 10\ \mathrm{ms}^{-2}, \mathrm{R} = 8.3\mathrm{kPadm}^3\ \mathrm{K}^{-1}\ \mathrm{mol}^{-1}$, density of solution $= 1000\ \mathrm{kg\ m}^{-3}$)

Correct Answer:
62

Solution:
Osmotic pressure, $\pi = \rho \mathrm{gh}$

$$= 1000 \times 10 \times 80 \times 10^{-3}$$
$$= 800\ \mathrm{Pa} \qquad (1)$$

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Let molar mass of haemoglobin $= \mathrm{Mg/mol}$

$$\text{conc. of haemoglobin } = \frac{20}{10^{-3}}\left(\frac{\text{mol}}{\text{m}^3}\right)$$

$$\pi = \text{ CRT} \quad \text{(S.I. units)}$$

$$\Rightarrow \pi = \left[\frac{20}{10^{-3}\text{M}}\right] \times 8.3 \times 300$$

$$= 800 \quad [\text{From (1)}]$$

Solving we get :

$$\text{M} = \frac{20 \times 8.3 \times 300}{0.8}\ \text{g/mol}$$

$$= 62.25\ \text{kg mol}^{-1}$$

$$\text{Ans } = 62$$

Q. 24 At 298 K , the molar conductivity of $\mathrm{x\%(w/w)MX}$ solution (aqueous) is $123.5\ \mathrm{S\ cm^2\ mol^{-1}}$. The conductance of same solution is $1.9 \times 10^{-3}\ \mathrm{S}$. The value of x is ____ $\times 10^{-2}$.
(Given : cell constant $= 1.3\ \mathrm{cm^{-1}}$; molar mass of MX is $75\ \mathrm{g\ mol^{-1}}$, density of aqueous solution of MX at 298 K is $1.0\ \mathrm{g\ mL^{-1}}$)

Correct Answer:
15

Solution:
Let molarity of MX solution = y M

$$\Rightarrow \Lambda_{\text{m}} = \text{c}(\ell/\text{A}) \times \frac{1000}{\text{y}}$$

$$\Rightarrow 123.5 = 1.9 \times 10^{-3} \times 1.3 \times \frac{1000}{\text{y}}$$

$$\Rightarrow \text{y} = 0.02\text{M}$$

Let molar mass of $\mathrm{MX} = \mathrm{Mgmol^{-1}}$
0.2 mol MX is dissolve in 1 L solution
$(0.02 \times 75)\mathrm{gmMX}$ is dissolved in 1000 g solution

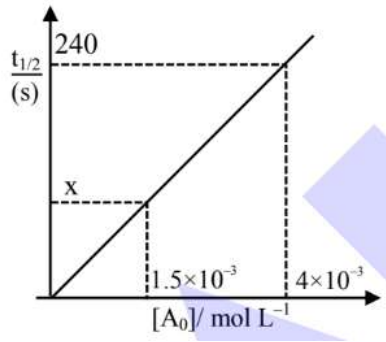
$$\%\text{w/w} = \text{x}\% = \frac{0.02 \times 75}{1000} \times 100$$

$$= 0.15\%$$

$$\text{Ans } = 15$$

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Q. 25 For a reaction $\mathrm{A} \rightarrow \mathrm{P}$ at T K , the half life $\left(\mathrm{t}_{1/2}\right)$ is plotted as a function of initial concentration $[\mathrm{A}]_0$ of A as given below :



The value of x in the given figure is ____ S
(Nearest integer)

Correct Answer:
90

Solution:
From the plot, $\mathrm{t}_{1/2} \propto [\mathrm{A}]_0$

$\propto [\mathrm{A}_0]^{1-\mathrm{N}}$

$1-\mathrm{N}=1$
$\mathrm{N}=0\ (\text{order})$
$\mathrm{t}_{1/2}=\dfrac{\mathrm{A}_0}{2\,\mathrm{K}}$
$240=\frac{4\times10^{-3}}{2\,\mathrm{K}}$ (1)

$\mathrm{x}=\frac{1.5\times10^{-3}}{2\,\mathrm{K}}$ (2)

Divide (1) / (2)

$\dfrac{240}{x}=\dfrac{4}{1.5}$
$x=90$

Maths

Q. 1 Let α, β be the roots of the equation $\mathrm{x}^2-\mathrm{x}+\mathrm{p}=0$ and γ, δ be the roots the equation $\mathrm{x}^2-4\mathrm{x}+\mathrm{q}=0$; $p, q \in Z$. If $\alpha, \beta, \gamma, \delta$ are in G.P., then $|p+q|$ equals:

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Option 1:
16

Option 2:
32

Option 3:
34

Option 4:
38

Correct Answer:
34

Solution:

$\text{Let } \alpha = a, \beta = ar, \gamma = ar^2, \delta = ar^3$

$a + ar = 1$

$a^2 + ar^3 = 4$

$\Rightarrow ar^2(1 + r) = 4$

$\Rightarrow r = 2, a = \frac{1}{3} \text{ (reject as } p \in z \text{)}$

$\text{or } r = -2, a = -1$

$\text{Now } |P + q| = \left|a(ar) + ar^2 - ar^3\right|$

$= \left|a^2 r\left(1 + r^4\right)\right|$

$|1 \times -2 \times 17| = 34$

Q. 2 Let $z_1, z_2 \in \mathrm{C}$ be the distinct solutions of the equation $z^2 + 4z - (1 + 12i) = 0$. Then $|z_1|^2 + |z_2|^2$ is equal to :

Option 1:
18

Option 2:
22

Option 3:
29

Option 4:
34

Correct Answer:
34

Solution:

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$Z^2 + 4Z - (1 + 12i) = 0$

$\Rightarrow Z = \dfrac{-4 \pm \sqrt{16 + 4(1 + 12i)}}{2}$

$\Rightarrow Z = -2 \pm \sqrt{5 + 12i}$

$\Rightarrow Z = -2 \pm (3 + 2i)$

$\Rightarrow Z = 1 + 2i, -5 - 2i$

$\therefore |Z_1|^2 + |Z_2|^2$

$= 5 + 29$

$= 34$

Q. 3 If $f : N \to Z$ is defined by $f(n) = \begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}$, $k \in N$, and $\sum_{n=1}^{k} f(n) = 98$, then k is equal to:

Option 1:
3

Option 2:
4

Option 3:
5

Option 4:
6

Correct Answer:
3

Solution:

$$f(n) = \begin{vmatrix} n & -1 & -5 \\ -2n^2 & 3(2k+1) & 2k+1 \\ -3n^3 & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix}$$

$$\sum_{n=1}^{k} f(n) = \begin{vmatrix} \frac{k(k+1)}{2} & -1 & -5 \\ -2\frac{k(k+1)(2k+1)}{6} & 3(2k+1) & 2k+1 \\ -3\frac{k^2(k+1)^2}{4} & 3k(2k+1) & 3k(k+2)+1 \end{vmatrix} = 98$$

$\Rightarrow \dfrac{k(k+1)(2k+1)}{2} \cdot \dfrac{7}{3} = 98$

$\Rightarrow k(k+1)(2k+1) = 84$

$\Rightarrow k = 3$

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Q. 4 Let M be a 3×3 matrix such that $\mathrm{M}\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix}, \mathrm{M}\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\1\\2\end{pmatrix}$ and $M\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\1\end{pmatrix}$. If $M\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1\\7\\11\end{pmatrix}$, then $x+y+z$ equals :

Option 1:
4

Option 2:
5

Option 3:
7

Option 4:
11

Correct Answer:
7

Solution:

Let $\mathrm{M} = \begin{bmatrix}\mathrm{a}_1 & \mathrm{a}_2 & \mathrm{a}_3\\ \mathrm{b}_1 & \mathrm{b}_2 & \mathrm{b}_3\\ \mathrm{c}_1 & \mathrm{c}_2 & \mathrm{c}_3\end{bmatrix}$

then $\mathrm{M}\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\2\\3\end{pmatrix} \Rightarrow \mathrm{a}_1 = 1,\ \mathrm{b}_1 = 2, \mathrm{c}_1 = 3$

$\mathrm{M}\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\1\\2\end{pmatrix} \Rightarrow \mathrm{a}_2 = 0,\ \mathrm{b}_2 = 1, \mathrm{c}_2 = 2$

$\&\mathrm{M}\begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\1\end{pmatrix} \Rightarrow \mathrm{a}_3 = -1,\ \mathrm{b}_3 = 1, \mathrm{c}_3 = 1$

Now $M\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}1\\7\\11\end{pmatrix}$

$\Rightarrow x + 0y - z = 1$
$2x + y + z = 7$
$3x + 2y + z = 11$

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on solving $x=2, y=2, z=1$

$\therefore x+y+z=5$

Q. 5 If the sum of the first 10 terms of the series $\frac{1}{1+1^4\times 4}+\frac{2}{1+2^4\times 4}+\frac{3}{1+3^4\times 4}+\frac{4}{1+4^4\times 4}+\ldots$ is $\frac{\mathrm{m}}{\mathrm{n}}, \gcd(\mathrm{m},\mathrm{n})=1$, then $\mathrm{m}+\mathrm{n}$ is equal to :

Option 1:
256

Option 2:
264

Option 3:
276

Option 4:
284

Correct Answer:
276

Solution:

$$\mathrm{T_r}=\frac{\mathrm{r}}{1+4\mathrm{r}^4}=\frac{\mathrm{r}}{4\mathrm{r}^4+4\mathrm{r}^2+1-4\mathrm{r}^2}=\frac{\mathrm{r}}{\left(2\mathrm{r}^2+1\right)^2-(2\mathrm{r})^2}$$

$$=\frac{\mathrm{r}}{\left(2\mathrm{r}^2+2\mathrm{r}+1\right)\left(2\mathrm{r}^2-2\mathrm{r}+1\right)}$$

$$\mathrm{S_{10}}=\frac{1}{4}\left(\frac{1}{1}-\frac{1}{221}\right)=\frac{55}{221}=\frac{\mathrm{m}}{\mathrm{n}}$$

$\therefore \mathrm{m}+\mathrm{n}=276$

Q. 6 Let $A_1, A_2, A_3, \ldots\ldots\ldots, A_{39}$ be 39 arithmetic means between the numbers 59 and 159 . Then the mean of $\mathrm{A_{25}},\ \mathrm{A_{28}},\ \mathrm{A_{31}}$ and $\mathrm{A_{36}}$ is equal to:

Option 1:
129

Option 2:
136

Option 3:
131.50

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Option 4:
134

Correct Answer:
134

Solution:
$59A_1A_2A_3 \ldots A_{39}159$

$d = \dfrac{159 - 59}{39 + 1} = \dfrac{5}{2}$

Now,

$A_{25} = 59 + 25d = 121.5$
$A_{28} = 59 + 28d = 129$
$A_{31} = 59 + 31d = 136.5$
$A_{36} = 59 + 36d = 149$

Mean of

$\mathrm{A}_{25},\ \mathrm{A}_{28},\ \mathrm{A}_{31},\ \mathrm{A}_{36} = \dfrac{121.5 + 129 + 136.5 + 149}{4}$

$= \dfrac{536}{4} = 134$

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Q. 7 The coefficient of x^2 in the expansion of $\left(2x^2 + \frac{1}{x}\right)^{10}, x \neq 0$, is :

Option 1:
33240

Option 2:
3360

Option 3:
3480

Option 4:
3600

Correct Answer:
3480

Solution:

$T_{r+1} = {}^{10}C_r \left(2x^2\right)^{10-r} (1/x)^r$
$= {}^{10}C_r 2^{10-r} x^{20-2r-r}$
$= {}^{10}C_r 2^{10-r} x^{20-3r}$
$\Rightarrow 20 - 3r = 2$
$r = 6$

So required coefficient is ${}^{10}C_6 2^4$

Q. 8 The probabilities that players A and B of a team are selected for the captaincy for a tournament are 0.6 and 0.4 , respectively. If A is selected the captain, the probability that the team wins the tournament is 0.8 and if B is selected the captain, the probability that the team wins the tournament is 0.7 . Then the probability, that the team wins the tournament, is :

Option 1:
0.74

Option 2:
0.76

Option 3:
0.72

Option 4:
0.78

Correct Answer:
0.76

Solution:
$P(\text{Win}) = P(A \cap \text{Win}) + P(B \cap \text{Win})$
$= P(A) \cdot P\left(\frac{\text{Win}}{A}\right) + P(B) \cdot P\left(\frac{\text{Win}}{B}\right)$
$= (0.6)(0.8) + (0.4)(0.7)$
$= 0.48 + 0.28 = 0.76$

Q. 9 A box contains 5 blue, 6 yellow and 4 red balls. The number of ways, of drawing 8 balls containing at least two balls of each colour, is:

Option 1:
4100

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Option 2:
4140

Option 3:
4230

Option 4:
4290

Correct Answer:
4100

Solution:
(But in this questions not mentioned that balls are distinct/alike)?
Blue-5 Yellow-6 Red-4
Reqd. No. of ways

$$= {}^{5}C_2 \times {}^{6}C_2 \times {}^{4}C_4 + {}^{5}C_2 \times {}^{6}C_3 \times {}^{4}C_3 + {}^{5}C_2 \times {}^{6}C_4 \times {}^{4}C_2$$
$$+ {}^{5}C_3 \times {}^{6}C_2 \times {}^{4}C_3 + {}^{5}C_3 \times {}^{6}C_3 \times {}^{4}C_2 + {}^{5}C_4 \times {}^{6}C_2 \times {}^{4}C_2$$
$$= 10 \times 15 \times 1 + 10 \times 20 \times 4 + 10 \times 15 \times 6 +$$
$$10 \times 15 \times 4 + 10 \times 20 \times 6 + 5 \times 15 \times 6$$
$$= 150 + 800 + 900 + 600 + 1200 + 450$$
$$= 4100$$

Q. 10 A variable X takes values $0, 0, 2, 6, 12, 20, \ldots\ldots n(n-1)$ with frequencies ${}^{n}C_0, {}^{n}C_1, {}^{n}C_2, {}^{n}C_3, {}^{n}C_4, {}^{n}C_5, \ldots\ldots {}^{n}C_n$, respectively. If the mean of this data is 60 , then its median is:

Option 1:
56

Option 2:
42

Option 3:
72

Option 4:
90

Correct Answer:
56

Solution:

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$$\bar{x} = \frac{0.{}^nC_0 + 0.{}^nC_1 + 2.{}^nC_2 + \ldots + n(n-1)\cdot nC_n}{{}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots + nC_n}$$

$$= \frac{\sum_{r=0}^{n} r(r-1){}^nC_r}{2^n}$$

$$= \frac{\sum_{r=1}^{n} n(r-1)^{n-1}C_{r-1}}{2^n} = \frac{\sum_{r=2}^{n} n(n-1)^{n-2}C_{r-2}}{2^n}$$

$$= \frac{n(n-1)2^{n-2}}{2^n} = \frac{n(n-1)}{4}$$

$$\Rightarrow \frac{\mathrm{n(n-1)}}{4} = 60 \Rightarrow \mathrm{n^2 - n - 240 = 0}$$

$$\Rightarrow \mathrm{n = 16, -15\ (Reject)}$$

$$\mathrm{N = \sum f_i = {}^{16}C_0 + {}^{16}C_1 + \ldots\ldots + {}^{16}C_{16} = 2^{16}}$$

$$\mathrm{N = 2^{16}\ (Even)}$$

$$\mathrm{Median\ = \frac{\left(\frac{N}{2}\right)^{th} + \left(\frac{N}{2}+1\right)^{th}}{2}\ term}$$

$$\mathrm{\frac{N}{2} = 2^{15} = 32768}$$

$\mathrm{x_i}$	$\mathrm{f_i}$
0	$\mathrm{{}^{16}C_0}$
0	$\mathrm{{}^{16}C_i}$
2	$\mathrm{{}^{16}C_2}$
6	$\mathrm{{}^{16}C_3}$
$\vdots$	$\vdots$
$\vdots$	$\vdots$
$\vdots$	$\vdots$
240	$\mathrm{{}^{16}C_{16}}$

$\mathrm{x_i}$	$\mathrm{f_i}$	CF
0	$\mathrm{{}^{16}C_0}$	1
0	$\mathrm{{}^{16}C_1}$	16
2	$\mathrm{{}^{16}C_2}$	137
6	$\mathrm{{}^{16}C_3}$	697
12	$\mathrm{{}^{16}C_4}$	2517
20	$\mathrm{{}^{16}C_5}$	6885
30	$\mathrm{{}^{16}C_6}$	14893
42	$\mathrm{{}^{16}C_7}$	26333
56	$\mathrm{{}^{16}C_8}$	$39203 \rightarrow \mathrm{\frac{N}{2}}$ & $\mathrm{\left(\frac{N}{2}+1\right)}$ lies

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$$\begin{array}{cc} 72 & {}^{16}C_9 \quad \frac{n(n-1)2^{n-2}}{2^n} \\ \vdots & \vdots \\ \vdots & \vdots \\ \vdots & \vdots \end{array}$$

$\Rightarrow$ median $= \frac{56+56}{2} = 56$

Q. 11 Let the point P be the vertex of the parabola $y = x^2 - 6x + 12$. If a line passing through the point P intersects the circle $x^2 + y^2 - 2x - 4y + 3 = 0$ at the points R and S , then the maximum value of $(\mathrm{PR} + \mathrm{PS})^2$ is:

Option 1:
10

Option 2:
20

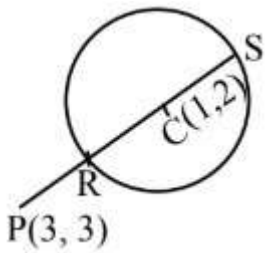
Option 3:
25

Option 4:
5

Correct Answer:
20

Solution:
Parabola $y = x^2 - 6x + 12 = (x-3)^2 + 3$

$y - 3 = (x-3)^2 \Rightarrow \text{Vertex } P(3,3)$



$x^2 + y^2 - 2x - 4y + 3 = 0$
$C(1,2); r = \sqrt{2}$

For max. of (PR + PS) ^{2}
RS should be diameter

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$$
\begin{aligned}
(\mathrm{PR}+\mathrm{PS})_{\max}^2 &= (\mathrm{PC}-\mathrm{r}+\mathrm{PC}+\mathrm{r})^2 \\
&= (2\mathrm{PC})^2 = 4(\mathrm{PC})^2 \\
&= 4\times 5 = 20
\end{aligned}
$$

Q. 12 Let the directrix of the parabola $P: y^2 = 8x$, cut x-axis at the point A. Let $B(\alpha, \beta), \alpha > 1$, be a point on P such that the slope of AB is $3/5$. If BC is a focal chord of P , then six times the area of $\triangle \mathrm{ABC}$ is:

Option 1:
80

Option 2:
160

Option 3:
174

Option 4:
192

Correct Answer:
160

Solution:

$$\text{Let B}\left(2\mathrm{t}^2, 4\mathrm{t}\right) \Rightarrow \mathrm{C}\left(\frac{2}{\mathrm{t}^2}, \frac{-4}{\mathrm{t}}\right); \mathrm{A}(-2,0)$$

$$\text{given } \mathrm{m}_{\mathrm{AB}} = \frac{4\mathrm{t}}{2\mathrm{t}^2+2} = \frac{3}{5}$$

$$20\mathrm{t} = 6\mathrm{t}^2 + 6$$

$$3\mathrm{t}^2 - 10\mathrm{t} + 3 = 0$$

$$\mathrm{t} = 3, \frac{1}{3} \text{ but } \mathrm{t} > 1 \text{ (given)}$$

$$\Rightarrow \mathrm{P}(18,12), \mathrm{Q}\left(\frac{2}{9}, \frac{-4}{3}\right)$$

$$\mathrm{A} = \left|\frac{1}{2}\right| \begin{vmatrix} -2 & 0 & 1 \\ 18 & 12 & 1 \\ \frac{2}{9} & \frac{-4}{3} & 1 \end{vmatrix} = \frac{80}{3}$$

$$\Rightarrow 6\ \mathrm{A} = 160$$

Q. 13 Let the eccentricity e of a hyperbola satisfy the equation $6\mathrm{e}^2 - 11\mathrm{e} + 3 = 0$. If the foci of the hyperbola are $(3,5)(3,-4)$, then the length of its latus rectum is:

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Option 1:
$\frac{11}{3}$

Option 2:
$\frac{17}{3}$

Option 3:
$\frac{15}{2}$

Option 4:
$\frac{17}{2}$

Correct Answer:
$\frac{15}{2}$

Solution:
$S_1(3,5) \& \ S_2(3,-4)$
$\because S_1 \ S_2 = 2ae$
$\Rightarrow 9 = 2ae$
and $\because 6e^2 - 11e + 3 = 0$
$\Rightarrow e = \frac{3}{2}, e = \frac{1}{3} < 1(\text{ rejected })$

From (1); $a = 3$

$\Rightarrow LR = 2 \cdot \frac{b^2}{a} = 2a\left(e^2 - 1\right) = 2.3\left(\frac{9}{4} - 1\right) = \frac{15}{2}$

Q. 14 Let a triangle PQR be such that P and Q lie on the line $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$ and are at a distance of 6 units from $R(1,2,3)$. If (α, β, γ) is the centroid of $\triangle PQR$, then $\alpha + \beta + \gamma$ is equal to:

Option 1:
4

Option 2:
5

Option 3:
6

Option 4:
8

Correct Answer:
6

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Solution:
Let the point P is $(8\lambda-3, 2\lambda+4, 2\lambda-1)$

$\mathrm{PR}=6 \Rightarrow \mathrm{PR}^2=36$
$(8\lambda-4)^2+(2\lambda+2)^2+(2\lambda-4)^2=36$
$(4\lambda-2)^2+(\lambda+1)^2+(\lambda-2)^2=9$
$\lambda^2-\lambda=0 \Rightarrow \lambda=0,1$
$\lambda=0 \Rightarrow \mathrm{P}(-3,4,-1)$
$\lambda=1 \Rightarrow \mathrm{P}(5,6,1)$
$\mathrm{R}(1,2,3)$
$\mathrm{G}(1,4,1)$
$=\alpha+\beta+\gamma=6$

Q. 15 If the distance of the point $(a, 2, 5)$ from the image of the point $(1, 2, 7)$ in the line $\frac{x}{1}=\frac{y-1}{1}=\frac{z-2}{2}$ is 4, then the sum of all possible values of a is equal to :

Option 1:
11

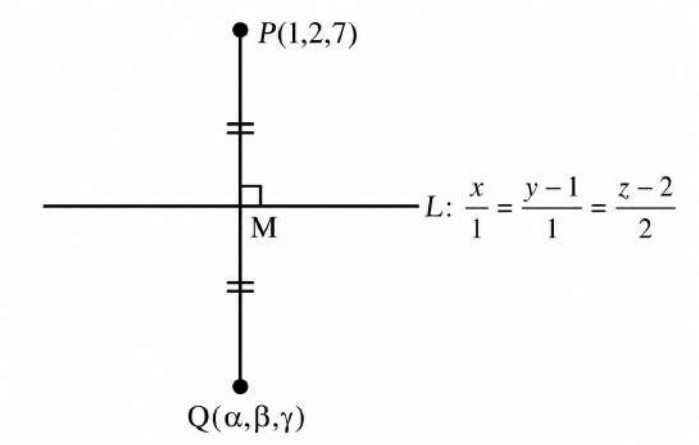
Option 2:
9

Option 3:
6

Option 4:
4

Correct Answer:
6

Solution:
$\mathrm{PQ} \perp \mathrm{L} \Rightarrow (\alpha-1)+(\beta-2)+2(\gamma-7)=0$
$\Rightarrow \alpha+\beta+2\gamma=17$



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M is mid point of PQ which will satisfy L

$$\frac{\alpha+1}{2}=\frac{\frac{\beta+2}{2}-1}{1}=\frac{\frac{\gamma+7}{2}-2}{2}$$
$$\Rightarrow \frac{\alpha+1}{2}=\frac{\beta}{2}=\frac{\gamma+3}{4}$$
$$\Rightarrow \alpha+1=\beta \quad \ldots (2)$$
$$\text{and } 2\beta=\gamma+3 \quad \ldots (3)$$
$$\Rightarrow \alpha=3, \beta=4, \gamma=5$$

Distance from $(a,2,5)$ is $=\sqrt{(a-3)^2+4+0}=4(a-3)^2+4=16 \Rightarrow a^2-6a-3=0 \Rightarrow$ sum of values of $a=6$

Q. 16

Let O be the origin, $\overrightarrow{\mathrm{OP}}=\vec{\mathrm{a}}$ and $\overrightarrow{\mathrm{OQ}}=\vec{\mathrm{b}}$. If R is the point on $\overrightarrow{\mathrm{OP}}$ such that $\overrightarrow{\mathrm{OP}}=5\overrightarrow{\mathrm{OR}}$, and M is the point such that $\overrightarrow{\mathrm{OQ}}=5\overrightarrow{\mathrm{RM}}$, then $\overrightarrow{\mathrm{PM}}$ is equal to:

Option 1:
$\frac{1}{5}(\vec{a}-4\vec{b})$

Option 2:
$\frac{1}{5}(\vec{b}-4\vec{a})$

Option 3:
$\frac{1}{5}(-\vec{a}+4\vec{b})$

Option 4:
$\frac{1}{5}(-\vec{b}+4\vec{a})$

Correct Answer:
$\frac{1}{5}(\vec{b}-4\vec{a})$

Solution:

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$$\overrightarrow{OR} = \frac{\overrightarrow{OP}}{5} = \frac{\vec{a}}{5}$$

$$\overrightarrow{RM} = \frac{\overrightarrow{OQ}}{5} = \frac{\vec{b}}{5}$$

$$\overrightarrow{OM} - \overrightarrow{OR} = \frac{\vec{b}}{5}$$

$$\overrightarrow{OM} = \frac{\vec{a} + \vec{b}}{5}$$

$$\overrightarrow{PM} = \overrightarrow{OM} - \overrightarrow{OP}$$

$$\frac{\vec{a} + \vec{b}}{5} - \vec{a}$$

$$= \frac{\vec{b} - 4\vec{a}}{5}$$

Q. 17 Let $f(x) = \lim_{y \to 0} \frac{(1-\cos(xy))\tan(xy)}{y^3}$. Then the number of solutions of the equation $f(x) = \sin x, x \in R$ is :

Option 1:
0

Option 2:
2

Option 3:
3

Option 4:
1

Correct Answer:
3

Solution:

$$f(x) = \lim_{y \to 0} \frac{1 - \cos(xy)}{(xy)^2} \cdot \frac{\tan xy}{xy} \frac{(xy)^3}{y^3}$$

$$f(x) = \frac{x^3}{2}$$

$$f(x) = \sin x$$

$$\frac{x^3}{2} = \sin x$$

3 points of intersection

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Q. 18 Let $\left(2^{1-a}+2^{1+a}\right), f(a), \left(3^{a}+3^{-a}\right)$ be a A. P. and α be the minimum value of $f(a)$. Then the value of the integral $\int_{\log_e(\alpha-1)}^{\log_a(\alpha)} \frac{dx}{(e^{2x}-e^{-2x})}$ is :

Option 1:
$\frac{1}{2}\log_e\left(\frac{4}{3}\right)$

Option 2:
$\frac{1}{4}\log_e\left(\frac{4}{3}\right)$

Option 3:
$\frac{1}{2}\log_e\left(\frac{8}{5}\right)$

Option 4:
$\frac{1}{4}\log_e\left(\frac{8}{5}\right)$

Correct Answer:
$\frac{1}{4}\log_e\left(\frac{4}{3}\right)$

Solution:

$$f(a)=\frac{1}{2}\left(2\left(2^{a}+2^{-a}\right)+\left(3^{a}+3^{-a}\right)\right)$$

$$a=\frac{1}{2}(2\times 2+2)=3$$

$$I=\int_{\ln(3-1)}^{\ln 3}\frac{e^{2x}}{e^{4x}-1}dx$$

$$\text{put } e^{2x}=t$$

$$I=\frac{1}{2}\int_{4}^{9}\frac{dt}{t^2-1}$$

$$I=\frac{1}{4}\left(\ln\left(\frac{t-1}{t+1}\right)\right)_4^9$$

$$I=\frac{1}{4}\ln\left(\frac{4}{3}\right)$$

Q. 19 Let $f:[1,\infty)\to R$ be a differentiable function defined as $f(x)=\int_1^x f(t)dt+(1-x)\left(\log_e x-1\right)+e.$

Then the value fo $f(f(1))$ is :

Option 1:
$(1+e^{e})$

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Option 2:
$(1+\mathrm{e})$

Option 3:
$(1+e+e^e)$

Option 4:
$1+2e$

Correct Answer:
$(1+e^e)$

Solution:

$$f(1)=\int_1^1 f(t)dt+(1-1)(\ln 1-1)+e$$

$$f(1)=e$$

$$f(x)=\int_1^x f(t)dt+(1-x)(\ln x-1)+e$$

diff. w.r.t. x

$$f'(x)=f(x)+\frac{1}{x}-\ln x$$

$$\text{I.F.}=e^{\int -1dx}=e^{-x}$$

$$f(x)e^{-x}=\int e^{-x}\left(\frac{1}{x}-\ell \ln x\right)dx$$

$$f(x)e^{-x}=e^{-x}\ln x+C$$

$$\text{put } x=1, y=e$$

$$\Rightarrow C=1$$

$$f(x)e^{-x}=e^{-x}\ln x+1$$

$$f(x)=e^x+\ln x$$

$$f(e)=e^e+1$$

Q. 20 Let $f(x)$ and $g(x)$ be twice differentiable functions satisfying $\mathrm{f''(x)=g''(x)}$ for all $\mathrm{x\in R}$, $f'(1)=2g'(1)=4$ and $g(2)=3f(2)=9$. Then $\mathrm{f(25)-g(25)}$ is equal to :

Option 1:
20

Option 2:
40

Option 3:
-20

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Option 4:
-40

Correct Answer:
40

Solution:
$\mathrm{f''(x) = g''(x)}$

Integrate

$\mathrm{f'(x) = g'(x) + C_1}$

Put $x = 1$

$f'(1) = g'(1) + C_1$
$4 = 2 + C_1$
$C_1 = 2$
$\Rightarrow f'(x) = g'(x) + 2$
$f'(x) = g(x) + 2x + C_2$

Put $x = 2$

$f(2) = g(2) + 4 + C_2$
$3 = 9 + 4 + C_2$
$C_2 = -10$
$f(x) = g(x) + 2x - 10$

Put $x = 25$

$f(x) - g(25) = 40$

Q. 21 Let $\mathrm{A} = \{1, 4, 7\}$ and $\mathrm{B} = \{2, 3, 8\}$. Then the number of elements, in the relation $R = \{((a_1, b_1), (\mathrm{a_2},\ \mathrm{b_2})) \in ((\mathrm{A} \times \mathrm{B}) \times (\mathrm{A} \times \mathrm{B})) : \mathrm{a_1 + b_2}$ divides $a_2 + b_1\}$ is ____

Correct Answer:
18

Solution:

A/B	2	3	8
1	3	4	9
4	6	7	12
7	9	10	15

These are possible sum of $\mathrm{a_i + b_j}$

Now $\mathrm{b_1 + a_2}$ divides $\mathrm{a_1 + b_2}$ So,

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$a_1 + b_2$	$a_2 + b_1$	No. of pairs
15	3, 15	2
12	3, 4, 6, 12	4
10	10	1
9	3, 9	6
7	7	1
6	3, 6	2
4	4	1
3	3	1
		18

Number of relations = 18

Q. 22 From the point $(-1, -1)$, two rays are sent making angles of 45° with the line $x + y = 0$. These rays get reflected from the mirror $x + 2y = 1$. If the equations of the reflected rays are $ax + by = 9$ and $cx + dy = 7, a, b, c, d \in Z$, then the value of $ad + bc$ is _____

Correct Answer:
7

Solution:

$$\tan 45^\circ = \left|\frac{m+1}{1-m}\right|$$

$$\Rightarrow \pm 1 = \frac{m+1}{1-m}$$

$\Rightarrow m = 0$ \& one line is perpendicular to x axis.

$L_1 : x = -1, L_2 : y = -1$

Now taking mirror image of $x + 1 = 0$ in $x + 2y = 1$

$$A' : \frac{x+1}{1} = \frac{y+1}{2} = \frac{-2(-1-2-1)}{1^2+2^2}$$

$$x = \frac{3}{5}, y = \frac{11}{5} A'\left(\frac{3}{5}, \frac{11}{5}\right)$$

$$\text{line } = 3x - 4y + 7 = 0$$

Now taking mirror image of $y + 1 = 0$ in $x + 2y = 1$

$$A'' : \frac{x+1}{1} = \frac{y+1}{2} = \frac{-2(-1-2-1)}{1^2+2^2}$$

$$x = \frac{3}{5}, y = \frac{11}{5}$$

$$A''\left(\frac{3}{5}, \frac{11}{5}\right)$$

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line : $(y+1)=\frac{\left(\frac{11}{5}+1\right)}{\left(\frac{3}{5}-3\right)}(x-3)$

$4x+3y=9$

comparing $4\mathrm{x}+3\mathrm{y}=9$ with $\mathrm{ax}+$ by $=9 \&$

$a=4, b=3$

by line

$-3x+4y=7$

comparing with

$cx+dy=7$
$\therefore c=-3, d=4$

So, $\mathrm{ad}+\mathrm{bc}$

$=4\times 4+3\times -3$
$=16-9=7$

Q. 23 If $S=\left\{\theta\in[-\pi,\pi]:\cos\theta\cos\frac{5\theta}{2}=\cos 7\theta\cos\frac{7\theta}{2}\right\}$, then $\mathrm{n(S)}$ is equal to _____

Correct Answer:
19

Solution:
$$\cos\theta\cos\frac{5\theta}{2}=\cos 7\theta\cos\frac{7\theta}{2}$$
$$\cos\frac{7\theta}{2}+\cos\frac{3\theta}{2}=\cos\frac{21\theta}{2}+\cos\frac{7\theta}{2}$$
$$\cos\frac{21\theta}{2}-\cos\frac{3\theta}{2}=0$$
$$-2\sin 6\theta\sin\frac{9\theta}{2}=0$$
$$\sin 6\theta=0$$
$$\theta=0,\pm\frac{\pi}{6},\pm\frac{2\pi}{6},\ldots\pm\frac{5\pi}{6},\pm\pi \quad (13 \text{ solutions})$$
$$\sin\frac{9\theta}{2}=0$$
$$\theta=\pm\frac{2\pi}{9},\pm\frac{4\pi}{9},\pm\frac{8\pi}{9} \quad (6 \text{ more solutions})$$
$$\text{Total } =19 \text{ solutions}$$

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Q. 24 Let $f : R \to R$ be a function such that $\mathrm{f(x)} + 3\mathrm{f}\left(\frac{\pi}{2} - \mathrm{x}\right) = \sin \mathrm{x}, \mathrm{x} \in \mathrm{R}$. Let maximum value of f on R be α. If the area of the region bounded by the curves $\mathrm{g(x)} = \mathrm{x}^2$ and $\mathrm{h(x)} = \beta \mathrm{x}^3$, $\beta > 0$, is α^2, then $30\beta^3$ is equal to .....

Correct Answer:
16

Solution:
$f(x) + 3f\left(\frac{\pi}{2} - x\right) = \sin x$

Put $x \to \frac{\pi}{2} - x$

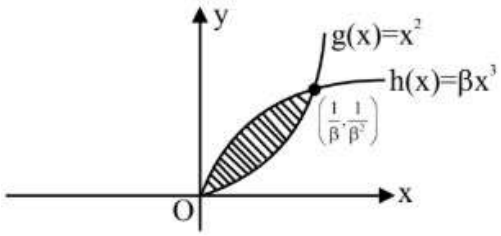
$\Rightarrow f\left(\frac{\pi}{2} - x\right) + 3f(x) = \cos x$

From (1) \& (2)

$f(x) = \frac{1}{8}(3\cos x - \sin x)$

$f_{\max} = \frac{\sqrt{10}}{8}$

$\mathrm{y = g(x) \& y = h(x)}$ intersect as shown in the figure



$\therefore$ Area bounded $= \Delta = \left|\int_0^{\frac{1}{\beta}} \left(\beta x^3 - x^2\right) dx\right|$

$= \frac{1}{12\beta^3} = \alpha^2$ (given)

$\Rightarrow 30\beta^3 = 16$

Q. 25 Let $y = y(x)$ be the solution of the differential equation $(\tan x)^{1/2} dy = \left(\sec^3 x - (\tan x)^{3/2} y\right) dx, 0 < x < \frac{\pi}{2}, y\left(\frac{\pi}{4}\right) = \frac{6\sqrt{2}}{5}$. If $y\left(\frac{\pi}{3}\right) = \frac{4}{5}\alpha$, then α^4 equals .....

Correct Answer:
48

Solution:

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$$\frac{dy}{dx}+y\tan x=\frac{\sec^3 x}{\sqrt{\tan x}}$$

$$\text{IF }=e^{\int \tan x dx}=\sec x$$

$$y\sec x=\int \frac{\sec^4 x}{\sqrt{\tan x}}dx$$

$$y\sec x=\int \frac{\left(1+\tan^2 x\right)}{\sqrt{\tan x}}\sec^2 x dx$$

$$\tan x=t$$

$$y\sec x=2\sqrt{\tan x}+\frac{2}{5}(\tan x)^{5/2}+c$$

$$y\left(\frac{\pi}{4}\right)=\frac{6\sqrt{2}}{5}$$

$$\frac{6\sqrt{2}}{5}\times\sqrt{2}=2+\frac{2}{5}+c\Rightarrow\frac{12}{5}=\frac{12}{5}+c$$

$$\Rightarrow c=0$$

$$y(\pi/3)=\frac{8}{5}\cdot 3^{1/4}\Rightarrow\alpha=2\times 3^{\frac{1}{4}}$$

$$\therefore \alpha^4=48$$

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JEE MAINS

2026

6th April

1st Shift

Question Paper

Physics

Q. 1 The density ρ of a uniform cylinder is determined by measuring its mass m , length l and diameter d . The measured values of m, l and d are $97.42 \pm 0.02\ \mathrm{g}, 8.35 \pm 0.05\ \mathrm{mm}$ and $20.20 \pm 0.02\ \mathrm{mm}$, respectively. Calculated percentage fractional error in ρ is _____

Option 1:
0.63 %

Option 2:
0.82%

Option 3:
0.72 %

Option 4:
25 %

Correct Answer:
0.82%

Solution:

$$\mathrm{m} = 97.42 \pm 0.02\ \mathrm{g}$$
$$\mathrm{l} = 8.35 \pm 0.05\ \mathrm{mm}$$
$$\mathrm{d} = 20.20 \pm 0.02\ \mathrm{mm}$$
$$\rho = \frac{\mathrm{m}}{\frac{\pi}{4}\,\mathrm{d}^2 \ell}$$
$$\Rightarrow \frac{\Delta\rho}{\rho} = \frac{\Delta \mathrm{m}}{\mathrm{m}} + \frac{2\Delta\ \mathrm{d}}{\mathrm{d}} + \frac{\Delta\ell}{\ell}$$
$$= \frac{0.02}{97.42} + 2 \times \frac{0.02}{20.20} + \frac{0.05}{8.35} = 0.0082$$
$$\%\ \text{error} = 0.82\%$$

Q. 2 The potential energy of a particle changes with distance x from a fixed origin as $V = \frac{A\sqrt{x}}{x+B}$, where A and B are constant with appropriate dimensions. The dimension of AB are _____

Option 1:
$\left[\mathrm{M}^1\ \mathrm{L}^{5/2}\ \mathrm{T}^{-2}\right]$

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Option 2:
$\left[\mathrm{M}^{3/2}\ \mathrm{L}^{5/2}\ \mathrm{T}^{-2}\right]$

Option 3:
$\left[\mathrm{M}^{1}\ \mathrm{L}^{2}\ \mathrm{T}^{-2}\right]$

Option 4:
$\left[\mathrm{M}^{1}\ \mathrm{L}^{7/2}\ \mathrm{T}^{-2}\right]$

Correct Answer:
$\left[\mathrm{M}^{1}\ \mathrm{L}^{7/2}\ \mathrm{T}^{-2}\right]$

Solution:
$[\mathrm{B}]=[\mathrm{x}]=\mathrm{L}$

$[\mathrm{U}]=\left[\frac{\mathrm{A}\sqrt{\mathrm{x}}}{\mathrm{B}+\mathrm{x}}\right] \Rightarrow \mathrm{ML}^{2}\ \mathrm{T}^{-2}=\frac{[\mathrm{A}]\mathrm{L}^{1/2}}{\mathrm{L}}$

$[\ \mathrm{A}]=\mathrm{ML}^{5/2}\ \mathrm{T}^{-2}$

Q. 3 The rain drop of mass 1 g , starts with zero velocity from a height of 1 km . It hits the ground with a speed of $5\ \mathrm{m/s}$. The work done by the unknown resistive force is _____ J . (take $\mathrm{g}=10\ \mathrm{m/s^2}$)

Option 1:
-8.75

Option 2:
-8.35

Option 3:
-9.55

Option 4:
-9.98

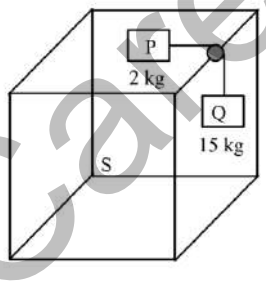
Correct Answer:
-9.98

Solution:

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$$w_g + w_{res} = \Delta k$$
$$mgh + w_{res} = \frac{1}{2}mv^2$$
$$\therefore w_{res} = \frac{1}{2}mv^2 - mgh$$
$$= 10^{-3}\left(\frac{25}{2} - 10^4\right)$$
$$= -9.98\ J$$

Q. 4 Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pully which is fixed on the edge of a cube (S), as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ. The cube (S) moves towards the right with acceleration of $\frac{g}{2}$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is _____ . (take $g = 10\ m/s^2$)



Option 1:
0.33

Option 2:
0.67

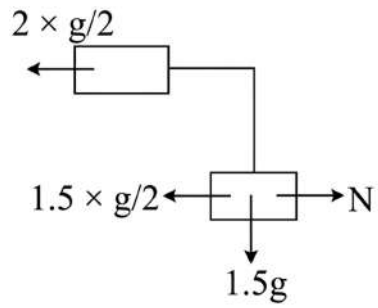
Option 3:
1

Option 4:
0.5

Correct Answer:
0.67

Solution:
w.r.t bigger block

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$f = \mu N = \mu \times \frac{1.5g}{2}$

$1.5g = g + \mu + \frac{1.5g}{2} \quad \Rightarrow 0.5 = \mu \times \frac{1.5}{2}$

$\left(\mu = \frac{2}{3}\right)$

Q. 5 A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is $4 \times 10^8\ \mathrm{N/m^2}$ and its radius is 4 mm , then maximum acceleration the lift can take is

____ $\mathrm{m/s^2}$. (take $\mathrm{g} = 10\ \mathrm{m/s^2}$ and $\pi = 3.14$)

Option 1:
2.56

Option 2:
3.89

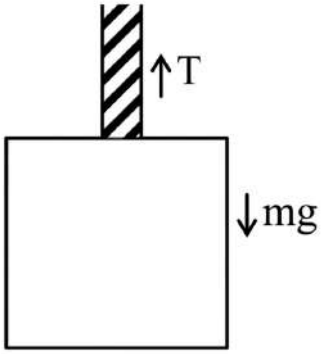
Option 3:
4.32

Option 4:
5.16

Correct Answer:
2.56

Solution:

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$T - mg = ma$

$T - mg = ma$

$(\sigma\pi r^2) - mg = ma$

$\Rightarrow 4 \times 10^8 \pi (4 \times 10^{-2})^2 - 16000 = 1600a$

$\Rightarrow a = 2.56\ m/s^2$

Q. 6 A solid sphere of radius 4 cm and mass 5 kg is rotating (rotation axis is passing through the centre of the sphere) with an angular velocity of 1200 rpm . It is brought to rest in 10 s by applying a constant torque. The torque applied and the number of rotations it made before it comes to rest are _____ and _____ respectively.

Option 1:
$0.128\pi Nm, 100$

Option 2:
$0.0128\pi Nm, 50$

Option 3:
$0.128\pi Nm, 50$

Option 4:
$0.0128\pi Nm, 100$

Correct Answer:
$0.0128\pi Nm, 100$

Solution:

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$$\omega_{\mathrm{i}}=1200 \mathrm{rpm}=1200 \times \frac{2 \pi}{60}=40 \pi \mathrm{rad} / \mathrm{sec}$$

$$\rightarrow \omega_{\mathrm{f}}=\omega_{\mathrm{i}}+\alpha \mathrm{t} \Rightarrow 0=40 \mathrm{p}+\alpha(10)$$

$$\alpha=-4 \pi \mathrm{rad} / \mathrm{sec}^2$$

$$\rightarrow \mathrm{I}=\frac{2}{5} \mathrm{MR}^2=\frac{2}{5}(5)(0.04)^2=2 \times 0.0016$$

$$\mathrm{I}=0.0032 \mathrm{Kg}-\mathrm{m}^2$$

$$\rightarrow \tau=\mathrm{I} \alpha=0.0032 \times 4 \pi$$

$$\tau=0.0128 \pi \mathrm{~N}-\mathrm{m}$$

$$\rightarrow \omega_{\mathrm{f}}^2=\omega_{\mathrm{i}}^2+2 \alpha \theta$$

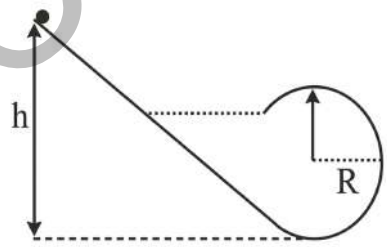
$$0=(40 \pi)^2+2(4 \pi) \theta$$

$$\theta=\frac{1600 \pi^2}{8 \pi}=200 \pi$$

$$\text{No. of revolution } =\frac{\theta}{2 \pi}=\frac{200 \pi}{2 \pi}$$

$$\text{No. of revolution } =100$$

Q. 7 A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height h as shown. If the body exerts a force of three times its weight on the plane at the highest point of circle then the height $h=\alpha R$. The value of α is



Option 1:
2

Option 2:
4

Option 3:
3

Option 4:
6

Correct Answer:
4

Solution:
at highest point

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$$mg + N = \frac{mv^2}{R}$$

$$4mg = \frac{mv^2}{R}$$

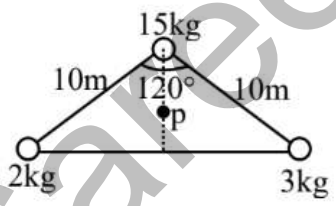
$$v = \sqrt{4gR}$$

Now energy conservation

$$mg(h - 2R) = \frac{1}{2} m \times 4gR$$

$$(h = 4R)$$

Q. 8 The position of center of mass of three masses2 kg, 3 kg and 15 kg placed with respect to midpoint (p) of normal bisector, as shown in the figure is ____



Option 1:
$\left(\frac{\sqrt{3}}{4}, 1.25\right)$

Option 2:
$\left(\frac{\sqrt{3}}{4}, 1.0\right)$

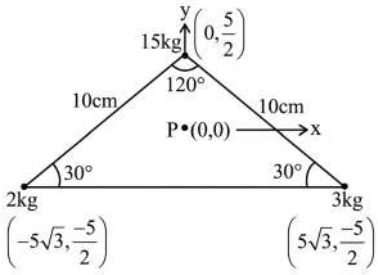
Option 3:
$(0, 0)$

Option 4:
$(1.25, 0)$

Correct Answer:
$\left(\frac{\sqrt{3}}{4}, 1.25\right)$

Solution:

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$$X_{cm}=\frac{2\times(-5\sqrt{3})+3(5\sqrt{3})+15\times 0}{20}=\frac{\sqrt{3}}{4}\ \text{cm}$$

$$Y_{cm}=\frac{2\times(-5/2)+3(5/2)+15\times(5/2)}{20}$$

$$Y_{cm}=1.25\ \text{cm}$$

Q. 9 The two wires A and B of equal cross-section but of different materials are joined together. The ratio of Young's modulus of wire A and wire B is 20/11. When the joined wire is kept under certain tension the elongations in the wires A and B are equal. If the length of wire A is 2.2 m , then the length of wire B is _____ m.

Option 1:
1.1

Option 2:
2.22

Option 3:
1.21

Option 4:
4.44

Correct Answer:
1.21

Solution:

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$$\Delta \ell = \frac{\mathrm{T}\ell}{\mathrm{YA}}$$
$$\ell \propto \mathrm{Y}$$
$$\frac{\ell_1}{\ell_2} = \frac{\mathrm{Y}_1}{\mathrm{Y}_2} = \frac{20}{11}$$
$$\because \ell_1 = 2.2\ \mathrm{m}$$
$$\therefore \ell_1 = 1.21\ \mathrm{m}$$

Q. 10 Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K . Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K . The final pressure in the vessels is
_____ kPa .

Option 1:
100

Option 2:
120

Option 3:
90

Option 4:
105

Correct Answer:
100

Solution:
Using mole conservation

$$\frac{2 \times 9 \times 10^4 \times V}{R \times 400} = \frac{P \times V}{R \times 400} + \frac{P \times V}{R \times 500}$$
$$P = 100\mathrm{kPa}$$

Q. 11 In interference experiment the path difference between two interfering waves at a point A on the screen is $\lambda/3$, where λ is the wavelength of these waves, and at another point B the path difference is $\lambda/6$. The ratio of intensities at points A and B is
_____ .

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Option 1:
3

Option 2:
4

Option 3:
$1/3$

Option 4:
$1/4$

Correct Answer:
$1/3$

Solution:

$$\Delta\phi_{\mathrm{A}} = \frac{2\pi}{\lambda} \times \frac{\lambda}{3} = \frac{2\pi}{3}$$

$$\therefore \mathrm{I_A} = \mathrm{I} + \mathrm{I} + 2\sqrt{\mathrm{I}} \cdot \sqrt{\mathrm{I}} \cdot \cos\left(\frac{2\pi}{3}\right) = \mathrm{I}$$

$$\Delta\phi_{\mathrm{B}} = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$$

$$\therefore \mathrm{I_B} = \mathrm{I} + \mathrm{I} + 2\sqrt{\mathrm{I}} \cdot \sqrt{\mathrm{I}} \cdot \cos\left(\frac{\pi}{3}\right) = 3\mathrm{I}$$

$$\frac{\mathrm{I_A}}{\mathrm{I_B}} = \frac{\mathrm{I}}{3\mathrm{I}} = \frac{1}{3}$$

Q. 12 A particle is executing simple harmonic motion. Its amplitude is A and time period is 5 sec . the time required by it to move from $x = A$ to $x = \frac{A}{\sqrt{2}}$ is _____ sec.

Option 1:
$\frac{1}{4}$

Option 2:
$5/4$

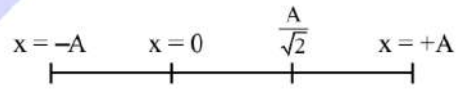
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Option 3:
$5/8$

Option 4:
$3/8$

Correct Answer:
$5/8$

Solution:



$x = 0 \rightarrow \theta = 0^{\circ}$

$x = \dfrac{A}{\sqrt{2}} \rightarrow \theta = \dfrac{\pi}{4}$

$\therefore$ phase covered $= \dfrac{\pi}{4}$

$\text{time} = \dfrac{\text{phase}}{w} = \dfrac{\pi}{4w} = \dfrac{\pi}{4} \times \dfrac{T}{2\pi} = \dfrac{T}{8}$

$\therefore t = \dfrac{5}{8}$ secs.

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Q. 13 A thin half ring of radius 35 cm is uniformly charged with a total charge of Q coulomb. If the magnitude of the electric field at centre of the half ring is $100\ \mathrm{V/m}$, then the value of Q is _____ nC . ($\epsilon_0 = 8.85 \times 10^{-12}\mathrm{C^2/Nm^2}$ and $\pi = 3.14$)

Option 1:
2.14

Option 2:
2.44

Option 3:
3.25

Option 4:
0.7

Correct Answer:
2.14

Solution:

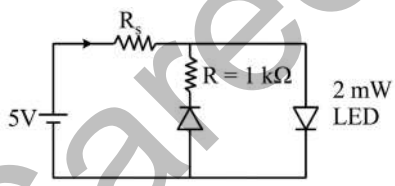
$$\mathrm{E}=\frac{2\mathrm{k}\lambda}{\mathrm{R}}\sin\left(\frac{\theta}{2}\right)$$

$$100=2\times\frac{9\times10^{9}\times\mathrm{Q}}{\pi\mathrm{R}^{2}}$$

$$\mathrm{Q}=2.14\times10^{-9}\mathrm{C}$$

Q. 14 The maximum rated power of the LED is 2 mW and it is used in the circuit with input voltage of 5 V as shown in the figure below. The current through resistance $\mathrm{R_s}$ is 0.5 mA .

The minimum value of the resistance of R_s, to ensure that the LED is not damaged is _____ $\mathrm{k\Omega}$.



Option 1:
6

Option 2:
2

Option 3:
4

Option 4:
5

Correct Answer:
2

Solution:
Diode in reverse bias so no current in it.

$$P_{bulb}=2\times10^{-3}=V_{bulb}$$
$$v_{bulb}=4V$$

So $5\ \mathrm{V}=\mathrm{V_{RS}}+\mathrm{V_{bulb}}$

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$V_{rs} = 5 - 4 = 1$
$1 = 0.5 \times 10^{-3} R_s \Rightarrow R_s = 2\text{k}\Omega$

Q. 15 A point light source emits E.M. waves in free space. A detector, placed at a distance of L m , measures the intensity as I_0. The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45°. The measured intensity at new location will be _____ .

Option 1:
$\frac{I_0}{4}$

Option 2:
I_0

Option 3:
$\frac{I_0}{\sqrt{2}}$

Option 4:
$\frac{I_0}{2}$

Correct Answer:
I_0

Solution:
For a point source emitting electromagnetic waves uniformly in free space, intensity at a distance rrr is given by

$I = \frac{P}{4\pi R^2}$

At the initial position of the detector,

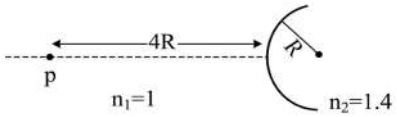
$I_0 = \frac{P}{4\pi L^2}$

The detector is shifted to another point on the same spherical surface, so its distance from the source remains L. Since intensity depends only on distance from the source and not on direction, the intensity remains unchanged even though the angle between the two positions is 45°.

$$\therefore I' = \frac{P}{4\pi L^2} = I_0$$

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Q. 16 A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of 4R in front of spherical interface. The magnitude of the magnification of point source image is _____ .



Option 1:
1.66

Option 2:
2.33

Option 3:
2.66

Option 4:
1.33

Correct Answer:
1.66

Solution:

$$m = \frac{n_1}{n_2}\frac{v}{u}$$

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

$$\frac{1.4}{v} + \frac{1}{4R} = \frac{1.4 - 1}{R}$$

$$\frac{1.4}{v} = \frac{4}{10R} - \frac{1}{4R}$$

$$v = \frac{56R}{6} = +\frac{28}{3}R$$

$$m = \frac{1}{1.4} \cdot \frac{28R/3}{(-4R)} = -1.67$$

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Q. 17 A small cube of side 1 mm is placed at the centre of a circular loop of radius 10 cm carrying a current of 2 A . The magnetic energy stored inside the cube is $\alpha \times 10^{-14}$ J. The value of α is ——— .

$\left(\mu_0 = 4\pi \times 10^{-7}\mathrm{Tm/A}, \pi = 3.14\right)$

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Option 1:
6.28

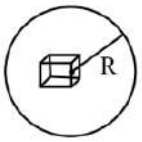
Option 2:
6.28×10^{-6}

Option 3:
628

Option 4:
6.28×10^{-4}

Correct Answer:
6.28

Solution:
Magnetic field at centre $B = \left(\frac{\mu_0}{4\pi}\right)\frac{2\pi i}{R}$

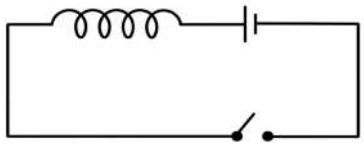


$= \frac{10^{-7}\times 2\times\pi\times 2}{10\times 10^{-2}}$

So energy stored

$$\mathrm{U} = \frac{\mathrm{B}^2}{2\mu_0} \times \mathrm{V}$$
$$= \frac{10^{-14} \times 4 \times \pi^2 \times 4 \times 100}{2 \times 4\pi \times 10^{-7}} \times \left(1 \times 10^{-9}\right)$$
$$= 2\pi \times 10^{-14}\ \mathrm{J}$$
$$= 6.28 \times 10^{-14}\ \mathrm{J}$$

Q. 18 An inductor of inductance 10 mH having resistance of 100Ω is connected to battery of E.M.F. 1.0 V through a switch as shown in the figure below. After switch is closed, the ratio of instantaneous voltages across the inductor when the current passing through it is 2 mA and 4 mA is _____ .



Option 1:
4/3

Option 2:
3/4

Option 3:
5/3

Option 4:
3/5

Correct Answer:
4/3

Solution:
Apply KVL

$$\mathrm{E}-\mathrm{iR}-\mathrm{V_L}=0$$
$$\mathrm{V_L}=\mathrm{E}-\mathrm{iR}=1-100\mathrm{i}$$
$$\text{at } \mathrm{i_1}=2\times10^{-3}\ \mathrm{A}$$
$$\mathrm{V_{L_1}}=0.8\ \mathrm{V}$$
$$\text{at } \mathrm{i_2}=4\times10^{-3}\ \mathrm{A}$$
$$\mathrm{V_{L_2}}=0.6\ \mathrm{V}$$
$$\therefore \frac{\mathrm{V_{L_1}}}{\mathrm{V_{L_2}}}=\frac{4}{3}$$

Q. 19 The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β. The possible values of α and β are :-

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Option 1:
27 and 20

Option 2:
3 and 16

Option 3:
5 and 36

Option 4:
20 and 27

Correct Answer:
20 and 27

Solution:

$$p = \frac{h}{\lambda}$$

$$\frac{1}{\lambda_1} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{1}{4} - \frac{1}{9}\right)$$

$$\frac{1}{\lambda_1} = R\frac{5}{36}$$

$$\frac{1}{\lambda_2} = R\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = R\left(\frac{1}{4} - \frac{1}{16}\right)$$

$$\frac{1}{\lambda_2} = \frac{3R}{16}$$

$$\frac{p_1}{p_2} = \frac{\lambda_2}{\lambda_1} = \frac{\frac{16}{3R}}{36} = \frac{20}{5R}$$

$$\alpha = 20 \& \beta = 27$$

Q. 20 A LCR series circuit driven with $E_{rms} = 90\ V$ at frequency $f_d = 30\ Hz$ has resistance $R = 80\Omega$, an inductance with inductive reactance $X_L = 20.0\Omega$ and capacitance with capacitive reactance $X_C = 80.0\Omega$. The power factor of the circuit is _____ .

Option 1:
0.8

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Option 2:
0.64

Option 3:
0.9

Option 4:
0.5

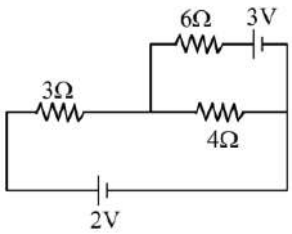
Correct Answer:
0.8

Solution:
Power factor is $\cos\phi = \frac{\mathrm{R}}{\mathrm{Z}}$

$$\mathrm{Z} = \sqrt{\mathrm{R}^2 + (\mathrm{X_L} - \mathrm{X_C})^2} = 100\Omega$$

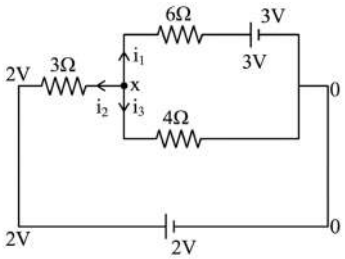
$$\cos\phi = \frac{80}{100} = 0.8$$

Q. 21 Refer to the circuit diagram given below. The heat generated across the 6Ω resistance in 100 second is $\frac{\alpha}{100}$ J. The value of α is _____ . (Nearest integer)



Correct Answer:
3477

Solution:



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$i_1 + i_2 + i_3 = 0$

$$\frac{x-0-3}{6} + \frac{x-2}{3} + \frac{x-0}{4} = 0$$

$$x = \frac{14}{9}V$$

Current across $6\Omega = \frac{3-(14/9)}{6} = \frac{13}{54}\ \mathrm{A}$

$$\mathrm{H} = \left(\frac{13}{54}\right)^2 \times 6 \times 100 = 34.77 = \frac{\alpha}{100}$$

$$\alpha = 3477$$

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Q. 22 An unpolarized light of intensity I_0 passes through polarizer and then through a certain optically active solution and finally it goes to analyser. If the angle between analyser and polariser is 0° and intensity of light emerged from analyser is $\frac{3}{8}\mathrm{I}_0$, the angle of rotation of the light by the solution with respect to analyser is _____ degrees.

Correct Answer:
30

Solution:

$$\underset{\text{(Unpolarised)}}{\mathrm{I}_0} \rightarrow \underset{\text{(Polarised)}}{\frac{\mathrm{I}_0}{2}} \xrightarrow{\text{Analyser}}$$

$$I = I' \cos^2\theta$$

$$\frac{3I_0}{8} = \frac{I_0}{2}\cos^2\theta$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ$$

Q. 23 The energy released when $\frac{7}{17.13}$ kg of $^7_3\mathrm{Li}$ is converted into $^4_2\mathrm{He}$ by proton bombardment is $\alpha \times 10^{32}\mathrm{eV}$. The value of α is _____ . (Nearest integer)
(Mass of $^7_3\mathrm{Li} = 7.0183\mathrm{u}$, mass of $^4_2\mathrm{He} = 4.004\mathrm{u}$, mass of proton $= 1.008u$ and $1u = 931\mathrm{MeV/c^2}$ and Avogadro number $= 6.0 \times 10^{23}$)

Correct Answer:
6

Solution:

$_{3}^{7}\mathrm{Li} + {}_{1}^{1}\mathrm{P} \rightarrow 2{}_{2}^{4}\mathrm{He}$
$\Delta\ \mathrm{m} = (7.018 + 1.008) - 2(4.004) = 0.0183\mathrm{u}$

Energy released per reaction $\mathrm{E} = \Delta\mathrm{mc}^2$
Number of moles of $\mathrm{Li} = \frac{7}{17.13} \times \frac{10^3}{7} = \frac{10^3}{17.13}$

$$\text{Number of molecules } = \frac{10^3}{17.13} \times 6 \times 10^{23}$$

Total energy released

$$\mathrm{E}_{\mathrm{Total}} = 0.0183 \times 931 \times \frac{6}{17.13} \times 10^{32}$$
$$\simeq 6 \times 10^{32}$$
$\alpha = 6$

Q. 24 A three coulomb charge moves from the point ($0, -2, -5$) to the point ($5, 1, 2$) in an electric field expressed as $\overrightarrow{\mathrm{E}} = 2x\hat{\mathrm{i}} + 3\mathrm{y}^2\hat{\mathrm{j}} + 4\hat{\mathrm{k}}\mathrm{N/C}$. The work done in moving the charge is _____ J.

Correct Answer:
186

Solution:
$\Delta U = -W_{\text{electrostatic}}$
$W_{\text{electrostatic}} = -q\Delta V$
$\Delta V = -\int 2x dx - \int 3y^2 dy - \int 4dz$
$= \left(-x^2 - y^3 - 4z\right)_{(0,-2,-5)}^{(5,1,2)}$
$= -62$
$W_{\text{electrostatic}} = 3(62) = 186\ \mathrm{J}$

Q. 25 A certain gas is isothermally compressed to $\left(\frac{1}{3}\right)^{\mathrm{rd}}$ of its initial volume ($\mathrm{V}_0 = 3$ litre) by applying required pressure. If the bulk modulus of the gas is $3 \times 10^5\ \mathrm{N/m^2}$, the magnitude of work done on the gas is _____ J.

Correct Answer:
989

Solution:
For isothermal process initially $\mathrm{B} = \mathrm{P}_0$.

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$$W = nRT\,n\left(\frac{V_2}{V_1}\right)$$

$$= P_0\,V_0 \ln\left(\frac{1}{3}\right)$$

$$W = BV_0 \ln\left(\frac{1}{3}\right)$$

$$W = -988.75\ J \approx 989\ J$$

Note : Bulk moldulus of gas for isothermal process is not constant. In the given solution we have considered it as initial bulk modulus.

Chemistry

Q. 1 An oxide of iron contains 69.9% iron, its empirica formula, is :
(Given : Molar mass of Fe and O are 56 anc $16\ g\ mol^{-1}$ respectively.)

Option 1:
FeO

Option 2:
Fe_2O_3

Option 3:
Fe_3O_4

Option 4:
FeO_3

Correct Answer:
Fe_2O_3

Solution:

	Fe	Oxygen
Moles :	$\frac{69.9}{56}$	$\frac{30.1}{16}$

Molar Ratio : 1 : 1.5

Molar Ratio : 2 : 3

Answer : Fe2O3

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Q. 2 If shortest wavelength of hydrogen atom in Lyman series is x , then longest wavelength in Balmer series of $\mathrm{He}^{+}$is :

Option 1:
$\frac{9x}{5}$

Option 2:
$\frac{36x}{5}$

Option 3:
$\frac{x}{4}$

Option 4:
$\frac{5x}{9}$

Correct Answer:
$\frac{9x}{5}$

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Solution:

$\frac{1}{\mathrm{x}} = \mathrm{R} \times 1^2 \times \left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$ (1)

$\frac{1}{\lambda} = \mathrm{R} \times 2^2 \times \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$ (2)

$$\frac{\mathrm{x}}{\lambda} = \mathrm{R} \times 4 \times \frac{5}{4 \times 9}$$

$$\frac{1}{\lambda} = \frac{1}{\mathrm{x}} \times \frac{5}{9}$$

$$\lambda = \frac{9\mathrm{x}}{5}$$

Q. 3 Match the List-I with List-II :

List-I Orbital		List-II Radial nodes and nodal plane	
A	2s	I	1 Radial node + two nodal planes
B	3s	II	1 Radial node + one nodal plane
C	3p	III	2 Radial nodes + No nodal plane
D	4d	IV	1 Radial node + No nodal plane.

Choose the correct answer from the options given below:

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Option 1:
A-IV, B-I, C-III, D-II

Option 2:
A-IV, B-II, C-III, D-I

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-IV, B-III, C-II, D-I

Correct Answer:
A-IV, B-III, C-II, D-I

Solution:
No. of Radial nodes $= n - \ell - 1$
No. of Angular nodes or nodal planes $= \ell$

$2\,\mathrm{s} \Rightarrow \mathrm{n} = 2, \ell = 0$

1 radial node + zero nodal plane

$3\,\mathrm{s} \Rightarrow \mathrm{n} = 3, \ell = 0$

2 radial node + zero nodal plane

$3\mathrm{p} \Rightarrow \mathrm{n}=3, \ell=1$

1 radial node +1 nodal plane

$4d \Rightarrow n=4, \ell=2$

1 radial node +2 nodal plane

Q. 4 The pairs among

$\mathrm{A}=\left[\mathrm{SO}_3^{2-}, \mathrm{CO}_3^{2-}\right], \mathrm{B}=\left[\mathrm{O}_2^{2-}, \mathrm{F}_2\right], \mathrm{C}=\left[\mathrm{CN}^{-}, \mathrm{CO}\right],$

$\mathrm{D}=\left[\mathrm{NH}_3, \mathrm{H}_3\mathrm{O}^{+}\right]$and $\mathrm{E}=\left[{\mathrm{MnO}_4}^{2-}, {\mathrm{CrO}_4}^{2-}\right]$ that do not have similar Lewis dot structure are :

Option 1:
A, B and E

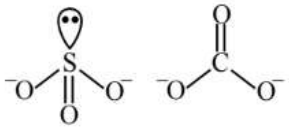
Option 2:
A and E

Option 3:
B, C and D

Option 4:
C and D

Correct Answer:
A and E

Solution:
A and E do not have similar Lewis structure



MnO_4^{2-} have are unpaired e but CrO_4^{2-} does not have unpaired e .
In $\mathrm{B}, \mathrm{C}, \mathrm{D}$ pair of species haved same no of e . They have similar Lewis dot structure.

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Q. 5 Arrange the following isothermal processes in order of the magnitude of the work $(p-V)$ involved between states 1 and 2 .

A. Expansion in single stage $\mathrm{w_A}$

B. Expansion in multi stages $\mathrm{w_B}$

C. Compression in single stage $\mathrm{w_C}$

D. Compression in multi stages $\mathrm{w_b}$

Option 1:

$|\mathrm{w_B}| > |\mathrm{w_A}| > |\mathrm{w_C}| > |\mathrm{w_D}|$

Option 2:

$|\mathrm{w_C}| > |\mathrm{w_D}| > |\mathrm{w_A}| > |\mathrm{w_B}|$

Option 3:

$|\mathrm{w_C}| > |\mathrm{w_D}| > |\mathrm{w_B}| > |\mathrm{w_A}|$

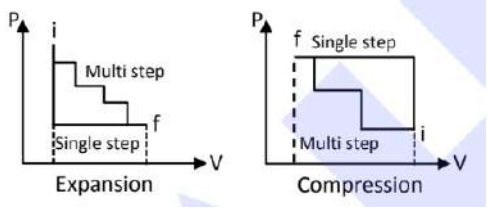
Option 4:

$|\mathrm{w_B}| > |\mathrm{w_A}| > |\mathrm{w_D}| > |\mathrm{w_C}|$

Correct Answer:

$|\mathrm{w_C}| > |\mathrm{w_D}| > |\mathrm{w_B}| > |\mathrm{w_A}|$

Solution:



$\therefore |w_A| < |w_B| < |w_D| < |w_C|$

Q. 6 When 0.25 moles of a non-volatile, non-ionizable solute was dissolved in 1 mole of a solvent the vapor pressure of solution was $x\%$ of vapor pressure of pure solvent. What is $\mathrm{x}\%$?

Option 1:

50%

Option 2:

60%

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Option 3:
70%

Option 4:
80%

Correct Answer:
80%

Solution:
$\mathrm{P_S} = \mathrm{P^0} \cdot \mathrm{X_A}$
$\frac{\mathrm{P_S}}{\mathrm{P^0}} = \mathrm{X_A} = \frac{1}{1+0.25} = 0.80$
$\% = 0.8 \times 100 = 80$

Q. 7 One mole each of He and $\mathrm{A(g)}$ are taken in a 10 L closed flask and heated to 400 K to establish the following equilibrium.

$$\mathrm{A(\ g) \rightleftharpoons B(\ g)}$$

$\mathrm{K_C}$ for this reaction at 400 K is 4.0. The partial pressures (in atm) of He and $\mathrm{B(g)}$ are respectively (at equilibrium).
(Assume $\mathrm{He}, \mathrm{A(g)}$ and $\mathrm{B(g)}$ behave as ideal gases) (Given : $\mathrm{R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}}$)

Option 1:
$3.28, 2.624$

Option 2:
$2.624, 3.28$

Option 3:
$3.28, 0.656$

Option 4:
$0.656, 6.56$

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Correct Answer:
$3.28, 2.624$

Solution:
$A \rightleftharpoons B$

$t = 0 \quad 1\ mol -$

$t_{eq^m} \quad 1 - x \quad x$

$$K_C = \frac{x/10}{\frac{1-x}{10}} = 4$$

$x = 0.8$

$$P_{He} = \frac{1 \times 0.082 \times 400}{10} = 3.28\ atm$$

$$P_B = \frac{0.8 \times 0.082 \times 400}{10} = 2.624\ atm$$

Q. 8 Consider the following data :

Electrolyte	$\Lambda_m^\circ\ (Scm^2\ mol^{-1})$
$BaCl_2$	X_1
H_2SO_4	X_2
HCl	X_3

$BaSO_4$ is sparingly soluble in water. If the conductivity of the saturated $BaSO_4$ solution is $x\ S\ cm^{-1}$ then the solubility product of $BaSO_4$ can be given as.
(Here $\wedge_m = \wedge_m^o$)

Option 1:
$\frac{10^6 x^2}{\alpha^2 (x_1 + x_2 - 2x_3)^2}$

Option 2:
$\frac{x^2}{(x_1 + x_2 - 2x_3)^2}$

Option 3:
$\frac{\alpha^2 (x_1 + x_2 - 2x_3)^2}{10^6 x^2}$

Option 4:
$\frac{x^2}{(x_1 + x_2 + 2x_3)^2}$

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Correct Answer:
$\frac{10^6 x^2}{\alpha^2(x_1+x_2-2x_3)^2}$

Solution:
$BaSO_4(s) \rightleftharpoons Ba^{+2}_{(aq)} + SO^{2-}_{4(aq)}$

$K_{sp} = S^2$

k of $BaSO_4 = xScm^{-1}$

Λ°_m of $BaSO_4 = (x_1 + x_2 - 2x_3)$

$\Lambda_m = \alpha\Lambda^\circ_M = \frac{k \times 1000}{S}$

$S = \frac{x \times 1000}{\alpha(x_1 + x_2 - 2x_3)}$

$k_{sp} = S^2 = \frac{1}{\alpha^2}\left[\frac{x \times 1000}{x_1 + x_2 - 2x_3}\right]^2$

Q. 9 Given below are two statements :
Statement-I : Aluminium is more electropositive than thallium as the standard electrode potential value of $E^\circ_{Al^{3+}/Al}$ is negative and $E^\circ_{Tl^{3+}/Tl}$ is positive.

Statement-II : The sum of first three ionization enthalpies of boron is very high when compared to that of aluminium. Due to this reason boron forms covalent compounds only and aluminium forms Al^{3+} ion.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Both Statement-I and Statement-II are true.

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Solution:
$\mathrm{E}^{\circ}_{\mathrm{Al}^{3+}/\mathrm{Al}} = -1.66\ \mathrm{V}$
$\mathrm{E}^{\circ}_{\mathrm{Tl}^{3+}/\mathrm{Tl}} = +1.26\ \mathrm{V}$

For B : IE $+IE_2 + IE_3 = 801 + 2427 + 3659$

$= 6887\ \mathrm{kJ}$

Both statements are true

Q. 10 The correct statements among the following are :
A. Basic vanadium oxide is used in the manufacture of $\mathrm{H_2SO_4}$.
B. The spin-only magnetic moment value of the transition metal halide employed in ZieglerNatta polymerization is 2.84 BM .
C. The p-block metal compound employed in Ziegler-Natta polymerization has the metal in +3 oxidation state.
D. The number of electrons present in the outer most ' d ' orbital of metal halide employed in Wacker process is 8 .
Choose the correct answer from the options given below :

Option 1:
A and B only

Option 2:
A, C and D only

Option 3:
C and D only

Option 4:
B, C and D only

Correct Answer:
C and D only

Solution:
A. $\mathrm{V_2O_5}$ is amphoteric. It is used in manufacture of $\mathrm{H_2SO_4}$.
B. Ziegler natta catalyst is $\underset{+4}{\mathrm{TiCl_4}} + \underset{+3}{\mathrm{AlMe_3}}$

$\mathrm{Ti}^{4+} : 3\ \mathrm{d}^0 4\ \mathrm{s}^0 \mathrm{n} = 0 \mu = 0$

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C. O.S. of Al is +3
D. $\frac{PdCl_2}{+2}$ is used in wacker process

$Pd^{2+} : 4\ d^8 5\ s^0$

C and D are correct.

Q. 11 Match the List-I with List-II

List-I Electronic configuration of tetrahedral metal ion		List-II Crystal Field Stabilization Energy (Δ_t)	
A	d^2	I	-0.6
B	d^4	II	-0.8
C	d^6	III	-1.2
D	d^8	IV	-0.4

Choose the correct answer from the options given below:

Option 1:
A-III, B-IV, C-II, D-I

Option 2:
A-III, B-I, C-IV, D-II

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-II, B-I, C-IV, D-III

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
In tetrahedral complex

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$$\text{CFSE} = \left[-\frac{3}{5}n_1 + \frac{2}{5}n_2\right]\Delta_t$$

$$d^2 : \underbrace{e^{1,1}}_{n_1=2}\underbrace{t_2^{0,0,0}}_{n_2=0};\ \text{C.F.S.E} = \left[\frac{-3}{5}(2)\right]\Delta t = -1.2\Delta t$$

$$d^4 : \underbrace{e^{1,1}}_{n_1=2}\underbrace{t^{1,1,0}}_{n_2=2};\ \text{C.F.S.E} = \left[\frac{-3}{5}(2) + \frac{2}{5}(2)\right]\Delta t = -0.4$$

Δt

$$d^6 : \underbrace{e^{2,1}}_{n_1=3}\underbrace{t_2^{1,1,1}}_{n_2=3};\ \text{C.F.S.E} = \left[\frac{-3}{5}(3) + \frac{2}{5}(3)\right]\Delta t = -0.6\Delta t$$

$$d^8 : \underbrace{e^{2,2}}_{n_1=4}\underbrace{t_2^{2,1,1}}_{n_2=4};\ \text{C.F.S.E} = \left[\frac{-3}{5}(4) + \frac{2}{5}(4)\right]\Delta t = -0.8\Delta t$$

A - III,B - IV, C - I, D - II

Q. 12 Which of the following are true about the energy of the given d-orbitals of a tetrahedral complex ?

A. $d_{xy} = d_{xz} > d_{x^2-y^2}$

B. $d_{xy} = d_{yz} > d_{z^2}$

C. $d_{x^2-y^2} > d_{z^2} > d_{xz}$

D. $d_{x^2-y^2} = d_{z^2} < d_{xz}$

Choose the correct answer from the options given below:

Option 1:
A, B and D only

Option 2:
A and B only

Option 3:
B and D only

Option 4:
B, C and D only

Correct Answer:
A, B and D only

Solution:

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Order of energy in tetrahedral splitting

$(\mathrm{dxy} = \mathrm{dyz} = \mathrm{dzx}) < \mathrm{d}_{x^2-y^2} = \mathrm{d}_{z^2}$

A, B and D are correct.

Q. 13 R_f value for 2-methylpropene in a solvent system (Ethyl acetate + ether) is 0.42 . 2-methylpropene is treated with dilute $\mathrm{H_2SO_4}$ to give major organic product (X). R_f value for (X) in the same solvent system under identical condition will be :

Option 1:
0.42

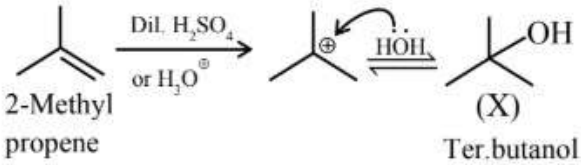
Option 2:
0.82

Option 3:
0.62

Option 4:
0.12

Correct Answer:
0.12

Solution:



Ter butanol (X) is more polar than 2-Methyl propene and $\mathrm{R_f} \propto \frac{1}{\text{polarity}}$ so $\mathrm{R_f}$ of ter.butanol is less than 2-Methylpropen

So. Ans is 0.12

Q. 14 Given below are two statements :

Statement-I : 2, 6-diethylcyclohexanone and 6-methyl-2-n-propylcyclohexanone are metamers.

Statement-II : 2, 2, 6, 6-tetramethylcyclohexanone exhibits keto-enol tautomerism.

In the light of the above statements, choose the correct answer from the options given below.

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Option 1:
Both Statement-I and Statement-II are true.

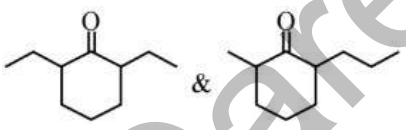
Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

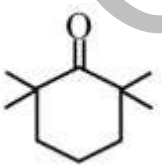
Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Statement-I is true but Statement-II is false.

Solution:
Statement-I is correct and Statement-II is incorrect



are metamers



No $\alpha - \mathrm{H}$

No Keto enol tautomerism

Q. 15 Given below are two statements :

Statement-I : Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of $\mathrm{CH_3MgBr}$ with water.

Statement-II : Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction.
In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement-I and Statement-II are true.

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Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

Option 4:
Statement-I is false but Statement-II is true.

Correct Answer:
Statement-I is false but Statement-II is true.

Solution:
Methane can not be prepared by Kolbe's electrolysis process. So, statement-1 is false

$$CH_3 - CH_3 + 2CO_2 + 2NaOH + H_2$$

Ethane
Methane also can not be prepared by Wurtz reaction so statement-2 is true

$$2R - X \xrightarrow[\text{Wurtz reaction}]{\text{Na.oxyether}} R - R + 2NaX$$

If $R = -CH_3$ then ethane is formed not methane.

Q. 16 Given below are two statements :

Statement-I : 3-phenylpropene reacts with HBr and gives secondary alkyl bromide having a chiral carbon atom as the major product.

Statement-II : Aryl chlorides and aryl cyanides can be prepared by Sandmeyer reaction as well as Gattermann reaction.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

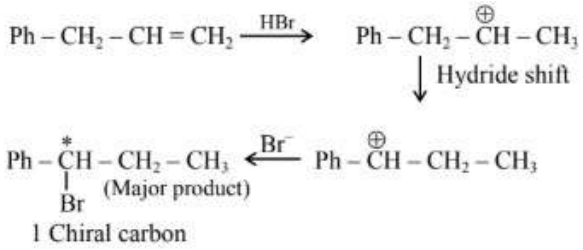
Option 4:
Statement-I is false but Statement-II is true.

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Correct Answer:

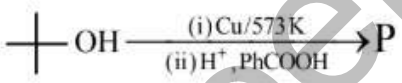
Statement-I is true but Statement-II is false.

Solution:



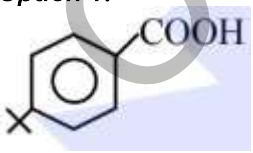
Aryl cyanide is not formed by Gattermann reaction.

Q. 17 Consider the following sequence of reactions

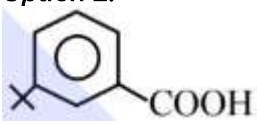


The major product P is :

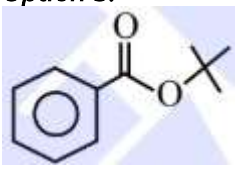
Option 1:



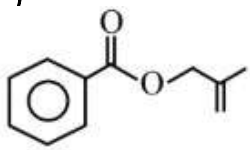
Option 2:



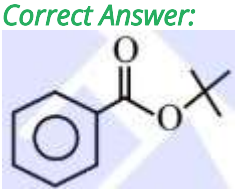
Option 3:



Option 4:

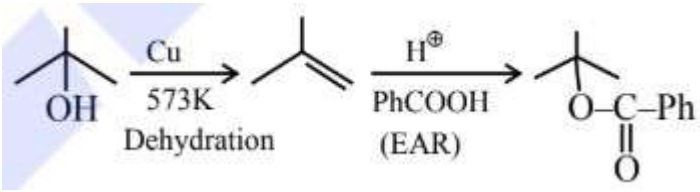


Correct Answer:



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Solution:



Q. 18 Arrange the following compounds according to increasing order of boiling points.

$n-\mathrm{C_4H_9OH(A)}, n-\mathrm{C_4H_9NH_2(B)}, n-\mathrm{C_4H_{10}(C)}$ and $\mathrm{C_2H_5NHC_2H_5(D)}$.

Option 1:
C < B < A < D

Option 2:
D < C < B < A

Option 3:
C < D < B < A

Option 4:
D < B < A < C

Correct Answer:
C < D < B < A

Solution:
Boiling point $\propto$ Intermolecular attraction

$\propto$ Number of H -bonding

n-butanol has strong H-bonding.

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Q. 19 Match List - I with List - II:

	List - I **Deficiency Disease**		**List - II** **Vitamin**
A	Scurvy	I.	Pyridoxine
B	Convulsions	II.	Vitamin A
C	Cheilosis	III.	Ascorbic Acid
D	Xerophthalmia	IV.	Riboflavin

Choose the correct answer from the options given below:

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Option 1:
A-I, B-III, C-II, D-IV

Option 2:
A-I, B-III, C-IV, D-II

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-III, B-I, C-II, D-IV

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:
Vitamin-A → Xerophthalmia
Vitamin-C (Ascorbic acid) → Scurvy
Vitamin-B $_2$ (Riboflavin) → Cheilosis
Vitamin- B_6 (Pyridoxine) → Convulsions

Q. 20 Match the List-I with List-II

List - I **Amino acid**	**List - II** **Positive reaction/Test for functional group present in side chain of amino acid**
A. Glutamine	I. Hinsberg;s test
B, Lysine	II. Neutral $FeCl_3$ test
C. Tyrosine	III. Ceric ammonium nitrate test
D. Serine	IV. Hoffman bromamide degradation

Choose the correct answer from the options given below:

Option 1:
A-IV, B-II, C-I, D-III

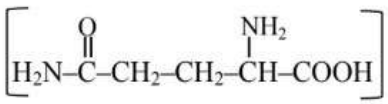
Option 2:
A-IV, B-I, C-II, D-III

Option 3:
A-III, B-II, C-I, D-IV

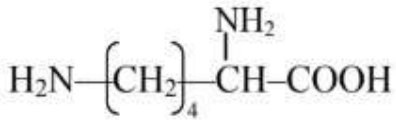
Option 4:
A-IV, B-I, C-III, D-II

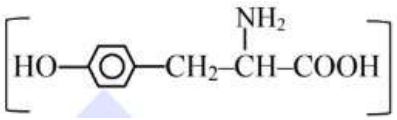
Correct Answer:
A-IV, B-I, C-II, D-III

Solution:

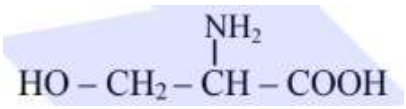
Glutamine 

has amide functional group inside chain. So it gives Hoffmann bromamide reaction.

Lysine  has 1° amine group. So it gives +ve Hinsberg test.

Tyrosine  has phenolic OH. So it gives +ve test with neutral $FeCl_3$.

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Serine  has alcohlic -OH group. So it gives +ve test with cerric ammonium nitrate.

Q. 21 First and second ionization enthalpies of lithium are $520\ \text{kJ mol}^{-1}$ and $7297\ \text{kJ mol}^{-1}$ respectively. Energy required to convert 3.5 mg lithium (g) into $\text{Li}^{2+}(\text{g})$ $\left[\text{Li}(\text{g}) \rightarrow \text{Li}^{2+}(\text{g})\right]$ is _____ kJmol^{-1}. (nearest integer)
[Molar mass of $\text{Li} = 7\ \text{g mol}^{-1}$]

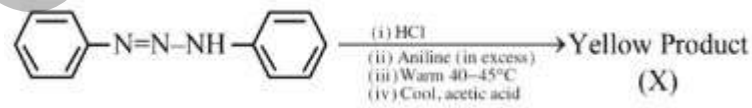
Correct Answer:
4

Solution:
$\text{Number of moles of Li} = \frac{3.5\times10^{-3}}{7} = 5 \times 10^{-4}$

Total energy needed to convert $\text{Li}(\text{g}) \rightarrow \text{Li}^{2+}(\text{g})$ is

$5 \times 10^{-4}(520 + 7297) = 3.9085\ \text{kJ} = 4\ \text{kJ}$

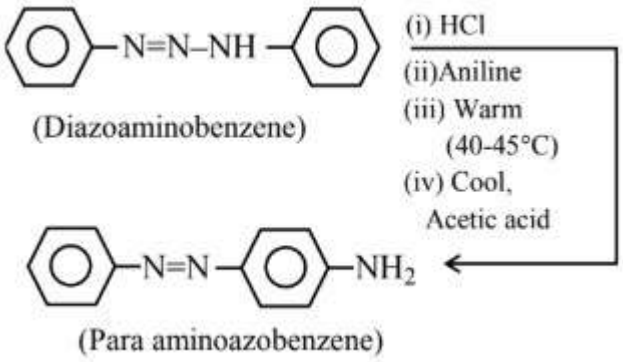
Q. 22 Consider the following sequence of reactions.



The percentage of nitrogen in the yellow product (X) formed is _____ %.
(Nearest Integer)
(Given Molar mass in $\text{gmol}^{-1}\text{H} : 1, \text{C} : 12,\ \text{N} : 14$)

Correct Answer:
21

Solution:



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- HCl catalysed hydrolysis.
- Aniline acts as a solvent and reactant.
- The mixture is gently heated to facilitate rearrangement.
- After the reaction is complete, the mixture is cooled and treated with acetic acid to neutralise excess acid and precipitate the final product.

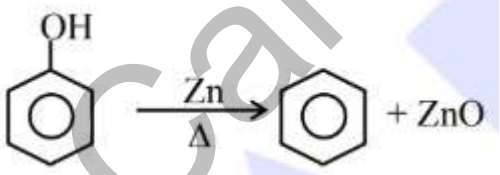
Mass percentage of nitrogen in final product

$= \frac{42}{197} \times 100 = 21.30\% \approx 21$

Q. 23 4.7 g of phenol is heated with Zn to give product X . If this reaction goes to 60% completion then the number of moles of compound X formed will be _____ $\times 10^{-2}$ (Nearest Integer)
(Given molar mass in gmol^{-1} : $\mathrm{H}: 1, \mathrm{C}: 12, \mathrm{O}: 16$)

Correct Answer:
3

Solution:



$\text{Moles of phenol } = \frac{4.7}{94.0} = 0.05 \text{ mol}$

Reaction is $1:1$ so 0.05 mole of phenol will give 0.05 mole of benzene on 100% yield

For 60% yield $= 0.05 \times 0.60$

$= 0.03 \text{ mol.}$

mol of ' X ' $= 3 \times 10^{-2} \text{ mol}$
So answer is (3)

Q. 24 Sucrose hydrolyses in acidic medium into glucose and fructose by first order rate law with $t_{1/2} = 3$ hour. The percentage of sucrose remaining after 6 hours is _____ . (Nearest integer)
(Given : $\log 2 = 0.3010$ and $\log 3 = 0.4771$)

Correct Answer:
25

Solution:

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$\text{Sucrose} + \mathrm{H_2O} \rightarrow \text{glucose} + \text{fructose}$

$t_{1/2} = 3\text{hr}$

For 1st order $\quad t_{75\%} = 2 \times t_{1/2}$

So $t = 6\text{hr}, \quad 75\%$ of sucrose decomposed

So 25% of sucrose remains after 6hr.

Q. 25 Consider the reaction $\mathrm{X} \rightleftharpoons \mathrm{Y}$ at 300 K . If $\Delta \mathrm{H}^{\theta}$ and K are $28.40\ \mathrm{kJ\ mol^{-1}}$ and 1.8×10^{-7} at the same temperature, then the magnitude of $\Delta \mathrm{S}^{\theta}$ for the reaction in $\mathrm{JK^{-1}\ mol^{-1}}$ is ____ .(Nearest Integer). (Given : $\mathrm{R} = 8.3\mathrm{JK^{-1}\ mol^{-1}}, \ln 10 = 2.3$, $\log 3 = 0.48, \log 2 = 0.30$

Correct Answer:
34

Solution:

$\mathrm{X} \rightleftharpoons \mathrm{Y} \mathrm{K_{eq}} = 1.8 \times 10^{-7}$

$$\ln\left(\mathrm{K_{eq}}\right) = \frac{\Delta_{\mathrm{r}} \mathrm{S}^{\circ}}{\mathrm{R}} - \frac{\Delta_{\mathrm{r}} \mathrm{H}^{\circ}}{\mathrm{R}}\left(\frac{1}{\mathrm{T}}\right)$$

$$2.3 \log\left(1.8 \times 10^{-7}\right) = \frac{\Delta_{\mathrm{r}} \mathrm{S}^{\circ}}{\mathrm{R}} - \frac{28400}{\mathrm{R}} \times \frac{1}{300}$$

$$\Delta_{\mathrm{r}} \mathrm{S}^{\circ} = 2.3 \times \mathrm{R} \log\left(1.8 \times 10^{-7}\right) + \frac{284}{3}$$

$$= 19.09[-8 + 2 \times 0.48 + 0.3] + \frac{284}{3}$$

$$= -128.66 + \frac{284}{3}$$

$$\Delta_{\mathrm{r}} \mathrm{S}^{\circ} = -33.99 \simeq -34\ \mathrm{J/K.mol}$$

Maths

Q. 1 Let $[\cdot]$ denote the greatest integer function. If the domain of the function $f(x) = \sin^{-1}\left(\frac{x+[x]}{3}\right)$ is $[\alpha, \beta)$, then $\alpha^2 + \beta^2$ is equal to:

Option 1:
2

Option 2:
5

Option 3:
10

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Option 4:
13

Correct Answer:
5

Solution:

$$-1 \leq \frac{x+[x]}{3} \leq 1$$
$$\Rightarrow -3 \leq x-1[x] \leq 3$$
$$\Rightarrow -3-[x] \leq x \leq 3-[x]$$
$$\text{if } [x]=-1 \Rightarrow x \in [-1,0)$$
$$\text{if } [x]=0 \Rightarrow x \in [0,1)$$
$$\text{if } [x]=1 \Rightarrow x \in [1,2)$$
$$\text{Hence } x \in [-1,2)$$
$$\Rightarrow \alpha=-1 \text{ and } \beta=2$$
$$\Rightarrow \alpha^2+\beta^2=5$$

Q. 2 Let one root of the quadratic equation in x :

$$\left(k^2-15k+27\right)x^2+9(k-1)x+18=0$$

be twice the other. Then the length of the latus rectum of the parabola $y^2=6\mathrm{kx}$ is equal to:

Option 1:
4

Option 2:
6

Option 3:
8

Option 4:
12

Correct Answer:
12

Solution:

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$$\alpha + 2\alpha = \frac{-9(k-1)}{k^2 - 15k + 27} = 3\alpha$$

$$\alpha \times 2\alpha = \frac{18}{k^2 - 15k + 27} = 2\alpha^2 \ldots$$

solving (1) & (2)

$k = 2$

$\Rightarrow$ parabola will be $y^2 = 6kx = 12x$

length of latus rectum $= 12$

Q. 3 Let e_1 and e_2 be two distinct roots of the equation $x^2 - ax + 2 = 0$. Let the sets $\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of hyperbolas $\} = (\alpha, \beta)$, and $\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of an ellipse and a hyperbola, respectively $\} = (\gamma, \infty)$.
The $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

Option 1:
18

Option 2:
22

Option 3:
26

Option 4:
34

Correct Answer:
26

Solution:

$e_1 + e_2 = a$

$e_1 e_2 = 2 \quad \Rightarrow e_2 = \frac{2}{e_1}$

Case 1 : Both are eccentricities of hyperbola

Now, $e_1 > 1$ and $e_2 > 1$

$\Rightarrow \frac{2}{e_1} > 1 \Rightarrow e_1 < 2 \Rightarrow e_1 \in (1, 2)$

Now, $a = e_1 + \frac{1}{e_1}$

Minimum occurs at $e_1 = \sqrt{2}$

$\Rightarrow a_{min} = 2\sqrt{2}$

if $e = 1, 2$ (end points)

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$a \to 2,3 \Rightarrow a \in (2\sqrt{2},3)$

$\Rightarrow \alpha = 2\sqrt{2}$ and $\beta = 3$

Case 2 : One ellipse, one hyperbola

(e_1)

(e_2)

$0 < e_1 < 1$ and $e_2 < 1$

$e_2 = \dfrac{2}{e_1} > 2$

Now $a = e_1 + \frac{2}{e_1}$

as $e_1 \to 1 \Rightarrow a \to 3$

$e_1 \to 0 \Rightarrow a \to \infty$

$\therefore (3,\infty) \Rightarrow r = 3$

Now $\alpha^2 + \beta^2 + \gamma^2 = 8 + 9 + 9 = 26$

Q. 4 Let the set of all values of $k \in R$ such that the equation $z(\bar{z} + 2 + i) + k(2 + 3i) = 0, z \in C$, has at least one solution, be the interval $[\alpha, \beta]$. Then $9(\alpha + \beta)$ is equal to:

Option 1:

-10

Option 2:

-8

Option 3:

$10\sqrt{13}$

Option 4:

$8\sqrt{13}$

Correct Answer:

-10

Solution:

Put $z = x + iy, \bar{z} = x - iy$

$\Rightarrow \bar{z} + 2 + i = x - iy + 2 + i$

$\Rightarrow z(\bar{z} + 2 + i) = (x + iy)(x + 2 + i(1 - y))$

$\Rightarrow$ Real part $= x(x + 2) - y(1 - y)$

Imaginary part $= x(1 - y) + y(x + 2)$

$\Rightarrow x^2 + y^2 + 2x - y + 2k = 0$

and $x + 2y + 3k = 0$

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eliminating x from (1) and (2)

$5y^2+(12k-5)y+9k^2-4k=0$
since $y \in R$
use $D \geq 0$
$(12k-5)^2-4.5\left(9k^2-4k\right) \geq 0$
$\Rightarrow 36k^2+40k-25 \leq 0$
$\Rightarrow \alpha+\beta=\frac{-10}{9}$
$\Rightarrow 9(\alpha+\beta)=-10$

Q. 5 The value of $1^3-2^3+3^3-\ldots .+15^3$ is:

Option 1:
1706

Option 2:
1856

Option 3:
1982

Option 4:
2403

Correct Answer:
1856

Solution:
$\left(1^3-2^3\right)+\left(3^3-4^3\right)+\ldots .13^3-14^3+15^3$
use $n^3-(n+1)^3=-3n^2-3n-1$
put $n=1,3,5,\ldots .13(7 \text{ terms })$
sum of pairs $=-1519$
add last term $=15^3$
$\Rightarrow 3375-1519=1856$

Q. 6 The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8 . If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:

Option 1:
$\frac{34}{9}$

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Option 2:
$\frac{34}{13}$

Option 3:
$\frac{32}{9}$

Option 4:
$\frac{32}{13}$

Correct Answer:
$\frac{34}{9}$

Solution:
Sum of 10 terms of A.P, $S_{10} = 160$

$5[2a + 9d] = 160$
$2a + 9d = 32$
$G.P. \Rightarrow d, da, da^2 \ldots$
$\Rightarrow d + da = 8$
$\Rightarrow \frac{d + d(32 - 9d)}{2} = 8$
$\Rightarrow 2d + 32d - 9d^2 = 8$
$\Rightarrow 9d^2 - 34d + 8 = 0$
$d_1 + d_2 = \frac{34}{9}$

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Q. 7 The number of 4 -letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is:

Option 1:
2670

Option 2:
2840

Option 3:
2920

Option 4:
3600

Correct Answer:
3600

Solution:

INCONSEQUENTIAL contains (I, E,O, A,U) and (N,C,S,Q,T,L) Total number of 4 letter words using two vowels and two consonants is

$= {}^5\mathrm{C}_2 \times {}^6\mathrm{C}_2 \times 4!$
$= 3600$

Q. 8 If the coefficients of the middle terms in the binomial expansions of $(1+\alpha x)^{26}$ and $(1-\alpha x)^{28}$, $\alpha \neq 0$, are equal, then the value of α is:

Option 1:
1

Option 2:
$\frac{14}{13}$

Option 3:
$\frac{27}{7}$

Option 4:
$\frac{7}{27}$

Correct Answer:
$\frac{7}{27}$

Solution:
$(1+\alpha \mathrm{x})^{26}$
$\text{Middle term, } \mathrm{T}_{14} = {}^{26}\mathrm{C}_{13}\alpha^{13}$
$\text{For } (1-\alpha \mathrm{x})^{28}$
$\text{Middle term } = {}^{28}\mathrm{C}_{14}\alpha^{14}$
$\Rightarrow {}^{26}\mathrm{C}_{13}\alpha^{13} = {}^{28}\mathrm{C}_{14}\alpha^{14}$
$\Rightarrow {}^{26}\mathrm{C}_{13} = {}^{28}\mathrm{C}_{14}\alpha$
$\Rightarrow {}^{26}\mathrm{C}_{13} = \frac{28}{14} \times \frac{27}{14} \cdot {}^{26}\mathrm{C}_{13}\alpha$
$\Rightarrow \alpha = \frac{7}{27}$

Q. 9 A data consists of 20 observations $x_1, x_2, \ldots\ldots, x_{20}$. If $\sum_{i=1}^{20}(x_i+5)^2 = 2500$ and $\sum_{i=1}^{20}(x_i-5)^2 = 100$, then the ratio of mean to standard deviation of this data is :

Option 1:
$2:1$

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Option 2:
$3:1$

Option 3:
$3:2$

Option 4:
$4:1$

Correct Answer:
$3:1$

Solution:

$\sum_{i=1}^{20}(x_i+5)^2=2500$

$\sum_{i=1}^{20}(x_i-5)^2=100$

$\text{on } (1)-(2) \Rightarrow 20\sum x_i=2400 \Rightarrow \sum x_i=120$

$\text{mean, } \bar{x}=6$

$(1)+(2) \Rightarrow 2\sum x_i^2+40\left(5^2\right)=2600$

$\sum x_i^2=800$

$\sigma^2=\frac{\sum x_i^2}{n}-(\bar{x})^2=\frac{800}{20}-36=4$

$\bar{x}:\sigma=6:2=3:1$

Ans. $3:1$

Q. 10 A bag contains $(\mathrm{N}+1)$ coins - N fair coins, and one coin with 'Head' on both sides. A coin is selected at random and tossed. If the probability of getting 'Head' is $\frac{9}{16}$, then N is equal to:

Option 1:
5

Option 2:
7

Option 3:
8

Option 4:
9

Correct Answer:
7

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Solution:
Total ($\mathrm{N}+1$) coins
$\mathrm{N} \Rightarrow$ fair coins
$1 \Rightarrow$ Head on both side

$$P\left(1^{+}\right)=\left(\frac{N}{N+1}\right)\left(\frac{1}{2}\right)+\left(\frac{1}{N+1}\right)1$$
$$=\frac{N+2}{2(N+1)}=\frac{9}{16}$$
$$8N+16=9N+9$$
$$\Rightarrow N=7$$

Q. 11 If the eccentricity e of the hyperbola $\frac{\mathrm{x}^2}{\mathrm{a}^2}-\frac{\mathrm{y}^2}{\mathrm{b}^2}=1$, passing through $(6,4\sqrt{3})$, satisfies $15\left(\mathrm{e}^2+1\right)=34\mathrm{e}$, then the length of the latus rectum of the hyperbola $\frac{\mathrm{x}^2}{\mathrm{b}^2}-\frac{\mathrm{y}^2}{2\left(\mathrm{a}^2+1\right)}=1$ is:

Option 1:
10

Option 2:
20

Option 3:
25

Option 4:
30

Correct Answer:
10

Solution:
$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$
$$\text{It passs through } (6,4\sqrt{3})$$
$$\Rightarrow \frac{36}{a^2}-\frac{48}{b^2}=1$$
$$\text{also } 15e^2-34e+15=0$$
$$\Rightarrow 15e^2-25e-9e+15=0$$
$$e=\frac{5}{3} \text{ or } \frac{3}{5} \Rightarrow e=\frac{5}{3}$$
$$1+\frac{b^2}{a^2}=\frac{25}{9} \Rightarrow \frac{b^2}{a^2}=\frac{16}{9} \cdots$$

using (1) and (2)

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$\Rightarrow \frac{36}{a^2}-\frac{48}{16a^2}\times 9=1\Rightarrow a=3,b=4$

Length of L.R.

$\frac{4\left(a^2+1\right)}{b}=\frac{4(10)}{4}=10$

Q. 12 Let chord PQ of length $3\sqrt{13}$ of the parabola $y^2=12x$ be such that the ordinates of points P and Q are in the ratio $1:2$. If the chord PQ subtends an angle α at the focus of the parabola, then $\sin\alpha$ is equal to :

Option 1:
$\frac{3}{5}$

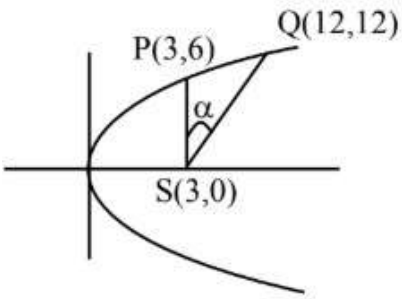
Option 2:
$\frac{4}{5}$

Option 3:
$\frac{5}{13}$

Option 4:
$\frac{12}{13}$

Correct Answer:
$\frac{3}{5}$

Solution:



$P\left(3t^2,6t\right),Q\left(12t^2,12t\right)$
$PQ=\sqrt{81t^4+36t^2}$
$81t^4+36t^2=117$
$\Rightarrow 9t^4+4t^2-13=0$
$\left(t^2-1\right)\left(9t^2+13\right)=0\Rightarrow t=1$
$P(3,6),Q(12,12)$
$\text{slope of } QS=\frac{4}{3}$
$\tan\alpha=\frac{3}{4},\sin\alpha=\frac{3}{5}$

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Q. 13 Let $0 < \alpha < 1, \beta = \frac{1}{3\alpha}$ and $\tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta) = \frac{\pi}{4}$. Then $6(\alpha+\beta)$ is equal to:

Option 1:
6

Option 2:
7

Option 3:
8

Option 4:
9

Correct Answer:
7

Solution:

$$\beta = \frac{1}{3\alpha}, \tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta) = \frac{\pi}{4}$$
$$\Rightarrow \tan\left(\tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta)\right) = 1$$
$$\Rightarrow \frac{1-\alpha+1-\beta}{1-(1-\alpha)(1-\beta)} = 1$$
$$\Rightarrow 2-\alpha-\beta = 1-(1-\alpha-\beta+\alpha\beta)$$
$$\Rightarrow 2-\alpha-\beta = \alpha+\beta-\alpha\beta$$
$$\Rightarrow 2(\alpha+\beta) = 2+\alpha\beta$$
$$\Rightarrow 2(\alpha+\beta) = 2+\frac{1}{3} = \frac{7}{3}$$
$$\alpha+\beta = \frac{7}{6}$$
$$6(\alpha+\beta) = 7$$

Q. 14 Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$. Then $\sum_{\theta \in S} \theta$ is equal to:

Option 1:
$-\frac{2\pi}{3}$

Option 2:
$-\frac{4\pi}{3}$

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Option 3:
$\frac{2\pi}{3}$

Option 4:
$\frac{4\pi}{3}$

Correct Answer:
$-\frac{4\pi}{3}$

Solution:
$\cos\theta + 1 = \sqrt{3}\sin\theta$

$$\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} + 1 = \sqrt{3}\left(\frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}\right)$$

$$2 = 2\sqrt{3}\tan\frac{\theta}{2}$$

$$\Rightarrow \tan\frac{\theta}{2} = \frac{1}{\sqrt{3}} \quad \theta \in (-2\pi, 2\pi), \frac{\theta}{2} \in (-\pi, \pi)$$

$$\frac{\theta}{2} = -\frac{5\pi}{6}, \frac{\pi}{6}$$

$$\theta = \frac{-5\pi}{3}, \frac{\pi}{3}$$

$$\text{Sum} = -\frac{-5\pi}{3} + \frac{\pi}{3} = \frac{-4\pi}{3}$$

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Q. 15 Let the image of the point $\mathrm{P}(1,6,a)$ in the line $\mathrm{L}: \frac{\mathrm{x}}{1} = \frac{\mathrm{y}-1}{2} = \frac{\mathrm{z}-\mathrm{a}+1}{\mathrm{b}}$, $\mathrm{b} > 0$, be $\left(\frac{\mathrm{a}}{3}, 0, \mathrm{a}+\mathrm{c}\right)$. If $\mathrm{S}(\alpha,\beta,\gamma), \alpha > 0$, is the point on L such that the distance of S from the foot of perpendicular from the point P on L is $2\sqrt{14}$, then $\alpha+\beta+\gamma$ is equal to:

Option 1:
19

Option 2:
20

Option 3:
21

Option 4:
22

Correct Answer:
21

Solution:
Let Q be the point $\perp^{\mathrm{r}}$ from $\mathrm{P}(1,6,\mathrm{a})$ to the line

$\mathrm{L}: \frac{\mathrm{x}}{1}=\frac{\mathrm{y}-1}{2}=\frac{\mathrm{z}-\mathrm{a}+1}{\mathrm{b}}$

The image of point P in line L is $\mathrm{P}'\left(\frac{\mathrm{a}}{3}, 0, \mathrm{a}+\mathrm{c}\right)$
$\because \mathrm{Q}$ is the mid point PP' :

$Q \equiv\left(\frac{1+\frac{a}{3}}{2}, \frac{6+0}{2}, \frac{a+a+c}{2}\right) \equiv\left(\frac{3+a}{6}, 3, \frac{2a+c}{2}\right)$

comparing x and y, $\frac{3+a}{6}=1 \Rightarrow a=3$

$$\therefore Q \equiv\left(1,3, \frac{6+c}{2}\right)$$

Direction vector of L is $\overrightarrow{\mathrm{v}}=(1,2, \mathrm{b})$ vector PQ is $\perp^{\prime}$ to $\overrightarrow{\mathrm{v}}$

$\therefore \overrightarrow{\mathrm{PQ}}=\left(1-1,3-6, \frac{6+\mathrm{c}}{2}-3\right)$
$=\left(0,-3, \frac{\mathrm{c}}{2}\right)$
$\overrightarrow{\mathrm{PQ}} \cdot \overrightarrow{\mathrm{v}}=0$
$(0) \cdot(1)+(-3)(2)+\left(\frac{\mathrm{c}}{2}\right)(\mathrm{b})=0$
$\mathrm{bc}=12$
Now comparing z with $\mathrm{Q}\left(1,3, \frac{6+\mathrm{c}}{2}\right)$

$\frac{3-1}{2}=\frac{\frac{6+c}{2}-3+1}{b}$
$2b-c=2$
solving $bc=12$ and $c=2b-2$
$b=3,-2$
$b=3 \quad(b>0)$
$c=4$
$\therefore Q(1,3,5)$

Now, any point S on the line L can be written as

$(k, 2k+1, 3k+2)$
$SQ=\sqrt{(k-1)^2+(2k-2)^2+(3k-3)^2}=2\sqrt{14}$
$k=3,-1$
for $k=3, S \equiv(3,7,11)$
$\therefore \alpha+\beta+\gamma=21$

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Q. 16 Let a line L be perpendicular to both the lines $L_1 : \frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $L_2 : \frac{x-2}{1} = \frac{y-4}{4} = \frac{z-6}{7}$.

If θ is the acute angle between the lines L and $L_3 : \frac{x-\frac{8}{7}}{2} = \frac{y-\frac{4}{7}}{1} = \frac{z}{2}$, then $\tan\theta$ is equal to:

Option 1:
$\frac{3}{2}\sqrt{2}$

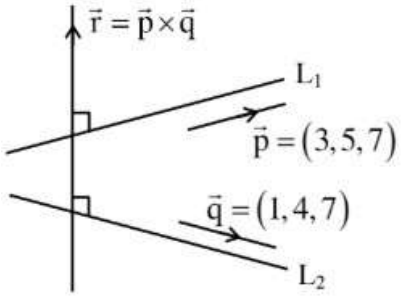
Option 2:
$\frac{5}{2}\sqrt{2}$

Option 3:
$\frac{5}{3}\sqrt{2}$

Option 4:
$\frac{4}{3}\sqrt{2}$

Correct Answer:
$\frac{5}{2}\sqrt{2}$

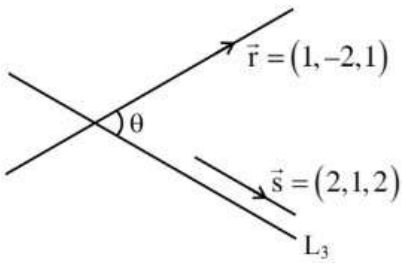
Solution:



$$\vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & 7 \\ 1 & 4 & 7 \end{vmatrix} = 7\hat{i} - 14\hat{j} + 7\hat{k}$$

$$\equiv 7(\hat{i} - 2\hat{j} + \hat{k})$$

Now,



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$$\because \cos\theta = \frac{2-2+2}{\sqrt{1+4+1}\sqrt{4+1+4}}$$

$$\Rightarrow \cos\theta = \frac{2}{3\sqrt{6}} \Rightarrow \tan\theta = \frac{5\sqrt{2}}{2}$$

Q. 17 The value of $\lim_{x\to 0}\left(\frac{x^2\sin^2 x}{x^2-\sin^2 x}\right)$

Option 1:
2

Option 2:
3

Option 3:
4

Option 4:
6

Correct Answer:
3

Solution:
$\lim_{x\to 0}\frac{x^2\sin^2 x}{x^2-\sin^2 x}$

Applying expansion

$$\lim_{x\to 0}\frac{x^2\sin^2 x}{x^2-\left(x-\frac{x^3}{3!}+\ldots\right)^2}$$

$$\lim_{x\to 0}\frac{x^2\sin^2 x}{x^2-x^2+\frac{2x^4}{3!}\ldots} = \frac{3!}{2} = 3$$

Q. 18 The value of the integral $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{32\cos^4 x}{1+e^{\sin x}}\right)dx$ is :

Option 1:
$4\pi+2$

Option 2:
$3\pi+8$

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Option 3:
$3\pi + 4$

Option 4:
$4\pi + 3$

Correct Answer:
$3\pi + 8$

Solution:
Apply (P-5)

$I = \int_{-\pi/4}^{\pi/4} \frac{32\cos^4\theta}{1+e^{-\sin\theta}} d\theta$

$$\text{Add } 2\text{I} = \int_{-\pi/4}^{\pi/4} 32\cos^4\theta \, \mathrm{d}\theta = 2\int_0^{\pi/4} 32\cos^4\theta \, \mathrm{d}\theta$$

$$\text{I} = 32\int_0^{\pi/4} \cos^4\theta \, \mathrm{d}\theta$$

$$= 8\int_0^{\pi/4} \left(2\cos^2\theta\right)^2 \mathrm{d}\theta = 8\int_0^{\pi/4} (1+\cos 2\theta)^2 \, \mathrm{d}\theta$$

$$= 8\int_0^{\pi/4} 1 + 2\cos 2\theta + \cos^2 2\theta \, \mathrm{d}\theta$$

$$= 8\int_0^{\pi/4} 1 + 2\cos 2\theta + \frac{1+\cos 4\theta}{2} \, \mathrm{d}\theta$$

$$= 8\left[\frac{3\theta}{2} + \frac{2\sin 2\theta}{2} + \frac{\sin 4\theta}{8}\right]_0^{\pi/4}$$

$$= 8\left[\left(\frac{3\pi}{8} + 1 + 0\right)\right] = 3\pi + 8$$

Q. 19 The area of the region $\{(x, y) : 0 \le y \le 6 - x, y^2 \ge 4x - 3, x \ge 0\}$ is :

Option 1:
8

Option 2:
9

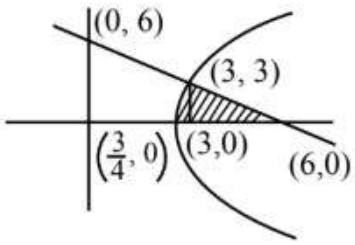
Option 3:
12

Option 4:
15

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Correct Answer:
9

Solution:
$A = \int_0^3 \left(6 - y - \frac{y^2+3}{4}\right) dy = 9$



Q. 20 Let e be the base of natural logarithm and let $f : \{1, 2, 3, 4\} \to \{1, e, e^2, e^3\}$ and $g : \{1, e, e^2, e^3\} \to \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\}$ be two bijective functions such that f is strictly decreasing and g is strictly increasing. If $\phi(x) = \left[f^{-1}\left\{g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x$, then the area of the region $R = \{(x, y) : x^2 \le y \le \phi(x), 0 \le x \le 1\}$ is :

Option 1:
$\frac{3-\log_e(2)}{3\log_e(2)}$

Option 2:
$\frac{1}{3\log_e(2)}$

Option 3:
$3 + \log_e(2)$

Option 4:
$\frac{3+\log_e(2)}{2+\log_e(3)}$

Correct Answer:
$\frac{3-\log_e(2)}{3\log_e(2)}$

Solution:
$f(x) = e^{x-1}$
$g(x) = \frac{1}{\log_e x + 1}$
$f^{-1}\left(g^{-1}(1/2)\right) = f^{-1}(e) = 2$
$\phi(x) = 2^x$
$\therefore \int_0^1 \left(2^x - x^2\right) dx = \left(\frac{2^x}{\ln 2} - \frac{x^3}{3}\right)_0^1 = \frac{1}{\ln 2} - \frac{1}{3}$
$= \frac{3 - \ln 2}{3 \ln 2}$

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Q. 21 Let $A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ satisfy

$$\mathrm{A}^2 + \alpha(\mathrm{adj}(\mathrm{adj}(\mathrm{A}))) + \beta(\mathrm{adj}(\mathrm{A})(\mathrm{adj}(\mathrm{adj}(\mathrm{A})))) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix} \text{ for}$$

Some $\alpha, \beta \in \mathbb{R}$.
Then $(\alpha - \beta)^2$ is equal to ____

Correct Answer:
4

Solution:
$|\mathrm{A}| = -1$
$\mathrm{adj}(\mathrm{adj}(\mathrm{A})) = |\mathrm{A}|^{n-2}\,\mathrm{A} = |\mathrm{A}|^{3-2} \cdot \mathrm{A} = |\mathrm{A}| \cdot \mathrm{A}$
$\mathrm{adj}(\mathrm{A})(\mathrm{adj}(\mathrm{adj}(\mathrm{A}))) = \mathrm{adj}(\mathrm{A})(|\mathrm{A}| \cdot \mathrm{A})$
$= |\mathrm{A}|(\mathrm{adj}(\mathrm{A}) \cdot \mathrm{A})$
since $\mathrm{adj}(\mathrm{A})\mathrm{A} = |\mathrm{A}|\mathrm{I}$,
then $\mathrm{adj}(\mathrm{A})(\mathrm{adj}(\mathrm{adj}(\mathrm{A}))) = |\mathrm{A}|^2\mathrm{I}$
or $\mathrm{adj}(\mathrm{A})(\mathrm{adj}(\mathrm{adj}(\mathrm{A}))) = \mathrm{I}$
solving the matrix equation

$$\mathrm{A}^2 + \alpha(-\mathrm{A}) + \beta(\mathrm{I}) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

$\mathrm{A}^2 - \alpha\mathrm{A} + \beta\mathrm{I} = \mathrm{M}$

$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \alpha \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} + \beta \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$\alpha = 1$ and $\beta = -1$
$\therefore (\alpha - \beta)^2 = 4$

Q. 22 Let the centre of the circle $x^2 + y^2 + 2gx + 2fy + 25 = 0$ be in the first quadrant and lie on the line $2x - y = 4$. Let the area of an equilateral triangle inscribed in the circle be $27\sqrt{3}$. Then the square of the length of the chord of the circle on the line $x = 1$ is ____ .

Correct Answer:
80

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Solution:

$x^2 + y^2 + 2gx + 2fy + 25 = 0$

$C \equiv (-g, -f)$ lies on line

$\therefore -2g + f = 4$

Also $g^2 + f^2 - 25 = r^2$

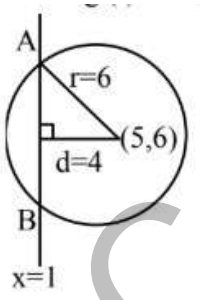
Area of equation triangle inside the circle

$= \frac{3\sqrt{3}}{4} r^2$

$27\sqrt{3} = \frac{3\sqrt{3}}{4} \times r^2$

$r^2 = 36$

Solving (i) and (ii) $g = -5, f = -6$



$\ell_{AB} = \sqrt{36 - 16} = 2\sqrt{20}$

$(\ell_{AB})^2 = 80$

Q. 23 If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{c}(\vec{a} - 2\vec{b})$ is equal to _____ .

Correct Answer:

3

Solution:

Using triple product identity with $\vec{a} \times \vec{c} = \vec{b}$

$\vec{a} \times (\vec{a} \times \vec{c}) = \vec{a} \times \vec{b}$

$(\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})\vec{c} = \vec{a} \times \vec{b}$

As $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{vmatrix} = -2\hat{i} + \hat{j} + \hat{k}$

Substituting given values

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$3\vec{a} - 3\vec{c} = -2\hat{i} + j + k$

$3(\hat{i} + \hat{j} + \hat{k}) + 2\hat{i} + \hat{j} + \hat{k} = 3\vec{c}$

$\vec{c} = \dfrac{1}{3}(5\hat{i} + 2\hat{j} + 2\hat{k})$

Now $\overrightarrow{c} \cdot (\overrightarrow{a} - 2\overrightarrow{b}) = \frac{1}{3}(5\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (\hat{i} - \hat{j} + 3\hat{k})$

$= \frac{1}{3}[5 - 2 + 6] = 3$

Q. 24 For the functions $f(\theta) = \alpha \tan^2\theta + \beta \cot^2\theta$, and $g(\theta) = \alpha \sin^2\theta + \beta \cos^2\theta, \alpha > \beta > 0$, let $\min_{0<\theta<\frac{\pi}{2}} f(\theta) = \max_{0<\theta<\pi} g(\theta)$. If the first term of a G.P. is $\left(\frac{\alpha}{2\beta}\right)$, its common ratio is $\left(\frac{2\beta}{\alpha}\right)$ and the sum of its first 10 terms is $\frac{m}{n}$, $\gcd(m, n) = 1$, then $m + n$ is equal to _____ .

Correct Answer:
1279

Solution:
$f(\theta)|_{\min} = 2\sqrt{\alpha\beta}(AM \geq GM)$
$g(\theta) = \alpha \sin^2\theta + \beta\left(1 - \sin^2\theta\right) = (\alpha - \beta)\sin^2\theta + \beta$
$\therefore g(\theta)|_{\max} = \alpha$

Equation $2\sqrt{\alpha\beta} = \alpha$

$4\alpha\beta = \alpha^2$
$\alpha = 4\beta(\alpha \neq 0)$

Now first term $\left(\frac{\alpha}{2\beta}\right) = 2$ and common ratio

$\left(\dfrac{2\beta}{\alpha}\right) = \dfrac{1}{2}$

$S_{10} = \dfrac{a\left(1 - r^{10}\right)}{1 - r} = \dfrac{1023}{256} = \dfrac{m}{n}$
$m + n = 1279$

Q. 25 Let $y = y(x)$ be the solution of the differential equation

$$\left(x^2 - x\sqrt{x^2 - 1}\right)dy + \left(y\left(x - \sqrt{x^2 - 1}\right) - x\right)dx = 0$$

$x \geq 1$. If $y(1) = 1$, then the greatest integer less than $y(\sqrt{5})$ is _____ .

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Correct Answer:
3

Solution:

$$\frac{dy}{dx}+\frac{y}{x}=\frac{1}{x-\sqrt{x^2-1}}$$

$$\frac{dy}{dx}+\frac{y}{x}=x+\sqrt{x^2-1}$$

$$\text{I.f } =e^{\ln x}=x$$

$$y\cdot x=\int\left(x+\sqrt{x^2-1}\right)xdx$$

$$=\frac{x^2}{2}+\frac{\left(x^2-1\right)^{3/2}}{3}+C$$

$$\text{Given } y(1)=1$$

$$1=\frac{1}{2}+C\Rightarrow C=\frac{1}{2}$$

$$[y(\sqrt{5})]=\left[\frac{\sqrt{5}}{2}+\frac{8}{3}+\frac{1}{2\sqrt{5}}\right]=3$$

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JEE MAINS

2026

6th April

2nd Shift

Question Paper

Physics

Q. 1 The percentage error in the calculated volume of a sphere, if there is 2% error in its diameter measurement, is _____ .

Option 1:
1

Option 2:
2

Option 3:
6

Option 4:
8

Correct Answer:
6

Solution:
$v = \frac{4}{3}\pi R^3$

$\frac{\Delta v}{v} = \frac{3\Delta R}{R}$

$= 6\%$

Q. 2 Match List-I with List-II.

	List-I		List-II.
A	Boltzmann constant	I	$\left[\mathrm{M}^{-1}\,\mathrm{L}^{3}\,\mathrm{T}^{-2}\right]$
B	Stefan's constant	II	$\left[\mathrm{ML}^{2}\,\mathrm{T}^{-1}\right]$
C	Planck's constant	III	$\left[\mathrm{ML}^{2}\,\mathrm{T}^{-2}\,\mathrm{K}^{-1}\right]$
D	Gravitational constant	IV	$\left[\mathrm{ML}^{0}\,\mathrm{T}^{-3}\,\mathrm{K}^{-4}\right]$

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Option 1:
A-I, B-II,C-III, D-IV

Option 2:
A-IV, B-III,C-II, D-I

Option 3:
A-III, B-IV,C-II, D-I

Option 4:
A-II, B-I,C-IV, D-III

Correct Answer:
A-III, B-IV,C-II, D-I

Solution:

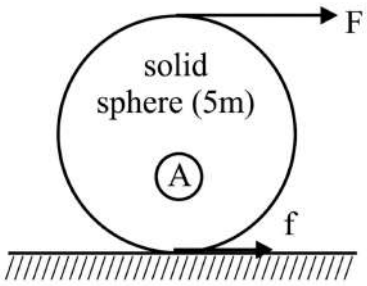
$(\mathrm{A}) \rightarrow \mathrm{ML}^2\,\mathrm{T}^{-2}\,\mathrm{K}^{-1}$ (III)

$(\mathrm{B}) \rightarrow \mathrm{MT}^{-3}\,\mathrm{K}^{-4}$ (IV)

$(\mathrm{C}) \rightarrow \mathrm{ML}^2\,\mathrm{T}^{-1}$ (II)

$(\mathrm{D}) \rightarrow \mathrm{M}^{-1}\,\mathrm{L}^3\,\mathrm{T}^{-2}$ (IV)

Q. 3 A solid sphere (A) of mass, 5 m and spherical shell (B) of mass m , both having same radius, are placed on a rough surface. When a force of same magnitude is applied tangentially at the highest points of A and B, they start rolling without slipping with an acceleration of $\mathrm{a_A}$ and $\mathrm{a_B}$, respectively. The ratio of $\mathrm{a_A}$ and $\mathrm{a_B}$ is _____ .



Option 1:
$5:21$

Option 2:
$6:10$

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Option 3:
$21 : 25$

Option 4:
$1 : 5$

Correct Answer:
$5 : 21$

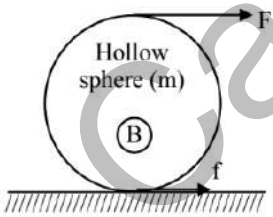
Solution:
$\mathrm{F}+\mathrm{f}=5\ \mathrm{ma_A}$

$\mathrm{F(R)}-\mathrm{f(R)}=\frac{2}{5}(5\ \mathrm{m})\mathrm{R}^2\cdot\alpha$

$\alpha=\mathrm{R}\alpha$

$2\ \mathrm{F}=7\ \mathrm{ma_A}$

$\mathrm{a_A}=\frac{2\ \mathrm{F}}{7\ \mathrm{m}}$



$\mathrm{F}+\mathrm{f}=\mathrm{ma_B}$

$\mathrm{F(R)}-\mathrm{f(R)}=\frac{2}{3}\mathrm{mR}^2\times\alpha$

$\mathrm{a}=\mathrm{R}\alpha$

$2\ \mathrm{F}=\frac{5\mathrm{ma_B}}{3}$

$\frac{\mathrm{a_A}}{\mathrm{a_B}}=\frac{2\ \mathrm{F}}{7\ \mathrm{m}}\times\frac{5\ \mathrm{m}}{6\ \mathrm{F}}=\frac{5}{21}$

Q. 4 A body of mass 1 kg moves along a straight line with a velocity $v=2\mathrm{x}^2$. The work done by the body during displacement from $\mathrm{x}=0$ to 5 m is _____ J.

Option 1:
0

Option 2:
250

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Option 3:
1250

Option 4:
1000

Correct Answer:
1250

Solution:
$$\begin{aligned}&\mathrm{w}\cdot\Delta=\Delta\mathrm{K}\cdot\mathrm{E}\\&=\frac{1}{2}\,\mathrm{m}\left(\mathrm{v}_{\mathrm{f}}^{2}-\mathrm{v}_{\mathrm{i}}^{2}\right)\\&=\frac{1}{2}\times1\left[\left(2(5)^{2}\right)^{2}-(2(0))^{2}\right]^{0}\\&=\frac{1}{2}\times50\times50=1250\ \mathrm{J}\end{aligned}$$

Q. 5 A cylinder with adiabatic walls is closed at both ends and is divided into two compartments by a frictionless adiabatic piston. Ideal gas is filled in both (left and right) the compartments at same $\mathrm{P}, \mathrm{V}, \mathrm{T}$. Heating is started from left side until pressure changes to $\frac{27\mathrm{P}}{8}$. If initial volume of each compartment was 9 litres then the final volume in right-hand side compartment is _____ litres. (for this ideal gas $\mathrm{C_p}/\mathrm{C_v}=1.5$)

Option 1:
3

Option 2:
4

Option 3:
14

Option 4:
9

Correct Answer:
4

Solution:
$P_{\text{Left}}=P_{\text{Right}}$ (final) for right chamber

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$$P_i\ V_i^{\gamma} = P_f\ V_f^{\gamma}$$
$$P(9)^{3/2} = \frac{275}{8} \times V_f^{3/2}$$
$$27 = \frac{27}{8}\ V_f^{3/2}$$
$$\therefore\ V_f = 4 \text{ litres}$$

Q. 6

For an electromagnetic wave propagating through vacuum, $\overrightarrow{k}, \overrightarrow{E}$ and ω represent propagation vector, electric field and angular frequency, respectively. The magnetic field associated with this wave is represented by :

Option 1:
$\frac{\vec{E}\times\vec{k}}{\omega}$

Option 2:
$\frac{\overrightarrow{k}\times\overrightarrow{E}}{\omega}$

Option 3:
$\omega(\overrightarrow{E} \times \overrightarrow{k})$

Option 4:
$\omega(\overrightarrow{k} \times \overrightarrow{E})$

Correct Answer:
$\frac{\overrightarrow{k}\times\overrightarrow{E}}{\omega}$

Solution:
$\overrightarrow{B} = \frac{1}{\omega}(\overrightarrow{k} \times \overrightarrow{E})$

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Q. 7 Two identical bodies A and B of equal masses have initial velocities $\overrightarrow{v_1} = 4\hat{i}\,m/s$ and $\overrightarrow{v_2} = 4\hat{j}m/s$ respectively. The body A has acceleration $\overrightarrow{a_1} = 6\hat{i} + 6\hat{j}\,m/s^2$ while the acceleration of the other body B is zero. The centre of mass of the two bodies moves in _____ path.

Option 1:
circular

Option 2:
parabolic

Option 3:
straight line

Option 4:
elliptical

Correct Answer:
straight line

Solution:

$$\overrightarrow{v_{cm}} = \frac{m_1\overrightarrow{v_1} + m_2\overrightarrow{v_2}}{m_1 + m_2} = \frac{m(4\hat{i}) + m(4\hat{j})}{2\,m} = 2\hat{i} + 2\hat{j}$$

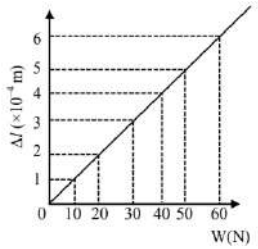
$$\overrightarrow{a_{cm}} = \frac{m_1\overrightarrow{a_1} + m_2\overrightarrow{a_2}}{m_1 + m_2} = \frac{m(6\hat{i} + 6\hat{j}) + m(0)}{2\,m} = 3\hat{i} + 3\hat{j} -$$

$\overrightarrow{v_{cm}} \| \overrightarrow{a_{cm}}$

so, path is straight line

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Q. 8 Figure represents the extension (Δl) of a wire of length 1 meter, suspended from the ceiling of the room at one end with load W connected to the other end. If the cross- sectional area of the wire is $10^{-5}\ \mathrm{m}^2$ then the Young's modulus of the wire is _____ $\mathrm{N/m^2}$.



Option 1:
1.0×10^{11}

Option 2:
2.0×10^{10}

Option 3:
1.0×10^{10}

Option 4:
2.0×10^{11}

Correct Answer:
1.0×10^{10}

Solution:

$$\frac{\Delta \ell}{\mathrm{w}} = \tan\theta = \frac{\Delta \mathrm{y}}{\Delta \mathrm{x}} = \frac{1 \times 10^{-4}}{10} = 10^{-5}$$

$$\frac{\mathrm{w}}{\Delta \ell} = 10^{5}$$

$$\Rightarrow \mathrm{y} = \frac{\mathrm{F/A}}{\Delta \ell / \ell} = \frac{\mathrm{F}\ell}{\mathrm{A}\Delta \ell} = \left(\frac{\mathrm{w}}{\Delta \ell}\right)\frac{\ell}{\mathrm{A}}$$

$$\Rightarrow \mathrm{y} = \frac{10^{5} \times 1}{10^{-5}} = 10^{10}\ \mathrm{N/m^2}$$

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Q. 9 A cylindrical vessel of 40 cm radius is completely filled with water and its capacity is $528\mathrm{dm}^3(\mathrm{dm}$: decimeter). The vessel is placed on a solid block of exactly same height as vessel. If a small hole is made at 70 cm below the top of water level, then horizontal range of water falling on the ground in the beginning is _____ cm .

Option 1:
$120\sqrt{2}$

Option 2:
$140\sqrt{2}$

Option 3:
$140\sqrt{3}$

Option 4:
$120\sqrt{3}$

Correct Answer:
$140\sqrt{2}$

Solution:

$\mathrm{v}=\pi \mathrm{r}^2 \mathrm{~h} \Rightarrow 528 \times 10^{-3}=\pi(0.4)^2 \mathrm{~h}$

$\mathrm{h}=\dfrac{528 \times 10^{-3} \times 7}{(0.4)^2 \times 22}=1.05 \mathrm{~m}=105 \mathrm{~cm}$

$\rightarrow \mathrm{t}=\sqrt{\dfrac{2 \mathrm{H}}{\mathrm{g}}}=\sqrt{\dfrac{2 \times(105+35) \times 10^{-2}}{10}}$

$=\sqrt{\dfrac{2(140) \times 10^{-2}}{10}} \sqrt{28} \times 10^{-1} \mathrm{sec}$

$\mathrm{R}=\mathrm{ut}=\sqrt{2 \times 10 \times 70 \times 10^{-2}} \times \sqrt{28} \times 10^{-1}$

$=(\sqrt{14})(\sqrt{28}) \times 10^{-1}=14 \sqrt{2} \times 10^{-1} \mathrm{~m}$

$\mathrm{R}=140 \sqrt{2} \mathrm{~cm}$

Q. 10 If 2 mole of an ideal monoatomic gas at temperature T , is maxed with 6 mole of another ideal monoatomic gas at temperature 2T then the temperature of mixture is :

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Option 1:
$\frac{5}{2}\ \mathrm{T}$

Option 2:
$\frac{5}{4}\ \mathrm{T}$

Option 3:
$\frac{7}{2}\ \mathrm{T}$

Option 4:
$\frac{7}{4}\ \mathrm{T}$

Correct Answer:
$\frac{7}{4}\ \mathrm{T}$

Solution:

$$\mathrm{T_{mix}} = \frac{\mathrm{n_1 C_1\ T_1 + n_2 C_2\ T_2}}{\mathrm{n_1 C_1 + n_2 C_2}}$$

$$= \frac{\left(2 \times \frac{3}{2}\mathrm{R} \times \mathrm{T}\right) + \left(6 \times \frac{3}{2}\mathrm{R} \times 2\ \mathrm{T}\right)}{\left(2 \times \frac{3}{2}\mathrm{R}\right) + \left(6 \times \frac{3}{2}\mathrm{R}\right)}$$

$$\mathrm{T_{mix}} = \frac{2\ \mathrm{T} + 12\ \mathrm{T}}{8} = \frac{7}{4}\ \mathrm{T}$$

Q. 11 A spring stretches by 2 mm when it is loaded with a mass of 200 g . From equilibrium position the mass is further pulled down by 2 mm and released. The frequency associated with the system and maximum energy in the spring are _____ Hz and _____ J , respectively. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Option 1:
$\frac{5\sqrt{50}}{\pi}$ and 8×10^{-3}

Option 2:
$\frac{5\sqrt{50}}{\pi}$ and 8

Option 3:
$10\sqrt{50}$ and 2×10^{-3}

Option 4:
$\frac{5\sqrt{50}}{\pi}$ and 16×10^{-3}

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Correct Answer:
$\frac{5\sqrt{50}}{\pi}$ and 8×10^{-3}

Solution:

$\mathrm{kx} = 0.2\ \mathrm{g};$

$\mathrm{k} \times 2 \times 10^{-3} = 2$

$\mathrm{k} = 1000\ \mathrm{N/m}$

$\mathrm{E_{max}} = \frac{1}{2}\mathrm{k} \quad \mathrm{x_{max}^2}$

$\frac{1}{2} \times 1000 \times \left(4 \times 10^{-3}\right)^2 = 8 \times 10^{-3}$ Joule

$\mathrm{f} = \frac{1}{2\pi}\sqrt{\frac{\mathrm{k}}{\mathrm{m}}}$

$\mathrm{f} = \frac{1}{2\pi}\sqrt{\frac{1000}{0.2}}$

$= \frac{1}{2\pi}\sqrt{5000} = \frac{5}{\pi}\sqrt{50}$

Q. 12 The electric potential as a function of x, y is given by $\mathrm{V} = 5\left(\mathrm{x}^2 - \mathrm{y}^2\right)\mathrm{V}$. The electric field at a point $(2, 3)\mathrm{m}$ is _____ V/m.

Option 1:
$(-20\hat{\mathrm{i}} + 30\hat{\mathrm{j}})$

Option 2:
$(20\hat{\mathrm{i}} - 30\hat{\mathrm{j}})$

Option 3:
$(20\hat{\mathrm{i}} + 45\hat{\mathrm{j}})$

Option 4:
$(-4\hat{\mathrm{i}} + 6\hat{\mathrm{j}})$

Correct Answer:
$(-20\hat{\mathrm{i}} + 30\hat{\mathrm{j}})$

Solution:

$V = 5\left(x^2 - y^2\right)V$

$\varepsilon_x = \dfrac{-dv}{dx} = -10x\hat{i}$

$\varepsilon_y = \dfrac{-dv}{dx} = +10y\hat{i} \Rightarrow \varepsilon(2, 3) = -20\hat{i} + 30\hat{j}$

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Q. 13 A current of 30 A each flows in opposite directions in two conducting wires, placed parallel to each other at a distance of 8 cm . The magnetic field at the mid point between the two wires is _____ $\mu\mathrm{T}$.

$\left(\frac{\mu_0}{4\pi} = 10^{-7}\ \mathrm{N/A^2}\right)$

Option 1:
30

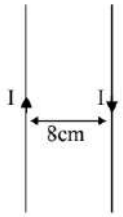
Option 2:
300

Option 3:
150

Option 4:
0.1

Correct Answer:
300

Solution:



$$\mathrm{B} = \frac{\mu_0 \mathrm{I}}{2\pi(4) \times 10^{-2}} \times 2$$
$$= 10^{-7} \times 30 \times 10^{2}$$
$$= 30 \times 10^{-5}\ \mathrm{T}$$
$$= 300\mu\ \mathrm{T}$$

Q. 14 A square loop of side 2 cm is placed in a time varying magnetic field with magnitude as $\mathrm{B} = 0.4\sin(300t)$ Tesla. The normal to the plane of loop makes an angle of 60° with the field. The maximum induced emf produced in the loop is _____ mV.

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Option 1:
12

Option 2:
18

Option 3:
21

Option 4:
24

Correct Answer:
24

Solution:

$$\mathrm{B}=0.4 \sin 300 \mathrm{t} \quad \theta=60^{\circ}$$

$$\mathrm{A}=(0.02)^2=4 \times 10^{-4} \mathrm{~m}^2$$

$$\phi=\mathrm{BA} \cos \theta$$

$$\phi=0.4 \sin (300 \mathrm{t}) \mathrm{A} \cos \theta$$

$$\text{emf } =\frac{\mathrm{d} \phi}{\mathrm{dt}}=0.4 \mathrm{~A} \cos \theta \cos (300 \mathrm{t}) \times 300$$

$$\begin{aligned}\max \text{ emf } &=0.4 \times 4 \times 10^{-4} \times 300 \times \frac{1}{2} \\ &=240 \times 10^{-4} \mathrm{~V} \\ &=24 \mathrm{mV}\end{aligned}$$

Q. 15 A sphere of capacitance 100 pF is charged to a potential of 100 V . Another identical undercharged metal sphere is brought in contact with the charged sphere, then the change in the total energy stored on these spheres, when they touch is $\alpha \times 10^{-7}$ J. The value of α is _____ . (combined capacitance of spheres is 200 pF).

Option 1:
5

Option 2:
$\frac{5}{2}$

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Option 3:
$\frac{7}{2}$

Option 4:
$\frac{9}{2}$

Correct Answer:
$\frac{5}{2}$

Solution:
$Q_1 = CV = 100 \times 100\text{pC} = 10^4\text{pC}$

$V_1 = V_2 = \frac{V}{2} = 50$

$\varepsilon_i = \frac{1}{2} \times 100 \times (100)^2\text{pJ} = \varepsilon$

$\varepsilon_f = 2 \times \frac{1}{2} \times 100 \times (50)^2 = \frac{\varepsilon}{2}$

$\Delta\varepsilon = \frac{\varepsilon}{2} = 100 \times (50)^2\text{pJ}$

$= 25 \times 10^4 \times 10^{-12} = 2.5 \times 10^{-7}$

Q. 16 The energy released if hydrogen atoms are combined to form $\frac{4}{2}\text{He}$ is _____ .
(Take binding energies per nucleon of $\frac{2}{1}\text{H}$ and $\frac{4}{2}\text{He}$ as 1.1 meV and 7.2 MeV , respectively).

Option 1:
6.1

Option 2:
24.4

Option 3:
26.6

Option 4:
5

Correct Answer:
24.4

Solution:

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$_1^2\mathrm{H} + _1^2\mathrm{H} \rightarrow _2^4\mathrm{He}$

$\mathrm{Q} = \mathrm{BE(RHS)} - \mathrm{BE(LHS)}$

$= 4 \times 7.2 - 4 \times 1.1$

$= 4 \times 6.1$

$\mathrm{Q} = 24.4\mathrm{MeV}$

Q. 17 Angle of minimum deviation is equal to the half of the angle of prism is an equilateral prism. The refreactive index of the prism is _____ .

Option 1:
1.5

Option 2:
$\sqrt{3}$

Option 3:
$\sqrt{2}$

Option 4:
1.65

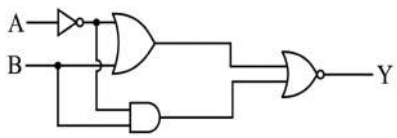
Correct Answer:
$\sqrt{2}$

Solution:

$$\mu_p = \mu_s \frac{\sin\left(\frac{A+\delta_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\frac{3A}{4}}{\sin\frac{A}{2}}$$

$$\mu_p = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2} = \sqrt{2}$$

Q. 18 Refer to the logic circuit given below. For two inputs ($\mathrm{A} = 1,\ \mathrm{B} = 1$) and ($\mathrm{A} = 0,\ \mathrm{B} = 1$), output (Y) will be



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Option 1:
1,0 respectively

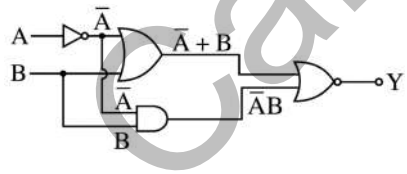
Option 2:
0,1 rsepectively

Option 3:
0,0 respectively

Option 4:
1,1 respectively

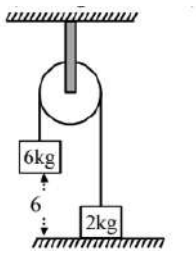
Correct Answer:
0,0 respectively

Solution:



$$y=\overline{\overline{A}+B+\overline{A}B}$$
$$=\overline{A}+B=\overline{\overline{A}}\cdot\overline{B}$$
$$=A\overline{B}$$
$$\text{for A}=1,\ \text{B}=1, \text{y}=0$$
$$\text{for A}=0,\ \text{B}=1, \text{y}=0$$

Q. 19 The velocity at which 6 kg mass (shown in figure) strikes the ground when it is released from a height of 6 m above the ground is _____ $\mathrm{m/s}$. Assume pulley is massless and string is light and inextensible (Take $\mathrm{g}=10\ \mathrm{m/s^2}$).



Option 1:
7.74

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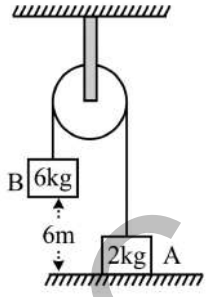
Option 2:
7.20

Option 3:
6.55

Option 4:
4.50

Correct Answer:
7.74

Solution:



$$W_G = \Delta K$$

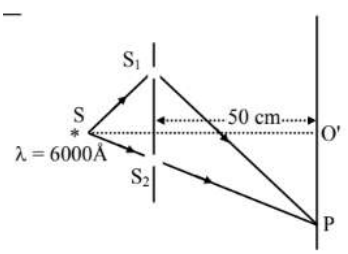
$$m_B gh - m_A gh = \frac{1}{2}(m_A + m_B)V^2$$

$$(6-2) \times 10 \times 6 = \frac{1}{2}(8)v^2$$

$$v = \sqrt{60}\ m/s$$

$$v = 7.746\ m/s$$

Q. 20 In a Young double slit experiment, the wavelength of incident of incident light is 6000 $\mathring{A}$, the separation between slits S_1 and 5 cm the distance between slits plane and screen is 50 cm , as shown in the figure below. If the resultant intensity at P is equal to the intensity due to individual slits, the path different between interfering waves is _____ Å.



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Option 1:
4000

Option 2:
3000

Option 3:
2000

Option 4:
1000

Correct Answer:
2000

Solution:

$$\mathrm{I_p} = \mathrm{I_0} = 4\mathrm{I_0}\cos^2\frac{\Delta\phi}{2}$$

$$\cos\frac{\Delta\phi}{2} = \frac{1}{2}$$

$$\frac{\Delta\phi}{2} = \frac{\pi}{3}$$

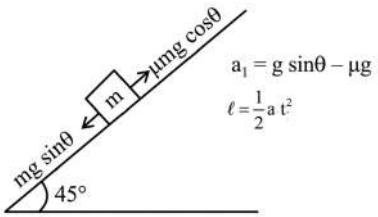
$$\frac{2\pi}{\lambda} = \Delta\mathrm{x} = \Delta\phi = \frac{2\pi}{3}$$

$$\Delta\mathrm{x} = \frac{\lambda}{3} = 2000\ \mathrm{A}^\circ$$

Q. 21 A block takes t time to slide down a plane inclined at 45° to the horizontal. If the surface is made smooth (frictionless), the block takes times $\frac{t}{2}$ to slide down the plane. The coeffcient of friction between the block and the inclined plane is $\left(\frac{\alpha}{100}\right)$. The value of α is _____ .

Correct Answer:
75

Solution:



for frictionless surface $a_2 = g\sin\theta$

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$$\ell = \frac{1}{2} a_2 \frac{t^2}{4}$$

$$a, t^2 = a_2 \frac{t^2}{4}$$

$$g\left(\sin 45 - \mu \cos 45^\circ\right) = \frac{g \sin 45}{4}$$

$$1 - \mu = \frac{1}{4}$$

$$\mu = \frac{3}{4} = \frac{\alpha}{100}$$

$$\alpha = 75$$

Q. 22 The de Broglie weave length for an electron accelerated through the potential difference V_1 volt is λ_1. When the potential difference is changed to V_2 volt, the associated de Broglie wavelength is increased by 50%. If $(\mathrm{V}_1/\mathrm{V}_2) = (9/\alpha)$, then the value of α is _____ .

Correct Answer:
4

Solution:

$$\lambda_1 = \frac{\mathrm{h}}{\mathrm{m}\nu} = \frac{\mathrm{h}}{\sqrt{2\mathrm{meV}_1}} = \lambda$$

$$\lambda_2 = \frac{\mathrm{h}}{\sqrt{2\mathrm{meV}_2}} = \frac{3\lambda}{2} = \frac{3}{2} \frac{\mathrm{h}}{\sqrt{2\mathrm{meV}_1}}$$

$$\mathrm{V}_2 = \frac{4}{9} \cdot \mathrm{V}_1$$

$$\frac{\mathrm{V}_1}{\mathrm{V}_2} = \frac{9}{4} \Rightarrow \alpha = 4$$

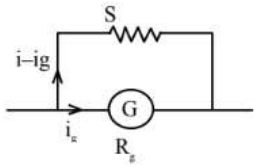
Q. 23 A moving coil of a golvanometer when shunted with 2Ω resistance gives a full scale deflection for a current of 500 mA . When a resistance of 470Ω is connected in series it gives a full scale deflection for 10 V potential applied on it. The value of resistance of galvanometer coil is _____ Ω.

Correct Answer:
50

Solution:

$$\mathrm{i_g R_g} = 2\left(\tfrac{1}{2} - \mathrm{ig}\right)$$

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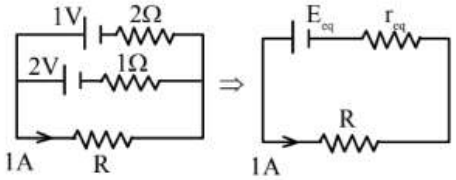


$$i_g = \frac{1}{R_g + 2}$$
$$10 = (R_g + 470)i_g$$
$$10 = \frac{R_g + 470}{R_g + 2}$$
$$10R_g + 20 = R_g + 470$$
$$R_g = 50\Omega$$

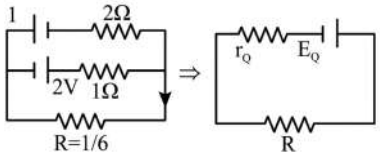
Q. 24 Two cells of emfs 1 V and 2 V and internal resistance 2Ω and 1Ω, respectively connected in parallel, gave current of 1A through an external resistance. If the polarity of one cell is reversed, then value of current through the external resistance will be $\frac{\alpha}{5}$ A. The value of α is _____ .

Correct Answer:
3

Solution:



$$E_{eq} = \frac{\frac{1}{2} + \frac{2}{1}}{\frac{1}{2} + \frac{1}{1}} = \frac{\frac{5}{2}}{\frac{3}{2}} = \frac{5}{3}$$
$$r_Q = \frac{3}{2}$$
$$i = \frac{E_{eq}}{r_{eq} + R} = \frac{\frac{5}{3}}{\frac{3}{2} + R} = 1$$
$$\frac{5}{3} = \frac{3}{2} + R$$
$$R = \frac{1}{6}\Omega$$



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$$r_{eq} = \frac{3}{2}$$

$$E_{eq} = \frac{\frac{2}{1} - \frac{1}{2}}{\frac{1}{1} + \frac{1}{2}} = 1$$

$$i = \frac{E_{eq}}{r_{eq} + R} = \frac{1}{\frac{3}{2} + \frac{1}{6}} = \frac{3}{5} \quad \Rightarrow \alpha = 3$$

Q. 25 A concave mirror of focal length 10 cm forms an image which is double the size of object when the object is placed at two different positions. The distance between the two positions of the object is
_____ cm.

Correct Answer:
10

Solution:
Case I $v = 2u$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-2u} + \frac{1}{-u} = \frac{1}{-f}$$

$$\frac{3}{2u} = \frac{1}{f}$$

$$u_1 = \frac{3f}{2}$$

Case $2v = -2u$.

$$\frac{1}{+2u} + \frac{1}{-u} = \frac{1}{-f}$$

$$\frac{-1}{2u} = \frac{1}{-f}$$

$$u_2 = \frac{f}{2}$$

$$\Delta u = f = 10 \text{ cm}$$

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Chemistry

Q. 1 Which of the following contain the same number of atoms?
(Given : Molar mass in gmol^{-1} of $\mathrm{H}, \mathrm{He}, \mathrm{O}$ and S are $1, 4, 16$ and 32 respectively)

A. 2 g of O_2 gas
B. 4 g fo SO_2 gas
C. 1400 mL of O_2 at STP
D. 0.05 L of He at STP
E. $0.0625\ \mathrm{mol}^{\circ}\mathrm{H}_2$ gas

Choose the correct answer from the options given below :

Option 1:
A and B only

Option 2:
B and C only

Option 3:
C and D only

Option 4:
A, C and E only

Correct Answer:
A, C and E only

Solution:

$$\mathrm{N}_{\text{atms}}\ \text{in}\ \mathrm{O}_2 = \frac{\mathrm{m}_{\mathrm{O}_2}}{\mathrm{M}_{\mathrm{O}_2}}\mathrm{N_A} = \frac{2}{32} \times 2\,\mathrm{N_A} = \frac{\mathrm{N_A}}{8} = 0.125\,\mathrm{N_A}$$

$$\mathrm{N}_{\text{atms}}\ \text{in}\ \mathrm{SO}_2 = \frac{\mathrm{m}_{\mathrm{SO}_2}}{\mathrm{M}_{\mathrm{SO}_2}} = 3\,\mathrm{N_A} = \frac{4}{64} \times 3\,\mathrm{N_A} = \frac{3\,\mathrm{N_A}}{16} = 0.1875\,\mathrm{N_A}$$

$$\mathrm{N}_{\text{atoms}}\ \text{in}\ \mathrm{O}_2\ \text{at STP} = \frac{\mathrm{V}_{\mathrm{O}_2}}{22.7\ \text{lit}}\mathrm{N_A} = \frac{1400\text{ml}}{22400\text{ml}} \times 2\,\mathrm{N_A} = 0.125\,\mathrm{N_A}$$

$$\mathrm{N}_{\text{atoms}}\ \text{in}\ \mathrm{He(g)}\ \text{at STP} = \frac{\mathrm{V}_{\mathrm{He}}}{22.4\ \text{lit.}}\mathrm{N_A} = \frac{0.05}{22.4}\,\mathrm{N_A} = 0.002\,\mathrm{N_A}$$

$$\mathrm{N}_{\text{atoms}}\ \text{in}\ \mathrm{H}_2(\mathrm{g}) = \text{moles} \times \text{atomicity} \times \mathrm{N_A} = 0.0625 \times 2\,\mathrm{N_A} = 0.125\,\mathrm{N_A}$$

Q. 2 The Bohr radius of a hydrogen like species is 70.53 pm . The species and the stationary state(n) are respectively
(Given: Hydrogen atom Bohr radius is 52.9 pm)

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Option 1:
$Li^{2+}, 3$

Option 2:
$He^{+}, 3$

Option 3:
$He^{+}, 2$

Option 4:
$Li^{2+}, 2$

Correct Answer:
$Li^{2+}, 2$

Solution:

$r_{species} = a_0 \frac{n^2}{Z}$

$70.53 = 52.9 \left(\frac{n^2}{Z}\right)$

$\frac{n^2}{Z} = \frac{70.53}{52.9} = 1.33 = \frac{4}{3}$

Data will satisfy if $n = 2$ and $z = 3$

$Li^{+2}, 2 \Rightarrow \frac{4}{3} = 1.33$

Q. 3 Given below are two statements:

Statement I: The number of compounds among SO_2, SO_3, SF_4, SF_6 and H_2S in which sulphur does not obey the Octet rule is 3 .

Statement II : Among $[H_2O, ClF_3, SF_4]$, $[NH_3, BrF_5, SF_4]$, $[BrF_5, ClF_3, XeF_4]$ and $[XeF_4, ClF_3, H_2O]$, the number of sets in which all the molecules have one lone pair of electrons on the central atom is 1 .

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

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Option 3:
Statement I is true Statement II is false

Option 4:
Statement I is false Statement II is true

Correct Answer:
Statement I is false Statement II is true

Solution:
$SO_2, SO_3, SF_6 \& SF_4$ do not follow octet rule

NH_3, BrF_5, SF_4 have one lone pair on central atom

ClF_3, XeF_4 \& H_2O have two lone pair on central atom.

Q. 4 Match List-I with List-II

Given V_1 and V_2 are initial and final volumes respectively:

List - I **(Isothermal process)**	**List - II** **(Expression)**
A. Reversible expansion	I. $q = 0$
B. Free expansion	II. $q = nRT \ln \frac{V_2}{V_1}$
C, Irreversible Compression	III. $w = -p_{ext}(V_1 - V_2)$
D. Cyclic reversible	IV. $\frac{q_{rev}}{T} = 0$

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-I, D-IV

Option 2:
A-II, B-I, C-IV, D-III

Option 3:
A-II, B-I, C-III, D-IV

Option 4:
A-I, B-II, C-III, D-IV

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Correct Answer:
A-II, B-I, C-III, D-IV

Solution:
(A) For isothermal process, $\Delta U = 0$

$$W = -nRT\ell n\left(\frac{V_2}{V_1}\right) \text{ so } q = nRT\ell n\left(\frac{V_2}{V_1}\right)$$

(B) Free expansion which is isothermal must be adiabatic, so $q = 0$
(C) Irreversible process $W = -P_{ext}\ [V_2 - V_1]$
(D) Cyclic process, change in state function (S) so change in entropy must be zero.

$$dS = \frac{dq_{rev}}{T}$$
$$\oint dS = 0$$

Q. 5 Given below are two statements :

Chamber 1	Semi-permeable membrane	Chamber 2
18 g glucose in 100 mL aqueous solution		30 g glucose in 250 mL aqueous solution

Statement I : H_2O molecules move from the chamber 1 to chamber 2 .

Statement II : The osmotic pressure of a solution prepared by dissolving 50 mg of potassium sulphate (molar mass $= 174\ g/mol$) in 2 L of water (at $27^\circ C$) is 0.0107 bar . (Given : $R = 0.083 dm^3$ bar $K^{-1}\ mol^{-1}$ and assume complete dissociation of electrolyte)

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

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Correct Answer:
Statement I is false but Statement II is true

Solution:
On side (I) $\mathrm{C}_1=\frac{18/180}{0.1}=1\mathrm{M}$
On side (II) $\mathrm{C}_2=\frac{30/180}{250/1000}=\frac{2}{3}\mathrm{M} \quad ; \quad \{\pi_1>\pi_2\}$
Hence water will flow from chamber-II to Chamber-I.
$\pi=\mathrm{iCRT} \quad ;\mathrm{i}=3$ for $\mathrm{K}_2\mathrm{SO}_4$
And $\left(\pi=3\times\frac{50\times10^{-3}/174}{2}\right)\times0.083\times300$
$=0.0107$ bar.

Q. 6 Given is a concentrated solution of a weak electrolyte $\mathrm{A}_x\ \mathrm{B}_y$ of concentration ' c ' and dissociation constant ' K '. the degree of dissociation is given by:

Option 1:
$\left[K\times\mathrm{c}^{x+y-1}\mathrm{x}^{\mathrm{x}}\mathrm{y}^{y}\right]^{x+y}$

Option 2:
$\left(\frac{K}{c^{x+y-1}x^{x}y^{y}}\right)^{\frac{1}{x+y}}$

Option 3:
$\left(\frac{\mathrm{c}^{\mathrm{x+y-1}}\mathrm{x}^{\mathrm{x}}\mathrm{y}^{\mathrm{y}}}{\mathrm{K}}\right)^{\mathrm{x+y}}$

Option 4:
$\left(\frac{\mathrm{c}^{\mathrm{x+y-1}}\mathrm{x}^{\mathrm{x}}\mathrm{y}^{\mathrm{y}}}{\mathrm{K}}\right)^{\frac{1}{\mathrm{x+y}}}$

Correct Answer:
$\left(\frac{K}{c^{x+y-1}x^{x}y^{y}}\right)^{\frac{1}{x+y}}$

Solution:
$\mathrm{A}_x\ \mathrm{B}_y\rightleftharpoons\mathrm{xA}^{y+}+\mathrm{yB}^{\mathrm{x}-}$
$\mathrm{C}(1-\alpha)\quad\mathrm{xC}\alpha\quad\mathrm{yC}\alpha$
$\mathrm{K}=\frac{(\mathrm{xC}\alpha)^{\mathrm{x}}(\mathrm{xC}\alpha)^{\mathrm{y}}}{\mathrm{C}(1-\alpha)}$

As $\alpha\ll1\Rightarrow(1-\alpha)\simeq1$

$K=x^{x}y^{y}C^{x+y-1}\alpha^{x+y}$

$\alpha=\left(\frac{K}{C^{x+y-1}x^{x}y^{y}}\right)^{\frac{1}{x+y}}$

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Q. 7 For a general redox reaction

Anode $\mathrm{Red_1 \rightarrow Ox_1}^{n_i^+} + n_1 e^-$

Cathode $\mathrm{Ox_2} + n_2 e^- \rightarrow \mathrm{Red_2}^{n_2-}$

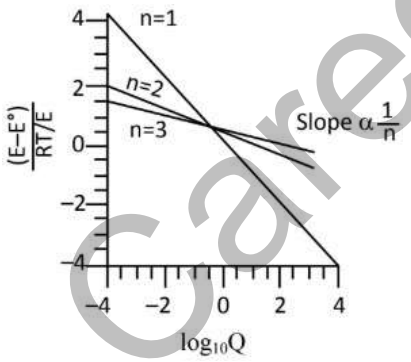
Which of the following statement is incorrect?

Option 1:

The overall reaction can be written as $n_2\,\mathrm{Red_1} + n_1\mathrm{Ox_2} \rightleftharpoons n_2\mathrm{Ox_1}^{n_1^+} + n_1\,\mathrm{Red_2}^{n_2-}$

Option 2:

The electrons do not appear in the overall reaction because electrons produced at the anode are consumed at the cathode.



Option 3:

Here n is the number of electrons transferred in redox reaction.

Option 4:

If the reaction is carried out reversibly, the electrical work done is equal to the ratio of charge and potential difference through which charge is moved.

Correct Answer:

If the reaction is carried out reversibly, the electrical work done is equal to the ratio of charge and potential difference through which charge is moved.

Solution:

Anode : $\left[\mathrm{Red_1 \rightarrow Ox_1}^{n_1^+} + n_1 e^-\right] \times n_2$

Cathode: $\left[\mathrm{Ox_2} + n_2 e^- \rightarrow \mathrm{Red_2}^{n_2-}\right] \times n_1$

Overall reaction :

$n_2\,\mathrm{Red_1} + n_1\mathrm{Ox_2} \rightarrow n_2\mathrm{Ox_1}^{n_1^+} + n_1\,\mathrm{Red_2}^{n_2^-}$

The electrons does not appear in overall reaction

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$$E = E^\circ - 2.303\frac{RT}{nF}\log Q$$

$$\frac{(E - E^\circ)}{RT/F} = -\frac{2.303}{n}\log Q$$

$$\text{Slope } = \frac{-2.303}{n}$$

Straight line passing through origin.
Electrical work = Charge × Potential difference

Q. 8 In a period, the first ionisation enthalpy of the element at extreme left and the negative electron gain enthalpy of the extreme right element, except noble gases, are respectively.

Option 1:
lowest and lowest

Option 2:
highest and lowest

Option 3:
lowest and highest

Option 4:
highest and highest

Correct Answer:
lowest and highest

Solution:
Left most element in a period will be of group 1 which will have lowest IE, while right most element will be of group 17 which will have most negative ΔH_{cg}.

Q. 9 Given below are two statements :

Statement I : $F_2O < H_2O < Cl_2O$ is the correct trend in terms of bond angle.

Statement II : SiF_4, SnF_4 and PbF_4 are ionic in nature.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

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Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is true but Statement II is false

Solution:

Specie	Bond angle
OCl_2	111°
H_2O	104.5°
OF_2	103°

SiF_4 is covalent.

Q. 10 The correct order of first $(\Delta_1 H_1)$ and second $(\Delta_1 H_2)$ ionisation enthalpy values of cr and Mn are :

A. $\Delta_i H_1 : Cr > Mn$
B. $\Delta_i H_2 : Cr > Mn$
C. $\Delta_i H_1 : Mn > Cr$
D. $\Delta_i H_2 : Mn > Cr$

Choose the correct answer from the options given below:

Option 1:
A and B only

Option 2:
B and C only

Option 3:
A and D only

Option 4:
C and D only

Correct Answer:
B and C only

Solution:

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$Cr - [Ar]3\,d^5 4\,s^1$
$Mn - [Ar]3\,d^5 4\,s^2$
$IE_1 - Mn > Cr$
$IE_2 - Mn < Cr$

Q. 11 Which of the following sequence of hybridisation, geometry and magnetic nature are correct for the given coordination compounds ?

A. $[NiCl_4]^{2-} - sp^3$, tetrahedral, paramagnetic
B. $[Ni(NH_3)_6]^{2+} - sp^3\,d^2$, octahedral, paramagnetic
C. $[Ni(CO)_4] - sp^3$, tetrahedral, paramagnetic
D. $[Ni(CN)_4]^{2-} - dsp^2$, square planar diamagnetic

Choose the correct answer from the options given below :

Option 1:
A, B, C and D

Option 2:
B, C and D only

Option 3:
A, C and D only

Option 4:
A, B and D only

Correct Answer:
A, B and D only

Solution:
$[Ni(CO)_4]$ is sp^3, tetrahedral and diamagnetic

Q. 12 Given below are two statements :

Statement I: A mixture of $C_{12}H_{22}O_{11}$ (sugar) and NaCl can be separated by dissolving sugar in alcohol, due to differential solubility

Statement II : Rose essence from rose petals is separated by steam distillation due to its high volatility and insolubility in H_2O.

In the light of the above statements, choose the correct answer from the options given below :

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Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

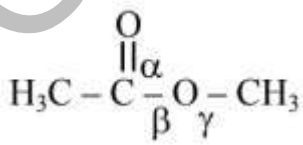
Option 3:
Statement I is true Statement II is false

Option 4:
Statement I is false Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:
Sugar is soluble in alcohol not NaCl
So they are separated by solvent extraction (or differential extraction)
Rose essence oil is steam volatile so it is separated by steam distillation.

Q. 13 Shown below is the structure of methyl acetate with three different α, β and γ carbon-oxygen bonds.



The correct order of bond length of these bonds is :

Option 1:
$\alpha > \beta > \gamma$

Option 2:
$\alpha < \beta < \gamma$

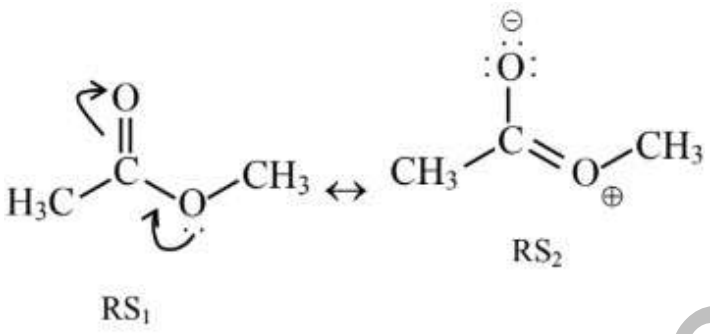
Option 3:
$\alpha = \beta = \gamma$

Option 4:
$\alpha < \beta = \gamma$

Correct Answer:
$\alpha < \beta < \gamma$

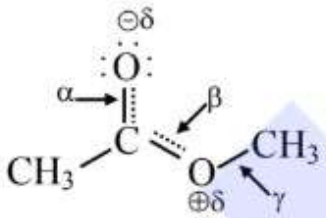
Solution:

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RS1

RS2



R.Hybrid

RS1 is more stable than RS2

∴ contribution of RS1 is more than RS2 in it's R.H.

∴ Order of bond length $\alpha < \beta < \gamma$

Q. 14 ' x ' is the product which is obtained by the hydrolysis of prop-1-yne in the presence of mercuric sulphate under dilute acidic medium at 333 K . ' y ' is the product which is obtained by the reaction of ethane nitrile with methyl magnesium bromide in dry ether followed by hydrolysis. IUPAC name of product obtained from ' x ' and ' y ' in the presence of barium hydroxide followed by heating is:

Option 1:
2-Methylpent-4-en-3-one

Option 2:
4-Methylpent-3-en-2-one

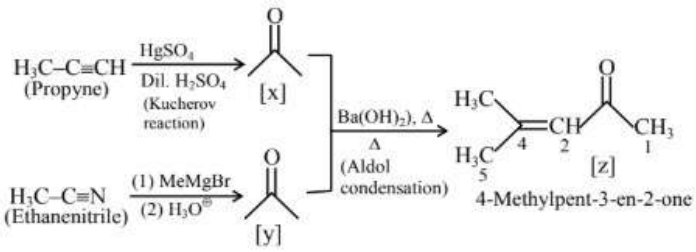
Option 3:
4-Methylpent-1-ene

Option 4:
2-Methylpent-3-one

Correct Answer:
4-Methylpent-3-en-2-one

Solution:

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Q. 15 An optically active alkyl bromide C_4H_9Br, reacts with ethanolic KOH to form major compound $[A]$ which reacts with bromine to give compound $[B]$. Compound $[B]$ reacts with ethanolic KOH and sodamide to give compound [C]. One molecule of water adds to compound [C] on warming with mercuric sulphate and dilute sulphuric acid at 333 K to form compound [D]. The functional group in compound D will be confirmed by:

Option 1:
Haloform test

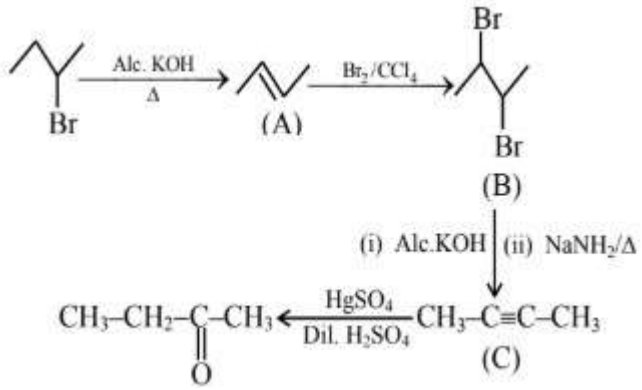
Option 2:
Lucas test

Option 3:
Silver mirror test

Option 4:
Benedict test

Correct Answer:
Haloform test

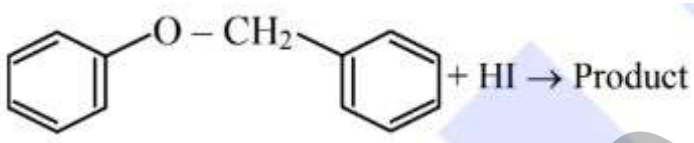
Solution:



Final product gives +ve haloform test

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Q. 16 Consider the following reaction.



Statement I: In the above reaction, product formed will be a mixture of benzyl alcohol and iodobenzene.

Statement II : In the above reaction, the $-O-CH_2-$ bond is cleaved to give the product. In the light of the above statements,

choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true

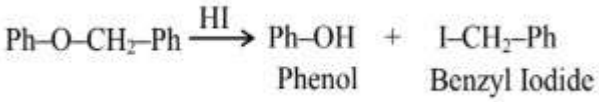
Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

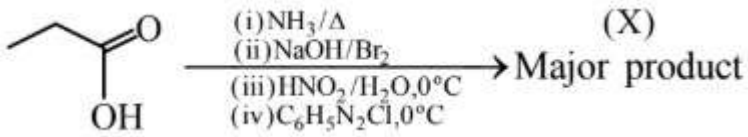
Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

Solution:



Q. 17 Consider the following organic reaction sequence. Choose the final product (X) from the following (consider the major product in all intermediate reactions)



Option 1:
Benzene

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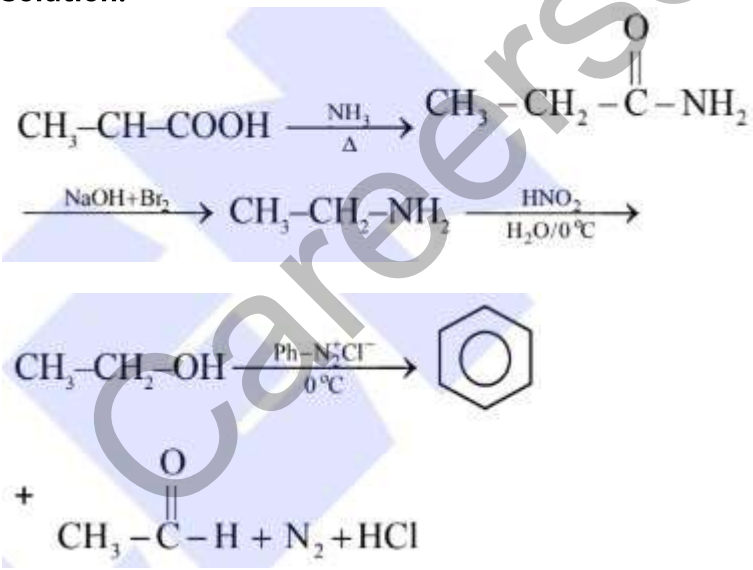
Option 2:
Phenol

Option 3:
Propanol

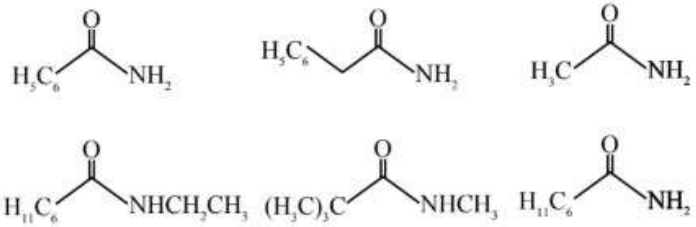
Option 4:
Chlorobenzene

Correct Answer:
Benzene

Solution:



Q. 18 The number of compounds from the following which can undergo reaction with $Br_2/KOH/$ (alcoholic) to give respective products and these respective products can also be obtained separately by Gabriel phthalimide reaction is:



Option 1:
5

Option 2:
4

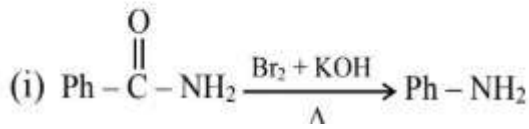
Option 3:
3

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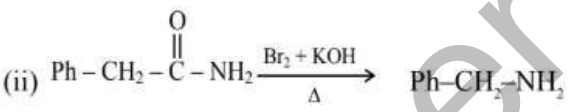
Option 4:
6

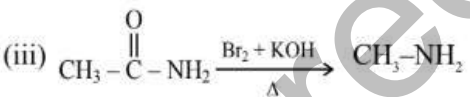
Correct Answer:
3

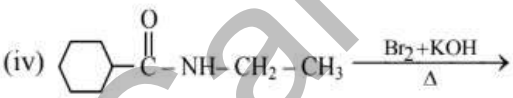
Solution:
(ii), (iii), (vi)



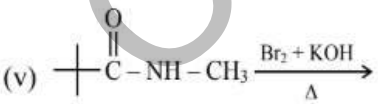
Aniline is not formed by Gabriel phtalimide synthesis.



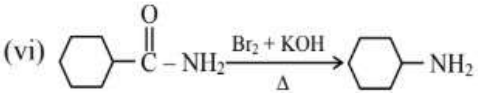




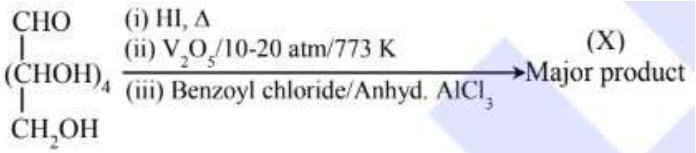
No 1° amine is form



No 1° amine is form



Q. 19 Consider the following reactions. Total number of electrons in the π bonds and lone pair of electrons in the product (X) is:



Option 1:
12

Option 2:
16

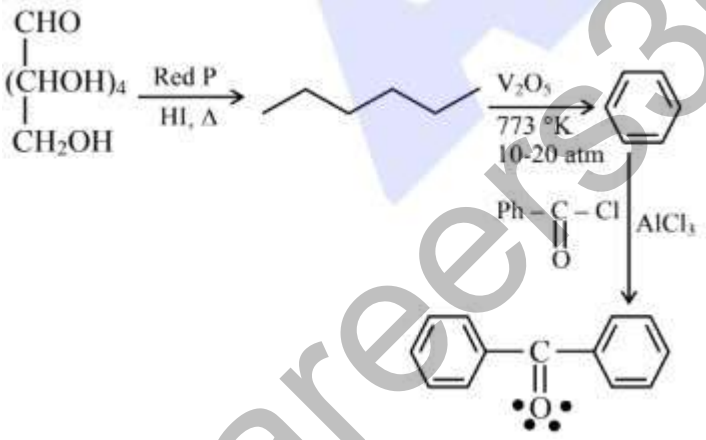
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Option 3:
14

Option 4:
18

Correct Answer:
18

Solution:



Total number of π and lone pair electrons in final product $= 18$

Q. 20 Treatment of a gas ' X ' with a freshly prepared ferrous sulphate solution gives a compound ' Y ' as a brown ring. The compounds X and Y are.

Option 1:
NO and $[\mathrm{Fe(NO)}]\mathrm{SO_4}$

Option 2:
$\mathrm{NO_2}$ and $[\mathrm{Fe\,(NO_2)}]\mathrm{SO_4}$

Option 3:
$\mathrm{N_2O}$ and $[\mathrm{Fe\,(N_2O)}]\mathrm{SO_4}$

Option 4:
$\mathrm{N_2O_4}$ and $[\mathrm{Fe\,(N_2O_4)}]\mathrm{SO_4}$

Correct Answer:
NO and $[\mathrm{Fe(NO)}]\mathrm{SO_4}$

Solution:

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$$NO + FeSO_4 \xrightarrow{\text{dil. } H_2SO_4} [Fe(NO)]SO_4$$

Q. 21 An excess of $AgNO_3$ is added to 100 mL of a 0.05 M solution of tetraaquadichloridochromium (III) chloride. The number of moles of AgCl precipitated will be _____ $\times 10^{-3}$. (Nearest integer)

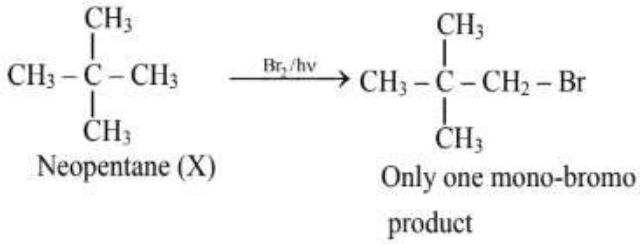
Correct Answer:
5

Solution:
$[Cr(H_2O)_4Cl_2]Cl + AgNO_3 \rightarrow AgCl\downarrow + [Cr(H_2O)_4Cl_2]NO_3$
$\Rightarrow$1 mole of complex will precipitate 1 mole of AgCl.
$\Rightarrow$ Given moles of complex $=$
$M \times V = (0.05 \times 100 \times 10^{-3})$ moles
$\Rightarrow$ Moles of $AgCl = 5 \times 10^{-3}$ moles
$\Rightarrow x = 5$

Q. 22 An alkane (Y) requires 8 moles of oxygen for complete combustion and on chlorination with Cl_2/hv, (Y) gives only one monochlorinated product (Z). The total number of primary carbon atoms in (Y) is _____ .

Correct Answer:
4

Solution:
$$C_nH_{2n+2} + \left(\frac{3n+1}{2}\right)O_2 \rightarrow nCO_2 + (n+1)H_2O$$
$$\frac{3n+1}{2} = 8$$
$$3n + 1 = 8 \times 2$$
$$n = 5 (\therefore C_5H_{12})$$



Neopentane (X)
Only one mono-bromo product

Number of primary carons = 4

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Q. 23 500 mL of $0.2 \mathrm{M} \mathrm{MnO}_4^-$ solution in basic medium when mixed with 500 mL of 1.5 M KI solution, oxidises iodide ions to liberate molecular iodine. This liberated iodine is then titrated with a standard x M thiosulphate solution in presence of starch till the end point. If 300 mL of thiosulphate was consumed, then the value of x is _____ .

Correct Answer:
1

Solution:
$3 \mathrm{e}^- + \mathrm{MnO}_4^- + 2 \mathrm{H}_2 \mathrm{O} \rightarrow \mathrm{MnO}_2 + 4 \mathrm{OH}^{\ominus}$

$2 \mathrm{I}^{\ominus} \rightarrow \mathrm{I}_2 + 2 \mathrm{e}^-$

$\mathrm{I}_2 + 2\ \mathrm{S}_2 \mathrm{O}_3^{-2} \rightarrow 2 \mathrm{I}^{\ominus} + \mathrm{S}_4 \mathrm{O}_6^{-2}$

$\text{gm equivalent of } \mathrm{MnO}_4^-$

$= 0.2 \times \frac{500}{1000} \times 3 = 0.3 \text{ (Limiting reagent)}$

$\text{gm equivalent of KI}$

$= 1.5 \times \frac{500}{1000} \times 1 = 0.75$

$\text{gm equivalent of } \mathrm{MnO}_4^-$

$=$ gm equivalent of hypo

$0.3 = \mathrm{x} \times \frac{300}{1000} \times 1$

$\mathrm{x} = 1$

Q. 24 In a closed flask at 600 K , one mole of $\mathrm{X}_2 \mathrm{Y}_4(\mathrm{~g})$ attains equilibrium as given below:

$$\mathrm{X}_2 \mathrm{Y}_4(\mathrm{~g}) \rightleftharpoons 2 \mathrm{XY}_2(\mathrm{~g})$$

At equilibrium, $75 \% \mathrm{X}_2 \mathrm{Y}_4(\mathrm{~g})$ was dissociated and the total pressure is 1 atm . The magnitude of $\Delta_{\mathrm{r}} \mathrm{G}^{\ominus}$ (in kJmol^{-1}) at this temperature is _____ . (Nearest Integer)
(Given: $\mathrm{R}=8.3 \mathrm{~J} \mathrm{~mol}^{-1} \mathrm{~K}^{-1} ; \ln 10=2.3$,

$\log 2=0.3, \log 3=0.48, \log 5=0.69, \log 7=0.84$)

Correct Answer:
8

Solution:

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$$X_2Y_4(g) \rightleftharpoons 2XY_2(g)$$

$t = 0 \quad P^\circ$

$t = t \quad P^\circ(1-\alpha) \quad 2P^\circ\alpha$

$P^\circ(1+\alpha) = 1$

$\alpha = \frac{3}{4}; P^\circ = \frac{4}{7}$

$k_P = \frac{4P_o\alpha^2}{1-\alpha^2} = \frac{36}{7}$

$\Delta G^\circ = -RT\ell n k_p$

$|\Delta G^\circ| = 8.314 \times 600 \ell n\left(\frac{36}{7}\right)$

$= 8169.5 \text{ J mol}^{-1}$

$= 8.169 \text{ kJ/mol}$

Q. 25 Decomposition of a hydrocarbon follows the equation $k = \left(5.5 \times 10^{11} \text{ s}^{-1}\right) e^{\frac{-28000 \text{ K}}{T}}$. The activation energy of reaction is _____ kJ mol^{-1}. (Nearest Integer)
Given : $R = 8.3 JK^{-1} \text{ mol}^{-1}$

Correct Answer:
232

Solution:

$K = \left(5.5 \times 10^{11} \text{ S}^{-1}\right) e^{-\frac{28000 \text{ K}}{T}}$

$K = Ae^{-E_0/RT}$

$\frac{E_a}{R} = 28000$

$E_a = \frac{28000 \times 8.3}{1000} \text{ kJ/mol} = 232.4 \text{ kJ/mol}$

Maths

Q. 1 Let $f : R \to R$ be defined as $f(x) = \frac{2x^2-3x+2}{3x^2+x+3}$.
Then f is :

Option 1:
both one-one and onto

Option 2:
one-one but not onto

Option 3:
onto but not one-one

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Option 4:
neither one-one nor onto

Correct Answer:
neither one-one nor onto

Solution:
$\frac{2x^2-3x+2}{3x^2+x+3}=y$
$(3y-2)x^2+(y+3)x+(3y-2)=0$
$D\geq 0$
$(7y-1)(5y-7)\leq 0$
$y\in\left[\frac{1}{7},\frac{7}{5}\right]$

Hence $f(x)$ is into and $f(x)$ is many-one because $\mathrm{f(x)}$ is non-monotonic

---

Q. 2 Consider the quadratic equation $\left(n^2-2n+2\right)x^2-3x+\left(n^2-2n+2\right)^2=0, n\in R$. Let α be the minimum value of the product of its roots and β be the maximum value of the sum of its roots. Then the sum of the first six terms of the G.P., whose first term is α and the common ratio is $\frac{\alpha}{\beta}$, is :

Option 1:
$\frac{61}{37}$

Option 2:
$\frac{121}{81}$

Option 3:
$\frac{364}{243}$

Option 4:
$\frac{1093}{729}$

Correct Answer:
$\frac{364}{243}$

Solution:
$\alpha=n^2-2n+2$
$\alpha=(n-1)^2+1$

$\therefore$ minimum value of α is 1
Similarly $\beta=\frac{3}{(n-1)^2+1}$
$\therefore$ maximum value of β is 3

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required G.P. is $1, \frac{1}{3}, \frac{1}{3^2}, \ldots$

$S_6 = \frac{1\left(1-\left(\frac{1}{3}\right)^6\right)}{1-\frac{1}{3}} = \frac{3}{2}\left[1-\left(\frac{1}{3}\right)^6\right] = \frac{364}{243}$

Q. 3 Let $S = \left\{z \in \mathbb{C} : z^2 + \sqrt{6}iz - 3 = 0\right\}$. Then $\sum_{z \in S} z^8$ is equal to :

Option 1:
162

Option 2:
184

Option 3:
262

Option 4:
324

Correct Answer:
162

Solution:

$$z = \frac{-\sqrt{6}i \pm \sqrt{-6+12}}{2}$$
$$= \frac{-\sqrt{6}i \pm \sqrt{6}}{2}$$
$$= \frac{\sqrt{6}}{2}(\pm 1 - i)$$
$$\alpha = \frac{-\sqrt{6}}{2}(1+i), \beta = \frac{\sqrt{6}}{2}(1-i)$$
$$\alpha = \frac{-\sqrt{6}}{2} \times \sqrt{2}e^{i\pi/4}, \beta = +\frac{\sqrt{6}}{2} \times \sqrt{2} - i\pi/4$$
$$\alpha = -\sqrt{3}e^{i\pi/4}, \beta = +\sqrt{3}e^{-i\pi/4}$$
$$\alpha^8 + \beta^8 = 81 \cdot e^{i2\pi} + 81 \cdot e^{i(-2\pi)}$$
$$= 81(1+1) = 162$$

Q. 4 The sum of all possible values of $\theta \in [0, 2\pi]$, for which the system of equations :

$$x\cos 3\theta - 8y - 12z = 0$$
$$x\cos 2\theta + 3y + 3z = 0$$
$$x + y + 3z = 0$$

has a non-trivial solution, is equal to :

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Option 1:
π

Option 2:
2π

Option 3:
3π

Option 4:
4π

Correct Answer:
4π

Solution:

$$\begin{vmatrix}\cos 3\theta & -8 & -12\\ \cos 2\theta & 3 & 3\\ 1 & 1 & 3\end{vmatrix}=0$$

$$\begin{vmatrix}\cos 3\theta & -8 & -4\\ \cos 2\theta & 3 & 1\\ 1 & 1 & 1\end{vmatrix}=0$$

$$(\mathrm{C}_1\rightarrow \mathrm{C}_1-\mathrm{C}_2)\&\mathrm{C}_2\rightarrow \mathrm{C}_2-\mathrm{C}_3$$

$$\begin{vmatrix}\cos 3\theta+8 & -4 & -3\\ \cos 2\theta-3 & 2 & 1\\ 0 & 0 & 1\end{vmatrix}=0$$

$$2\cos 3\theta+16+4\cos 2\theta-12=0$$

$$\left(4\cos^3\theta-3\cos\theta\right)+2\left(2\cos^2\theta-1\right)+2=0$$

$$4\cos{}^3\theta+4\cos^2\theta-3\cos\theta=0$$

$$\cos\theta\left(4\cos^2\theta+4\cos\theta-3\right)=0$$

$$\cos\theta\left(2\cos^{\theta}+3\right)(2\cos\theta-1)=0$$

$$\cos\theta=0,\frac{1}{2},-\frac{3}{2}(\text{ rejected })$$

$$\theta=\frac{\pi}{2},\frac{3\pi}{2},\frac{\pi}{3},\frac{5\pi}{3}$$

$$\text{Sum }=4\pi$$

Q. 5

Let $\mathrm{A}=\begin{bmatrix}1 & 0 & 0\\ 3 & 1 & 0\\ 9 & 3 & 1\end{bmatrix}$ and $\mathrm{B}=[\mathrm{b_{ij}}],1\leq \mathrm{i},\mathrm{j}\leq 3.$

If $B=A^{99}-I$, then the value of $\frac{b_{31}-b_{21}}{b_{32}}$ is :

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Option 1:
99

Option 2:
199

Option 3:
149

Option 4:
159

Correct Answer:
149

Solution:

$$A = P + I$$

$$A = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 9 & 3 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = P + I \text{ where } P^3 = 0$$

$$A^{99} = I + 99P + {}^{99}C_2 P^2$$

$$A^{99} - I = 99 \begin{bmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 9 & 3 & 0 \end{bmatrix} + {}^{99}C_2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 9 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 99 \times 3 & 0 & 0 \\ 44550 & 99 \times 3 & 0 \end{bmatrix}$$

$$b_{31} = 44550,\ b_{21} = 99 \times 3,\ b_{32} = 99 \times 3$$

$$\frac{44550 - 297}{297} = 149$$

Q. 6 The sum $1 + \frac{1}{2}\left(1^2 + 2^2\right) + \frac{1}{3}\left(1^2 + 2^2 + 3^2\right) + \ldots\ldots\ldots$ upto 10 terms is equal to :

Option 1:
130

Option 2:
155

Option 3:
$\frac{315}{2}$

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Option 4:
$\frac{325}{2}$

Correct Answer:
$\frac{315}{2}$

Solution:

$$T_n = \frac{1}{n}\left(1^2 + 2^2 + \ldots + n^2\right)$$

$$T_n = \frac{n(n+1)(2n+1)}{6n}$$

$$T_n = \frac{1}{6}\left[2n^2 + 3n + 1\right]$$

$$S_n = \Sigma T_n = \frac{1}{6}\left[2\Sigma n^2 + 3\Sigma n + \Sigma 1\right]$$

$$\Sigma T_n = \frac{1}{6}\left[2 \cdot \frac{n(n+1)(2n+1)}{6} + \frac{3n(n+1)}{2} + n\right]$$

$$S_{10} = \frac{1}{6}\left[\frac{2 \times 10 \times 11 \times 21}{6} + \frac{3}{2} \times 10 \times 11 + 10\right]$$

$$= \frac{10}{6}\left[77 + \frac{33}{2} + 1\right] = \frac{315}{2}$$

Q. 7 A building has ground floor and 10 more floors. Nine persons enter in a lift at the ground floor. The lift goes up to the 10th floor. The number of ways, in which any 4 persons exit at a floor and the remaining 5 persons exit at a different floor, if the lift does not stop at the first and the second floors, is equal to :

Option 1:
2184

Option 2:
3064

Option 3:
7056

Option 4:
11340

Correct Answer:
7056

Solution:
Form group of 4 and 5 person $\frac{9!}{4!5!} = 126$ choose two floor out of 8 floor ${}^8\mathrm{C}_2 = 28$
Total ways in which 4 \& 5 persons can leave at different floor $126 \times 28 \times 2 = 7056$

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Q. 8 Let the mean and the variance of seven observations $2, 4, \alpha, 8, \beta, 12, 14, \alpha < \beta$, be 8 and 16 respectively. Then the quadratic equation whose roots are $3\alpha + 2$ and $2\beta + 1$ is :

Option 1:
$x^2 - 35x + 306 = 0$

Option 2:
$x^2 - 41x + 420 = 0$

Option 3:
$x^2 - 45x + 506 = 0$

Option 4:
$x^2 - 37x + 342 = 0$

Correct Answer:
$x^2 - 41x + 420 = 0$

Solution:

$$\frac{2 + 4 + \alpha + \beta + 8 + 12 + 14}{7} = 8$$

$$\alpha + \beta = 56 - 40 = 16 \Rightarrow \alpha + \beta = 16 \ldots\ldots \text{ (i)}$$

$$\text{Variance } = \frac{\sum x_i^2}{n} - \left(\frac{\sum x_i}{n}\right)^2$$

$$16 = \frac{2^2 + 4^2 + \alpha^2 + \beta^2 + 8^2 + 12^2 + 14^2}{7} - 64$$

$$560 = \alpha^2 + \beta^2 + 424$$

$$\Rightarrow \alpha^2 + \beta^2 = 136$$

From (i) \& (ii) $\alpha = 6 \& \beta = 10$

$$x_1 = 3\alpha + 2 = 20; x_2 = 2\beta + 1 = 21$$

$$\text{Q.E. } : x^2 - 41x + 420 = 0$$

Q. 9 A bag contains 6 blue and 6 green balls. Pairs of balls are drawn without replacement until the bag is empty. The probability that each drawn pair consists of one blue and one green ball is :

Option 1:
$\frac{63}{925}$

Option 2:
$\frac{17}{231}$

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Option 3:
$\frac{16}{231}$

Option 4:
$\frac{64}{925}$

Correct Answer:
$\frac{16}{231}$

Solution:
Probability $= \frac{\lfloor 6 \times \lfloor 6}{\left(\frac{\lfloor 12}{(\lfloor 2)^6 \lfloor 6}\right) \times \lfloor 6} = \frac{16}{231}$

Q. 10 Let C be a circle having centre in the first quadrant and touching the x -axis at a distance 3 units from the origin. If the circle C has an intercept of length $6\sqrt{3}$ on y-axis, then the length of the chord of the circle C on the line $x - y = 3$ is :

Option 1:
8

Option 2:
6

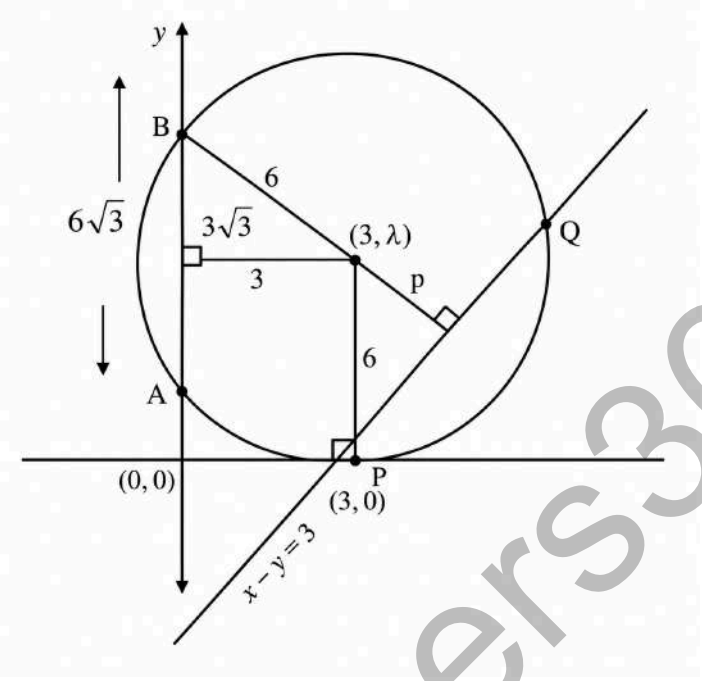
Option 3:
$6\sqrt{2}$

Option 4:
$8\sqrt{2}$

Correct Answer:
$6\sqrt{2}$

Solution:
Centre of circle is $(3, 6)$ and radius $= 6$ units.

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$PQ = 2\sqrt{6^2 - p^2} \quad \therefore p = \dfrac{6}{\sqrt{2}}$

$p = \dfrac{|3 - 6 - 3|}{\sqrt{2}}$

$PQ = 2\sqrt{36 - \dfrac{36}{2}}$

$= 2 \times \dfrac{6}{\sqrt{2}} = 6\sqrt{2}$

Q. 11 The eccentricity of an ellipse E with centre at the origin O is $\frac{\sqrt{3}}{2}$ and its directrices are $x = \pm\frac{4\sqrt{6}}{3}$. Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be a hyperbola whose eccentricity is equal to the length of semi-major axis of E , and whose length of latus rectum is equal to the length of minor axis of E . Then the distance between the foci of H is :

Option 1:
$\frac{4\sqrt{2}}{\sqrt{7}}$

Option 2:
$\frac{4\sqrt{2}}{7}$

Option 3:
$\frac{4}{\sqrt{7}}$

Option 4:
$\frac{8}{7}$

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Correct Answer:
$\frac{8}{7}$

Solution:

$e = \frac{\sqrt{3}}{2}$

$\frac{a}{e} = \frac{4\sqrt{6}}{3}$

$a = 2\sqrt{2}$

$(ae)^2 = a^2 - b^2$

$b^2 = 2$

so $E : \frac{x^2}{8} + \frac{y^2}{2} = 1$

Now $e_H = 2\sqrt{2} \Rightarrow b^2 = 7a^2$

$\frac{2b^2}{a} = 2\sqrt{2} \Rightarrow a = \frac{\sqrt{2}}{7}$

$(ae_H)^2 = a^2 + b^2 = 8a^2 = \frac{16}{49}$

$ae_H = \frac{4}{7}$

Distance between focii $= 2ae_H = \frac{8}{7}$

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Q. 12 Let $x = 9$ be a directrix of an ellipse E, whose centre is at the origin and eccentricity is $\frac{1}{3}$. Let $P(\alpha, 0), \alpha > 0$, be a focus of E and AB be a chord passing through P . Then the locus of the mid point of AB is :

Option 1:
$9y^2 = 8x(1 - x)$

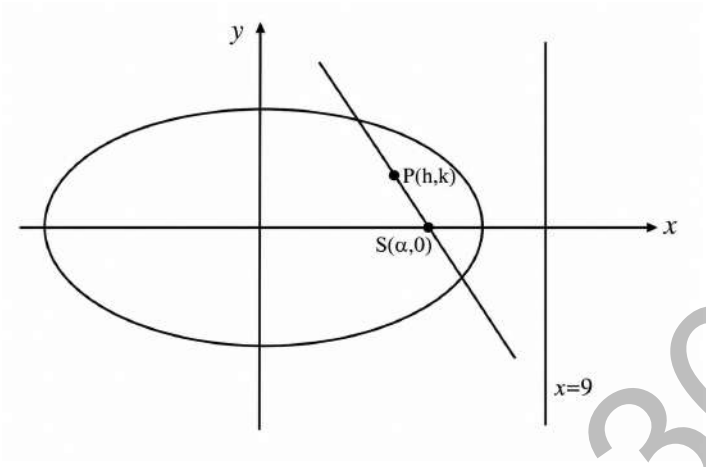
Option 2:
$3y^2 = 4x(1 - x)$

Option 3:
$9y^2 = 8x(x - 1)$

Option 4:
$3y^2 = 4x(x - 1)$

Correct Answer:
$9y^2 = 8x(1 - x)$

Solution:



$\frac{a}{e}=9 \Rightarrow a=3$

$b^2=a^2\left(1-e^2\right)=9\left(1-\frac{1}{9}\right)=8$

$\frac{x^2}{9}+\frac{y^2}{8}=1$

$S(\alpha, 0) \equiv S(1,0)$

$T=S_1 \Rightarrow \frac{hx}{9}+\frac{ky}{8}=\frac{h^2}{9}+\frac{k^2}{8}$

$(1,0) \Rightarrow \frac{h}{9}+0=\frac{h^2}{9}+\frac{k^2}{8}$

$h=h^2+\frac{9}{8}k^2$

$9y^2=8x(1-x)$

Q. 13 If $\sin\left(\tan^{-1}(x\sqrt{2})\right)=\cot\left(\sin^{-1}\sqrt{1-x^2}\right), x \in (0,1)$, then the value of x is :

Option 1:
$\frac{1}{2}$

Option 2:
$\frac{1}{3}$

Option 3:
$\frac{2}{3}$

Option 4:
$\frac{5}{8}$

Correct Answer:
$\frac{1}{2}$

Solution:

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$$\frac{x\sqrt{2}}{\sqrt{2x^2+1}} = \frac{1}{\sqrt{1-x^2}}$$
$$2\left(1-x^2\right) = \left(2x^2+1\right)$$
$$2-2x^2 = 2x^2+1$$
$$x^2 = \frac{1}{4}$$
$$x = \frac{1}{2}$$

Q. 14 The shortest distance between the lines $\frac{x-4}{1} = \frac{y-3}{2} = \frac{z-2}{-3}$ and $\frac{x+2}{2} = \frac{y-6}{4} = \frac{z-5}{-5}$ is:

Option 1:
$\frac{5\sqrt{6}}{6}$

Option 2:
$2\sqrt{5}$

Option 3:
$3\sqrt{5}$

Option 4:
$4\sqrt{5}$

Correct Answer:
$3\sqrt{5}$

Solution:

$$\text{SD} = \frac{\begin{vmatrix} 4-(-2) & 3-6 & 2-5 \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}}$$
$$\text{SD} = \frac{|6(-10+12)-(-3)(-5+6)-3(4-4)|}{|2\hat{i}-\hat{j}|}$$
$$\text{SD} = \left|\frac{12+3}{\sqrt{5}}\right|$$
$$= \frac{15}{\sqrt{5}}$$
$$\text{SD} = 3\sqrt{5}$$

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Q. 15 Let $\vec{a}=2\hat{i}+3\hat{j}+3\hat{k}$ and $\vec{b}=6\hat{i}+3\hat{j}+3\hat{k}$. Then the square of the area of the triangle with adjacent sides determined by the vectors $(2\vec{a}+3\vec{b})$ and $(\overrightarrow{\mathrm{a}}-\overrightarrow{\mathrm{b}})$ is :

Option 1:
450

Option 2:
900

Option 3:
1800

Option 4:
2400

Correct Answer:
1800

Solution:

$2\vec{a}+3\vec{b}=2(2\hat{i}+3\hat{j}+3\hat{k})+3(6\hat{i}+3\hat{j}+3\hat{k})$

$\Rightarrow 22\hat{i}+15\hat{j}+15\hat{k}$

$\vec{a}-\vec{b}=-4\hat{i}$

$\text{Area }=\frac{1}{2}|(2\vec{a}+3\vec{b})\times(\vec{a}-\vec{b})|$

$=\frac{1}{2}|-60\hat{j}+60\hat{k}|=\frac{1}{2}\sqrt{(60)^2\times 2}$

$A=\frac{60}{\sqrt{2}}$

Square of area is $\mathrm{A}^2=1800$

Q. 16 Let $\lim_{x\to 2}\frac{(\tan(x-2))\left(rx^2+(p-2)x-2p\right)}{(x-2)^2}=5$ for some $r,p\in R$. If the set of all possible values of q, such that the roots of the equation $\mathrm{rx}^2-\mathrm{px}+\mathrm{q}=0$ lie in $(0,2)$, be the interval $(\alpha,\beta]$, then $4(\alpha+\beta)$ equals.

Option 1:
11

Option 2:
13

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Option 3:
17

Option 4:
21

Correct Answer:
17

Solution:

$$\lim_{x\to 2}\frac{\tan(x-2)}{x-2}\cdot\frac{\left[x^2+(p-2)x-2p\right]}{x-2}=5$$

$$\Rightarrow \lim_{x\to 2}\frac{r^2-2x+p(x-2)}{x-2}=5$$

$$\because D^r\to 0\therefore N^r\to 0\Rightarrow r=1$$

$$\Rightarrow \lim_{x\to 2}\frac{x(x-2)+p(x-2)}{x-2}=5$$

$$\Rightarrow 2+p=5$$

$$\Rightarrow p=3$$

Now, quadratic equation is $x^2-3x+q=0$ its both roots lie in $(0,2)$



$$D\geq 0,\quad 0<\frac{-b}{2a}<2,\quad f(0)>0,\quad f(2)>0$$

$$\Rightarrow 9-4q\geq 0 \text{ and } q>2$$

$$\Rightarrow q\in\left(2,\frac{9}{4}\right]$$

$$\alpha=2,\beta=\frac{9}{4}\Rightarrow\left(2+\frac{9}{4}\right)=17$$

Q. 17 Let $\mathrm{A}=\begin{vmatrix}1 & 3 & -1\\ 2 & 1 & \alpha\\ 0 & 1 & -1\end{vmatrix}$ be a singular matrix. Let $f(x)=\int_0^x\left(t^2+2t+3\right)dt, x\in[1,\alpha]$. If M and m are respectively the maximum and the minimum values of f in $[1,\alpha]$ then $3(\mathrm{M}-\mathrm{m})$ is equal to :

Option 1:
64

Option 2:
68

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Option 3:
72

Option 4:
76

Correct Answer:
68

Solution:
$|A| = 0$
$\Rightarrow 1.(-1-\alpha) - 3(-2-0) - 1(2-0) = 0$
$\Rightarrow \alpha = 3$
$f(x) = \int_0^x \left(t^2 + 2t + 3\right) dt; x \in [1, 3]$
$f(x) = \frac{x^3}{3} + x^2 + 3x$
$f'(x) = x^2 + 2x + 3 \quad \text{where } D < 0$
$\Rightarrow f(x) \text{ is strictly increasing}$
$M = f(3) = 9 + 9 + 9 = 27$
$m = f(1) = \frac{1}{3} + 1 + 3 = \frac{13}{3}$
$3(M - m) = 3\left(27 - \frac{13}{3}\right) = 68$

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Q. 18 Let $f : R \to R$ be such that $f(xy) = f(x)f(y)$, for all $x, y \in R$ and $f(0) \neq 0$. Let $g : [1, \infty) \to R$ be a differentiable function such that

$x^2 g(x) = \int_1^x \left(t^2 f(t) - \text{tg}(t)\right) dt$

Then $\text{g}(2)$ is equal to :

Option 1:
$\frac{13}{8}$

Option 2:
$\frac{11}{16}$

Option 3:
$\frac{15}{32}$

Option 4:
$\frac{17}{64}$

Correct Answer:
$\frac{15}{32}$

Solution:

$\because f(xy) = f(x)f(y)$

$\therefore f(x) = x^n$ or $f(x) = k$

But $f(0) \neq 0 \Rightarrow f(x) \neq x^n$

For $f(x) = k \Rightarrow k = k \cdot k \Rightarrow k = 1 (\because f(0) \neq 0)$

$\Rightarrow f(x) = 1$

$$x^2\, g(x) = \int_1^x \left(t^2(1) - tg(t)\right) \cdot dt$$

diff w/r to x

$$\Rightarrow x^2\, g'(x) + g(x) \cdot 2x = x^2 - xg(x)$$

$$\Rightarrow xg'(x) + 3\, g(x) = x$$

$$\Rightarrow\ g'(x) + \frac{3}{x}\, g(x) = 1$$

$IF = e^{\int \frac{3}{x} dx} = e^{3\ln(x)} = x^3$

Solution of DE :

$$\Rightarrow g(x) \cdot x^3 = \int x^3 dx$$

$$\Rightarrow x^3 \cdot g(x) = \frac{x^4}{4} + C$$

$$\Rightarrow C = -\frac{1}{4} \{\because\ g(1) = 0\}$$

$$g(x) \cdot x^3 = \frac{x^4}{4} - \frac{1}{4}$$

Put $x = 2$

$$(8)\, g(2) = 4 - \frac{1}{4}$$

$$g(2) = \frac{15}{32}.$$

Q. 19 The area of the region $\left\{(x, y) : x^2 - 8x \leq y \leq -x\right\}$ is:

Option 1:

$\frac{343}{6}$

Option 2:

$\frac{637}{6}$

Option 3:

$\frac{437}{6}$

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Option 4:

$\frac{523}{6}$

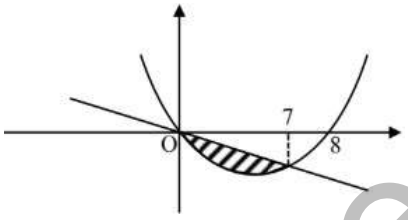
Correct Answer:

$\frac{343}{6}$

Solution:

$$\text{Required area } = \int_0^7 (7x - x^2)\,dx$$

$$= \left[\frac{7x^2}{2} - \frac{x^3}{3}\right]_0^7 = \frac{343}{6}$$



Q. 20 The value of the integral $\int_{-1}^{1} \left(\frac{x^3+|x|+1}{x^2+2|x|+1}\right)dx$ is equal to

Option 1:

$3\log_e 2$

Option 2:

$2\log_e 2$

Option 3:

$5\log_e 3$

Option 4:

$3\log_e 3$

Correct Answer:

$2\log_e 2$

Solution:

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$$I = \int_{-1}^{1} \frac{x^3 + |x| + 1}{x^2 + 2|x| + 1} \cdot dx$$
$$I = \int_{-1}^{1} \frac{x^3}{x^2 + 2|x| + 1} \cdot dx + \int_{-1}^{1} \frac{|x| + 1}{x^2 + 2|x| + 1} \cdot dx$$
$$I = 0 + \int_{-1}^{1} \frac{|x| + 1}{|x|^2 + 2|x| + 1} \cdot dx$$
$$2\int_{0}^{1} \frac{1}{x + 1} \cdot dx$$
$$I = 2[\ln|x + 1|]_0^1$$
$$I = 2\ln(2)$$

Q. 21 Let $\mathrm{R} = \{(\mathrm{x}, \mathrm{y}) \in \mathrm{N} \times \mathrm{N} : \log_e(\mathrm{x} + \mathrm{y}) \leq 2\}$. Then the minimum number of elements, required to be added in R to make it a transitive relation, is

____ .

Correct Answer:
15

Solution:
$x + y \leq e^2 \left(e^2 \simeq 7.29\right)$
$R = \{(1,1), (1,2), (1,4), (1,5), (1,6)\}$
$(2,1), (2,2), (2,3), (2,4), (2,5)$
$(3,1), (3,2), (3,3), (3,4)$
$(4,1), (4,2), (4,3)$
$(5,1), (5,2)$
$(6,1)$

For R to be transitive, we have to add

$(6,2), (6,3), (6,4), (6,5), (6,6)$
$(5,3), (5,4), (5,5), (5,6)$
$(4,4), (4,5), (4,6)$
$(3,5), (3,6)$
$(2,3)$
$= 15$ elements

Q. 22 If $\left(1 - x^3\right)^{10} = \sum_{\mathrm{r}=0}^{10} \mathrm{a_r x^r (1 - x)^{30-2r}}$, then $\frac{9\mathrm{a_9}}{\mathrm{a_{10}}}$ is equal to ____

Correct Answer:
30

Solution:

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$(1-x)^3 = 1 - x^3 - 3x(1-x)$
$(1-x)^3 = 1 - x^3 - 3x + 3x^2$
$(1-x)^3 + 3x - 3x^2 = 1 - x^3$
$\left(1-x^3\right)^{10} = \left[(1-x)^3 + 3x(1-x)\right]^{10}$
$\Rightarrow {}^{10}C_r (3x(1-x))^r \cdot \left((1-x)^3\right)^{10-r}$
$\left(1-x^3\right)^{10} = {}^{10}C_r \cdot 3^r \cdot x^r \cdot (1-x)^{30-2r}$
$\therefore a_r = {}^{10}C_r \cdot 3^r$
$\frac{a_9}{a_{10}} = \frac{{}^{10}C_9 3^9}{{}^{10}C_{10} \cdot 3^{10}}$
$\frac{a_9}{a_{10}} = \frac{10}{3}$
$\therefore \frac{9a_9}{a_{10}} = 30$

Q. 23 Let the line $x - y = 4$ intersect the circle $\mathrm{C} : (\mathrm{x}-4)^2 + (\mathrm{y}+3)^2 = 9$ at the points Q and R . If $\mathrm{P}(\alpha, \beta)$ is a point on C such that $\mathrm{PQ} = \mathrm{PR}$, then $(6\alpha + 8\beta)^2$ is equal to _____ .

Correct Answer:
18

Solution:
$x - y = 4$
$(x-4)^2 + (y+3)^2 = 9$

(α, β) lies on circle

$\therefore (\alpha - 4)^2 + (\beta + 3)^2 = 9$

eq. (1) \& (2) $\Rightarrow y^2 + (y+3)^2 = 9 \Rightarrow 2y^2 + 6y = 0$

$\Rightarrow \mathrm{y} = 0, -3$
$\therefore \mathrm{x} = 4, 1$
$\mathrm{Q}(4,0), \mathrm{R}(1,-3)$

now

$(\mathrm{PQ})^2 = (\mathrm{PR})^2$
$\Rightarrow (\alpha - 4)^2 + \beta^2 = (\alpha - 1)^2 + (\beta + 3)^2$
$\Rightarrow \alpha + \beta = 1$

eq. (3) \& (4)

$\Rightarrow \alpha = 4 + \frac{3}{\sqrt{2}}, \beta = -3 - \frac{3}{\sqrt{2}}$

$\therefore (6\alpha + 8\beta)^2 = 18$

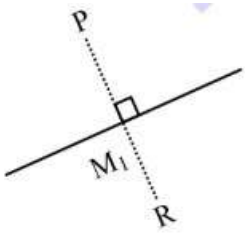
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Q. 24 Let the image of the point $\mathrm{P}(0,-5,0)$ in the line $\frac{x-1}{2}=\frac{y}{1}=\frac{z+1}{-2}$ be the point R and the image of the point $\mathrm{Q}\left(0, \frac{-1}{2}, 0\right)$ in the line $\frac{x-1}{-1}=\frac{y+9}{4}=\frac{z+1}{1}$ be the point S. Then the square of the area of the parallelogram PQRS is

____ .

Correct Answer:
162

Solution:
$\mathrm{P}(0,-5,0)$



$\frac{\mathrm{x}-1}{2}=\frac{\mathrm{y}}{1}=\frac{\mathrm{z}+1}{-2}=\mathrm{k}$

Let $\mathrm{R}\left(\mathrm{x}_1, \mathrm{y}_1, \mathrm{z}_1\right)$

$2\mathrm{k}+1, \mathrm{k},-2\mathrm{k}-1$

$(2\mathrm{k}+1)(2)+(\mathrm{k}+5)(1)+(-2\mathrm{k}-1)(-2)=0$

$\Rightarrow 9\mathrm{k}+9=0 \Rightarrow \mathrm{k}=-1$

$\therefore \quad \mathrm{M}_1 \Rightarrow(-1,-1,1)$

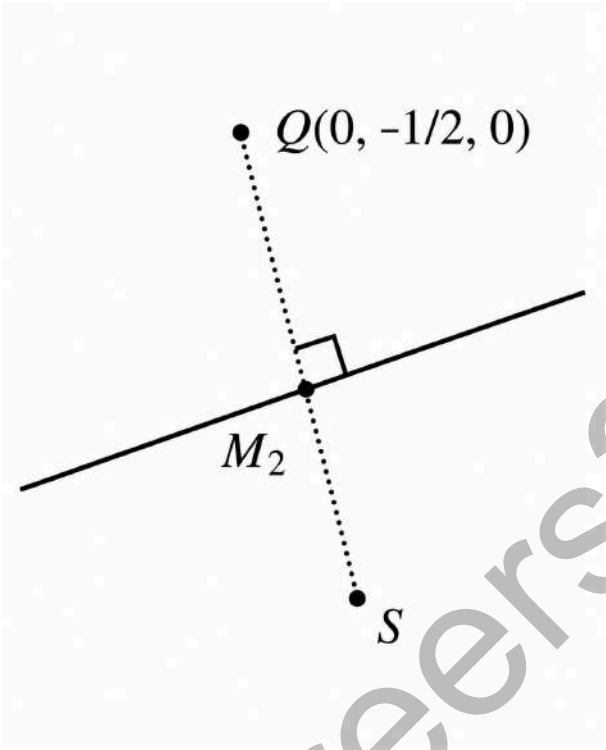
$\mathrm{R} \Rightarrow(-2,3,2)$

$\mathrm{Q}(0,-1/2,0)$

$$\frac{\mathrm{x}-1}{-1}=\frac{\mathrm{y}+9}{4}=\frac{\mathrm{z}+1}{1}=\lambda$$

$(-\lambda+1, 4\lambda-9, \lambda-1)$

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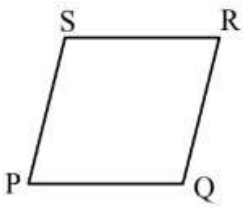
$S\left(x_2, y_2, z_2\right)$

$(-\lambda+1)(-1)+\left(4\lambda-\frac{17}{2}\right)(4)+(\lambda-1)(1)=0$

$\Rightarrow 18\lambda-36=0 \Rightarrow \lambda=2$

$M_2 \Rightarrow (-1,-1,1)$

$S \Rightarrow (-2,-3/2,2)$



$\mathrm{PR}=\sqrt{4+64+4}=\sqrt{72}=6\sqrt{2}$

$\mathrm{QS}=\sqrt{4+1+4}=\sqrt{9}=3$

$\mathrm{Ar.}=\frac{1}{2} \cdot 6\sqrt{2} \cdot 3=9\sqrt{2}$

$(\mathrm{Ar.})^2=(9\sqrt{2})^2=162 \text{ Sq. unit}$

Q. 25

Let $f(x)=\begin{cases}x^3+8 & x<0 \\ x^2-4 & x \geq 0\end{cases}$ and

$g(x)=\begin{cases}(x-8)^{1/3}; & x<0 \\ (x+4)^{1/2}; & x \geq 0\end{cases}$

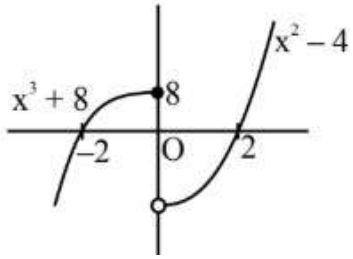
Then the number of points, where the function gof is discontinuous, is _____ .

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Correct Answer:
3

Solution:

$$g(f(x)) = \begin{cases} (f(x)-8)^{1/3} & f(x) < 0 \\ (f(x)+4)^{1/2} & f(x) \geq 0 \end{cases}$$



$$g(f(x)) = \begin{cases} \left(x^3\right)^{1/3}, & x < -2 \\ \left(x^2-12\right)^{1/3}, & 0 < x < 2 \\ \left(x^3+12\right)^{1/2}, & -2 \leq x \leq 0 \\ \left(x^2\right)^{1/2}, & x \geq 2 \end{cases}$$

$$g(f(x)) = \begin{cases} x, & x < -2 \\ \left(x^3+12\right)^{1/2}, & -2 \leq x \leq 0 \\ \left(x^2-12\right)^{1/3}, & 0 < x < 2 \\ x, & x \geq 2 \end{cases}$$

Number of points of discontinuity is equal to 3 .

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JEE MAINS

2026

8th April

2nd Shift

Question Paper

Physics

Q. 1 A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth in terms of α units if light takes 6 min. 40 s to cover this distance?

Option 1:
200α

Option 2:
400α

Option 3:
300α

Option 4:
500α

Correct Answer:
400α

Solution:
$\alpha = 3 \times 10^8 \text{ m}$

$\text{Distance } = (400) \times 3 \times 10^8 \text{ m} = 400\alpha$

Q. 2 Consider the equation $H = \frac{x^p \epsilon^q E^r}{t^s}$
Where H = magnetic field; E = electric field; $\in$= permittivity, x = distance, t = time. The values of $\mathrm{p}, \mathrm{q}, \mathrm{r}$ and s respectively are :

Option 1:
$1, 1, 1, 1$

Option 2:
$-1, 1, 2, 1$

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Option 3:
$1, -1, -2, 1$

Option 4:
$-1, -2, -2, 1$

Correct Answer:
$1, 1, 1, 1$

Solution:
As per the question, considering H as magnetic field

$$[\mathrm{H}] = \frac{\mathrm{MLT}^{-2}}{\mathrm{LT}^{-1}\mathrm{IT}^{-}} = \mathrm{MI}^{-1}\,\mathrm{T}^{-2}$$
$$[\mathrm{E}] = \frac{\mathrm{MLT}^{-2}}{\mathrm{IT}} = \mathrm{MLI}^{-1}\,\mathrm{T}^{-3}$$
$$[\epsilon] = \frac{\mathrm{I}^2\,\mathrm{T}^2}{\mathrm{MLT}^{-2}\,\mathrm{L}^2} = \mathrm{M}^{-1}\,\mathrm{L}^{-3}\mathrm{I}^2\,\mathrm{T}^4$$
$$\mathrm{MI}^{-1}\,\mathrm{T}^{-2} = \mathrm{M}^{-q+\mathrm{r}}\,\mathrm{L}^{\mathrm{p}-3\mathrm{q}+\mathrm{r}}\mathrm{I}^{2\mathrm{q}-\mathrm{r}}\,\mathrm{T}^{4\mathrm{q}-3\mathrm{r}-\mathrm{s}}$$
$$-\mathrm{q}+\mathrm{r}=1$$
$$\mathrm{P}-3\mathrm{q}+\mathrm{r}=0$$
$$2\mathrm{q}-\mathrm{r}=-1$$
$$4\mathrm{q}-3\mathrm{r}-\mathrm{S}=-2$$

Solving we get

$q = 0, r = 1$
$p = -1, S = -1$

No option matching
Considering H as magnetic field intensity

$$\mathrm{H} = \frac{\mathrm{B}}{\mu_0} = \mathrm{Ni} \Rightarrow \text{ current per unit length } \Rightarrow \mathrm{IL}^{-1}$$

$$-\mathrm{q}+\mathrm{r}=0 \qquad \mathrm{q}=1$$
$$\mathrm{p}-3\mathrm{q}+\mathrm{r}=-1 \qquad \mathrm{r}=1$$
$$2\mathrm{q}-\mathrm{r}=1 \qquad \mathrm{p}=+1$$
$$4\mathrm{q}-3\mathrm{r}-\mathrm{s}=0 \qquad \mathrm{s}=+1$$

Option (1) is matching
Note : As per the question if we consider H as magnetic field, no option is matching but if we consider H as magnetic field intensity then option (1) is matching.

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Q. 3 A car moving with a speed of $54\ \mathrm{km/h}$ takes a turn of radius 20 m . A simple pendulum is suspended from the ceiling of the car. Determine the angle made by the string of the pendulum with the vertical during the turning. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

Option 1:
$\tan^{-1}(0.5)$

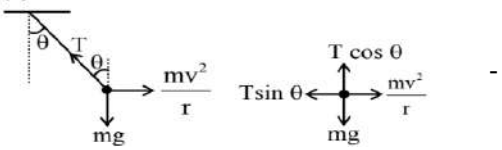
Option 2:
$\tan^{-1}(0.75)$

Option 3:
$\tan^{-1}(1.125)$

Option 4:
$\tan^{-1}(0.25)$

Correct Answer:
$\tan^{-1}(1.125)$

Solution:



$$\tan\theta = \frac{\mathrm{v}^2}{\mathrm{rg}} \quad \left(\mathrm{v} = 54 \times \frac{5}{18} = 15\ \mathrm{m/s}\right)$$
$$\tan\theta = \frac{225}{20 \times 10}$$
$$\tan\theta = \frac{9}{8}$$
$$\theta = \tan^{-1}(1.125)$$

Q. 4 A gas balloon is going up with a constant velocity of $10\ \mathrm{m/s}$. When this balloon reached a height of 75 m , a stone is dropped from it and balloon keeps moving up with the same velocity. The height of the balloon when the stone hits the ground is _____ m. (Take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

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Option 1:
85

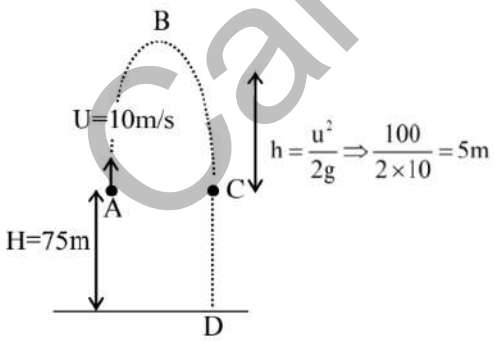
Option 2:
150

Option 3:
129

Option 4:
125

Correct Answer:
125

Solution:



$$-75 = 10 \times t - \frac{1}{2} \times 10 \times t^2$$
$$t^2 - 2t - 15 = 0$$
$$(t-5)(t+3) = 0 \Rightarrow \quad t = 5$$
$$\text{Height } = 75 + 10 \times 5 = 125 \text{ m}$$

Q. 5 A thin biconvex lens is prepared from the glass ($\mu = 1.5$) both curved surfaces of which have equal radii of 20 cm each. Left side surface of the lens is silvered from outside to make it reflecting. To have the position of image and object at the same place, the object should be placed, from the lens at a distance of _____ cm .

Option 1:
10

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Option 2:
12.5

Option 3:
13

Option 4:
13.5

Correct Answer:
10

Solution:
Combination of lens + mirror

$$f_m = \frac{R}{2} = -10 \text{ cm}; f_L = \frac{R}{2(\mu - 1)} = 20 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{f_m} - \frac{2}{f_L}$$

$$\frac{1}{\mathrm{f}} = \frac{-1}{10} - \frac{2}{20} = -\frac{1}{5}$$

$$\mathrm{f} = -5 \text{ cm}$$

To form image at the position of object, it should be placed at center of curvature of mirror + lens combination.

$$\therefore \mathrm{u} = 2\mathrm{f} = 2 \times 5 = 10 \text{ cm}$$

Q. 6 Two identical bodies, projected with the same speed at two different angles cover the same horizontal range R . If the time of flight of these bodies are 5 s and 10 s , respectively, then the value of R is _____ m . (Take $\mathrm{g} = 10 \text{ m/s}^2$)

Option 1:
250

Option 2:
25

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Option 3:
500

Option 4:
125

Correct Answer:
250

Solution:
For range to be same, angle of projection must be complementary angles.

$$T_1 = \frac{2u\sin\theta}{g} = 10, T_2 = \frac{2u\cos\theta}{g} = 5$$

$$\text{Range } = \frac{2(u\sin\theta)(u\cos\theta)}{g}$$

$$= \frac{2(50)(25)}{10}m = 250m$$

Q. 7 A solid cylinder having radius R and length L is slipping on a rough horizontal plane. At time $\mathrm{t} = 0$ the cylinder has a translational velocity $\mathrm{v_0} = 49\ \mathrm{m/s}$, perpendicular to its axis and a rotational velocity $\mathrm{v_0/4R}$ about the centre. The time taken by the cylinder to start rolling is _____ seconds. (coefficient of kinetic friction $\mu_{\mathrm{K}} = 0.25$ and $\mathrm{g} = 9.8\ \mathrm{m/s^2}$)

Option 1:
15

Option 2:
5

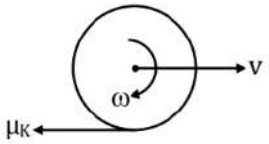
Option 3:
10

Option 4:
7.5

Correct Answer:
5

Solution:

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$$a = \frac{\mu_K mg}{m} = \mu_K g$$

$$\tau = I\alpha \Rightarrow \mu_K mgR = \frac{MR^2}{2} \times \alpha$$

$$\Rightarrow \alpha = \frac{2\mu_K g}{R}$$

$\therefore v = v_0 - \mu_k gt$ (i)

Also, $\omega = \omega_0 + \frac{2\mu_K gt}{R}$ (ii)

$\frac{v}{R} = \frac{v_0}{4R} + \frac{2\mu_k gt}{R}$

Solving Eq. (i) \& (ii)

$$\frac{v_0}{4} + 2\mu gt = v_0 - \mu_k gt$$

$$3\mu gt = v_0 - \frac{v_0}{4}$$

$$3\mu gt = \frac{3v_0}{4}$$

$$\Rightarrow t = \frac{v_0}{4\mu_K\, g} = \frac{49}{4 \times \frac{1}{4} \times 9.8} = 5\text{sec.}$$

Q. 8 A liquid of density $600\ \text{kg/m}^3$ flowing steadily in a tube of varying cross-section. The cross-section at a point A is $1.0\ \text{cm}^2$ and that at B is $20\ \text{mm}^2$. Both the points A and B are in same horizontal plane, the speed of the liquid at A is $10\ \text{cm/s}$. The difference is pressures at A and B points is

_____ Pa .

Option 1:
18

Option 2:
144

Option 3:
36

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Option 4:
72

Correct Answer:
72

Solution:
Using continuity equation,

$$A_1V_1 = A_2V_2$$
$$1 \times 10 = \frac{1}{5} \times V_2$$
$$\Rightarrow V_2 = 50 \text{ cm/s}$$

Using Bernoulli's theorem,

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$$
$$P_1 - P_2 = \frac{1}{2}\rho \left[V_2^2 - V_1^2\right]$$
$$= \frac{1}{2} \times 600 \left[(50)^2 - (10)^2\right] \times 10^{-4} = 72 \text{ Pa}$$

Q. 9 A spherical liquid drop of radius R acquires the terminal velocity v_1 when falls through a gas of viscosity η. Now the drop is broken into 64 identical droplets and each droplet acquires terminal velocity v_2 falling through the same gas. The ratio of terminal velocities v_1/v_2 is ____ .

Option 1:
4

Option 2:
0.25

Option 3:
32

Option 4:
16

Correct Answer:
16

Solution:

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Using volume conservation,

$\frac{4}{3}\pi R_1^3 = 64 \times \frac{4}{3}\pi R_2^3 \Rightarrow R_2 = \frac{R_1}{4}$

Terminal velocity $\mathrm{v} \propto \mathrm{R}^2$

$\frac{\mathrm{v}_1}{\mathrm{v}_2} \Rightarrow \frac{\mathrm{R}_1^2}{\mathrm{R}_2^2} \Rightarrow (4)^2 = 16$

Q. 10 One mole of diatomic gas having rotational modes only is kept is a cylinder with a piston system. The cross-section area of the cylinder is $4\ \mathrm{cm}^2$. The gas is heated slowly to raise the temperature by $1.2^{\circ}\mathrm{C}$ during which the piston moves by 25 mm . The amount of heat supplied to the gas is
_____ J. (Atmospheric pressure $= 100\mathrm{kPa}$, $\mathrm{R} = 8.3\ \mathrm{J/mol.\ K}$) (Neglect mass of the piston)

Option 1:
24.8

Option 2:
25

Option 3:
15.04

Option 4:
29.98

Correct Answer:
25

Solution:
I st Method

$$\mathrm{Q} = \Delta\mathrm{U} + \mathrm{W}$$
$$\mathrm{Q} = \frac{\mathrm{f}}{2}\mathrm{nR}\Delta\ \mathrm{T} + \mathrm{nR}\Delta\ \mathrm{T} = \left(\frac{\mathrm{f}}{2} + 1\right)\mathrm{nR}\Delta\ \mathrm{T}$$

if $\mathrm{f} = 2 \Rightarrow \mathrm{Q} = 2\mathrm{nR}\Delta\mathrm{T} \Rightarrow 2 \times 1 \times 8.3 \times 1.2 = 19.92$
(No option is matching)
if $\mathrm{f} = 5 \Rightarrow \mathrm{Q} = 3.5\mathrm{nR}\Delta\mathrm{T}$

$\Rightarrow 3.5 \times 1 \times 8.3 \times 1.2 = 34.86$

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(No option is matching)

Method-2

$$\mathrm{Q}=\Delta \mathrm{U}+\mathrm{W}$$
$$\mathrm{Q}=\frac{\mathrm{f}}{2} \mathrm{nR} \Delta \mathrm{T}+\mathrm{P} \Delta \mathrm{V}$$

if $\mathrm{f}=2$

$$\Rightarrow Q=nR\Delta T+P\Delta V$$
$$=1 \times 8.3 \times 1.2+10^{5} \times 25 \times 10^{-3} \times 4 \times 10^{-4}$$
$$=10.96$$

(No option is matching)
if $\mathrm{f}=5$

$$\Rightarrow Q=2.5nR\Delta T+P\Delta V$$
$$2.5 \times 1 \times 8.3 \times 1.2+10^{5} \times 25 \times 10^{-3} \times 4 \times 10^{-4}$$
$$=25.9$$

(closest is 25)
Note : As $\mathrm{P} \Delta \mathrm{V} \neq \mathrm{nR} \Delta \mathrm{T}$, given state is inconsistent
Hence we can opt the most suitable option as (2).

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Q. 11 Initial pressure and volume of a monoatomic ideal gas are P and V . The change in internal energy of this gas in adiabatic expansion to volume $\mathrm{V}_{\text{final}}=27 \mathrm{~V}$ is _____ J.

Option 1:
$-2 \mathrm{PV}(3 \sqrt{3}-1)$

Option 2:
$\frac{4}{3} \mathrm{PV}$

Option 3:
$-\frac{4}{3} \mathrm{PV}$

Option 4:
$\frac{3}{4} \mathrm{PV}$

Correct Answer:
$-\frac{4}{3}\mathrm{PV}$

Solution:
$\mathrm{P}_1\ \mathrm{V}_1^{\gamma} = \mathrm{P}_2\ \mathrm{V}_2^{\gamma}$
$\gamma = \frac{5}{3}$
$\mathrm{PV}^{5/3} = \mathrm{P}_2 \times (27\ \mathrm{V})^{5/3}$
$\mathrm{P}_2 = \frac{\mathrm{P}}{(27)^{5/3}} = \frac{\mathrm{P}}{3^5}$
$\Delta \mathrm{U} \Rightarrow \mathrm{nC_v}\Delta \mathrm{T} = \frac{\mathrm{nR}\Delta \mathrm{T}}{(\gamma - 1)}$
$= \frac{\mathrm{P}_2\ \mathrm{V}_2 - \mathrm{P}_1\ \mathrm{V}_1}{(\gamma - 1)}$
$\Rightarrow \frac{\frac{\mathrm{P}}{3^5} \times 3^3.\mathrm{V} - \mathrm{PV}}{\frac{5}{3} - 1} \Rightarrow \frac{-\mathrm{PV}\left\{1 - \frac{1}{9}\right\}}{\frac{2}{3}}$
$= -\mathrm{PV} \times \frac{8}{9} \times \frac{3}{2}$
$\Rightarrow -\frac{4}{3}\mathrm{PV}$

Q. 12 The frequency of oscillation of a mass m suspended by spring is v_1. If the length of the spring is cut to half, the same mass oscillates with frequency v_2. The value of $\frac{v_2}{v_1}$ is _____ .

Option 1:
1

Option 2:
2

Option 3:
$\sqrt{2}$

Option 4:
$\sqrt{3}$

Correct Answer:
$\sqrt{2}$

Solution:

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For a spring, $k\ell =$ constant

$$k_1\ell = k_2\frac{\ell}{2} \Rightarrow k_2 = 2k_1$$

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

$$\Rightarrow \frac{f_1}{f_2} = \sqrt{\frac{k_1}{k_2}}$$

$$\Rightarrow \frac{f_1}{f_2} = \sqrt{\frac{1}{2}} \Rightarrow \frac{f_2}{f_1} = \sqrt{2}$$

Q. 13 A monochromatic source of light operating at 15 kW emits 2.5×10^{22} photons/s. The region of an electromagnetic spectrum to which the emitted electromagnetic radiation belongs to _____ (Take $h = 6.6 \times 10^{-34}$ J. s and $c = 3 \times 10^{8}$ m/s).

Option 1:
Microwave

Option 2:
Infrared

Option 3:
Visible

Option 4:
Ultraviolet

Correct Answer:
Ultraviolet

Solution:

$$\frac{hC}{\lambda} = \frac{15 \times 10^3}{2.5 \times 10^{22}}$$

$$\therefore \lambda = \frac{6.6 \times 10^{-34} \times 3 \times 10^8 \times 2.5 \times 10^{22}}{15 \times 10^3}$$

$$\lambda = 3.3 \times 10^{-7}$$

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Q. 14 A current carrying circular loop of radius 2 cm with unit normal $\hat{n}=\frac{\hat{k}+\hat{i}}{\sqrt{2}}$ is placed in a magnetic field, $\overrightarrow{B}=B_0(3\hat{i}+2\hat{k})$. If $B_0=4\times10^{-3}\ T$ and current $I=100\sqrt{2}\ A$, the torque experienced by the loop is _____ Wb.A. $(\pi=3.14)$

Option 1:
$16\times10^{-5}\hat{k}$

Option 2:
$5024\times10^{-7}\hat{k}$

Option 3:
$5024\times10^{-7}\hat{i}$

Option 4:
$5024\times10^{-7}\hat{j}$

Correct Answer:
$5024\times10^{-7}\hat{j}$

Solution:

$\vec{\tau}=\overrightarrow{M}\times\overrightarrow{B}$

$\Rightarrow 100\sqrt{2}\times3.14\times4\times10^{-4}\left[\left(\frac{\hat{k}+\hat{i}}{\sqrt{2}}\right)\times(3\hat{i}+2\hat{k})\right]B_0$

$\Rightarrow 16\times3.14\times[+3\hat{j}-2\hat{j}]\times10^{-5}$

$\Rightarrow 50.24\times10^{-5}\hat{j}$

Q. 15 A 30 cm long solenoid has 10 turns per cm and area of $5\ cm^2$. The current through the solenoid coil varies from 2 A to 4 A is 3.14 s . The e.m.f. induced in the coil is $\alpha\times10^{-5}\ V$. The value α is _____ .

Option 1:
60

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Option 2:
12

Option 3:
120

Option 4:
34

Correct Answer:
12

Solution:

$$\varepsilon = -\frac{\mathrm{Ldi}}{\mathrm{dt}} = -\mu_0 \mathrm{n}^2\ \mathrm{A}\ell\frac{\mathrm{di}}{\mathrm{dt}}$$

$$= -4\pi \times 10^{-7} \times 10^6 \times 5 \times 10^{-4} \times 0.3 \times \frac{2}{\pi}$$

$$\Rightarrow -12 \times 10^{-5}$$

Q. 16 Two point charges $\mathrm{q}_1 = 3\mu\mathrm{C}$ and $\mathrm{q}_2 = -4\mu\mathrm{C}$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ respectively. Force on charge q_2 is _____ N.

$$\left(\text{Take } \tfrac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI Units}\right)$$

Option 1:
$(12\hat{\mathrm{i}} + 24\hat{\mathrm{j}} + 24\hat{\mathrm{k}}) \times 10^{-3}$

Option 2:
$(4\hat{\mathrm{i}} + 8\hat{\mathrm{j}} + 8\hat{\mathrm{k}}) \times 10^{-3}$

Option 3:
$(3\hat{i} + 6\hat{j} + 6\hat{k}) \times 10^{-3}$

Option 4:
$(-4\hat{\mathrm{i}} - 8\hat{\mathrm{j}} - 8\hat{\mathrm{k}}) \times 10^{-3}$

Correct Answer:
$(4\hat{\mathrm{i}} + 8\hat{\mathrm{j}} + 8\hat{\mathrm{k}}) \times 10^{-3}$

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Solution:

$\overrightarrow{r} = \overrightarrow{r_2} - \overrightarrow{r_1} \Rightarrow -\hat{i} - 2\hat{j} - 2\hat{k}$

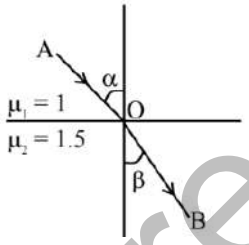
$\text{Force} \Rightarrow \frac{kq_1q_2\overrightarrow{r}}{r^3}$

$\Rightarrow \frac{9 \times 10^9 \times (3)(+4) \times 10^{-12}}{27}(\hat{i} + 2\hat{j} + 2\hat{k})$

$\Rightarrow 4 \times 10^{-3}[\hat{i} + 2\hat{j} + 2\hat{k}]$

Q. 17

Light ray incident along a vector $\overrightarrow{AO}(\overrightarrow{AO} = 2\hat{i} - 3\hat{j})$ emerges out along vector $\overrightarrow{OB}(\overrightarrow{OB} = C\hat{i} - 4\hat{j})$ as shown in the figure below. The value of C is _____ .



Option 1:
1.6

Option 2:
0.16

Option 3:
11.6

Option 4:
16

Correct Answer:
1.6

Solution:

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$1(\mathrm{AO} \times \hat{\mathrm{j}}) \Rightarrow (\mathrm{OB} \times \hat{\mathrm{j}}) \times 1.5$

$\frac{2\hat{\mathrm{k}}}{\sqrt{13}} = \frac{3}{2}\left[\frac{\mathrm{C}\hat{\mathrm{k}}}{\sqrt{\mathrm{C}^2+16}}\right]$

$\therefore \frac{4}{13} = \frac{9\mathrm{C}^2}{4\left(\mathrm{C}^2+16\right)} \Rightarrow 16\mathrm{C}^2 + 256 = 117\mathrm{C}^2$

$256 = 101\mathrm{C}^2$

$\mathrm{C} = 1.59 \approx 1.6$

Q. 18 K_1 and K_2 be the maximum kinetic energies of photoelectrons emitted from a surface of a given material for the light of wavelength λ_1 and λ_2, respectively. If $\lambda_1 = 2\lambda_2$ then the work function of material is given by :

Option 1:
$\mathrm{K}_2 + 2\,\mathrm{K}_1$

Option 2:
$2\,\mathrm{K}_2 - \mathrm{K}_1$

Option 3:
$\mathrm{K}_1 - 2\,\mathrm{K}_2$

Option 4:
$\mathrm{K}_2 - 2\,\mathrm{K}_1$

Correct Answer:
$\mathrm{K}_2 - 2\,\mathrm{K}_1$

Solution:
$2\,\mathrm{K}_1 = \frac{2\mathrm{hC}}{\lambda_1} - 2\phi$ (i)

$\mathrm{K}_2 = \frac{2\mathrm{hC}}{\lambda_1} - \phi$ (ii)

Solving (i) \& (ii), we get

$\mathrm{K}_2 - 2\,\mathrm{K}_1 = \phi$

Q. 19 Two radioactive substances A and B of mass numbers 200 and 212 respectively, shows spontaneous α-decay with same Q value of 1 MeV . The ratio of energies of α-rays produced by A and B is _____ .

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Option 1:
$\frac{2548}{2650}$

Option 2:
$\frac{2706}{2646}$

Option 3:
$\frac{2597}{2600}$

Option 4:
$\frac{2862}{2499}$

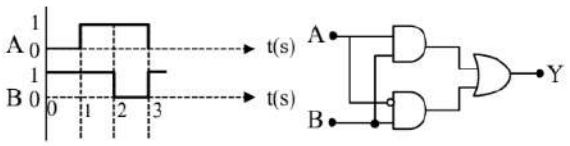
Correct Answer:
$\frac{2597}{2600}$

Solution:

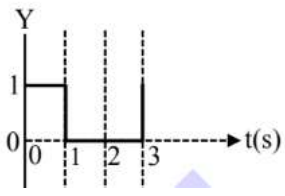
$$k_\alpha = \left(\frac{A-4}{A}\right)Q$$

$$\therefore \frac{k_{\alpha_1}}{k_{\alpha_2}} = \frac{196}{200} \times \frac{212}{208} = \frac{2597}{2600}$$

Q. 20 The output Y for the given inputs A and B to the circuit is :

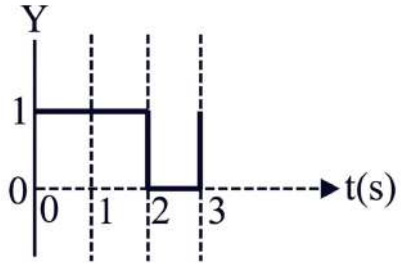


Option 1:

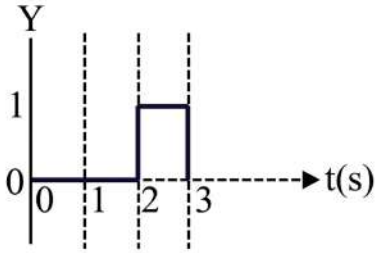


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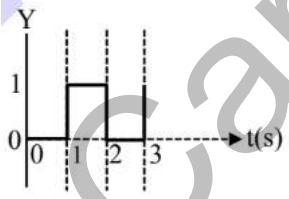
Option 2:



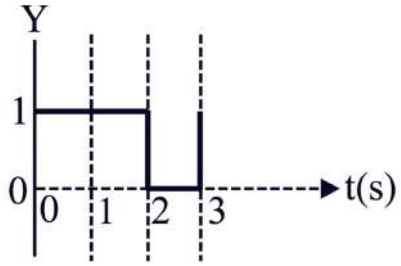
Option 3:



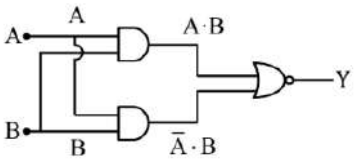
Option 4:



Correct Answer:



Solution:



$$A \cdot B + \overline{A} \cdot B \Rightarrow (A + \overline{A}) \cdot B = B$$

Note : In lower gate, input is taken from A as $\overline{A}$ due to node given in the figure.

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Q. 21 A parallel plate capacitor is having separation between plates 0.885 mm . It has a capacitance of $1\mu\ \mathrm{F}$ when the space between the plates is filled with an insulating material of resistivity $1 \times 10^{13}\Omega\ \mathrm{m}$ and resistance $17.7 \times 10^{14}\Omega$. Relative permittivity of the insulating material is $\alpha \times 10^{7}$. The value of α is _____ .
$\left(\text{Take permittivity of free space} = 8.85 \times 10^{-12}\ \mathrm{F/m}\right)$

Correct Answer:
2

Solution:
$\mathrm{RC} = \rho \mathrm{K} \in_0$

$$\mathrm{K} = \frac{\mathrm{RC}}{\rho\varepsilon_0} = \frac{17.7 \times 10^{14} \times 10^{-6}}{10^{13} \times 8.85 \times 10^{-12}} = 2 \times 10^{7}$$

Q. 22 Some distance star is to be observed by some telescope of diameter of objective lens a , at an angular resolution of 3.0×10^{-7} radian. If the wavelength of light from the star reaching the telescope is 500 nm , the minimum diameter of the objective lens of the telescope is _____ cm .
(Nearest integer)

Correct Answer:
203

Solution:
$$\theta = \frac{1.22\lambda}{b}$$

$$3 \times 10^{-7} = \frac{1.22 \times 5 \times 10^{-7}}{\mathrm{b}} \Rightarrow\ \mathrm{b} = 203.33\ \mathrm{cm}$$

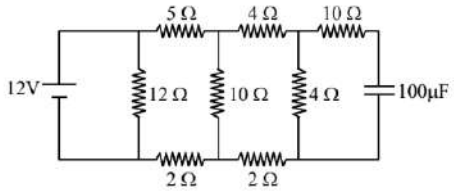
Q. 23 A 5 mg particle carrying a charge of $5\pi \times 10^{-6}\mathrm{C}$ is moving with velocity of $(3\hat{i} + 2\hat{k}) \times 10^{-2}\ \mathrm{m/s}$ in a region having magnetic field $\overrightarrow{\mathrm{B}} = 0.1\hat{\mathrm{k}}\mathrm{Wb/m^2}$. It moves a distance of α meter along $\hat{\mathrm{k}}$ when it completes 5 revolutions. The value of α is _____ .

Correct Answer:
2

Solution:
$$5\left(\frac{2\pi\ \mathrm{m}}{\mathrm{qB}}\right)\mathrm{v} \Rightarrow \frac{5\times2\times\pi\times5\times10^{-6}\times2\times10^{-2}}{5\pi\times10^{-6}\times0.1} = 2\ \mathrm{m}$$

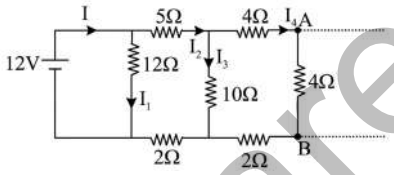
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Q. 24 The stored charge in the capacitor in steady state of the following circuit is _____ $\mu\mathrm{C}$.



Correct Answer:
200

Solution:
At steady state,



$$I = \frac{E}{R_{eq}} = \frac{12}{6} = 2\ \mathrm{A}$$

$$I_1 = I_2 = 1\ \mathrm{A}$$

$$I_3 = I_4 = \frac{1}{2}\ \mathrm{A}$$

$$V_{AB} = \frac{1}{2} \times 4 = 2\ \mathrm{V}$$

Charge on capacitor

$$Q = CV_{AB} = 100 \times 2 = 200\mu C$$

Q. 25 Two masses of 3.4 kg and 2.5 kg are accelerated from an initial speed of $5\ \mathrm{m/s}$ and $12\ \mathrm{m/s}$, respectively. The distances traversed by the masses in the 5th second are 104 m and 129 m , respectively. The ratio of their momenta after 10 s is $\frac{x}{8}$. The value of x is _____ .

Correct Answer:
9

Solution:

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$S = u + \frac{a}{2}(2n-1) \Rightarrow 104 = 5 + \frac{a}{2}(9) \Rightarrow a = 22\ \mathrm{m/s^2}$

$129 = 12 + \frac{a}{2} \times 9 \Rightarrow a = 26\ \mathrm{m/s^2}$

$\frac{p_1}{p_2} = \frac{(5 + 22 \times 10)3.4}{(12 + 26 \times 10)(2.5)} = \frac{225 \times (3.4)}{272 \times 2.5} = \frac{9}{8}$

Chemistry

Q. 1 Match the List - I with List - II

List-I **Mass of substance**	**List-II** **Number of atoms**
A. 1.8 mg water	I. $2 \times 10^{-4} \times \mathrm{N_A}$
B. 9.8 mg sulphuric acid	II. $1.5 \times 10^{-4} \times \mathrm{N_A}$
C. 1.8 mg carbon	III. $3 \times 10^{-4} \times \mathrm{N_A}$
D. 5.85 mg salt (NaCl)	IV. $7 \times 10^{-4} \times \mathrm{N_A}$

Choose the correct answer from the options given below:

Option 1:
A-IV, B-III, C-I, D-II

Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-III, B-IV, C-II, D-I

Option 4:
A-III, B-IV, C-I, D-II

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:
(A) No. of atoms in 1.8 mg water

$= \frac{1.8 \times 10^{-3}}{18} \times \mathrm{N_A} \times 3 = 3 \times 10^{-4}\ \mathrm{N_A}$

(B) No. of atoms in $9.8\mathrm{mg}\mathrm{H_2SO_4}$

$= \frac{9.8 \times 10^{-3}}{98} \times N_A \times 7 = 7 \times 10^{-4} \times N_A$

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(C) No. of atoms in 1.8 mg carbon

$= \frac{1.8\times10^{-3}}{12} \times \mathrm{N_A} = 1.5 \times 10^{-4} \times \mathrm{N_A}$

(D) No. of atoms in 5.85 mg NaCl

$= \frac{5.85\times10^{-3}}{58.5} \times \mathrm{N_A} \times 2 = 2 \times 10^{-4}\ \mathrm{N_A}$

Correct match : A-III, B-IV, C-II, D-I

Q. 2 Given below are two statements :

Given : Molar mass of $\mathrm{C, H, O, Cl}$ are $12, 1, 16$ and $35.5\ \mathrm{g\ mol^{-1}}$, respectively.

Statement I: In $30\%(\mathrm{w/w})$ solution of methanol in $\mathrm{CCl_4}$ (at T K), the mole fraction of $\mathrm{CCl_4}$ is equal to 0.33 .

Statement II : Mixture of methanol and $\mathrm{CCl_4}$ shows positive deviation deviation from Raoult's law.
In the light of the above statements, choose the correct answer from the option given below :

Option 1:
Both Statement I and Statement II are ture

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II are false

Option 4:
Statement I is false but Statement II are ture

Correct Answer:
Both Statement I and Statement II are ture

Solution:
(i) Let wt of solution $= 100\ \mathrm{g}$
wt of methanol $= 30\ \mathrm{g}$
wt of $\mathrm{CCl_4} = 100 - 30 = 70\ \mathrm{g}$

$$\text{mole fraction of } \mathrm{CCl_4} = \frac{\mathrm{n_{CCl_4}}}{\mathrm{n_{CCl_4} + n_{CH_3OH}}} = \frac{\frac{70}{154}}{\frac{70}{154} + \frac{30}{32}}$$

$$= 0.3267 \approx 0.33$$

(ii) $\mathrm{CCl_4}$ \& methanol is mixture of positive deviation from Raoult's law.

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So, statement (I) \& statement (II) are correct.

Q. 3 Bromine trifluoride autoionizes to form $\mathrm{BrF}_2^{\oplus}$ and $\mathrm{BrF}_4^{\ominus}$. The shapes of the cation and anion are respectively _____ , and _____ .

Option 1:
bent, square planar

Option 2:
linear, square planar

Option 3:
bent, see-saw

Option 4:
linear, tetrahedral

Correct Answer:
bent, square planar

Solution:
$\mathrm{BrF}_2^{\oplus} \Rightarrow \mathrm{sp}^3$

[Bond pair = 2, lone pair = 2]
⇒ bent

$\mathrm{BrF}_2^{\oplus} \Rightarrow \mathrm{sp}^3\,\mathrm{d}^2$

[Bond pair = 4, lone pair = 2]
⇒ square planar

Q. 4 Which of the following are not correct?
A. For water, magnitude of $\mathrm{K_b}$ is more than the magnitude of $\mathrm{K_f}$
B. The elevation in boiling point of water when a non-volative solute is added to it is larger in magnitude than its depression in freezing point.
C. Osmotic pressure measurement is preferred over any other colligative property to determine molar mass of proteins and polymers.
D. The dimerised form of benzoic acid in benzene is

is 

Choose the correct answer from the options given below:

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Option 1:
A and B only

Option 2:
A and D only

Option 3:
A, B and D only

Option 4:
A, C and D only

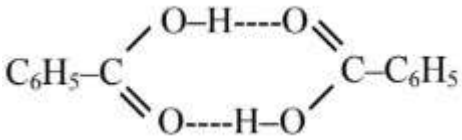
Correct Answer:
A, B and D only

Solution:
(A) $\mathrm{K_b}$ for water $= 0.512 \frac{^\circ\mathrm{C-Kg}}{\mathrm{mol}}$

$\mathrm{K_f}$ for water $= 1.86 \frac{^\circ\mathrm{C-Kg}}{\mathrm{mol}}$

So, $\mathrm{K_b} < \mathrm{K_f}$ for water

(B) Since, $\mathrm{K_b} < \mathrm{K_f}$ so $\Delta \mathrm{T_b} < \Delta \mathrm{T_f}$ [same equimolal solution]
(C) Osmotic pressure is used to determine molar mass of protiens \& polymers
(D) dimerised form of benzoic acid in benzene



option (A), (B) & (D) are incorrect

Q. 5 Consider the following reactions in which all the reactants and products are present in gasesous state

$$2xy \rightleftharpoons x_2 + y_2 \quad K_1 = 2.5 \times 10^5$$
$$xy + \frac{1}{2}z_2 \rightleftharpoons xyz \quad K_2 = 5 \times 10^{-3}$$

The value of $\mathrm{K_3}$ for the equilibrium $\frac{1}{2}x_2 + \frac{1}{2}y_2 + \frac{1}{2}z_2 \rightleftharpoons xyz$ is :

Option 1:
2.5×10^{-3}

Option 2:
2.5×10^{3}

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Option 3:
1.0×10^{-5}

Option 4:
5×10^{-3}

Correct Answer:
1.0×10^{-5}

Solution:
$x_2 + y_2 \rightleftharpoons 2xy, \quad K' = \frac{1}{K_1} = \frac{1}{2.5 \times 10^5} = \frac{1}{25 \times 10^4}$

$\frac{1}{2}x_2 + \frac{1}{2}y_2 \rightleftharpoons xy, K'' = \left(\frac{1}{25 \times 10^4}\right)^{1/2} = \frac{1}{5 \times 10^2} \ldots\ldots$

$xy + \frac{1}{2}z_2 \rightleftharpoons xyz, K_2 = 5 \times 10^{-3}$

On adding equation (1) \& (2)

$\frac{1}{2}x_2 + \frac{1}{2}y_2 + \frac{1}{2}z_2 \rightleftharpoons xyz$

$K_3 = \frac{1}{5 \times 10^2} \times 5 \times 10^{-3} = 1 \times 10^{-5}$

Q. 6 Given at 298 K :

$\mathrm{E}^{\ominus}_{\mathrm{Fe}^{2+}/\mathrm{Fe}} = \mathrm{X\ Volt}$

$\mathrm{E}^{\ominus}_{\mathrm{Fe}^{3+}/\mathrm{Fe}} = \mathrm{Y\ Volt}$

The $\mathrm{E}^{\ominus}_{\mathrm{Fe}^{3+}/\mathrm{Fe}^{2+}}$ in Volt at 298 K is given by :

Option 1:
$2X - 3Y$

Option 2:
$3Y - 2X$

Option 3:
$3Y + 2X$

Option 4:
Y + X

Correct Answer:
$3Y - 2X$

Solution:

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$(i)\ Fe^{+2}(aq) + 2e^- \rightarrow Fe(s) \ldots\ldots\ldots \Delta G_1^\circ$

$(ii)\ Fe^{+3}(aq) + 3e^- \rightarrow Fe(s) \ldots\ldots\ldots \Delta G_2^\circ$

$(iii)\ Fe^{+3}(aq) + e^- \rightarrow Fe^{+2}(aq) \ldots\ldots\ldots \Delta G_3^\circ$

$iii = ii - i$

$-1 \times F \times E^\circ_{Fe^{3+}/Fe^{2+}} = -3 \times F \times y - (-2 \times F \times x)$

$E^\circ_{Fe^{3+}/Fe^{2+}} = 3y - 2x$

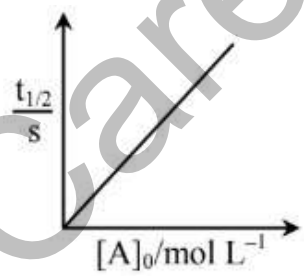
Option (2) is correct

Q. 7 Given below are two statements :

$R = 8.314\ J\ K^{-1}\ mol^{-1}$ and $1cal = 4.2\ J$

Statement I : When $Ea = 12.6 kcal/mol$, the room temperature rate constant is doubled by a $10^\circ C$ increase in temperature (298 K to 308 K)

Statement II : For a first order reaction $A \rightarrow B$,



Here $[A]_0$ is the initial concentration of A and $t_{1/2}$ is half-life of reaction.

In the light of the above statements, choose the correct answer from the option given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II are false

Option 4:
Statement I is false but Statement II are true

Correct Answer:
Statement I is true but Statement II are false

Solution:

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$$\text{S-I}: \ln\left(\frac{K_2}{K_1}\right) = \frac{12.6 \times 10^3}{\left(\frac{8.314}{4.2}\right)}\left[\frac{10}{298 \times 308}\right]$$

$$\Rightarrow \ln\left(\frac{K_2}{K_1}\right) = 0.693$$

$$\Rightarrow \frac{K_2}{K_1} = 2$$

$$\text{S-II}: \text{For } 1^{\text{st}} \text{ order}$$

$$t_{1/2} = \frac{\ln 2}{K}; t_{1/2} \propto [A]^0 = \text{constant.}$$

Statement-I is true but statement-II is false.

Q. 8 Match List - I with List - II

List-I **Electronic configuration of neutral atom (where $n = 2$)**	**List - II** **1st Ionization Energy** $\left(kJmol^{-1}\right)$
A. ns^2	I. 2080
B. ns^2np^1	II. 899
C. ns^2np^3	III. 800
D. ns^2np^6	IV. 1402

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-II, C-IV, D-I

Option 4:
A-III, B-II, C-I, D-IV

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:

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(A) $2\ s^2 \Rightarrow Be$
(B) $2\ s^2 2p^1 \Rightarrow\ B$
(C) $2s^2 2p^3 \Rightarrow\ N$
(D) $2\ s^2 2p^6 \Rightarrow Ne$

IE order : $B < Be < N < Ne$

A-II, B-III, C-IV, D-I

Q. 9 Find the correct statements related to group 15 hydrides.

A. Reducing nature increases from NH_3 to BiH_3
B. Tendency to donate lone pair of electrons decreases from NH_3 to BiH_3
C. The stability of hydrides decreases from NH_3 to BiH_3
D. HEH bond angle decreases from NH_3 to SbH_3 (E = Elements of group 15)
Choose the correct answer from the options given below :

Option 1:
A and B only

Option 2:
B and C only

Option 3:
A, B, C and D

Option 4:
A, C and D only

Correct Answer:
A, B, C and D

Solution:
(1) Reducing nature : $NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$
(2) Lone pair definition tendency :

$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

(3) Stability of Hydrides:

$NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

(4) HEH bond angle : $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$

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Q. 10 Given below are two statements :

Statement I : The number of pairs among $\left[\mathrm{Ti}^{4+}, \mathrm{V}^{2+}\right]$, $\left[\mathrm{V}^{2+}, \mathrm{Mn}^{2+}\right]$, $\left[\mathrm{Mn}^{2+}, \mathrm{Fe}^{3+}\right]$ and $\left[\mathrm{V}^{2+}, \mathrm{Cr}^{2+}\right]$ in which both ions are coloured is 3 .

Statement II : The number pairs among $\left[\mathrm{La}^{3+}, \mathrm{Yb}^{2+}\right]$, $\left[\mathrm{Lu}^{3+}, \mathrm{Ce}^{4+}\right]$ and $\left[\mathrm{Ac}^{3+}, \mathrm{Lr}^{3+}\right]$ ions in which both are diamagnetic is 3 .

In the light of the above statements, choose the correct answer from the option given below :

Option 1:
Both Statement I and Statement II are correct

Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is correct but Statement II are incorrect

Option 4:
Statement I is incorrect but Statement II are correct

Correct Answer:
Both Statement I and Statement II are correct

Solution:
Ti^{+4} is colourless due to d^0 configuration
$\therefore$ statement 1 is correct
$\mathrm{La}^{+3}, \mathrm{Yb}^{+2}, \mathrm{Lu}^{+3}, \mathrm{Ce}^{+4}, \mathrm{Al}^{+3} \& \mathrm{Lr}^{+3}$ all are diamagnetic
$\therefore$ Statement 2 is correct.

Q. 11 Given below are two statements for catalytic properties of transition metals :

Statement I : First row transition metals which act as catalyst utilise their 3d electrons only for formation of bonds between reactant molecules and atoms on the surface of catalyst.

Statement II : There is increase in the concentration of reactants on the surface of catalyst which strengthens the bonds in reacting molecules.
In the light of the above statements, choose the correct answer from the option given below :

Option 1:
Both Statement I and Statement II are correct

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Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is correct but Statement II are incorrect

Option 4:
Statement I is incorrect but Statement II are correct

Correct Answer:
Both Statement I and Statement II are incorrect

Solution:
First row transition metals utilize 3 d and 4 s electrons for bonding on the catalyst surface.
∴ Statement 1 is incorrect
Adsorption increases reactant concentration on the catalyst surface and weakens bonds facilitating formation of activated complex and lowering activation energy.
∴ Statement 2 is incorrect

Q. 12 Given below are two statements :

Statement I: Vapours of the liquid with higher boiling point condense before vapours of the liquid with lower boiling points in fractional distillation.

Statement II : The vapours rising up in the fractionating column become richer in high boiling component of the mixture.
In the light of the above statements, choose the correct answer from the option given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II are false

Option 4:
Statement I is false but Statement II are true

Correct Answer:
Statement I is true but Statement II are false

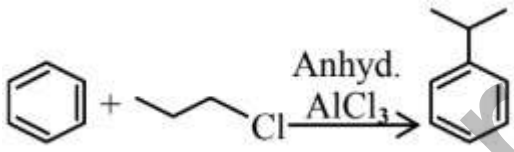
Solution:

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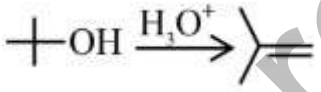
Statement I : True
Statement II : False
Reason - As vapours rise, they become richer in the lower-boiling component, not high-boiling component.

Q. 13 The major product of which of the following reaction is not obtained by rearrangement reaction?

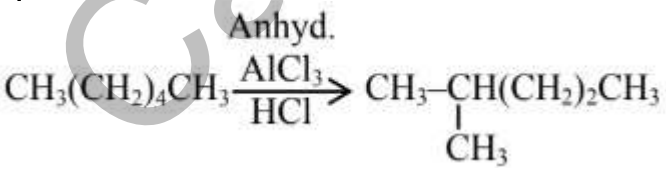
Option 1:



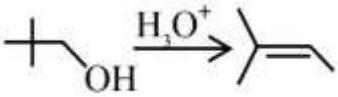
Option 2:



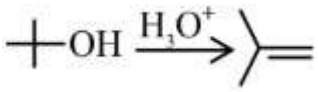
Option 3:



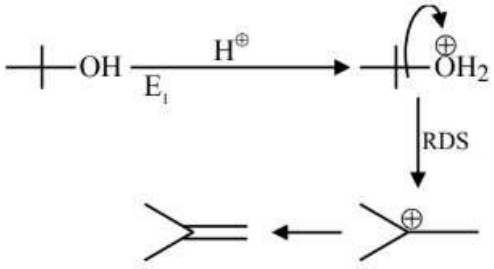
Option 4:



Correct Answer:



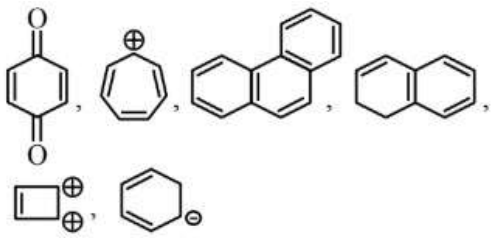
Solution:



Product obtained without rearrangement.

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Q. 14 The total number of aromatic compounds/species from the following is



Option 1:
6

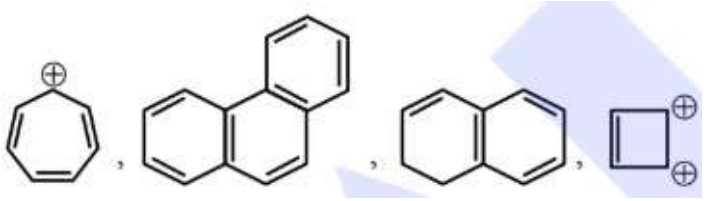
Option 2:
4

Option 3:
3

Option 4:
5

Correct Answer:
4

Solution:
Aromatic compounds are



Q. 15 n-Butane on monochlorination under photochemical conditions gives an optically active compound "P". "P" on further chlorination gives dichloro compounds.

The number of dichloro compounds obtained (ignore stereoisomers) is :

Option 1:
3

Option 2:
4

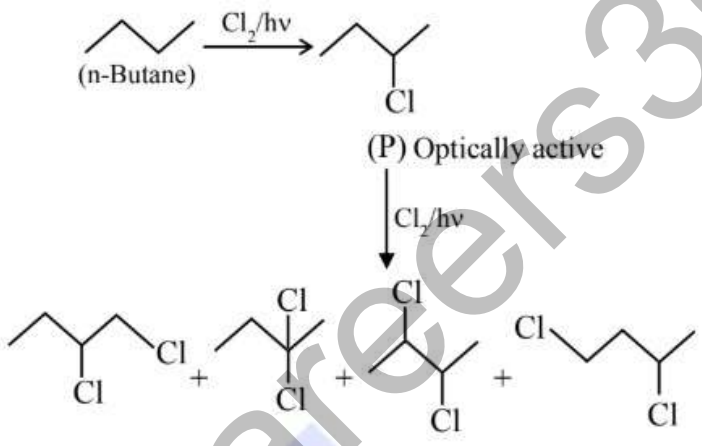
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Option 3:
5

Option 4:
6

Correct Answer:
4

Solution:



Total dichloro products (excluding stereoisomers)= 4

Q. 16 Given below are two statements :

Statement-I : Due to increase in van der Waals forces, the order of boiling points is $CH_3CH_2CH_2I > CH_3CH_2I > CH_3I$.

Statement - II As  is more symmetric ,its melting point is higher than 

however its boiling point is lower than 

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement-I and Statement-II are true.

Option 2:
Both Statement-I and Statement-II are false.

Option 3:
Statement-I is true but Statement-II is false.

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Option 4:
Statement-I is false but Statement-II is true.

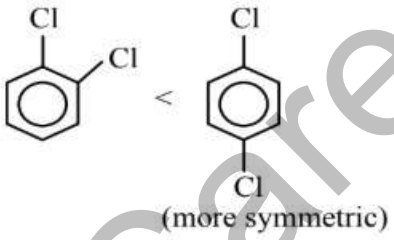
Correct Answer:
Both Statement-I and Statement-II are true.

Solution:
Statement I : True
Order of boiling point :

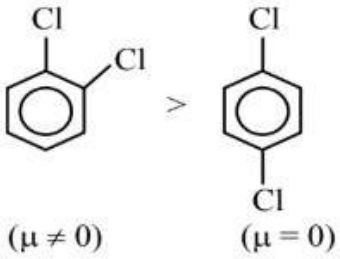
$$CH_3CH_2CH_2I > CH_3CH_2I > CH_3I$$

Vanderwaals forces increases due to increase in molecular weight
Statement 2: True

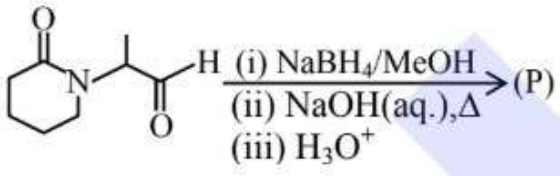
Order of melting point :-



Boiling point order :-
(Boiling point $\propto \mu$)

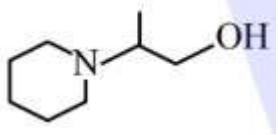


Q. 17 Consider the following reaction :



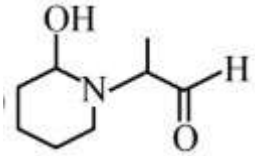
The major product (P) formed is :

Option 1:

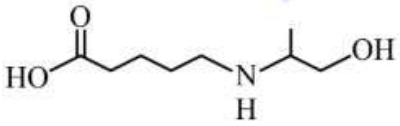


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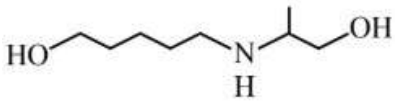
Option 2:



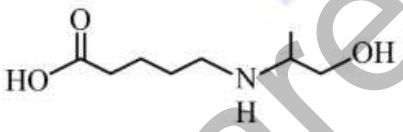
Option 3:



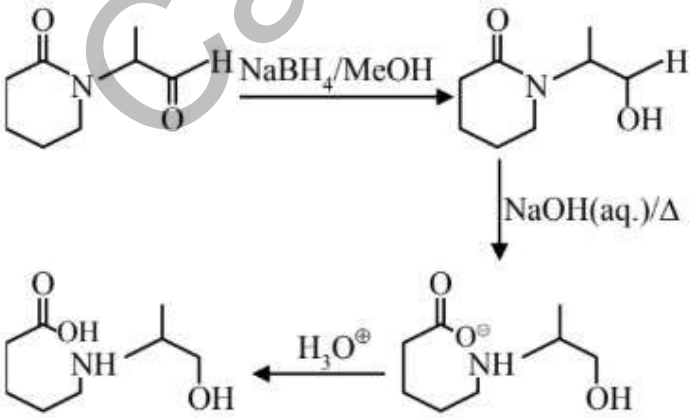
Option 4:



Correct Answer:



Solution:



Q. 18 Which statements are True?

A. In Hoffmann bromamide degradation, 4 moles of NaOH and 2 moles of $\mathrm{Br_2}$ are consumed per mole of an amide
B. Hoffmann bromamide reaction is not given by alkyl amides.
C. Primary amines can be synthesized by Hoffmann bromamide degradation.
D. Secondary amide on reaction with $\mathrm{Br_2}$ and NaOH will give secondary amine.
E. The by-products of Hoffmann degradation are $\mathrm{Na_2CO_3}$, NaBr and $\mathrm{H_2O}$.
Choose the correct answer from the options given below :

Option 1:
A, C and E only

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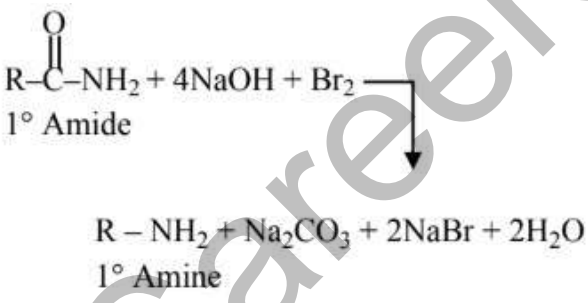
Option 2:
B, C and D only

Option 3:
C and E only

Option 4:
C, D and E only

Correct Answer:
C and E only

Solution:
Hoffmann bromamide degradation reaction -



Statement (A)-False
Statement (B) - False
Statement (C) - True
Statement (D) - False
Statement (E) - True

Q. 19 The incorrect statement from the following with respect to carbohydrates is :

Option 1:
All monosaccharides are reducing sugars.

Option 2:
The monosaccharide units obtained from hydrolysis of oligosaccharides are always the same.

Option 3:
Starch and cellulose are typical examples of polysaccharides, which are very high molecular weight compounds of more than ten monosaccharide units.

Option 4:
Open chain and cyclic structures co-exist at equilibrium that are responsible for certain properties as in the case of D-(+)-glucose.

Correct Answer:
The monosaccharide units obtained from hydrolysis of oligosaccharides are always the same.

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Solution:
The monosaccharide units obtained from hydrolysis of oligosaccharides may be or may not be same.

Example :

$$\text{Lactose} \xrightarrow{H_3O^{\oplus}} \underbrace{\text{Glucose} + \text{Galactose}}_{\text{different monosaccharide units}}$$

$$\text{Maltose} \xrightarrow{H_3O^{\oplus}} \underbrace{\text{Glu cose} + \text{Glucose}}_{\text{same monosaccharide units}}$$

Q. 20 Which of the following amino acid will give violet coloured complex with neutral ferric chloride solution?

Option 1:
Threonine

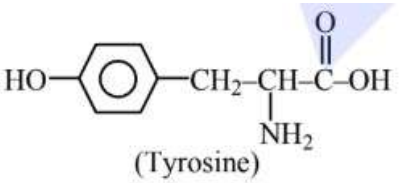
Option 2:
Serine

Option 3:
Tyrosine

Option 4:
Cysteine

Correct Answer:
Tyrosine

Solution:



(Tyrosine)

Tyrosine amino acid will give violet coloured complex with neutral $FeCl_3$ solution because it contains phenolile group.

Q. 21 Number of paramagnetic complexes among the following is _____ .

$[MnBr_4]^{2-}$, $[NiCl_4]^{2-}$, $[Ni(CN)_4]^{2-}$, $[Ni(CO)_4]$,
$[CoF_6]^3$, $[Fe(CN)_6]^4$, $[Mn(CN)_6]^{3-}$,
$[Cu(H_2O)_6]^{2+}$, $[Ti(CN)_6]^{3-}$,
$(C_2O_4)_3]^{3-}$

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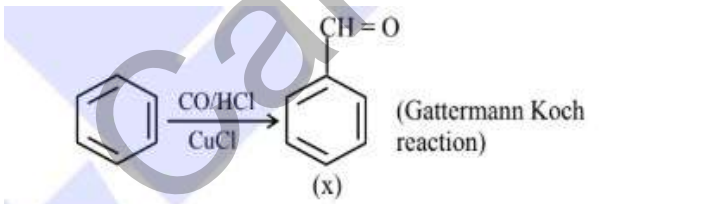
Correct Answer:
6

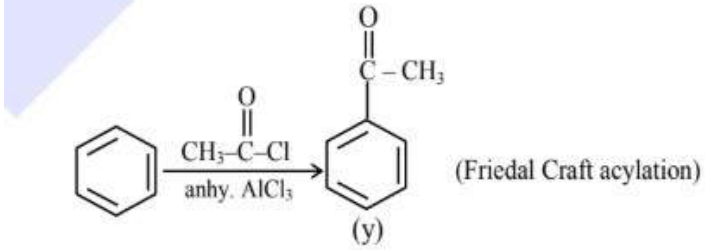
Solution:
$[MnBr_4]^2$, $[NiCl_4]^2$, $[CoF_6]^{3-}$, $[Mn(CN)_6]^{3-}$, $[Ti(CN)_6]^{3-}$, $[Cu(H_2O)_6]^{2+}$ are paramagnetic.

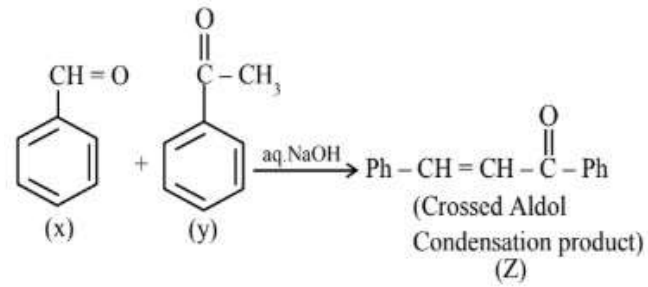
Q. 22 ' x ' is the product which is obtained from benzene by reacting it with carbon monoxide and hydrogen chloride in the presence of cuprous chloride. ' y ' is the major product obtained from the benzene by reacting it with ethanoyl chloride in the presence of anhydrous $AlCl_3$. Product (major) obtained by heating x and y in the presence of alkali is z. Total number of π (pi) electrons in z is _____ .

Correct Answer:
16

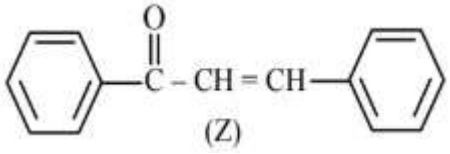
Solution:







Total number of pi (π) electrons in ' z ' $= 16\pi$ electrons



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Q. 23 Consider two radiations of wavelengths :

1. $\lambda_1 = 2000$
2. $\lambda_2 = 6000$

The ratio of the energies of these two radiations $\left(\frac{E_1}{E_2}\right)$ is _____ . (Nearest integer)

Correct Answer:
3

Solution:

$$\mathrm{E_{photon}} = \frac{\mathrm{hc}}{\lambda}$$
$$\Rightarrow \frac{\mathrm{E_1}}{\mathrm{E_2}} = \frac{\lambda_2}{\lambda_1}$$
$$\Rightarrow \frac{\mathrm{E_1}}{\mathrm{E_2}} = \frac{6000}{2000}$$
$$\Rightarrow \frac{\mathrm{E_1}}{\mathrm{E_2}} = 3$$

Q. 24 Consider the reaction :

$$2\mathrm{H_2\ S(\ g)} + 3\mathrm{O_2(\ g)} \rightarrow 2\mathrm{H_2O}(l) + 2\mathrm{SO_2(\ g)}$$

The magnitude of enthalpy change for the reaction in $\mathrm{kJmol^{-1}}$ is _____ . (Nearest integer)
Given : $\Delta_f\mathrm{H}^{\ominus}\ (\mathrm{H_2\ S}) = -20.1\ \mathrm{kJ\ mol^{-1}}$

$\Delta_f\mathrm{H}^{\ominus}\ (\mathrm{H_2O}) = -286.0\ \mathrm{kJ\ mol^{-1}}$
$\Delta_f\mathrm{H}^{\ominus}\ (\mathrm{SO_2}) = -297.0\ \mathrm{kJ\ mol^{-1}}$

Correct Answer:
1126

Solution:

$\Delta_\mathrm{r}\mathrm{H}^\circ = \Sigma\Delta\mathrm{H}_\mathrm{f}^\circ(\ \mathrm{product}\ ) - \Sigma\Delta\mathrm{H}_\mathrm{f}^\circ(\ \mathrm{Reactant}\ )$
$\Delta_\mathrm{r}\mathrm{H}^\circ = 2 \times \Delta\mathrm{H}_\mathrm{f}^\circ\ (\mathrm{SO_2},\ \mathrm{g}) + 2 \times \Delta\mathrm{H}_\mathrm{f}^\circ\ (\mathrm{H_2O}, \ell)$
$\quad -2 \times \Delta\mathrm{H}_\mathrm{f}^\circ\ (\mathrm{H_2\ S},\ \mathrm{g})$
$\Delta_\mathrm{r}\mathrm{H}^\circ = 2 \times (-297) + 2 \times (-286) - 2(-20.1)$
$= -594 - 572 + 40.2$
$\Delta_\mathrm{r}\mathrm{H}^\circ = -1125.8\ \mathrm{kJ/mol.}$
$|\Delta_\mathrm{r}\mathrm{H}^\circ| = 1125.8\ \mathrm{kJ/mol.}$

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Q. 25 Solid carbon, CaO and $CaCO_3$ are mixed and allowed to attain equilibrium at T K .

$CaCO_3 \rightleftharpoons CaO(s) + CO_2(g) \quad Kp_1 = 0.08\ atm$
$C(s) + CO_2(g) \rightleftharpoons 2CO(g) \quad Kp_2 = 2\ atm$

The partial pressure of CO is ____ $\times 10^{-1}$ atm.

Correct Answer:
4

Solution:
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g), K_{P_1} = 0.08\ atm$
$K_{P_1} = P_{CO_2} = 0.08\ atm$
$C(s) + CO_2(g) \rightleftharpoons 2CO(g), K_{P_2} = 2$

$$K_{P_2} = \frac{P_{CO}^2}{P_{CO_2}} = 2$$

$$\frac{P_{CO}^2}{0.08} = 2$$

$$P_{CO}^2 = 16 \times 10^{-2}$$

$$P_{CO} = 4 \times 10^{-1}$$

Maths

Q. 1 Consider the relation R on the set $\{-2, -1, 0, 1, 2\}$ defined by $(a, b) \in R$ if and only if $1 + ab > 0$. Then among the statements :

I. The number of elements in R is 17

II. R is an equivalence relation

Option 1:
Only I is true

Option 2:
Only II is true

Option 3:
Both I and II are true

Option 4:
Neither I nor II is true

Correct Answer:
Only I is true

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Solution:
Reflexive If $(a, a) \in R$

$\Rightarrow 1+a^2>0$

⇒ It is reflexive relation
Symmetric If $1+ab>0 \Rightarrow 1+ba>0$
i.e. If $(a, b) \in R$ and $(b, a) \in R$
Therefore it is symmetric relation
Transitive $\because (-2, 0) \in R \& (0, 2) \in R$

But $(-2, 2) \notin \mathrm{R}$
Hence it is not transitive
⇒ It is not an equivalence relation.
Check Number of elements in R

$\because 1+ab>0$

$\mathrm{a} \& \mathrm{b}$ each can be filled by 5 ways
$\therefore$ total elements $=25$ But out of these

$(-2,-1),(-2,2),(-1,2),(1,-2),(2,-2),(2,-1)$

$(-1,1) \& (1,-1)$ are not in relation.

Q. 2 The number of values of $z \in \mathbb{C}$, satisfying the equations $|z-(4+8i)|=\sqrt{10}$ and $|z-(3+5i)|+|z-(5+11i)|=4\sqrt{5}$, is :

Option 1:
0

Option 2:
2

Option 3:
1

Option 4:
4

Correct Answer:
2

Solution:
$|z-(3+5i)|+|z-(5+11i)|=4\sqrt{5}$

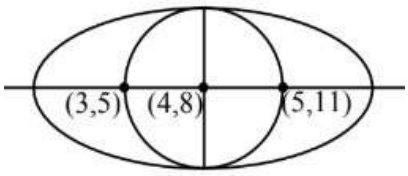
So, $\mathrm{S}_1(3,5)$ \& $\mathrm{S}_2(5,11)$ are two foci of ellipse, where $2ae=S_1S_2=2\sqrt{10} \& 2a=4\sqrt{5}$

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$\Rightarrow a=2\sqrt{5}, e=\frac{1}{\sqrt{2}} \& \ b^2=a^2\left(1-e^2\right)$ gives $b^2=10$

\& center of an ellipse is mid point of both focus
$\therefore$ center $(4,8)$
Equation of circle is $|Z-(4+8i)|=\sqrt{10}$

⇒ Ellipse \& given circle are concentric \& radius of given circle is equal to length of minor axis of given ellipse



⇒ Two points which are common to both.

Q. 3 If the system of linear equations :

$$x+y+z=6$$
$$x+2y+5z=10$$
$$2x+3y+\lambda z=\mu$$

has infinitely many solutions, then the value of $\lambda+\mu$ equals.

Option 1:
12

Option 2:
16

Option 3:
22

Option 4:
28

Correct Answer:
22

Solution:

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$$D=\begin{vmatrix}1 & 1 & 1\\ 1 & 2 & 5\\ 2 & 3 & \lambda\end{vmatrix}=\lambda-6$$

$$D_1=\begin{vmatrix}6 & 1 & 1\\ 10 & 2 & 5\\ \mu & 3 & \lambda\end{vmatrix}=2\lambda+3\mu-60$$

$$D_2=\begin{vmatrix}1 & 6 & 1\\ 1 & 10 & 5\\ 2 & \mu & \lambda\end{vmatrix}=4(\lambda-\mu+10)$$

$$D_3=\begin{vmatrix}1 & 1 & 6\\ 1 & 2 & 10\\ 2 & 3 & \mu\end{vmatrix}=\mu-16$$

For infinite many solution

$D=D_1=D_2=D_3=0$
$\Rightarrow \lambda=6 \& \mu=16$
$\Rightarrow \lambda+\mu=22$

Q. 4

Let $A=\begin{bmatrix}\alpha & 1 & 2\\ 2 & 3 & 0\\ 0 & 4 & 5\end{bmatrix}$ and

$$B=\begin{bmatrix}1 & 0 & 0\\ 0 & -5\alpha & 0\\ 0 & 4\alpha & -2\alpha\end{bmatrix}+\text{adj}(A).\ \text{If}\ \det(B)=66$$

then $\det(\text{adj}(\text{A}))$ equals :

Option 1:
289

Option 2:
361

Option 3:
441

Option 4:
529

Correct Answer:
441

Solution:

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$$A=\begin{bmatrix}\alpha & 1 & 2\\ 2 & 3 & 0\\ 0 & 4 & 5\end{bmatrix} \&|A|=5\alpha+6$$

Now $\text{adj } A=\begin{bmatrix}15 & 3 & -6\\ -10 & 5\alpha & 4\\ 8 & -4\alpha & 3\alpha-2\end{bmatrix}$

$$\therefore B=\begin{bmatrix}1 & 0 & 0\\ 0 & -5\alpha & 0\\ 0 & 4\alpha & -2\alpha\end{bmatrix}+\text{adj}(A)$$

$$B=\begin{bmatrix}16 & 3 & -6\\ -10 & 0 & 4\\ 8 & 0 & \alpha-2\end{bmatrix}$$

$\&|B|=6(5\alpha+6)=66$ (given in question)

$\Rightarrow \alpha=1$

$\Rightarrow |A|=21$

$\therefore \det(\text{adj } A)=|A|^{(n-1)^2}$

Here $n=3$

$=21^2$

$=441$

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Q. 5 Let $\alpha=3+4+8+9+13+14+\ldots\ldots$ upto 40 terms. If $(\tan\beta)^{\frac{\alpha}{1020}}$ is a root of the equation $x^2+x-2=0, \beta\in\left(0,\frac{\pi}{2}\right)$, then $\sin^2\beta+3\cos^2\beta$ is equal to :

Option 1:
2

Option 2:
$\frac{7}{4}$

Option 3:
$\frac{5}{2}$

Option 4:
$\frac{3}{2}$

Correct Answer:
2

Solution:

$a = (3 + 8 + 13 \ldots \text{ upto 20 terms }) + (4 + 9 + 14 +$
$\ldots \text{ upto 20 terms })$
$= \frac{20}{2}[6 + 19 \times 5] + \frac{20}{2}(8 + 19 \times 5)$
$= 2040$
$(\tan \beta)^{\frac{2040}{1020}} = \tan^2 \beta$
$x^2 + x - 2 = 0 \int_{-2}^{1}$
$\Rightarrow \tan^2 \beta = 1; \sin^2 \beta = \frac{1}{2} \& \cos^2 \beta = \frac{1}{2}$
$\sin^2 \beta + 3\cos^2 \beta = \frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2$

Q. 6 A candidate has to go to the examination centre to appear in an examination. The candidate uses only one means of transportation for the entire distance out of bus, scooter and car. The probabilities of the candidate going by bus, scooter and car, respectively, are $\frac{2}{5}$, $\frac{1}{5}$ and $\frac{2}{5}$. The probabilities that the candidate reaches late at the examination centre are $\frac{1}{5}$, $\frac{1}{3}$ and $\frac{1}{4}$ if the candidate uses bus, scooter and car, respectively. Given that the candidate reached late at the examination centre, the probability that the candidate travelled by bus is :

Option 1:
$\frac{11}{37}$

Option 2:
$\frac{12}{37}$

Option 3:
$\frac{13}{37}$

Option 4:
$\frac{14}{37}$

Correct Answer:
$\frac{12}{37}$

Solution:

$$\begin{array}{l|l} \mathrm{P}(\text{ Bus }) = \mathrm{P}(\text{ B}) = \frac{2}{5} & \mathrm{P}\left(\frac{\text{late}}{\mathrm{B}}\right) = \frac{1}{5} \\ \mathrm{P}(\text{ Scooter }) = \mathrm{P}(\text{ S}) = \frac{1}{5} & \mathrm{P}\left(\frac{\text{late}}{\mathrm{S}}\right) = \frac{1}{3} \\ \mathrm{P}(\text{ Car }) = \mathrm{P}(\mathrm{C}) = \frac{2}{5} & \mathrm{P}\left(\frac{\text{late}}{\mathrm{C}}\right) = \frac{1}{4} \end{array}$$

$P\left(\frac{B}{\text{late}}\right) = \frac{\frac{2}{5} \times \frac{1}{5}}{\frac{2}{5} \times \frac{1}{5} \times \frac{1}{5} \times \frac{1}{3} \times \frac{2}{5} \times \frac{1}{4}} = \frac{12}{37}$

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Q. 7 A set of four observations has mean 1 and variance 13. Another set of six observations has mean 2 and variance 1. Then, the variance of all these 10 observations is equal to :

Option 1:
5.96

Option 2:
6.14

Option 3:
6.04

Option 4:
6.24

Correct Answer:
6.04

Solution:
Given $\bar{x}=1, \sigma_1^2=13$

$\bar{y}=2, \sigma_2^2=1$

Combined variance $=\frac{n_1\sigma_1^2+n_2\sigma_2^2}{n_1+n_2}+\frac{n_1n_2}{(n_1+n_2)^2}(\bar{x}-\bar{y})^2$

$=\frac{4\times13+6\times1}{10}+\frac{4\times6}{10^2}(2-1)^2$

$\frac{58}{10}+\frac{24}{100}=\frac{580+24}{100}=\frac{604}{100}=6.04$

Q. 8 If

$$26\left(\frac{2^3}{3}\left({}^{13}\mathrm{C}_2\right)+\frac{2^5}{5}\left({}^{12}\mathrm{C}_4\right)+\frac{2^7}{7}\left({}^{12}\mathrm{C}_6\right)+\ldots.+\frac{2^{13}}{13}\left({}^{12}\mathrm{C}_{12}\right)\right)$$

$=3^{13}-\alpha$, then α is equal to :

Option 1:
45

Option 2:
48

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Option 3:
51

Option 4:
54

Correct Answer:
51

Solution:

$(1+x)^{12} = {}^{12}C_0 + {}^{12}C_1 x + {}^{12}C_2 x^2 + \ldots\ldots\ldots + {}^{12}C_{12} x^{12}$

integrating both sides

$\left(\frac{(1+x)^{13}}{13}\right)_{-2}^{2} = \left({}^{12}C_0 x + {}^{12}C_1 \frac{x^2}{2} + {}^{12}C_2 \frac{x^3}{3} + \ldots + {}^{12}C_{12} \frac{x^{13}}{13}\right)_{-2}^{2}$

$\frac{3^{13}}{13} + \frac{1}{13} = \left({}^{12}C_0 2 + {}^{12}C_1 \frac{2^2}{2} + {}^{12}C_2 \frac{2^3}{3} + \ldots + {}^{12}C_{12} \frac{2^{13}}{13}\right)$

$- \left(-{}^{12}C_0 \cdot 2 + {}^{12}C_1 \frac{2^2}{2} - {}^{12}C_2 \cdot \frac{2^3}{3} + \ldots + {}^{12}C_{12} \cdot \frac{2^{13}}{13}\right)$

$= 2\left(2 \cdot {}^{12}C_0 + {}^{12}C_2 \cdot \frac{2^3}{3} + {}^{12}C_4 \cdot \frac{2^5}{5} + \ldots + {}^{12}C_{12} \cdot \frac{2^{13}}{13}\right)$

$3^{13} + 1 = 26\left(2 + {}^{12}C_{12} \cdot \frac{2^3}{3} + {}^{12}C_4 \cdot \frac{2^5}{5} + \ldots + {}^{12}C_{12} \cdot \frac{2^{13}}{13}\right)$

$3^{13} - 51 = 26\left({}^{12}C_2 \cdot \frac{2^3}{3} + {}^{12}C_4 \cdot \frac{2^5}{5} + \ldots + {}^{12}C_{12} \cdot \frac{2^{13}}{13}\right)$

$\therefore \alpha = 51$

Q. 9 A person has three different bags and four different books. The number of ways, in which he can put these books in the bags so that no bag is empty, is

Option 1:
18

Option 2:
36

Option 3:
39

Option 4:
72

Correct Answer:
36

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Solution:

B_1	B_2	B_3
1	1	2

$$\text{No. of ways } = \frac{4!}{2!2!} \times 3! = 6 \times 6 = 36$$

Q. 10 If a straight line drawn through the point of intersection of the lines $4x + 3y - 1 = 0$ and $3x + 4y - 1 = 0$, meet the co-ordinate axes at the points P and Q , then the locus of the mid point of PQ is:

Option 1:
$x + y - 7 = 0$

Option 2:
$x + y - 14xy = 0$

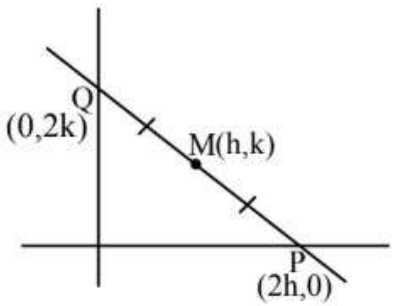
Option 3:
$2x + y + 14xy = 0$

Option 4:
$x + 2y - 14xy = 0$

Correct Answer:
$x + y - 14xy = 0$

Solution:
Point of intersection of the lines

$3x + 4y = 1$ and $4x + 3y = 1$ is $\left(\frac{1}{7}, \frac{1}{7}\right)$



Let the line be $\frac{x}{2h} + \frac{y}{2k} = 1$
Satisfy $\left(\frac{1}{7}, \frac{1}{7}\right)$

$$\frac{1}{14h} + \frac{1}{14k} = 1$$
$$\frac{1}{x} + \frac{1}{y} = 14$$

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Q. 11 Let O be the vertex of the parabola $y^2 = 4x$ and its chords OP and OQ are perpendicular to each other. If the locus of the mid-point of the line segment PQ is a conic C , then the length of its latus rectum is :

Option 1:
1

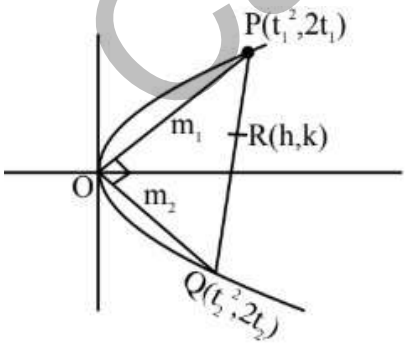
Option 2:
2

Option 3:
4

Option 4:
8

Correct Answer:
2

Solution:



$t_1 t_2 = -4$
$2h = t_1^2 + t_2^2 \quad 2k = 2(t_1 + t_2)$
$2h = k^2 - 2t_1 t_2$
$2h = k^2 + 8$
$k^2 = 2h - 8$
$= 2((h - 4)$
$\text{L.R.} = 2$

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Q. 12 Let $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, where inverse trigonometric functions take only the principal values.

Given below are two statements:

Statement I : $\cos(\alpha + \beta) > 0$

Statement II : $\cos(\alpha) < 0$

In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:

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$$\frac{1}{2} < \frac{6}{11} < \frac{1}{\sqrt{2}}$$

$$\sin^{-1}\left(\frac{1}{2}\right) < \sin^{-1}\left(\frac{6}{11}\right) < \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\frac{\pi}{6} < 3\sin^{-1}\left(\frac{6}{11}\right) < \frac{\pi}{4}$$

$$\frac{\pi}{2} < \alpha < \frac{3\pi}{4} \quad \therefore \cos\alpha < 0$$

$$0 < \frac{4}{9} < \frac{1}{2}$$

$$\frac{\pi}{3} < \cos^{-1}\left(\frac{4}{9}\right) < \frac{\pi}{2}$$

$$\pi < 3\cos^{-1}\left(\frac{4}{9}\right) < \frac{3\pi}{2}$$

$$\pi < \beta < \frac{3\pi}{2}$$

$$\text{Now } \frac{3\pi}{2} < \alpha + \beta < \frac{9\pi}{4}$$

$$\therefore \cos(\alpha+\beta) > 0$$

Q. 13 For the function $f(x) = e^{\sin|x|} - |x|, x \in \mathbf{R}$, consider the following statements:

Statement I : f is differentiable for all $x \in \mathbf{R}$.

Statement II : f is increasing in $\left(-\pi, -\frac{\pi}{2}\right)$.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

Solution:
$f(x) = e^{|\sin x|} - |x|$ is

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non-differentiable at $x = \pi$
and for $x \in \left(-\pi, \frac{-\pi}{2}\right)$

$f(x) = e^{-\sin x} + x$
$f'(x) = -e^{-\sin x}\cos x + 1$
for $x \in \left(-\pi, \frac{-\pi}{2}\right)$ $\cos x < 0$
$\therefore f'(x) > 0$
$\therefore f(x)$ is increasing

Q. 14 Let $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}, \vec{b} = 10\hat{i} + 2\hat{j} - \hat{k}$ and a vector $\vec{c}$ be such that $2(\vec{a} \times \vec{b}) + 3(\vec{b} \times \vec{c}) = \overrightarrow{0}$. If $\vec{a} \cdot \vec{c} = 15$, then $\overrightarrow{c} \cdot (\hat{i} + \hat{j} - 3\hat{k})$ is equal to :

Option 1:
-6

Option 2:
-5

Option 3:
-4

Option 4:
-3

Correct Answer:
-5

Solution:

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$2(\vec{a}\times\vec{b})-3(\vec{c}\times\vec{b})=\overrightarrow{0}$

$(2\vec{a}-3\vec{c})\times\vec{b}=\overrightarrow{0}$

$\vec{b}\|(2\vec{a}-3\vec{c})$

$2\vec{a}-3\vec{c}=\lambda\vec{b}$

$\vec{c}=\dfrac{2\vec{a}-\lambda\vec{b}}{3}$

$\vec{a}\cdot\vec{c}=15$

$\left(\dfrac{2\vec{a}-\lambda\vec{b}}{3}\right)\cdot\vec{a}=15$

$\dfrac{2|a|^2-\lambda\vec{b}\cdot\vec{a}}{3}=15$

$2(26)-\lambda(40-2-3)=45$

$\lambda=\dfrac{1}{5}$

$\Rightarrow\vec{c}\cdot(\hat{i}+\hat{j}-3\hat{k})=\dfrac{\left(2(4\hat{i}-\hat{j}+3k)-\frac{1}{5}(10\hat{i}+2\hat{j}-k)\cdot(\hat{i}+\hat{j}-3\hat{k})\right)}{3}$

$=\dfrac{2(4-1-9)-\frac{1}{5}(10+2+3)}{3}=-5$

Q. 15 Let the foot of perpendicular from the point $(\lambda,2,3)$ on the line $\frac{x-4}{1}=\frac{y-9}{2}=\frac{z-5}{1}$ be the point $(1,\mu,2)$. Then the distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+4}{6}$ and $\frac{x-\lambda}{2}=\frac{y-\mu}{3}=\frac{z+5}{6}$ is equal to

Option 1:
$\frac{12}{7}$

Option 2:
$\frac{\sqrt{145}}{7}$

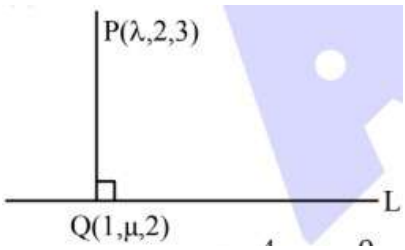
Option 3:
$\frac{\sqrt{146}}{7}$

Option 4:
$\frac{\sqrt{143}}{7}$

Correct Answer:
$\frac{\sqrt{146}}{7}$

Solution:

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$\frac{x-4}{1}=\frac{y-9}{2}=\frac{z-5}{1}$

$Q(1,\mu,2)$ satisfies the line L

$\therefore \frac{1-4}{1}=\frac{\mu-9}{2}=\frac{2-5}{1}$

$\Rightarrow -3=\frac{\mu-9}{2}=-3 \quad \Rightarrow \mu=3$

D.R. of $\mathrm{PQ}: \lambda-1, 2-\mu, 1$

Since $\mathrm{PQ} \perp \mathrm{L} \Rightarrow (\lambda-1)\cdot 1+2(2-\mu)+1.1=0$

$\Rightarrow \lambda-2\mu+4=0 \quad \therefore \lambda=2$

Now lines : $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z+4}{6}$ \&

$\frac{x-2}{2}=\frac{y-3}{3}=\frac{z+5}{6}$

i.e. $\vec{r}=(\hat{i}+2\hat{j}-4\hat{k})+t_1(2\hat{i}+3\hat{j}+6\hat{k})$

$\&\overrightarrow{\mathrm{r}}=(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}-5\hat{\mathrm{k}})+\mathrm{t}_2(2\hat{\mathrm{i}}+3\hat{\mathrm{j}}+6\hat{\mathrm{k}})$

Distance

$$=\frac{|((\hat{i}+2\hat{j}-4\hat{k})-(2\hat{i}+3\hat{j}-5\hat{k}))\times(2\hat{i}+3\hat{j}+6\hat{k})|}{|2\hat{i}+3\hat{j}+6\hat{k}|}$$

$$=\frac{|-9\hat{i}+8\hat{j}-\hat{k}|}{7}=\frac{\sqrt{146}}{7}$$

Q. 16 The value of the integral $\int_0^2 \frac{\sqrt{x(x^2+x+1)}}{(\sqrt{x+1})\left(\sqrt{x^4+x^2+1}\right)}dx$ is equal to :

Option 1:
$\frac{1}{3}\log_e(3-2\sqrt{2})$

Option 2:
$\frac{2}{3}\log_e(4+\sqrt{2})$

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Option 3:
$\frac{2}{3}\log_e(3+2\sqrt{2})$

Option 4:
$\frac{1}{3}\log_e(1+6\sqrt{2})$

Correct Answer:
$\frac{2}{3}\log_e(3+2\sqrt{2})$

Solution:

$$I=\int_0^2\sqrt{\frac{x\left(x^2+x+1\right)}{(x+1)\left(x^2-x+1\right)\left(x^2+x+1\right)}}dx$$

$$=\int_0^2\sqrt{\frac{x}{x^3+1}}dx$$

$$\left(x^{3/2}=t\Rightarrow\frac{3}{2}x^{1/2}dx=dt\right)$$

$$=\frac{2}{3}\int_0^{2^{3/2}}\frac{dt}{\sqrt{t^2+1}}$$

$$=\frac{2}{3}\left(\ln\left(t+\sqrt{t^2+1}\right)\right)_0^{2^{3/2}}$$

$$=\frac{2}{3}\ln\left(2^{3/2}+3\right)$$

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Q. 17 Let $y=y(x)$ be the solution of the differential equation

$$x\sqrt{1-x^2}dy+\left(y\sqrt{1-x^2}-x\cos^{-1}x\right)dx=0,$$

$x\in(0,1),\lim_{x\to1^-}y(x)=1$. Then $y\left(\frac{1}{2}\right)$ equals :

Option 1:
$3-\frac{\pi}{\sqrt{3}}$

Option 2:
$4-\sqrt{3}\pi$

Option 3:
$4-\frac{2\pi}{\sqrt{3}}$

Option 4:
$3-\frac{\pi}{2\sqrt{3}}$

Correct Answer:
$3-\frac{\pi}{\sqrt{3}}$

Solution:

$x\sqrt{1-x^2}dy+\left(y\sqrt{1-x^2}-x\cos^{-1}x\right)dx=0$

$\Rightarrow \frac{dy}{dx}+\frac{y}{x}=\frac{\cos^{-1}x}{\sqrt{1-x^2}}$

it is L.D.E.

$\therefore I.F.=e^{\int\frac{1}{x}dx}=e^{\ell\ln x}=x$

$\therefore$ solution is $y\cdot x=\int\frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx+C$

$y\cdot x=I_1+C$

$I_1=\int\frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx$

Let $\cos^{-1}\mathrm{x}=\mathrm{t}\Rightarrow \mathrm{x}=\mathrm{cost}$

$-\frac{1}{\sqrt{1-x^2}}dx=dt$

$I_1=\int -t\cdot\cot dt$

$=-(t\sin t+\cos t)$

$I_1=-\left(\sqrt{1-x^2}\cdot\cos^{-1}x+x\right)$

$\therefore y\cdot x=-\left(\sqrt{1-x^2}\cdot\cos^{-1}x+x\right)+C$

$\because \lim\limits_{x\to1^-}y(x)=1$

$\therefore 1=-(0+1)+\mathrm{C}$

$\Rightarrow \mathrm{C}=2$

$\therefore \mathrm{y}\cdot\mathrm{x}=-\sqrt{1-\mathrm{x}^2}\cdot\cos^{-1}\mathrm{x}-\mathrm{x}+2$

$\left.\text{put } \mathrm{x}=\frac{1}{2}\right)$

$\mathrm{y}\left(\frac{1}{2}\right)\times\frac{1}{2}=-\frac{\sqrt{3}}{2}\times\frac{\pi}{3}-\frac{1}{2}+2$

$\Rightarrow \mathrm{y}\left(\frac{1}{2}\right)=3-\frac{\pi}{\sqrt{3}}$

Q. 18 Let $\mathrm{f}:(1,\infty)\to\mathbf{R}$ be function defined as $f(x)=\frac{x-1}{x+1}$. Let $f^{i+1}(x)=f\left(f^i(x)\right)$, $\mathrm{i}=1,2,\ldots,25$, where $\mathrm{f}'(\mathrm{x})=(\mathrm{x})$. If $\mathrm{g}(\mathrm{x})+\mathrm{f}^{26}(\mathrm{x})=0$, $x\in(1,\infty)$, then the area of the region bounded by the curves $\mathrm{y}=\mathrm{g}(\mathrm{x}),2\mathrm{y}=2\mathrm{x}-3,\mathrm{y}=0$ and $\mathrm{x}=4$ is:

Option 1:

$\frac{1}{8}+\log_e 2$

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Option 2:
$\frac{1}{4}+\log_e 2$

Option 3:
$\frac{5}{6}+3\log_e 2$

Option 4:
$\frac{5}{6}+\log_e 2$

Correct Answer:
$\frac{1}{8}+\log_e 2$

Solution:

$$f^2(x)=f(f(x))=\frac{\frac{x-1}{x+1}-1}{\frac{x-1}{x+1}+1}=\frac{-1}{x}$$

$$f^3(x)=f\left(f^2(x)\right)=f\left(\frac{-1}{x}\right)=\frac{1+x}{1-x}$$

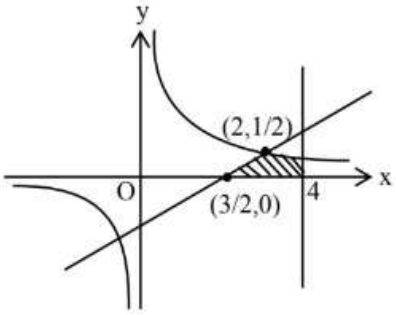
$$f^4(x)=f\left(f^3(x)\right)=f\left(f\left(\frac{-1}{x}\right)\right)=x$$

$$f^6(x)=f\left(f^4(x)\right)=f(x)$$

$$f^6(x)=f\left(f^6(x)\right)=f(f(x))=\frac{-1}{x}$$

$$f^{26}(x)=\frac{-1}{x}$$

$$g(x)=\frac{1}{x}$$



$$x-3/2=1/x\Rightarrow 2x^2-3x-2=0$$

$$\Rightarrow x=-1/2,2$$

$$\text{Req. area }=\int_2^4\frac{1}{x}dx+\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$$

$$=\frac{1}{8}+(\ell\text{nx})_2^4=\frac{1}{8}+\ell\text{n}2$$

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Q. 19 Let $f(x)=\begin{cases}\frac{1}{3} & , \quad x \leq \pi/2 \\ \frac{b(1-\sin x)}{(\pi-2x)^2} & , \quad x>\pi/2\end{cases}$, If f is continuous at $\mathrm{x}=\pi/2$, then the value of $\int_0^{3\,\mathrm{b}-6}\left|\mathrm{x}^2+2\mathrm{x}-3\right|\mathrm{dx}$ is :

Option 1:
5

Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
4

Solution:

$\lim_{x\to\frac{\pi}{2}}\frac{b\left(1-\cos\left(\frac{\pi}{2}-x\right)\right)}{4\left(\frac{\pi}{2}-x\right)^2}=\frac{1}{3}\Rightarrow b=\frac{8}{3}$

$\int_0^2\left|\mathrm{x}^2+2\mathrm{x}-3\right|\mathrm{dx}=\int_0^2|(\mathrm{x}+3)(\mathrm{x}-1)|\mathrm{dx}$

put $\mathrm{x}-1=\mathrm{t}$

$\int_{-1}^1|(t+4)t|dt=\int_{-1}^0\left(-t^2-4t\right)dt+\int_0^1\left(t^2+4t\right)dt$

$\left(\frac{-t^3}{3}-2t^2\right)_{-1}^0+\left(\frac{t^3}{3}+2t^2\right)_0^1=4$

Q. 20 Let $\frac{x^2}{f(a^2+7a+3)}+\frac{y^2}{f(3a+15)}=1$ represent an ellipse with major axis along y-axis, where f is a strictly decreasing positive function on $\mathbf{R}$. If the set of all possible values of a is $\mathbf{R}-[\alpha,\beta]$, then $\alpha^2+\beta^2$ is equal to :

Option 1:
28

Option 2:
40

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Option 3:
61

Option 4:
24

Correct Answer:
40

Solution:
Given ellipse is vertical

$\therefore \mathrm{f}(3\mathrm{a}+15)>\mathrm{f}\left(\mathrm{a}^2+7\mathrm{a}+3\right)$
$\because \mathrm{f}(\mathrm{x})$ is decreasing $\forall \mathrm{x} \in \mathrm{R}$
$\Rightarrow 3\mathrm{a}+15<\mathrm{a}^2+7\mathrm{a}+3$
$\Rightarrow \mathrm{a}^2+4\mathrm{a}-12>0$
$\Rightarrow (\mathrm{a}+2)^2>16$
$\Rightarrow |(\mathrm{a}+2)|>4$
$\Rightarrow \mathrm{a}>2$ or $\mathrm{a}<-6$
$\Rightarrow \mathrm{a} \in \mathrm{R}-[-6,2]$
$\therefore \alpha=-6, \beta=2$
$\therefore \alpha^2+\beta^2=36+4=40$

Q. 21 The sum of squares of all real solution of the equation $\log_{(x+1)}\left(2x^2+5x+3\right)=4-\log_{(2x+3)}\left(x^2+2x+1\right)$ is equal to _____ .

Correct Answer:
2

Solution:
$\log_{(x+1)}(2x+3)(x+1)+\log_{(2x+3)}(x+1)^2=4$
$1+\log_{(x+1)}(2x+3)+2\log_{(2x+3)}(x+1)=4$
$\log_{x+1}(2x+3)=t \quad t+\dfrac{2}{t}=3$
$t^2-3t+2=0 \Rightarrow t=1,2$
for $t=1 \Rightarrow \log_{(x+1)}(2x+3)=1 \Rightarrow 2x+3=x+1$
$\Rightarrow x=-2$ (rejected)
for $t=2 \Rightarrow 2x+3=x^2+2x+1 \Rightarrow x^2=2$
$\Rightarrow x=\pm\sqrt{2}$
rejecting $x=-\sqrt{2}$, we get $x=\sqrt{2}$

Q. 22 If $\int_{\pi/6}^{\pi/4}\left(\cot\left(x-\frac{\pi}{3}\right)\cot\left(x+\frac{\pi}{3}\right)+1\right)dx=\alpha\log_e(\sqrt{3}-1)$, then $9\alpha^2$ is equal to _____ .

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Correct Answer:
12

Solution:

$$\int_{\pi/6}^{\pi/4}\left(\left(\cot\left(x-\frac{\pi}{3}\right)\cot\left(x+\frac{\pi}{3}\right)+1\right)dx$$

$$\text{using } \cot(A-B)=\frac{\cot B\cot A+1}{\cot B-\cot A}$$

$$\Rightarrow\int_{\pi/6}^{\pi/4}-\frac{1}{\sqrt{3}}\left[\cot\left(x-\frac{\pi}{3}\right)-\cot\left(x+\frac{\pi}{3}\right)\right]dx$$

$$\Rightarrow\frac{-1}{\sqrt{3}}\left[\ln\left|\frac{\sin\left(x-\frac{\pi}{3}\right)}{\sin\left(x+\frac{\pi}{3}\right)}\right|\right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$$

$$=\frac{-1}{\sqrt{3}}\left[\ln\left|\left(\frac{\sin 15^\circ}{\sin 105^\circ}\right)\right|-\ell n\left|\frac{\sin 30^\circ}{\sin 90^\circ}\right|\right]$$

$$=\frac{-1}{\sqrt{3}}\left[\ln\left(\tan 15^\circ\right)-\ell n\left(\frac{1}{2}\right)\right.$$

$$=\frac{-1}{\sqrt{3}}[\ln(2-\sqrt{3})+\ell n2]$$

$$\Rightarrow\frac{-1}{\sqrt{3}}\ln(4-2\sqrt{3})=\frac{-2}{\sqrt{3}}\ln(\sqrt{3}-1)$$

$$\therefore\alpha=\frac{-2}{\sqrt{3}}\therefore 9\alpha^2=12$$

Q. 23 Let a line L_1 pass through the origin and be perpendicular to the lines
$L_2:\vec{r}=(3+t)\hat{i}+(2t-1)\hat{j}+(2t+4)\hat{k}$ and
$L_3:\overrightarrow{\mathbf{r}}=(3+2s)\hat{i}+(3+2s)\hat{j}+(2+s)\hat{k}, t, s\in\mathbf{R}$. If $(a,b,c), a\in\mathbf{Z}$, is the point on L_3 at a distance of $\sqrt{17}$ from the point of intersection of L_1 and L_2, then $(\mathrm{a}+\mathrm{b}+\mathrm{c})^2$ is equal to _____ .

Correct Answer:
4

Solution:

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$L_2 : \frac{x-3}{1} = \frac{y+1}{2} = \frac{z-4}{2} = t$

$L_5 : \frac{x-3}{2} = \frac{y-3}{2} = \frac{z-2}{1} = s$

vector $\perp^{ar}$ to $L_2 \& L_3$

$\Rightarrow \begin{vmatrix} i & j & k \\ 1 & 2 & 2 \\ 2 & 2 & 1 \end{vmatrix} = \hat{i}(-2) - \hat{j}(-3) + \hat{k}(-2)$

$\Rightarrow -2\hat{i} + 3\hat{j} - 2\hat{k}$

$\Rightarrow -(2\hat{i} - 3\hat{j} + 2\hat{k})$

Now, $\mathrm{L}_1 : \frac{\mathrm{x}}{2} = \frac{\mathrm{y}}{-3} = \frac{\mathrm{z}}{2} = \ell$ intersection of L_1 \& L_2

$\Rightarrow \mathrm{t}+3 = 2\ell, 2\mathrm{t}-1 = -3\ell, 2\mathrm{t}+4 = 2\ell$

$\mathrm{t}+3 = 2\mathrm{t}+4$

$\mathrm{t} = -1$

$\Rightarrow \mathrm{P} \equiv (2, -3, 2)$

A point on L_3 is $\mathrm{Q}(2\,\mathrm{s}+3, 2\,\mathrm{s}+3,\ \mathrm{s}+2)$

$\Rightarrow PQ^2 = (2s+1)^2 + (2s+6)^2 + s^2 = 17$

$\Rightarrow 9s^2 + 28s + 20 = 0$

$\Rightarrow (9s+10)(s+2) = 0 \Rightarrow s = -2$

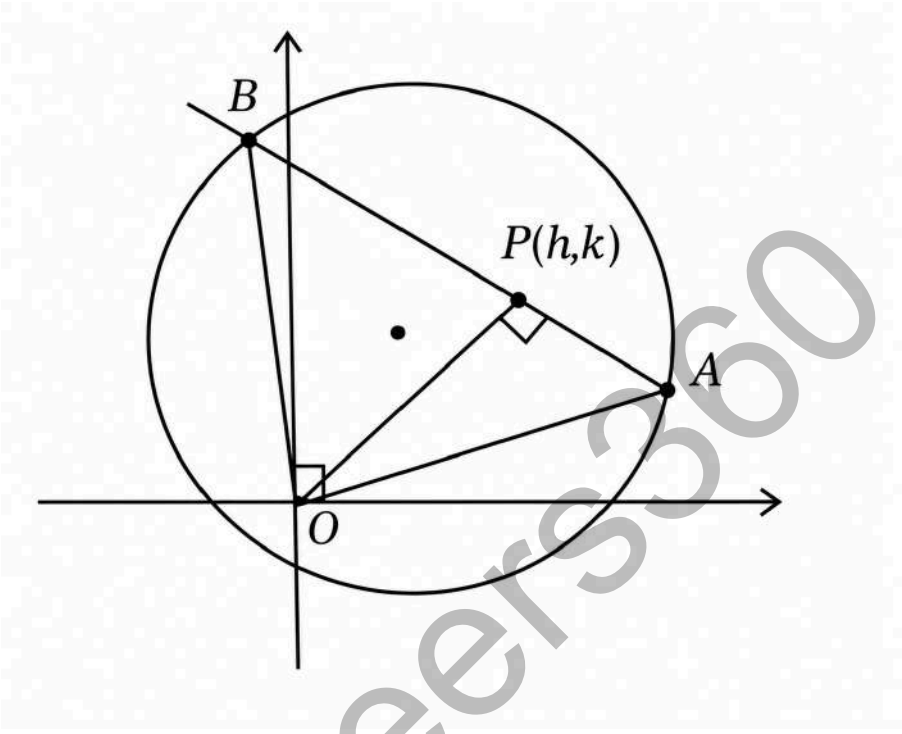
$\therefore Q \equiv (-1, -1, 0) \equiv (a, b, c) \Rightarrow (a+b+c)^2 = 4$

Q. 24 Consider the circle $C : x^2 + y^2 - 6x - 8y - 11 = 0$. Let a variable chord AB of the circle C subtend a right angle at the origin. If the locus of the foot of the perpendicular drawn from the origin on the chord AB is the circle $x^2 + y^2 - \alpha x - \beta y - \gamma = 0$, then $\alpha + \beta + 2\gamma$ is equal to _____ .

Correct Answer:
18

Solution:

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$C : x^2 + y^2 - 6x - 8y - 11 = 0$

$\mathrm{E^{qn}}$ of chord AB

$y - k = -\frac{h}{k}(x - h)$

$hx + ky = h^2 + k^2$

Homogenising the $\mathrm{eq^n}$ of the circle

$\mathrm{x^2 + y^2 - 6x\left(\frac{hx+ky}{h^2+k^2}\right) - 8y\left(\frac{hx+ky}{h^2+k^2}\right) - 11\left(\frac{hx+ky}{h^2+k^2}\right)^2 = 0}$

coeff. of x^2+ coeff of $y^2 = 0$

$1 - \frac{6h}{h^2+k^2} - 11 \cdot \frac{h^2}{(h^2+k^2)^2} + 1 - \frac{8k}{h^2+k^2} - 11 \cdot \frac{k^2}{(h^2+k^2)^2} = 0$

$2\left(h^2+k^2\right) - 6h - 8k - 11 = 0$

$x^2 + y^2 - 3x - 4y - \frac{11}{2} = 0$

$\therefore \alpha + \beta + 2\gamma = 3 + 4 + 11 = 18$

Q. 25 Let f be a polynomial function such that $\log_2(\mathrm{f(x)}) = \left(\log_2\left(2 + \frac{2}{3} + \frac{2}{9} + \ldots\ldots\right)\right).$ $\log_3\left(1 + \frac{f(x)}{f(1/x)}\right), x > 0$ and $f(6) = 37$. Then $\sum_{n=1}^{10} f(n)$ is equal to _____ .

Correct Answer:
395

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Solution:

$$\log_2 f(x) = \log_2 \left(\frac{2}{1-\frac{1}{3}} \right) \cdot \log_3 \left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right)$$

$$\log_2 f(x) = \log_2 3 \cdot \log_3 \left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)} \right)$$

$$\Rightarrow f(x) = 1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}$$

$$\Rightarrow f(x) f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

$$\Rightarrow f(x) = 1 \pm x^n \quad \Rightarrow f(6) = 37$$

$$\Rightarrow 1 \pm 6^n = 37 \quad \Rightarrow 6^n = 36$$

$$\Rightarrow n = 2$$

$$\therefore f(x) = 1 + x^2$$

$$\sum_{n=1}^{10} \left(1 + n^2\right) = 10 + \frac{10.11.21}{6} = 395$$

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